



Business Finance

FOR STUDENTS AT RMIT

M. Gangemi



Business Finance

Michael Gangemi

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First printing, February 2015

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ISBN: 978-0-9942767-2-8



Contents

1	Introduction to Business Finance	9
1.1	What is Business Finance?	9
1.2	Financial Market	9
1.3	Corporate Decisions and Objectives	10
1.3.1	The Corporate Objective	10
1.3.2	Measuring Owners' Wealth	10
1.3.3	Maximising Market Value	11
1.4	Key Decisions In Business Finance	11
1.4.1	The Dividend Decision	11
1.4.2	The Investment Decision	11
1.4.3	The Financing Decision	11
1.5	Factors in Financial Decision-Making	12
1.5.1	Cash-Flow Analysis	12
1.5.2	Cash-Flow Timing	14
1.5.3	Cash-Flow Risk	14
1.6	Valuation of a Firm	14
1.6.1	An Alternative Approach to Valuation	15
1.7	Finance vs. Accounting	15
2	Financial Mathematics - Introduction	17
2.1	Time Value of Money	17

2.2	Simple Interest	18
2.2.1	Compounding and Discounting	19
2.3	Compound Interest	19
2.3.1	Future Value of a Single Sum	20
2.3.2	Adjusting “r” and “n”	21
2.4	Present Value of a Single Sum	23
2.5	Unknown “n” and “r”	26
2.6	Multiple Uneven Cash-Flows	27
2.7	Revision Problems	29
3	Financial Mathematics - Annuities	31
3.1	Annuities	31
3.1.1	Types of Annuities	31
3.2	Ordinary Annuities	33
3.2.1	Future Value of an Ordinary Annuity	33
3.2.2	Present Value of an Ordinary Annuity	34
3.2.3	Unknown PMT for an Ordinary Annuity	35
3.3	Annuity Due	36
3.3.1	Future Value of an Annuity Due	37
3.3.2	Present Value of an Annuity Due	38
3.3.3	Finding PMT for an Annuity Due	39
3.4	Deferred Annuity- Present Value	40
3.5	Perpetuity– Present Value	41
3.6	Growing Perpetuity– Present Value	42
3.7	Effective Interest Rates	43
3.8	Loans	45
3.8.1	Amortised Loan – Calculations	45
3.9	Revision Problems	50
3.10	Financial Calculator Steps and Keystrokes	53
4	Bond Valuation	57
4.1	Debt Financing	57
4.1.1	Government Financing	58
4.1.2	Cost of Debt	58

4.2	Bond	60
4.2.1	Bond Pricing Model	62
4.2.2	Cost of Debt and Yield To Maturity	62
4.2.3	Bond Price Sensitivity to YTM	63
4.3	Interest Rate Risk	65
4.3.1	Yield Curve	66
4.3.2	Explaining the Shape of the Yield Curve	67
4.4	Revision Problems	73
5	Share Valuation	75
5.1	Equity Financing	75
5.1.1	Cost of Equity	76
5.2	Share Valuation Models	76
5.3	Dividend Valuation Model	76
5.3.1	Constant Growth Stock	77
5.3.2	Dividend Yield, Capital Gains Yield and Total Return	78
5.3.3	Zero Growth in Dividends	80
5.3.4	Required Rate of Return	80
5.3.5	Negative Dividend Growth	80
5.3.6	Modified Constant Growth	81
5.3.7	Short-Term vs. Long-Term Performance	84
5.3.8	Sensitivity to the Dividend Growth Rate	84
5.3.9	Sensitivity to Required Rate of Return	84
5.4	Price/Earnings Model of Share Valuation	85
5.5	Revision Problems	89
6	Capital Budgeting	91
6.1	Capital Budgeting	91
6.2	Methods of Project Evaluation	91
6.2.1	Non-Discounted Cash-Flow Analysis	92
6.2.2	Discounted Cash-Flow Analysis	96
6.3	Detailed NPV Analysis	106
6.3.1	NPV Investment Evaluation Process	106
6.3.2	Incremental Cash-Flows Analysis	111
6.4	Evaluating Projects with Different Lifespans	118
6.5	Depreciation, Inflation & Effective Write-off	120

6.6	Capital Rationing	122
6.6.1	Profitability Index (Benefit-Cost Ratio)	122
6.7	Revision Problems	124
7	Risk and Return	129
7.1	Rate of Return	129
7.1.1	Historical Return	129
7.1.2	The Expected Return	130
7.2	Measuring Risk	131
7.2.1	The Risk-Free Asset	131
7.2.2	Company Risk	131
7.2.3	Distribution of Returns	132
7.2.4	Variance and Standard Deviation	133
7.2.5	Coefficient of Variation	134
7.3	Portfolio Expected Return	135
7.4	Portfolio Diversification	136
7.4.1	Covariance and Correlation Coefficient	137
7.5	Portfolio Variance and Standard Deviation	138
7.6	Portfolio with more than two assets	139
7.7	Opportunity Set and Efficient Frontier	140
7.7.1	Rational Investing	141
7.7.2	Risk-Return Preference	142
7.8	Revision Problems	144
8	Asset Pricing Models	147
8.1	Capital Market Line	147
8.1.1	Borrowing and Lending at the Risk-Free Rate	149
8.2	Capital Asset Pricing Model	152
8.2.1	Systematic vs Non-Systematic Risk	152
8.2.2	Model Setup	153
8.2.3	Security Market Line	155
8.3	Calculation of Systematic Risk	155
8.4	CAPM and Equilibrium	157
8.5	Efficient Market Hypothesis	158
8.5.1	Types of Market Efficiency (EMH)	158
8.5.2	Informational Efficiency	159

8.5.3	Requirements for Informational Efficiency	159
8.5.4	EMH and Information Sets	160
8.5.5	Market Anomalies – ‘How To Beat The Market’	161
8.6	Revision Problems	164
9	Weighted Average Cost of Capital	167
9.1	Introduction	167
9.2	Capital and the Cost of Capital	167
9.3	Simple formulation of WACC	169
9.3.1	Interest Payments and Tax	170
9.3.2	Market Value of the Firm	170
9.3.3	Cost of Ordinary Shares	171
9.3.4	Cost of Preference Shares	171
9.3.5	Cost of Debt	171
9.4	Caveats in Use of the WACC	172
9.5	Detailed WACC Problem	173
9.6	Revision Problems	179
10	Capital Structure	181
10.1	Capital Structure	181
10.2	Optimal Capital Structure	181
10.2.1	Debt and capital structure	182
10.3	Capital Structure, WACC, and the Firm’s Value	186
10.4	Traditional Approach to Capital Structure	187
10.5	Modigliani and Miller’s Approach	189
10.5.1	M&M Proposition 1	190
10.5.2	M&M Proposition 2	190
10.5.3	Debt and Risk	192
10.5.4	M&M Proposition 3	193
10.6	Introducing Market Imperfections	193
10.7	All Debt Capital Structures	193
10.7.1	Financial Distress Costs	193
10.7.2	Conflict of Interest Costs	194
10.8	Capital Structures of Australian Firms	194
10.9	Revision Problems	196

What is Business Finance?

Financial Market

Corporate Decisions and Objectives

- The Corporate Objective
- Measuring Owners' Wealth
- Maximising Market Value

Key Decisions In Business Finance

- The Dividend Decision
- The Investment Decision
- The Financing Decision

Factors in Financial Decision-Making

- Cash-Flow Analysis
- Cash-Flow Timing
- Cash-Flow Risk

Valuation of a Firm

- An Alternative Approach to Valuation

Finance vs. Accounting

1. Introduction to Business Finance

In this chapter you will learn about the nature of business and corporate finance; the nature of financial markets; the corporate objective; corporate financial decisions; the critical factors in financial decision-making; valuation of a firm and real and financial assets; and some key differences between finance and accounting.

1.1 What is Business Finance?

Business finance is also referred to as corporate finance and is a body of knowledge that focuses on explaining and interpreting financial markets. It provides an analytical framework to guide managers of firms and to assist them to evaluate corporate financial decisions. The two key elements of the definition of corporate finance are that it focuses on financial markets and that it assists in addressing corporate financial decisions.

1.2 Financial Market

The first of the above two key elements is to define what a financial market is. A financial market is a medium for the issue and exchange of financial assets. The Australian financial market includes

- the “big four” banks and all other banks;
- building societies, as well as savings and loans operating in Australia;
- the Australian Stock Exchange (ASX); and
- the Australian government and corporate bond markets.

The Financial market brings together the buyers and sellers of debt and equity securities and is made up of the money market, which involves the trading of short-term debt securities, i.e. debt securities that mature (i.e. have to be repaid) in 12 months or less, and the capital market, which involves the trading of long-term debt securities, i.e. debt securities that mature in a period of greater than 12 months, and equity securities, i.e. shares. In turn, the money market and the capital market are each made up of a primary market, which involves the original sale of securities (IPOs) to raise finance, and a secondary market, which involves the continual buying and selling of already issued securities by investors for profit or loss.

Highlight 1.1 — Initial Public Offering. An initial public offering (IPO), also referred to as a public offering, is the first sale of a security by a company or organisation to the public in order to raise finance. In an IPO the issuer obtains assistance from an underwriting firm who will help determine what type of security to issue, for example, ordinary shares or preference shares, the initial listing price (or the par value), and the best time to offer the securities on the market.

1.3 Corporate Decisions and Objectives

The second of the two key elements is to address the corporate financial decisions. However, before we address corporate financial decisions we must first address what is the main corporate objective.

1.3.1 The Corporate Objective

What do you think is the main goal or objective of a company? Normally, the main objective of most companies, whether they are a large, multi-billion dollar corporation listed on the stock exchange or a small, mum and dad local store, is to maximise profits. Why else would you go into business other than to maximise your profit or income from the business? However, to say that maximisation of profit is the main objective of a firm from a corporate finance perspective is actually incorrect.

From a business or corporate finance perspective, the main objective of a firm is actually to maximise the market value of the company or maximise shareholders' wealth. This means that management of the company must use the scarce resources (land, labour and capital) of the company in the most efficient manner possible and should make decisions that maximise the wealth of the company's owners, which is synonymous with increasing the value of the firm.

1.3.2 Measuring Owners' Wealth

The wealth of the owners of a corporation listed on the stock exchange, i.e. of a publicly-listed company, is measured by the market capitalisation of the securities issued by the company. So, for shares, this is the total market value of all of the shares of a company on issue.

Business Snapshot 1.1 BHP Billiton is an Anglo-Australian multinational mining, metals and petroleum company headquartered in Melbourne, Australia. It is the world's largest mining company measured by 2013 revenues.

The Australia-registered BHP Billiton Limited, which has equal financial share in the company, has a primary listing on the Australian Securities Exchange and is the largest company in Australia measured by market capitalisation. The UK-registered BHP Billiton Plc has a primary listing on the London Stock Exchange and is a constituent of the FTSE 100 Index.

Number of shares on issue = 5.348billion

Sale price (as at 24 February 2015) = \$33.06

Market capitalisation of ordinary shares = 5.348 billion shares $\times$ \$33.06
= \$176.80.00 billion

1.3.3 Maximising Market Value

Using the standard economic model, we can explain why it is that the management of a company should aim to maximise the market value of the company and, hence, the wealth of the owners of the company.

The *standard economic model* (SEM) states that individuals are utility (satisfaction or happiness) maximisers, that utility is a function of consumption, that consumption is a function of wealth; thus, individuals will wish to maximise their wealth. In general, individuals hold part of their wealth in the form of shares in companies, and that the market value of a company is represented by its share price multiplied by the number of shares on issue. Therefore, the greater the market value of a company's shares, the greater the company's market value, and the greater will be the wealth of its shareholders, and, therefore, the greater will be their utility.

1.4 Key Decisions In Business Finance

Now, we are better prepared to address the key decisions in corporate finance. The three key decisions faced by corporate finance managers in achieving the corporate objective are the investment decision, the financing decision, and the dividend decision. In this textbook, we will address the first two of the key decision areas, namely, the investment decision and the financing decision. However, it is worth, very briefly, discussing the third decision area, which is the dividend decision.

1.4.1 The Dividend Decision

The *dividend decision* of a company relates to the form in which returns generated by the company are passed on to the shareholders of the company. In making the dividend decision senior management of the company will decide on whether and how much of the after-tax profit and/or retained earnings of the company will be passed on to shareholders in the form of dividends.

1.4.2 The Investment Decision

The *investment decision* of a corporation relates to the ways in which the capital or funds of the company will be used in productive activities, with the main objective being to generate a return for investors by maximising the market value of the company. The investment decisions of a firm are dealt with under the heading 'Capital Budgeting' or 'Project Evaluation' and these topics will be discussed in more detail in Chapter 6.

The Investment decision of a company deals with the evaluation of investment opportunities in real assets, in terms of the size of future cash flows to come from the real assets, the timing of these future cash flows, and the risks associated with these future cash flows. The question to be asked, in terms of the investment decision, is which real assets should the firm invest in in order to maximise its market value. In terms of accounting, these investment activities are represented by the asset side of the balance sheet, i.e. the left-hand side.

1.4.3 The Financing Decision

The *financing decision* of a corporation relates to the mix of funding obtained from capital markets in terms of the proportional holdings of debt and equity. The financing decisions of a firm are dealt with under the heading *Capital Structure* and are summarised by the company's debt-equity ratio, and will be discussed in more detail in Chapter 10.

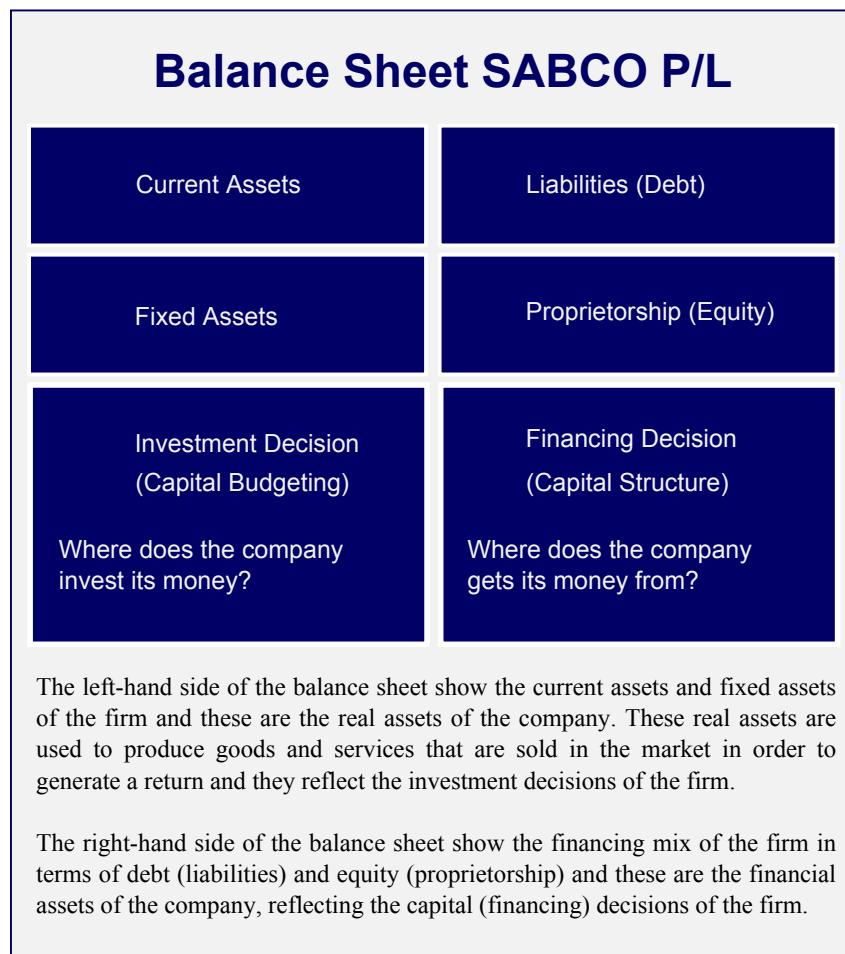


Figure 1.1: Accounting Balance Sheet

The financing decision of the firm deal with the determination of the firm's capital structure. How should the firm finance the investment in real assets in order to maximise its market value? Further, is it possible to create wealth on the financing side of the balance sheet? Can the value of the firm be affected by the way it is financed?

1.5 Factors in Financial Decision-Making

We now know what the main goal of the firm is maximisation of market value and owner wealth, and we also now know the key decisions that have to be made to achieve the corporate objective, the investment decision and the financing decision (along with the dividend decision). However, we must also be aware that there are three critical factors that need to be taken into account when making the key corporate finance decisions and these three critical factors are cash, time, and risk.

1.5.1 Cash-Flow Analysis

In corporate or business finance, the focus is always on cash-flows, not accounting earnings. So, while most accounting systems are accrual in nature, in corporate finance we are focused on cash-flow analysis, i.e. we focus on cash flowing into and out of the business in the period under consideration.

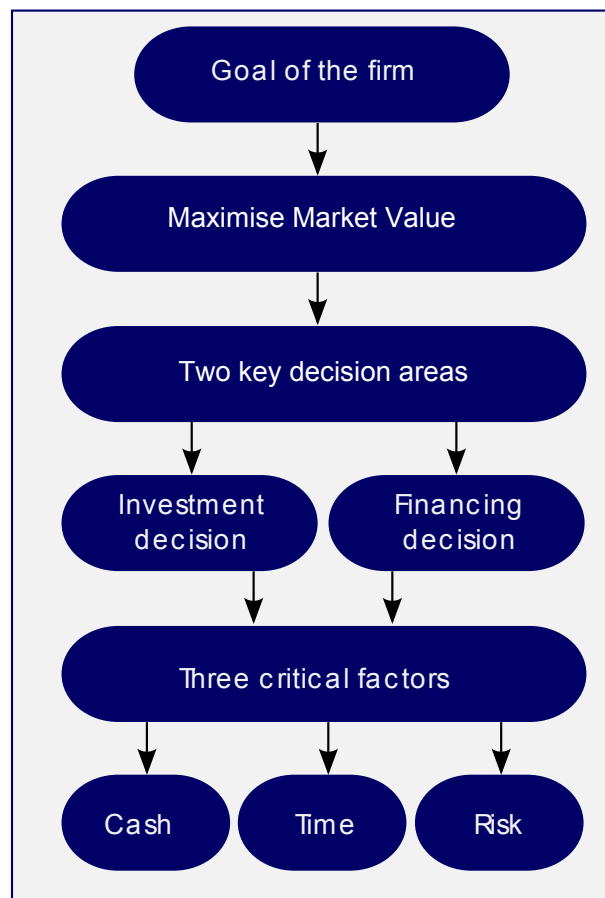


Figure 1.2: Critical Factors In Financial Decision-Making

Highlight 1.2 — Accrual Accounting. Accrual accounting measures the performance and position of a company by recognizing economic events regardless of when cash transactions occur by matching revenues to expenses (the matching principle), at the time in which the transaction occurs rather than when payment is made (or received). Accrual accounting is standard accounting practice for most companies.

In corporate finance, because we focus mainly on cash-flow analysis we are in effect using a cash accounting approach, which recognizes transactions only when there is an exchange of cash.

■ Example 1.1 — Accrual Accounting v. Cash-Flow Analysis.

Accrual Accounting Example

If in 2014 our business has \$1 million worth of credit sales for which we will receive payment in 2015, then these sales, because they occurred in 2014, will be recorded in the accounts of 2014. Similarly, if we had \$1 million of credit purchases in 2013, which we pay for in 2014, then these purchases, because they were made in 2013, would be recorded in the accounts of 2013. The key with accrual accounting is when does the transaction occur, not when does payment or receipt take place.

Cash-Flow Analysis Example

If in 2014 our business has \$1 million worth of credit sales for which we will receive payment in 2015, then these sales (or the cash-inflow from these sales) will be recorded in the accounts

of 2015 because that is when we will receive the money from the sales. Similarly, if we had \$1 million of credit purchases in 2013, which we pay for in 2014, then these purchases (or the cash-outflow for these purchases) will be recorded in the accounts of 2014 because that is when the payment for the purchases will be made. The key with cash-flow analysis is when does the payment or receipt take place. ■

1.5.2 Cash-Flow Timing

Money has a time value, so decision-making in corporate finance must take account of the timing of the cash-flows – a dollar today is worth more than a dollar at some future date (*ceteris paribus*). In finance, there is a trade-off between the size of an investment's cash-flows and when the cash-flows are received. For example, it may be the case that, once the time value of money has been taken into account, a \$3 million dollar investment return to be received in two years' time is better than a \$5 million investment return to be received in four years' time.

1.5.3 Cash-Flow Risk

Risk refers to variability of a cash-flow stream, i.e. how likely is it that the cash-flows we will receive from an investment will be different to what we expect? The greater the chance that the cash-flows we will receive will be different to what we expect, the greater the variability of the cash-flows and the greater the risk of the investment. In corporate finance adjustments must be made to cash-flows to take account of the differing degrees of variability of various cash-flows, and the risk-return relationship must always be kept in mind. For example, it may be the case that, once the variability or risk of the respective cash-flows is taken into account, a low-risk project, with a projected return of \$1 million, may be better than a high-risk project, with a projected return of \$3 million. The corporate finance manager and department is to deal with and account for the uncertainty or risk associated with investment decisions, and assessing the risk associated with expected future cash-flows is a critical component of the firm's investment decisions.

1.6 Valuation of a Firm

By now, we know that the main corporate objective is the maximisation of the market value of the company in order to maximise the wealth of the owners (shareholders) of the company. In maximising the market value of the company, we should aim to maximise the market value of the firm's assets.

From the discussion above about the main corporate finance decisions, we know that a firm's assets can be classified as real assets, which are those assets that can be put to productive use to generate a return, e.g. machinery and equipment that is used to produce goods and services, which are then sold in the market to generate cash-inflows to the business, with these real assets represented by the left-hand side of the balance sheet. Alternatively, a firm will have financial assets, which are those assets that represent a claim to a series of cash flows against the company, such as shares, a bank account, bond or debenture. These financial assets are represented by the right-hand side of the balance sheet, i.e. by the liabilities (debt) and proprietorship (equity) of the company.

In valuing a firm we know that the firm is a collection of real assets (e.g. plant, equipment, tools, stock, buildings, land, intellectual capital) that generate cash-flows. Therefore, one natural approach in valuing a firm is to value the real assets of the firm. However, there is a major problem with this approach as the real assets of a firm are not frequently traded, i.e. it is not often that all of the real assets of the firm will be sold in the market. Since the real assets of the

firm are not frequently traded, it is difficult to determine their true value, i.e. their market value, the value that you would get for them in the market today if they were sold. It is quite easy to determine the book-value of an asset, but to get the market value the asset must be sold.

Highlight 1.3 — Real Asset vs Financial Asset.

Real assets are value-generating properties and commodities owned by the business. They are essentially physical items, such as land, buildings, inventory, precious metals and oil, over which a business can attach intrinsic value. Intrinsic value is the exact value of an asset as determined by factors such as location, functionality and acquisition costs. Some types of real assets, such as land, buildings and inventories, are used to facilitate production processes.

Financial assets are cash or transactional instruments that are readily convertible into cash. Cash reserves; trade receivables; notes receivable; shares and bonds are some of the common types of financial assets. These liquid assets actually represent claims on the underlying value of other business possessions, such as real assets and properties. For example, an ordinary share represents a claim against the assets of a company that remain after the full payment of debts.

1.6.1 An Alternative Approach to Valuation

An alternative to valuing a company by its real assets, is to value the financial assets of the company, i.e. to find the current value of the debt and equity of those having a claim on the income produced by the real assets of the firm by having provided financing to the firm. Examples of financial assets include shares, which are a claim against a company that produces a cash-flow to the holder or investor made up of a dividend and a sale price, as well as bonds and debentures, which produce a stream of cash-flows to the lenders for interest and principal payments. It must be remembered that if a company is being valued by way of its financial assets, it is still important to use current market values or costs in finding the values of the financing items, e.g. for shares we would use the current price of the shares on the stock exchange, and for debt financing we would use current costs, i.e. current interest rates or yields, to find current market values.

1.7 Finance vs. Accounting

We have already discussed some of the differences between corporate finance and accounting. It is also important to note that, generally, accounting, with its emphasis on review and compliance, has a historical outlook. While finance, with its emphasis on valuation and decision-making, generally has a focus on the future.

The primary focus of accounting is stewardship and compliance. Accounting statements, such as the profit and loss statement and the balance sheet, must be prepared to comply with the law. The tax department requires a P&L statement to be prepared to determine how much tax it can collect from the company. The stock exchange requires P&L statements and balance sheets to be prepared so that investors can be fully informed about the performance and prospects of a company. However, accounting standards give discretion in the selection of accounting procedures, which can cause comparability problems when analysing reports of different companies, or even the same company at different points in time. This enables deliberate manipulation of accounting reports, i.e. allows for creative accounting or window dressing.

For example, it is often the case that a company can, legally, use different methods of depreciation or valuation of assets, as well as bad and doubtful debts, which means that it may be difficult to work out what is really going on with the company over time. Also, different

companies may, legally, use different accounting methods, meaning that it can be difficult to accurately compare one company to another when making an investment decision.

Another issue with accounting, from a corporate finance perspective, is in regards to accounting profit. Profit may be referred to as the dollar amount of returns from assets or a percentage return on an investment. For example, is it the case that a \$10 million profit from Project A is better than a \$1 million profit from Project B? Well, it may actually be the case that Project A costs \$1 billion and generates an accounting profit of \$10 million, meaning the percentage return is just 1%, while Project B has a cost of only \$1 million and generates a profit of \$1 million, giving a (much higher) percentage return of 100%. Also, in regards to accounting profit, we must ask whether the figure being quoted is before-tax or after-tax, as taking into account the tax or differing tax rates on profits may significantly alter their after-tax values.

Another problem with accounting profit from a corporate finance perspective is the neglect of time, since accounting profit ignores the time value of money concept. As an example, would a \$10 million profit over eight years be better than a \$3 million profit over two years? Which of these profits is better when the time-value of money is taken into consideration? Additionally, accounting profits ignore risk and are not adjusted for risk. As an example, we could ask whether a \$10 million accounting profit from Project A is better than a \$1 million accounting profit from Project B? It may be the case that Project A's \$10 million profit is generated from a very high-risk project and there is a big chance that we could lose all our money on Project A, while Project B's \$1 million profit is coming from a very low-risk project with little chance of losing our money. Once we take the relative risks into account, the higher profit of Project A may not look so good.

Accounting profit is also problematic from a corporate finance perspective, because accounting statements generally neglect cash-flows and are calculated on an accrual basis, not cash-flow basis. Finally, we also now know that with accounting standards there is discretion to use arbitrary and varying allocations, approaches, and standards in, for example, depreciation regimes, asset valuations, recording of revenues and expenses, and provisions for bad and doubtful debts, which can lead to creative accounting and window-dressing (and losses for investors, as the Enron scandal showed).

Business Snapshot 1.2 — Creative Accounting Exposed –The Enron Scandal. Enron was a US energy-trading and utilities company that is associated with one of the biggest accounting frauds in history, which came to light in 2001. Enron's executives, in conjunction with their accounting firm Arthur Andersen, employed accounting practices that falsely and grossly inflated the company's revenues, which, just prior to the scandal breaking, valued Enron as the seventh-largest corporation in the US. However, once the scandal broke, Enron filed for bankruptcy and was de-listed from the New York Stock Exchange (NYSE), while the accounting division of Arthur Andersen was wiped-out. Other large corporations to be involved in accounting scandals include WorldCom and Tyco International.

Time Value of Money

Simple Interest

Compounding and Discounting

Compound Interest

Future Value of a Single Sum

Adjusting "r" and "n"

Present Value of a Single Sum

Unknown "n" and "r"

Multiple Uneven Cash-Flows

Revision Problems

2. Financial Mathematics - Introduction

In this chapter you will learn about the time value of money concept, simple interest, and compound interest. In particular, we discuss the calculations for the present value and future value of a single amount for both one period and multiple periods, as well as the present value and future value of multiple uneven cash-flows.

2.1 Time Value of Money

Time value of money (TVM) is the concept that money available at the present time is worth more than the same amount in the future due to its potential earning capacity. TVM is an important principle of finance that is based on the premise that as long as money can earn interest, any amount of money is worth more the sooner it is received. Money deposited in a savings account will earn interest, so receiving \$1 today is worth more than \$1 in the future, holding all other factors constant (i.e. *ceteris paribus*). That is because you could invest the \$1 in a bank savings account today and in the future you will have more than \$1. The opportunity cost of \$1 in the future is the interest that could have been earned on \$1 if received earlier. So at the most basic level, TVM demonstrates that, all things being equal, it is better to have money now rather than later.

For a single sum time value problem (a single sum problem will be explained fully below) there are four variables that have to be taken into account, these being:

- n : the number of interest paying time or investment periods between a present value and a future value,
- r : the rate of interest for discounting or compounding (discounting and compounding will be explained in this chapter),
- PV : the present value or the price/value of the asset/investment now (at time period zero (T_0)), and
- FV_n : the future value or the price/value of the asset/investment at some future specified time (T_n).

It is important to note that when dealing with a time value problem n and r need to be consistent. For instance, if interest (r) is paid monthly the number of periods n has to be worked out in terms of months. So, all single sum time value questions involve four values - PV , FV , r

and n . Therefore, given any three of the values it is always possible to calculate the unknown fourth value.

Future Value

The *future value* (FV) of an investment is the value of an asset or cash at a specified date in the future that is equivalent in value to a specified sum today. Written as FV_n , it measures the nominal future sum of money that a given sum of money is worth at a specified time in the future assuming a certain interest rate, rate of return, or compounding rate.

Present Value

The *present value* (PV) of an investment is the current worth or value today of a future sum of money or stream of cash flows given a specified rate of return, where the future cash flows are discounted at an appropriate discount rate to find the value or price of the investment or asset in today's dollars.

2.2 Simple Interest

If a bank pays you simple interest on a deposit, the interest payment each period will be the same and will be the interest rate times the initial amount deposited. Simple interest refers to interest earned only on the original capital investment amount, the amount invested in the bank account at time-period zero (T_0). The formula for the future value (FV) of a single sum calculated with simple interest is

$$FV_n = PV(1 + r \times n).$$

The numerical example below will explain how to calculate the FV of a single sum with simple interest. Before we get to the numerical example it is timely to explain what is meant by the term single sum.

Highlight 2.1 — Single sum. A single sum investment is one where a single amount of money is invested for a certain number of periods. This means that over the life of the investment only one amount is invested, at the beginning of the investment, i.e. at T_0 (time-period zero). This amount is left to accrue a return over the life of the investment.

■ **Example 2.1 — Future Value of a Single Sum with Simple Interest.** Returning to the simple interest numerical example, we will assume that we have a single amount of \$100, which is our present value at time-period zero (PV_0) that we will invest in a bank deposit for three years and on which we will earn simple interest of 10% p.a. To find the future value of this single sum investment we would use the formula

$$FV_n = PV(1 + r \times n).$$

which would be re-specified as

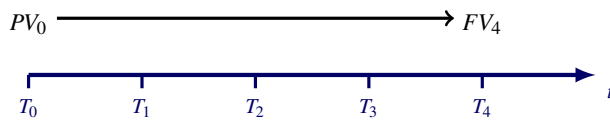
$$FV_3 = \$100(1 + 0.10 \times 3) = \$130.00.$$

Therefore, the interest earned each year on this investment would be $\$100 \times 0.10 = \10 , which is earned each year for three years, giving total interest earned of \$30 over three years. It is important to remember that with simple interest the dollar amount of interest earned each period is fixed, so in the numerical example the interest earned in each of the three years is always \$10, found as the original investment or PV_0 of \$100 time 10% per period. ■

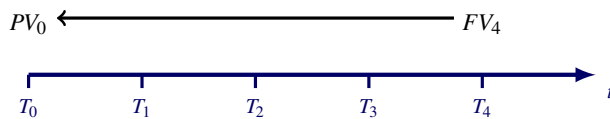
2.2.1 Compounding and Discounting

Compounding involves translating an amount invested today (PV_0) into its equivalent future value, FV_n . In order to find the future value we compound forward the present value to find FV_n . Discounting, on the other hand, involves translating an amount to be received in the future, FV_n , into its equivalent present value or value today (PV_0). In order to find a present value we discount back the future value to find PV_0 . We can see graphically in the timeline how the concepts of compounding and discounting work.

PV_0 is compounded four periods to find its future value at T_4 of FV_4

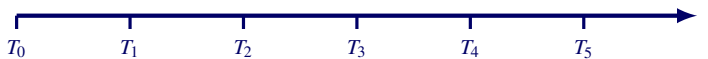


FV_4 is discounted four periods to find its present value at T_0 of PV_0



Timeline

Timelines are often used in finance, and are used extensively throughout this book, in order to graphically demonstrate cash-flows. For example, with future value and present value questions. They can aid in gaining a better understanding of how to solve finance-related problems.



Here we have a five-period timeline, indicating that the investment period in question is over five periods, for example, five years. T_0 is time period zero, also known as the present time or now, which is when the investment is made. T_1 is the end of the first period and beginning of the second period, T_2 is the end of the second period and beginning of the third period, and etc.

2.3 Compound Interest

Compound interest refers to where, for an investment such as a bank deposit, interest is calculated on the initial principal invested and also on the accumulated interest earned in previous periods. For example, if a bank is paying you compound interest on your bank deposits then you will receive interest payments not just on the initial amount you invest but also on previous interest payments made to you, so that interest is earned on both the initial capital investment and on the interest reinvested from prior periods, i.e. you earn interest on interest plus on the original amount invested. Compound interest will make a deposit or loan grow at a faster rate than with simple interest, which is interest calculated only on the principal amount. In finance, interest or returns are usually calculated on a compounding basis. In the Business Finance Course, unless a question states otherwise, you can assume thus we are using compound interest.

2.3.1 Future Value of a Single Sum

We have seen previously how to calculate the future value of a single sum with simple interest as $FV_n = PV_0(1 + (r \times n))$. However, generally in finance it is compound interest that is used. To find the future value of a single sum with compounding of interest the formula used is:

$$FV_n = PV_0(1 + r)^n.$$

Highlight 2.2 — Simple interest vs compound interest. Notice the key difference between compound interest and simple interest is in the bracketed term $(1 + r)$. With compound interest, the bracketed term is to the power of n , while in simple interest n is multiplied by the interest rate. The following example will help to explain how we calculate FV_n with compounding of interest for a single sum problem.

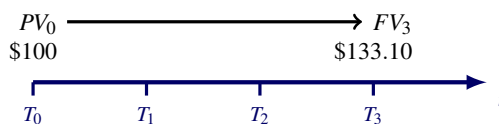
■ **Example 2.2 — Future Value of a Single Sum with compound Interest.** Assume you are a personal investment adviser and a client, John, comes to see you to invest \$100 in a savings account at ANZ Bank. John will earn a compounded return of 10% p.a. on the investment for three years. John wants to know how much money he will have in the account at the end of three years. This is a single sum question and John is really asking you to work out the future value of his single sum after three years. Since John will earn compound interest on his deposit, you use the future value of a single sum formula to find the value of John's deposit after three years as follows

$$FV_n = PV_0(1 + r)^n.$$

Note, the expression $(1 + r)^n$ is the future value interest factor (FVIF) for a single sum. Multiply this by PV_0 to get FV_n .

$$\begin{aligned} FV_3 &= \$100(1 + 0.10)^3 \\ &= \$100(1.331) = \$133.10 \end{aligned}$$

Here we have worked out for John that at the end of three years he will have \$133.10 in the bank account, meaning that he will earn total interest over the three years of $\$33.10 = \$133.10 - \$100$. The following timeline illustrates how John's initial investment of \$100, PV_0 , grows to FV_3 over three periods to be worth \$133.10 at the end of Year 3.



Here it is important to notice the difference in interest earned, where we are assuming compound interest, as compared to Example 2.1 where we assumed simple interest was being earned.

In Example 2.1 an amount of \$100 was invested for three years at a simple interest rate of 10% p.a. Using the future value of a single sum formula with simple interest of $FV_n = PV_0(1 + (r \times n))$, we found that at the end of three years FV_3 was \$130, meaning that in total \$30 had been earned

in interest over the three years. However, in Example 2.2 with compound interest we have found that given the same basic information of an initial deposit of \$100 for three years and the same annual interest rate of 10% being earned, FV_3 is slightly higher at \$133.10, meaning that over the three years the total interest earned with compound interest is \$33.10, which is \$3.10 higher than with simple interest. This is because with compound interest, over the life of an investment you are not only earning interest on the initial amount invested, but also on any interest earned in previous periods. Therefore, you always end up with a higher future value than with the same investment earning only simple interest.

2.3.2 Adjusting “r” and “n”

It is important to note that with both compounding (finding a FV_n) and discounting (finding a PV_0) if the number of compounding periods is more than once a year. r and n must be adjusted accordingly, with r to be divided by the number of compounding periods per year and n to be the number of compounding periods per year times the investments or loans maturity period in years.

For example, if you are told that an individual has deposited a single amount of \$1,000 into a bank account for four years at an annual interest rate of 6% and with interest calculated monthly, i.e. 12 times per year, there are 12 investment periods each year. Then r , the interest rate per investment period per year, would be

$$\frac{0.06}{12} \times 100 = 0.0050 \times 100 = 0.50\%$$

Further, n , the total number of investment periods, would be 48, i.e. four years times 12 investment periods per year. To find the future value of the single amount of \$1,000 at the end of four years we would specify the future value of a single sum formula as

$$FV_4 = \$1,000(1.0050)^{48}$$

where we are compounding forward the amount of \$1,000 at an interest rate of 0.50% per month for 48 months. The following is another numerical example of a future value of a single sum problem.

■ **Example 2.3** Assume that you are a personal investment adviser and a client, Stacey, comes to see you for advice. Stacey plans to invest a single sum of \$1,000 in a deposit with NAB Bank for five years and she will earn interest of 12% p.a. compounded annually, i.e. interest will be calculated once per year. Stacey needs to know how much money will she have in the deposit in five years’ time.

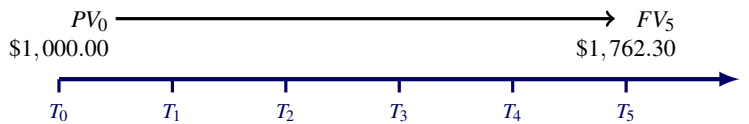
Since you know that you are dealing with a single sum problem and must find a value in the future, you need to use the future value of a single sum formula as follows:

$$FV_n = PV_0(1 + r)^n$$

$$FV_5 = \$1,000(1.12)^5$$

$$FV_5 = \$1,000(1.7623)$$

$$FV_5 = \$1,762.30.$$



Now assume that instead of earning 12% p.a. compounded annually for five years on her money, alternatively Stacey could earn 12% p.a. compounded monthly on the deposit, and would like to know whether that is a better option than 12% p.a. compounded annually.

Again, you know that you are dealing with a single sum problem and the only thing that has changed is the frequency of compounding per year. In Stacey's first example where interest was compounded once per year at 12% for five years, r was 0.12 and n was 5. In Stacey's second example we must remember to adjust r and n since we are given an annual interest rate (12%) but are also told that interest is calculated more than once per year for five year (12 times per year).

We find the monthly rate, as the annual interest rate of 12% (written as 0.12) divided by the number of times interest is calculated each year. Thus,

$$r = 0.12/12 = 0.01.$$

As for n , we have a five-year investment and interest is compounded 12 times per year. Then,

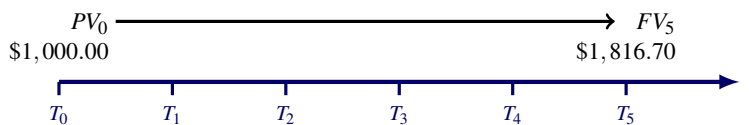
$$5 \times 12 = 60.$$

Therefore, the future value in Stacey's second example is as follows

$$\begin{aligned} FV_{60} &= \$1,000(1.01)^{60} \\ &= \$1,000(1.8167) \\ &= \$1,816.70. \end{aligned}$$

Therefore, we can tell Stacey that she is better off with the second option of 12% p.a. compounded monthly as with this option her future value after five years will be higher by \$54.40, i.e. \$1,816.70 minus \$1,762.30.

The timeline below shows how Stacey's initial investment of \$1,000.00, PV_0 , grows to FV_5 over five periods to be worth \$1,816.70 at the end of Year 5, i.e. at T_5 .



■

It is important at this stage to explain more fully why, in the example above, Stacey will have a higher future value after five years when interest is calculated monthly, as compared to annually. This is because when interest is calculated monthly, as compared to yearly, compounding occurs more often over the entire investment period, i.e. 60 times in the second example versus five

times in the first example. This means interest on interest is earned more often, therefore, even though the interest rate per investment period will be lower with more frequent compounding (1% per month in example 2 versus 12% per year in example 1), future value will be higher, *ceteris paribus*. It is also important to note that the future value of an investment also depends critically on the assumed interest rate, and that the higher the annual interest rate, the greater the future value, *ceteris paribus*. These two concepts of more frequent compounding and a higher interest rate impacting on the future value of an investment are highlighted and explained in the following table.

Future value of \$100				
Time periods	<i>r</i> =5%	<i>r</i> =10%	<i>r</i> =15%	<i>r</i> =20%
1	\$105.00	\$110.00	\$115.00	\$120.00
2	\$110.25	\$121.00	\$132.25	\$144.00
3	\$115.76	\$133.10	\$152.09	\$172.80
4	\$121.55	\$146.41	\$174.90	\$207.36
5	\$127.63	\$161.05	\$201.14	\$248.83

Here we are assuming a single sum of \$100 is invested from one period to five periods with interest rates per period of 5%, 10%, 15%, and 20%. If we look at row 1 of the table we have 1 investment period and we can compare the future value of the \$100 earning 5% versus 20% interest. From the table we see that after one period if \$100 earns 5% interest it amounts to \$105, while if it earns 20%, after one period it amounts to \$120, highlighting a key concept in finance.

Highlight 2.3 — Future value and time. For a given number of investment periods, the higher the interest rate the higher the future value.

If we look at column 1 of the table we have 5% interest being earned on the \$100 over each of the five investment periods, one period up to five periods, and we can compare the future value of the \$100 earning 5% over one period versus over five periods. From the table we see that after one period \$100 earning 5% interest amounts to \$105, while if it earns 5% over five periods it amounts to \$127.63, highlighting another key concept in finance. For a given interest rate, the more compounding periods the greater the future value.

2.4 Present Value of a Single Sum

The present value (*PV*) of a single sum is the the value or price in today's dollars of an amount of money that is to be received in the future discounted back to today at *r*% for *n* periods, and is found as follows

$$PV_0 = FV_n(1 + r)^{-n},$$

or

$$PV_0 = \frac{FV_n}{(1 + r)^n},$$

where the expression $(1 + r)^{-n}$ is the present value interest factor (PVIF) for a single sum. In both formulae above we are assuming compound interest, rather than simple interest. In Business Finance, we always assume compound interest is being used, unless the question states otherwise.

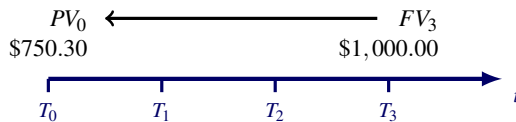
The following example will help to explain how we calculate PV_0 with compound interest for a single sum problem.

■ **Example 2.4** Assume that you are a personal investment adviser and a client, Steven, who will receive an amount of \$1,000 in three years' time, wants to know what is the value today of this sum if his opportunity cost of capital is 10% p.a.?

This is a single sum question, as Steven is to receive only one single amount in three years time. To calculate PV_0 use the present value of a single sum formula to find the value of the single amount of \$1,000 discounted back three periods to T_0 as follows

$$\begin{aligned} PV_0 &= FV_n(1+r)^{-n} \\ &= \$1,000(1.10)^{-3} \\ &= \$1,000(0.7513) = \$751.30. \end{aligned}$$

The following timeline shows how the amount that Steven will receive in three years' time, $FV_3 = \$1,000.00$, is discounted back three periods to T_0 to find $PV_0 = \$751.30$.

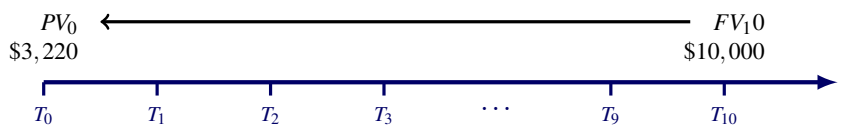


Here, we have worked out for Steven that the present value of the single amount of \$1,000 that he will receive in three years time, discounted at 10% p.a., is \$751.30. In other words, in order to receive \$1,000 in three years time the price that must be paid today is \$751.30, assuming an interest rate of 10% p.a.

The PV_0 tells us the amount that must be invested today in order to receive a certain amount in the future, or the price today of the future amount. ■

■ **Example 2.5** Assume that you are personal investment adviser. One of your clients, Mary, who is promised by her grandmother to receive \$10,000 in 10 years time, comes to see you for advice. Mary's required rate of return or discount rate is 12% p.a. Mary wants to know what the future amount of \$10,000 is worth in today's terms.

$$\begin{aligned} PV_0 &= FV_n(1+r)^{-n} \\ &= \$10,000(1.12)^{-10} \\ &= \$10,000(0.3220) = \$3,220 \end{aligned}$$



In this example, we have worked out for Mary that the value in today's dollars of the single amount of \$10,000 that she will receive in 10 years time discounted back to T_0 at 12% p.a. is \$3,220.00. Another way of expressing this is, if someone wishes to receive \$10,000 in 10 years' time the price that must be paid today is \$3,220.00. This means that for an investor to receive \$10,000 in 10 years' time, he or she has to invest \$3,220.00 today in, say, a bank account or some other asset, which earns a return of 12% p.a. ■

We saw in the previous section that with the future value of a single sum that changes in r or n can alter the future value of an investment, and that is also true for the present value of an investment. Like the future value of an investment, the present value of an investment depends critically on the assumed interest rate and the number of discounting periods. In particular, a higher annual interest rate lowers the present value, *ceteris paribus*. Also, a higher number of discounting periods also reduces the present value of an investment, *ceteris paribus*. These two concepts of a higher discount rate and more frequent discounting impacting on the present value of an investment are highlighted and explained in the table below.

Future value of \$100				
Time periods	$r=5\%$	$r=10\%$	$r=15\%$	$r=20\%$
1	\$95.24	\$90.91	\$86.96	\$83.33
2	\$90.70	\$82.64	\$75.61	\$69.44
3	\$86.38	\$75.13	\$65.75	\$57.87
4	\$82.27	\$68.30	\$57.18	\$48.23
5	\$78.35	\$62.09	\$49.72	\$40.19

In this table we are assuming a single sum of \$100 is to be received from one period to five periods in the future and with interest rates per period of 5%, 10%, 15%, and 20%. If we look at row 1 of the table we have 1 investment period and we can compare the present value of the future value of \$100 discounted back to T_0 at an interest rate of 5% versus 20%. From the table we see that after one period of discounting at 5% the future value of \$100 has a present value of \$95.24, while if the discount rate is 20% then after one period of discounting the future value of \$100 has a PV_0 of \$83.33. This highlights the key concept in finance that for a given number of investment periods, the higher the interest rate the lower the present value.

If we look at column 1 of the table we have a 5% interest rate being applied to the future value of \$100 over each of the five investment periods, one period up to five periods. We can compare the present value of the \$100 being discounted at 5% over one period versus over five periods. From the table we see that after one period \$100 discounted back to T_0 at a 5% discount rate has a PV_0 of \$95.24, while if the future value of \$100 is discounted back to T_0 at 5% p.a. over five periods it has a present value of \$78.35. This highlights another key concept in finance, which is for a given interest rate, the more discounting periods the lower the present value.

Here, we have discussed two very important concepts in financial mathematics, namely 1) the higher the interest rate, the lower the present value and 2) the more frequent the discounting, the lower the present value. Another way of thinking about these two concepts is to imagine you go to a department store to buy a pair of sports shoes. Assume that originally the shoes were priced at \$100. On the morning of the day that you visit the department store, you see that the shoes are discounted by 5%, meaning their price is \$95. In the afternoon of the same day, you return to the store and see the discount on the shoes has been increased to 20%, ie new price is \$80. So, the higher the discount rate on a product or an investment, the lower is its current price, i.e. the lower is its PV_0 .

Now, imagine you go to the same department store on another day to buy a sports bag which is priced at \$50. Assume that on the morning of the day you visit the department store the bag is discounted by 5%, meaning the price is \$47.50. Then assume that when you return to the store in the afternoon the bag has been discounted again by another 5%, meaning that it would then be priced lower at about \$45. So, the more often that a product or an investment is discounted, the lower is its current price, i.e. the lower is its PV_0 .

2.5 Unknown “n” and “r”

In general the problems that students will confront in Business Finance will involve working out either present values or future values. However, it is of course possible to also want to work out r given n , PV and FV . For example, it is quite a common question to want to know what the rate of return on an asset is when it grows from PV to FV over a given period of time. It is, also, possible to work out n if given PV , FV and r . In order to explain how to work-out an unknown r or n for a single sum problem we will use numerical examples.

■ **Example 2.6 — Solving for unknown r .** Assume that you are an investment adviser and a client, Alvin, comes to see you and says that he currently has \$10,000 available for investment for a 21 year period and he would like you to tell him at what annual interest rate must he invest this amount in order for it to be worth \$50,000 at maturity? Now, remembering that if you are given any three factors in the present value or future value of a single sum formula the fourth factor can be solved. Realising that Alvin is investing a single amount at T_0 and knows its FV_n , you tell him that you can use either the present value of a single sum formula or the future value of a single sum formula to give him the answer, as follows:

PV method

$$PV_0 = FV_n(1+r)^{-n}$$

$$\$10,000 = \$50,000(1+r)^{-21}$$

$$\frac{\$10,000}{\$50,000} = (1+r)^{-21}$$

$$0.20 = (1+r)^{-21}$$

$$0.20^{-\frac{1}{21}} = (1+r)^{-\frac{21}{21}}$$

$$0.20^{-0.04762} = 1+r$$

$$1.0797 = 1+r$$

$$r = 0.0797 = 7.97\%$$

FV method

$$FV_n = PV_0(1+r)^n$$

$$\$50,000 = \$10,000(1+r)^{21}$$

$$\frac{\$50,000}{\$10,000} = (1+r)^{21}$$

$$5 = (1+r)^{21}$$

$$5^{\frac{1}{21}} = (1+r)^{\frac{21}{21}}$$

$$5^{0.04762} = 1+r$$

$$1.0797 = 1+r$$

$$r = 0.0797 = 7.97\%$$

■

The following is another example that you can try yourself using either the PV or FV of a single sum formula, as above. Assume that an investor client of yours, Joanne, plans to sell a small piece of land for \$119,330 (FV_n), that she bought as an investment five years ago (n) for \$50,000 (PV_0). She would like you to tell her the annual rate of return on this investment. Using the same methods as in the previous example, you should find that the rate of return (r) is equal to approximately 19% p.a. (0.19).

■ **Example 2.7 — Solving for unknown n .** Assume that you are an investment adviser and a client, Thanh, comes to see you and says that he intends to invest an amount of \$10,000 into a bank account that pays interest of 9.6% p.a. compounded monthly. He would like you to tell him how long it will take (in months) for the account to grow to \$50,000. Remembering that if you are given any three factors in the present value or future value of a single sum formula the fourth factor can be solved. Realising that Thanh is investing a single amount at T_0 and knows its FV_n , you tell him that you can use either the present value of a single sum formula or the future value of a single sum formula to give him the answer, as follows

Note 1: $r = 0.096/12 = 0.008$ (i.e. 0.8% per month or 9.6% p.a. compounded monthly).

Note 2: To find the unknown n you will have to use the natural logarithm function, i.e. use the ‘ln’ button on a scientific calculator.

PV method

$$PV_0 = FV_n(1+r)^{-n}$$

$$\$100 = \$500(1.008)^{-n}$$

$$\frac{\$100}{\$500} = (1.008)^{-n}$$

$$0.20 = (1.008)^{-n}$$

$$\ln(0.20) = -n \ln(1.008)$$

$$-1.6094 = -n(0.007968)$$

$$n = \frac{1.6094}{0.007968}$$

$$n = 202 \text{ months}$$

FV method

$$FV_n = PV_0(1+r)^n$$

$$\$500 = \$100(1.008)^n$$

$$\frac{\$500}{\$100} = (1.008)^n$$

$$5 = (1.008)^n$$

$$\ln(5) = n \ln(1.008)$$

$$1.6094 = n(0.007968)$$

$$n = \frac{1.6094}{0.007968}$$

$$n = 202 \text{ months}$$

■

Highlight 2.4 In conclusion, for the present value and future value of single sum problems, it is important to remember that there are only four variables involved, these being FV_n , PV_0 , r , and n . You will always be given three of the variables and asked to solve for the unknown fourth, and this hint makes solving single sum time-value problems much easier.

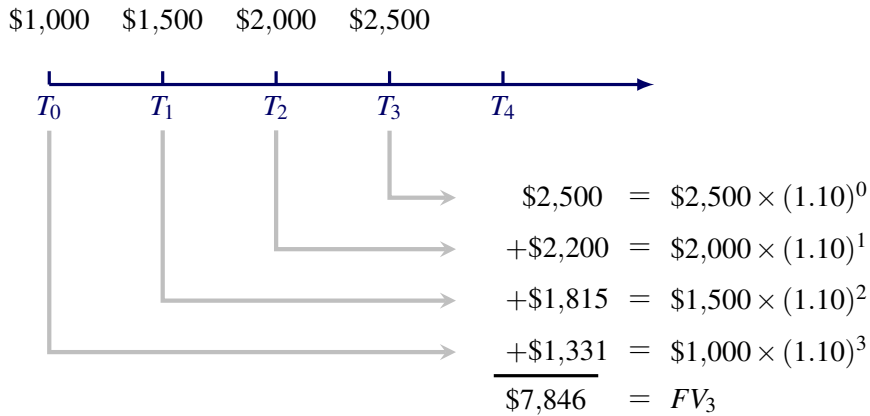
2.6 Multiple Uneven Cash-Flows

With single sum problems we deal with only one single amount, PV_0 , being invested at T_0 and growing to a value at some time in the future, FV_n . However, it is possible with present value and future value questions to have more than one cash-flow involved. In this case we might have multiple even cash-flows, ie cash-flows with the same values, or multiple uneven cash-flows, i.e cash-flows with different values. For example, if you are an investment adviser a client may ask you to calculate the balance of their account at a future time, if they deposit varying amounts of money into a bank over the time.

In such a situation, when we are dealing with multiple uneven cash-flows, in order to find the future value or present value of these cash-flows, because each of the cash-flows is assumed to be of a different value, the principles involved are the same as with single sum problems. This means that to find the future value of a stream of multiple uneven cash-flows we would apply the future value of a single sum formula to each of the cash-flows to find the FV_n of each of the cash-flows. We would then sum up these individual future values to find the total FV_n . The same principle applies to finding the present value of a stream of uneven cash-flows. We would apply the present value of a single sum formula to each of the individual cash-flows to find their PV_0 , then sum these present values to get the total PV_0 .

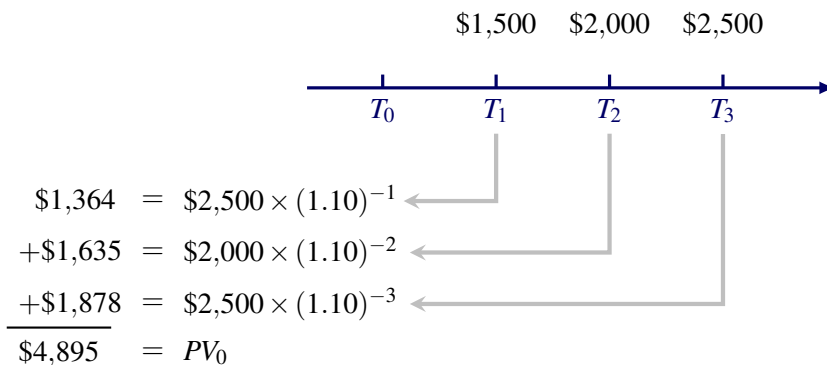
The following examples will help explain how to find the future value and present value of multiple uneven cash-flows.

■ **Example 2.8 — Future value of multiple uneven cash-flows.** Assume that an investor client, Sarah, comes to see you and says that she will deposit \$1,000 today, \$1,500 in one year, \$2,000 in two years, and \$2,500 in three years into a bank account that pays interest of 10% p.a. Sarah would like to know how much she will have in the account in total in three years time. Knowing that Sarah is asking for FV_n of a multiple uneven cash-flow stream, you apply the future value of a single sum formula to each of the cash-flows to find their FV_n , then sum these individual future values to find the total FV_n as follows



In finding the total FV_n of \$7,846, we have applied the future value of a single sum formula to each of the deposits and compounded Deposit 1 of \$1,000, which is made today, forward three periods to the end of Year 3 at 10% p.a. Then compounded Deposit 2 of \$1,500, which will be made in one year's time, forward two periods to the end of Year 3 at 10% p.a. Then compounded Deposit 3 of \$2,000, which will be made in two years time, forward one period to the end of Year 3 at 10%. For Deposit 4 of \$2,500, which will be made in three years time, i.e. at the end of Year 3, we do not compound it forward as it is made right at the very end of the total investment period, so does not earn a return. We then sum these individual future values to find FV_3 of \$7,846. ■

■ **Example 2.9** Assume that an investor client, Raju, comes to see you and says that he will deposit \$1,500 in one year, \$2,000 in two years, and \$2,500 in three years into a bank account that pays interest of 10% p.a. Raju would like to know the value today of these three deposits. Knowing that Raju is asking for PV_0 of a multiple uneven cash-flow stream, you apply the present value of a single sum formula to each of the cash-flows to find their PV_0 , then sum these individual present values to find the total PV_0 as follows



In finding the total PV_0 of \$4,895 we have applied the present value of a single sum formula to each of the deposits and discounted Deposit 1 of \$1,500, which is made in one years time, back one period to T_0 at 10% p.a. Then discounted Deposit 2 of \$2,000, which will be made in two years time, back two periods to T_0 at 10% p.a. Then discounted deposit 3 of \$2,500, which will be made in three years' time, back three periods to T_0 at 10%. We then sum these individual present values to find the total PV_0 of \$4,895. ■

2.7 Revision Problems

Problem 2.1 A person places \$25,000 in a term deposit with a fixed term and interest rate of five years and 8% p.a., respectively. If the interest is compounded on a weekly basis what is the value of the investment at the end of the five years?

- (a) \$27,081
- (b) \$52,774
- (c) \$37,284
- (d) \$41,005

Problem 2.2 What is the present value of \$500 to be received at the end of each of the next three years assuming a discount rate of:

- (a) 4 per cent i
- (b) 25 per cent?

Problem 2.3 What quarterly payment is necessary to accumulate \$1.5 million over 15 years if the annual interest rate is 6.75% compounded quarterly? Assume payments are made at the end of each quarter.

- (a) \$10,703
- (b) \$14,637
- (c) \$18,534
- (d) \$24,748

Problem 2.4 The present value of a stream of two annual cash-flows of \$100 each beginning in one year's time where the interest rate is 5% p.a. for the first year and 8% p.a. for the second year is?

Problem 2.5 What sum would Susan have to receive in five years time to make her indifferent between that sum and \$1,000 in 10 years time? Susan is able to invest at an interest rate of 10% p.a.

- (a) \$621
- (b) \$500
- (c) \$710
- (d) \$645

Problem 2.6 AMP Life Insurance offers a policy known as the 'Pension Creator Six Pay'. Typically, the policy is bought by a parent or grandparent for a child at the child's birth. The details of the policy are as follows: The purchaser (parent or grandparent) makes the following six payments on behalf of the child to AMP:

1st birthday	\$730	4th birthday	\$855
2nd birthday	\$730	5th birthday	\$855
3rd birthday	\$730	6th birthday	\$855

After the child's sixth birthday no more payments are made. When the child reaches age 65 he or she receives a payout of \$143,723 from AMP. If the relevant interest rate for a bank deposit is 6 % p.a. for the first six years and 7 % p.a. for all subsequent years, as the family's investment adviser advise them as to whether the policy is worth buying?

Annuities

Types of Annuities

Ordinary Annuities

Future Value of an Ordinary Annuity
Present Value of an Ordinary Annuity
Unknown PMT for an Ordinary Annuity

Annuity Due

Future Value of an Annuity Due
Present Value of an Annuity Due
Finding PMT for an Annuity Due

Deferred Annuity- Present Value

Perpetuity- Present Value

Growing Perpetuity- Present Value

Effective Interest Rates

Loans

Amortised Loan – Calculations

Revision Problems

Financial Calculator Steps and Keystrokes

3. Financial Mathematics - Annuities

3.1 Annuities

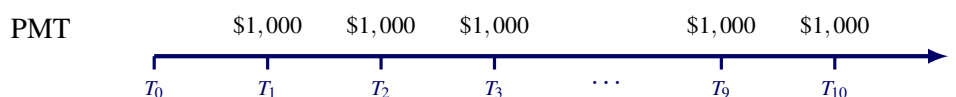
An *annuity* is a financial product, usually sold by financial institutions such as banks, superannuation, and insurance companies. It is generally comprised of regular periodic contributions made by an investor and then, upon annuitization, provides a stream of periodic payments to the individual some time in the future. Put more simply, an annuity is a series of constant cash-flows (payments or receipts) occurring at regular intervals; for example, superannuation payments.

3.1.1 Types of Annuities

There are five main types of annuities, these being an *ordinary annuity*, an *annuity due*, a *deferred annuity*, a *perpetuity*, and a *growing perpetuity*.

Ordinary Annuity

An ordinary annuity is a series of constant (fixed value) cash-flows occurring at the end of each period for some fixed number of periods and commencing at the end of the first period, i.e. commencing at T_1 . Generally, ordinary annuity payments are made monthly, quarterly, semi-annually, or annually. Examples of an ordinary annuity include coupon payments on bonds and superannuation payments (both receipt annuities), as well as mortgage repayments (a payment annuity).

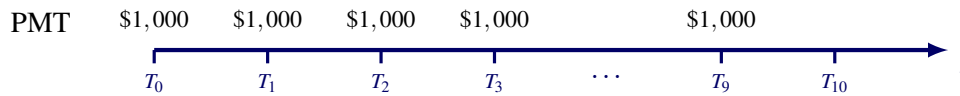


As shown in the diagram above, with an ordinary annuity each payment is of a fixed value and is at the end of each period, and the first payment is at the end of the first period (i.e. at T_1).

Annuity Due

An annuity due is a series of constant cash-flows occurring at the start of each period for some fixed number of periods and commencing at the beginning of the first period, i.e. commencing at T_0 . Examples of an annuity due include most lease payments, rental payments, and the paying of

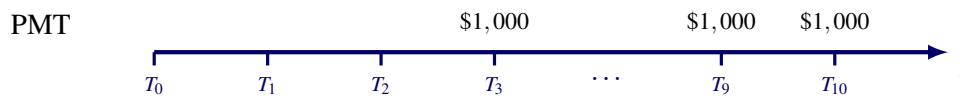
university fees in advance at the beginning of each semester.



The above diagram shows that with an annuity due, each payment is of a fixed value and is at the beginning of each period, and the first payment is at the beginning of the first period, T_0 .

Deferred Annuity

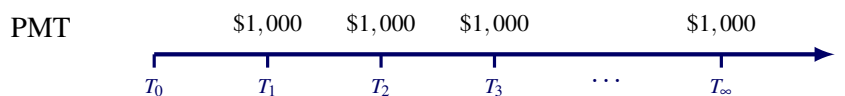
A deferred annuity is a series of constant cash-flows occurring at the end of each period for some fixed number of periods and commencing at some future period after period one. For example, a deferred annuity may commence at the end of the third period, T_3 . Deferred annuities have two phases, a savings phase, whereby regular periodic payments are made by the annuity owner into the annuity account, and an income phase, whereby regular periodic payments are made to the annuity owner. An example of a deferred annuity is superannuation payments.



With a deferred annuity each payment is of a fixed value and is made at the end of each period. The first payment is made sometime after the end of period 1; for example, at T_3 in the above example.

Perpetuity

A perpetuity is a series of constant cash-flows occurring at the end of each period that goes on indefinitely. The concept of a perpetuity is used often in financial theory, such as in the dividend valuation model of shares, which will be explored further in Chapter 5. An example of a perpetuity is an academic scholarship, such as a Rhodes Scholarship¹, which provides scholarships to university students perpetually.

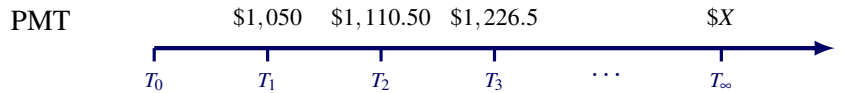


With a perpetuity each payment is of a fixed value and is made at the end of each period. The first payment is made at the end of the first period and the payment stream goes on forever, i.e. into perpetuity.

¹Rhodes Scholarships were first established in 1902 and are postgraduate awards supporting exceptional students from around the world to study at the University of Oxford, England. Established in the will of Cecil Rhodes, the Rhodes Scholarship is the oldest and perhaps the most prestigious international scholarship programme in the world. Eighty-three Rhodes Scholars are selected annually from 14 countries or groupings of countries around the world, and the scholarship is usually awarded for two years, although can be awarded for one or three years. With the scholarship through the Rhodes Trust University, college fees are paid and the scholar receives a monthly maintenance stipend (payment) to cover accommodation and living expenses. <http://www.rhodeshouse.ox.ac.uk/>

Growing Perpetuity

A growing perpetuity is a series of cash-flows occurring at the end of each period that goes on forever and grows by a fixed percentage each period. As with perpetuities, the concept of a growing perpetuity is used often in financial theory, such as in the dividend valuation model of shares, which is covered in Chapter 5.



With a growing perpetuity, each payment is made at the end of each period and the first payment is made at the end of the first period. The payment stream goes on forever, i.e. into perpetuity, and the payments increase by a fixed percentage each period (by 5% in the timeline example above).

3.2 Ordinary Annuities

3.2.1 Future Value of an Ordinary Annuity

As stated above, an ordinary annuity involves a series of constant cash-flows occurring at the end of each period for some fixed number of periods and commencing at the end of the first period, T_1 . In order to find the future value of an ordinary annuity we use the future value of an ordinary annuity formula as follows

$$FV_n = PMT \left[\frac{(1+r)^n - 1}{r} \right]$$

where PMT is the fixed value annuity payment and the the compounding term (square bracketed term) is called the future value interest factor of the annuity (FVIFA is what the annuity payment (PMT) is multiplied by to get the future value). It is important to note that with the future value of an ordinary annuity formula future value is given at the time of the last annuity payment. For example, if the last annuity payment is at the end of the fourth period, then using the future value of an ordinary annuity formula will give us FV_4 , which is future value at the end of the fourth period.

The following numerical example will help to explain how to calculate the future value of an ordinary annuity.

■ **Example 3.1** Assume that you are an investment adviser and that one of your clients, Peter, comes to see you and says that he intends to invest \$1,000 at the end of each of the next three years in a bank account earning 8% p.a. interest. Peter wants to know how much he will have in the account at the end of three years. Since the three deposits that Peter will make are all of the same value and each deposit is made at the end of each annuity period, you tell him that you are dealing with an ordinary annuity stream. You can calculate the future value using the future

value of an ordinary annuity formula as follows:

$$FV_n = PMT \left[\frac{(1+r)^n - 1}{r} \right]$$

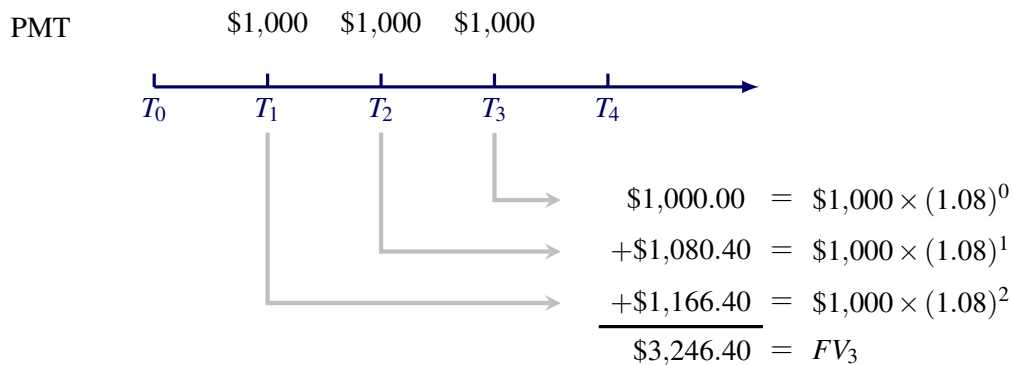
$$FV_3 = \$1,000 \left[\frac{(1+0.08)^3 - 1}{0.08} \right]$$

$$= \$1,000 \times 3.2464$$

$$= \$3,246.40.$$

At T_3 , which is the end of the third year, Peter will have in the account \$3,246.40.

Here, the timeline shows how Peter's three deposits of \$1,000 each grow to be worth a total FV_3 of \$3,246.40 at the end of Year 3 (i.e. at T_3).



Notice that Deposit 1 at T_1 is compounded twice at 8% p.a., to get its future value at T_3 of \$1,166.40. Deposit 2 at T_2 is compounded once at 8% to get its future value at T_3 of \$1,080.00, and Deposit 3 at T_3 is not compounded at all, i.e. does not earn a return or any interest, as it is made right at the end of the total investment period of T_3 . Also notice that the total future value FV_3 of \$3,246.40 is the future value at the time of the last annuity payment.

■

3.2.2 Present Value of an Ordinary Annuity

In order to find the present value of an ordinary annuity, we use the present value of an ordinary annuity formula as follows

$$PV_0 = PMT \left[\frac{1 - (1+r)^{-n}}{r} \right]$$

where PMT is the fixed value annuity payment and the the compounding term (square bracketed term) is called the *present value interest factor* of the annuity (PVIFA is what the annuity payment (PMT) is multiplied by to get the present value). It is important to note that with the present value of an ordinary annuity formula, present value is given at time-period zero, T_0 , which is one period before the first annuity payment, T_1 .

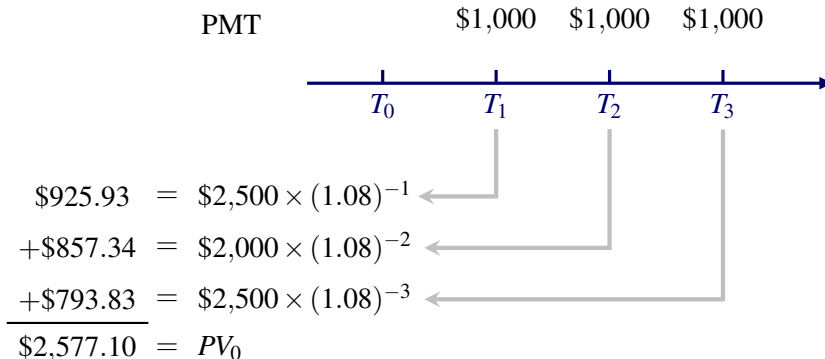
The following numerical example will explain how to calculate the present value of an ordinary annuity.

■ **Example 3.2** Assume that you are an investment adviser and that one of your clients, Mary, would like to receive a payment of \$1,000 at the end of each of the next three years. Mary is able to earn a return of 8% p.a. on her money. She then asks you to calculate how much she would need to invest today so that she is able to receive the three payments of \$1,000 each. Since the payments that Mary will receive are all of the same value and each payment will be received at the end of each annuity period, you are dealing with an ordinary annuity stream and can determine the amount that must be invested today, i.e. the total *PV* of the annuity stream, using the present value of an ordinary annuity formula as follows

$$\begin{aligned}
 PV_0 &= PMT \left[\frac{1 - (1 + r)^{-n}}{r} \right] \\
 &= \$1,000 \left[\frac{1 - (1 + .08)^{-3}}{0.08} \right] \\
 &= \$1,000 \times 2.5771 \\
 &= \$2,577.10.
 \end{aligned}$$

At T_0 , which is the beginning of the first year, Mary will have to invest \$2,577.10 at a return of 8% p.a. to be able to receive the three payments of \$1,000. PV_0 of \$2,577.10 is the price or cost of the annuity stream that Mary wishes to receive.

Here, the timeline shows how the three payments of \$1,000 are discounted back at 8% p.a. to T_0 , which will give Mary a combined present value (PV_0) of \$2,577.10.



■

3.2.3 Unknown PMT for an Ordinary Annuity

In the two previous numerical examples we were given the total number of annuity payments or investment periods (n), the interest rate per investment period (r), and the regular periodic annuity payment (PMT). Subsequently, we were required to calculate the future value and present value of ordinary annuities. However, it is common to want to know PMT , the regular periodic annuity payment, if given n , r and FV or PV . This is particularly so in instances of trying to work out the regular periodic payments on a loan.

In order to see how we are able to work out the unknown PMT for an ordinary annuity we will use a numerical example, based on the same information that was used for calculating the present value in Example 3.2.

■ **Example 3.3** Suppose that we know that the present value of the annuity stream is $PV_0 =$

\$2,577.10, n is 3 and that r is 8% p.a., The unknown PMT can be worked out using the present value of an ordinary annuity formula as follows

$$\begin{aligned}
 PV_0 &= PMT \left[\frac{1 - (1 + r)^{-n}}{r} \right] \\
 \$2577.1 &= PMT \left[\frac{1 - (1 + .08)^{-3}}{0.08} \right] \\
 \$2577.1 &= PMT \times 2.5771 \\
 PMT &= \frac{\$2577.1}{2.5771} \\
 &= \$1,000.
 \end{aligned}$$

which we know is correct, since the annuity payment we assumed in Example 3.2 to find the PV_0 of \$2,577.10 was \$1,000. ■

Therefore, to find the unknown PMT in a present value of an ordinary annuity question we simply re-arrange the formula and divide the present value interest factor of the ordinary annuity (PVIFA; the square-bracketed term) into the given PV_0 . Similarly, to find the unknown PMT in a future value of an ordinary annuity question we simply re-arrange the formula and divide the future value interest factor of the ordinary annuity (FVIFA; the square-bracketed term) into the given FV_n . The following example is based on the information provided in Example 3.1.

■ **Example 3.4** Assume that we know that the future value of the annuity stream is $FV_3 = \$3,246.40$, n is 3 and that r is 8% p.a., then the unknown PMT can be worked out using the future value of an ordinary annuity formula as follows

$$\begin{aligned}
 FV_n &= PMT \left[\frac{1 - (1 + r)^{-n}}{r} \right] \\
 \$3,246.40 &= PMT \left[\frac{(1 + .08)^3 - 1}{0.08} \right] \\
 \$3,246.40 &= PMT \times 3.2464 \\
 PMT &= \frac{\$3,246.40}{3.2464} \\
 &= \$1,000.
 \end{aligned}$$

which, again, we know is correct since the annuity payment we assumed in Example 3.1 to find the FV_3 of \$3,246.40 was \$1,000. ■

3.3 Annuity Due

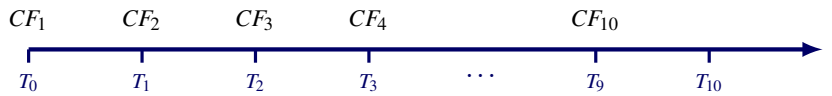
An *annuity due* involves a series of constant cash-flows occurring at the beginning of each period for some fixed number of periods and commencing at the beginning of the first period, T_0 . Hence, the key difference between an annuity due and an ordinary annuity is that with an annuity due the cash-flows or annuity payments occur at the beginning of each period, rather than at the end.

The following time-lines show the difference in the timing of annuity payments between an annuity due and an ordinary annuity. In both time-lines we have 10 cash flows, denoted by $CF_1, CF_2, \dots, CF_{10}$.

With an ordinary annuity each cash-flow (CF) or annuity payment is transacted at the end of the time period. Therefore, on the time-line the subscript of the cash-flow matches the subscript of the time period.



On the other hand, with an annuity due each cash-flow (CF) or annuity payment is transacted at the beginning of the time period.



3.3.1 Future Value of an Annuity Due

In order to find the future value of an annuity due we use the future value of an annuity due formula as follows.

$$FV_n = PMT \left[\frac{(1+r)^n - 1}{r} \right] (1+r)$$

The following numerical example will help to explain how to calculate the future value of an annuity due.

■ **Example 3.5** Assume that you are an investment adviser and that one of your clients, Albert, comes to see you and says that he will deposit three payments of \$1,000 each at the beginning of each year into a savings account yielding 12% p.a. interest compounded annually. Albert would like to know how much he will have in the account in three years' time. Since the deposits that Albert will make are all of the same value and each deposit is to be made at the beginning of each annuity period, you are dealing with an annuity due stream of cash-flows. You can determine the future value of the annuity stream in three years' time of FV_3 , i.e. total future value at the end of the third year, using the future value of an annuity due formula as follows

$$\begin{aligned} FV_3 &= \$1,000 \left[\frac{(1+0.12)^3 - 1}{0.12} \right] (1+0.12) \\ &= \$1,000 \times 3.3744 \times 1.12 \\ &= \$3,374.4 \times 1.12 \\ &= \$3,779.33. \end{aligned}$$

Remember, the FV of an ordinary annuity formula gives the FV at the time the last payment is made, which in this example is at the beginning of the third (last) period. Since the question is asking for the FV at T_3 , the FV at the beginning of the third period must be compounded forward to the end of the third period, i.e. compounded forward one period, as a single sum to get the total future value FV_3 in three years' time, i.e. total future value at T_3 . ■

We can see above that at T_2 , the time of the last deposit, the total value of Albert's account of FV_2 will be \$3,374.40. However, this is not our final answer as Albert has asked for FV_3 , which

is the total future value of his three deposits at the end of Year 3, so we must take the value FV_2 and compound it forward one period as a single sum by multiplying it by the future value interest factor of a single sum of $(1 + r)$ to get FV_3 of \$3,779.33.

Highlight 3.1 — Ordinary annuity vs annuity due. With an annuity due the future value will always be higher than for an equivalent ordinary annuity because with the annuity due compounding occurs more frequently or more often. As we learned earlier for future values, the more often compounding occurs, the higher the future value, *ceteris paribus*.

3.3.2 Present Value of an Annuity Due

In order to find the present value of an annuity due we use the present value of an annuity due formula as follows

$$PV_0 = PMT + PMT \left[\frac{1 - (1 + r)^{-(n-1)}}{r} \right]$$

Note that in the formula above, the first cash-flow (the first PMT in the formula) with an annuity due, which will occur at T_0 , is not discounted, while the remaining annuity payments are discounted. Also, note that the negative n in the formula will be one less than the total number of payments, i.e. it will be $-(n - 1)$, as the first payment at T_0 is not discounted.

The following numerical example will explain how to calculate the present value of an annuity due.

■ **Example 3.6** Assume that you are an investment adviser and that one of your clients, Phuong, wishes to receive three payments of \$1,000 each, with each payment to be received at the beginning of each year and the first payment to be received immediately. Phuong is able to earn a return on her money of 12% p.a. and would like to know how much she must invest today in order to be able to receive the three payments. Since the payments that Phuong will receive are all of the same value and each payment is received at the beginning of each annuity period, you are dealing with an annuity due stream of cash-flows. Determine the present value of the annuity stream now of PV_0 using the present value of an annuity due formula as follows

$$\begin{aligned} PV_0 &= \$1,000 + \$1,000 \left[\frac{1 - (1 + 0.12)^{-(3-1)}}{0.12} \right] \\ &= \$1,000 + \$1,000 \times 1.6901 \\ &= \$2,690.10. \end{aligned}$$

The first payment that Phuong is to receive of \$1,000 immediately is not discounted (it is the first \$1,000 in the formula and is not multiplied by the PVIFA). Additionally, the number of payments that are discounted is only two, that is $n - 1 = 3 - 1 = 2$, as the first of the three payments, which occurs at T_0 , is not discounted. ■

We can see from the example above that at T_0 , when Phuong receives her first payment of \$1,000, the total present value of Phuong's annuity stream will be \$2,690.10.

At T_0 , which is the beginning of the first year, Phuong will have to invest \$2,690.10 at a return of 12% p.a. to be able to receive the three payments of \$1,000. So, this PV_0 of \$2,690.10 is the price or cost of the annuity stream that Phuong wishes to receive.

Highlight 3.2 — Ordinary annuity vs annuity due. An annuity due will always have a higher present value than an equivalent ordinary annuity. This is because with the annuity due discounting occurs less frequently or less often, and as we learned earlier for present values, the less (more) often discounting occurs, the higher (lower) the present value, *ceteris paribus*.

3.3.3 Finding PMT for an Annuity Due

In the two previous numerical examples we were given the total number of annuity payments or investment periods (n), the interest rate per investment period (r), and the regular periodic annuity payment (PMT). Subsequently, we were required to calculate the future value and present value of the annuities due. However, in some practical examples, we might need to calculate PMT , given n , r and FV (or PV), for example when trying to work out the regular periodic payments on a mortgage loan.

This is best explained using a numerical example, based on the same information used in Examples 3.5 and 3.6.

■ **Example 3.7** The present value of the annuity due stream is $PV_0 = \$2,690.10$, n is 3 and r is 12% p.a. The unknown PMT can be worked out using the present value of an annuity due formula as follows

$$\begin{aligned}
 PV_0 &= PMT + PMT \left[\frac{1 - (1 + r)^{-(n-1)}}{r} \right] \\
 \$2,690.10 &= PMT + PMT \left[\frac{1 - (1.12)^{-(3-1)}}{0.12} \right] \\
 \$2,690.10 &= PMT + PMT \times 1.6901 \\
 PMT &= \frac{\$2,690.10}{2.6901} \\
 &= \$1,000.
 \end{aligned}$$

Therefore, to find the unknown PMT in a present value of an annuity due question we simply re-arrange the formula and divide 1 plus the present value interest factor of the annuity due (PVIFA; the square-bracketed term) into the given PV_0 .

Similarly, to find the unknown PMT in a future value of an annuity due question we simply re-arrange the formula and divide the future value interest factor of the annuity due (FVIFA; the square-bracketed term) multiplied by the future value interest factor of the single sum into the given FV_n , as follows

$$\begin{aligned}
 \$3,779.33 &= PMT \left[\frac{(1.12)^3 - 1}{0.12} \right] (1.12) \\
 \$3,779.33 &= PMT \times 3.3744 \times 1.12 \\
 \$3,779.33 &= PMT \times 3.7793 \\
 PMT &= \frac{\$3,779.33}{3.7793} \\
 &= \$1,000.
 \end{aligned}$$

■

3.4 Deferred Annuity- Present Value

As previously stated, a deferred annuity is a series of constant cash-flows occurring at the end of each period for some fixed number of periods, commencing at a future time, for example T_3 . The calculations for the present value of a deferred annuity are comprised of two phases, namely

1. a *savings phase*, whereby regular periodic payments are made by the annuity owner into the annuity account, and
2. an *income phase*, whereby regular periodic payments are made to the annuity owner.

The following numerical example will explain how to calculate the present value of a deferred annuity.

■ **Example 3.8** Assume that you are an investment adviser and that one of your clients, Jing, comes to see you and says that he wishes to invest a sum of money today that will yield him three equal instalments of \$1,000 each, with the first instalment to be received three years from today. If Jing is able to earn a return on his money of 12% p.a., what amount must he invest today in order to be able to receive the three payments of \$1,000, i.e. what is the *PV* of this investment?

Since the payments that Jing will receive are all of the same value, each payment is to be received at the end of each annuity period, and the first payment will be received in three years' time, you are dealing with a deferred annuity stream of cash-flows. You can determine the present value of the annuity stream now of PV_0 using, firstly, the present value of an ordinary annuity formula, then, secondly, the present value of a single sum formula, as follows

Step 1: *PV* of an ordinary annuity formula is used

$$PV_n = PMT \left[\frac{1 - (1 + r)^{-n}}{r} \right]$$

$$PV_2 = \$1,000 \left[\frac{1 - (1 + 0.12)^{-3}}{0.12} \right]$$

$$= \$1,000 \times 2.4018$$

$$= \$2,401.83.$$

Step 2: *PV* of a single sum formula is used

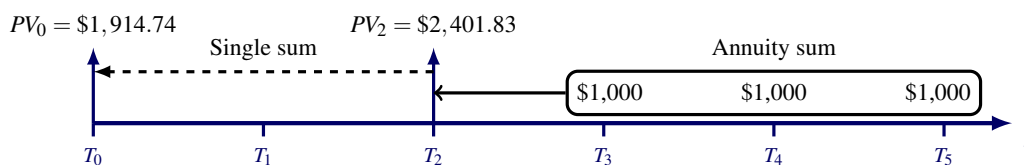
$$PV_0 = PV_n (1 + r)^{-n}$$

$$= \$2,401.83 \times (1.12)^{-2}$$

$$= \$2,401.83 \times 0.7972$$

$$= \$1,914.74$$

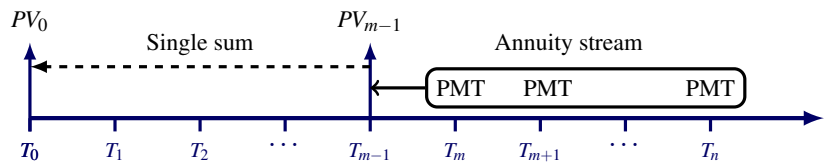
The timeline below can be used to help explain what needs to be done in order to find the present value at time period zero of a deferred annuity stream.



In order to find the present value at time-period zero (PV_0) of a deferred annuity stream of cash-flows, a two-step process is involved.

Firstly, treat the annuity stream as an ordinary annuity and use the present value of an ordinary annuity formula to find the present value of the annuity stream at the end of the period before the first annuity payment is made. Remember, with the present value of an ordinary annuity formula you are always given the present value one period before the first annuity payment takes place. So, in the example above, since the first of the three annuity payments of \$1,000 occurs at the end of Year 3, applying the PV of an ordinary annuity formula to the annuity stream must give us PV_2 . That is exactly one period before the first annuity payment takes place. Secondly, we take the present value, PV_2 , of the annuity stream and treat this as a single sum. Then, we discount PV_2 back to time-period zero using the present value of a single sum formula.

In general, when we have a n -period annuity stream that is deferred until the end of time-period m , we apply T_m . Apply the PV of an ordinary annuity formula to the n annuity payments. This would give us PV_{m-1} , which we will discount to time zero as follows.



Then the formula becomes

$$\begin{aligned}
 PV_0 &= PV_{m-1} \times (1 + r)^{-(m-1)} \\
 &= PMT \left[\frac{1 - (1 + r)^{-n}}{r} \right] \times (1 + r)^{-(m-1)}.
 \end{aligned}$$

3.5 Perpetuity– Present Value

As previously stated, a perpetuity is a series of constant cash-flows occurring at the end of each period that goes on indefinitely.

It should be noted that the future value of a perpetuity cannot be calculated as the cash flows are infinite. However, even though the cash-flows associated with a perpetuity go on forever, it is possible to work out the present value of a perpetuity, using the following formula

$$PV_0 = \frac{PMT}{r}$$

The following numerical example will explain how to calculate the present value of a perpetuity, also known as a perpetual annuity cash-flow stream.

■ **Example 3.9** You are asked to work out the present value of a \$500 perpetuity discounted back to the present at 8% p.a.

$$PV_0 = \frac{\$500}{0.08} = \$6,250.00$$

Even though this annuity stream of \$500 per period goes on forever, we have been able to work out its present value at T_0 of \$6,250.00. ■

You may be asking yourself at this stage how is it possible to work out the present value at time-period zero of a payment stream that goes on forever. Although this seems impossible, the present value of a perpetuity is a very simple concept.

In the above example, we know that the payment that is to be received each year is \$500 and that the relevant interest rate is 8% p.a. This means that in order to be able to receive the perpetual stream of \$500 a year, you simply must invest \$6,250 into the bank account and collect 8% of \$6,250, that is $0.08 \times \$6,250 = \500 , each year.

In general, the present value of a perpetuity can be worked out and is simply the amount of money that must be invested at T_0 at a certain interest rate in order to earn a fixed amount of interest to give a fixed payment each time-period. As long as the original amount of money is left in the investment at the original interest rate, then the fixed annuity payment will be received forever.

3.6 Growing Perpetuity– Present Value

As stated above, a growing perpetuity is a series of cash-flows occurring at the end of each period that goes on forever and grows by a fixed percentage each period, and the present value of a growing perpetuity formula is as follows

$$PV_0 = \frac{PMT_0(1+g)}{r-g},$$

where PMT_0 is the most recently paid annuity payment and g is the constant growth rate in PMT .

The following numerical example will explain how to calculate the present value of a growing perpetuity stream of cash-flows.

■ **Example 3.10** A company has just made an annuity payment of \$10,000 to its investors and this payment is expected to grow at a rate of 3% per year and the required rate of return of the company's investors is 8% p.a. Calculate the present value (current price) of this perpetual payment stream.

We know that PMT_0 in this case (the most recent annuity payment) is \$10,000. We also know that r (the required rate of return of the investors) is 8% p.a., and g (the growth rate in the annuity payment) is 3% p.a. Therefore, PV_0 is found as follows

$$\begin{aligned} PV_0 &= \frac{PMT_0(1+g)}{r-g} \\ &= \frac{\$10,000(1.03)}{0.08 - 0.03} \\ &= \frac{\$10,300}{0.05} \\ &= \$206,000. \end{aligned}$$

Therefore, the current price of this investment (growing perpetuity stream) is \$206,000. ■

One of the very important applications for growing perpetuities is in the *dividend valuation model* for company share valuations, which is discussed extensively in Chapter 5. Following, we study a simple example, in order to introduce the concept.

■ **Example 3.11** A company has just paid a dividend to its shareholders of \$12 per share. This dividend is expected to grow at the rate of 4% per year and the required rate of return of the company's shareholders is 9% p.a. What is the current market price of this share? Equivalently, what is the present value of this perpetual payment stream?

This is an example of a growing perpetuity stream, the first payment of which is D_0 , the most recently paid dividend.

$$\begin{aligned} PV_0 &= \frac{D_0(1+g)}{r-g} \\ &= \frac{\$12(1.04)}{0.09-0.04} \\ &= \frac{\$12.48}{0.05} \\ &= \$249.60 \end{aligned}$$

Therefore, the current price of this growing perpetuity stream (share dividend) is \$249.60. ■

Highlight 3.3 — Single sums, multiple uneven cash-flows, and annuities. As we have come to the end of our discussion of single sums, multiple uneven cash-flows, and annuities for the moment, it is worthwhile to give you a quick reminder about working out the solutions to questions involving these types of cash-flows.

- First, draw a timeline— this can give you a clearer visual and mental picture of what you need to work out.
- Second, identify the class of the problem – are you dealing with a single sum, multiple uneven cash-flow, or annuity question? If you are dealing with an annuity question, is it an ordinary annuity, an annuity due, a deferred annuity, a perpetuity, or a growing perpetuity?
- Third, determine what unknown the problem involves – is the unknown the PV , FV , PMT , r , or n ?
- Finally, recognise any 'traps' in the problem – are you given an annual interest rate and more than one compounding period per year? If yes, then you must adjust r and n .

3.7 Effective Interest Rates

The effective interest rate is an investment's annual rate of interest when compounding occurs more often than once a year, and is calculated as follows

$$EFF = \left[1 + \frac{NIR}{m} \right]^m - 1$$

where EFF is the effective interest rate, NIR is the nominal, or annual, interest rate, and m is the number of times interest is compounded or calculated each year. When given an annual or nominal interest rate and told that interest is compounded more than once per year, the effective interest rate is the equivalent interest rate when interest is compounded once per year. Additionally, whenever interest is compounded more frequently than once per year, $m > 1$ and EFF will be greater than NIR .

■ **Example 3.12** The nominal interest rate (NIR) is 12% p.a. and it is compounded every quarter ($m = 4$), which means r , the interest rate per compounding period, is $3\% = 12\%/4$. Then, the effective interest rate can be found as follows

$$\begin{aligned}
 EFF &= \left[1 + \frac{0.12}{4}\right]^4 - 1 \\
 &= 1.03^4 - 1 \\
 &= 1.1255 - 1 \\
 &= 12.55\%
 \end{aligned}$$

This tells us that 12% p.a. compounded quarterly is equivalent to 12.55% p.a. compounded once per year. In other words, 12% p.a. compounded quarterly is effectively the same as 12.55% p.a. compounded annually. Therefore, if you took out a loan from a bank and were charged an interest rate of 12% p.a. compounded quarterly, in effect you would be paying 12.55% p.a. on the loan. ■

■ **Example 3.13** Assume that you are an investment adviser and that one of your clients, Vineetha, comes to see you and says that she plans to borrow a sum of money from HSBC Bank and has been offered two options in terms of the interest rate that will be charged, the first being 8.25% p.a. compounded daily, and the second being 8.5% p.a. compounded annually. Vineetha says that it seems, on face-value, that 8.25% is a lower interest rate than 8.5%. However, she does ask you to work out for her which option is best in terms of saving her money.

Since, with the first option, Vineetha is being offered a loan where an annual interest rate is being charged but that interest is compounded more than once per year, you will need to work out the effective interest rate on the first option so that she is able to see the interest rate she will be paying in reality.

For the first option, where Vineetha is being offered the loan at 8.25% p.a. compounded daily, you need to work out the effective interest rate as follows (Note that *NIR* will be 8.25% and *m* will be 365 as interest is calculated daily).

$$\begin{aligned}
 EFF &= \left[1 + \frac{0.0825}{365}\right]^{365} - 1 \\
 &= (1.000226)^{365} - 1 \\
 &= 1.0860 - 1 \\
 &= 8.60\%,
 \end{aligned}$$

with option 1, in reality or in effect, Vineetha will be paying interest on the loan of 8.60% p.a.

With option 2 Vineetha will be charged 8.50% p.a. with annual compounding, meaning that *m*, the number of compounding periods per year, is 1. Therefore, as *m* = 1 with option 2, the effective interest rate on this loan is exactly the same as the nominal interest rate, meaning that since the *NIR* is 8.50% p.a., then the *EFF* is 8.50% p.a. We can prove this using the effective interest rate formula as follows

$$\begin{aligned}
 EFF &= \left[1 + \frac{0.085}{1}\right]^1 - 1 \\
 &= (1.0850)^1 - 1 \\
 &= 1.0850 - 1 \\
 &= 8.50\%.
 \end{aligned}$$

If you are given an annual interest rate and told that interest is compounded annually, *m* = 1, then the effective interest rate will be equal to the nominal annual interest rate.

Given that you we have now calculated the effective interest rate on both loan options for Vineetha, to save money and pay less interest on her loan she should choose option 2. ■

In the above example, we have assumed that a loan is being taken out and the borrower would choose the option that gives the lowest effective interest rate. However, you must remember that if money is being invested then the best option, holding all other factors constant (such as the risk of the investments), is to choose the option that has the highest effective interest rate. This will ensure a higher rate of return on the sum invested and will give a higher value in the future.

Being able to calculate effective interest rates will become more important when we cover Weighted Average Cost of Capital in Chapter 9.

3.8 Loans

There are three main types of loans that can be taken out:

Pure discount loan

These are loans where the borrower receives money today and repays a single lump sum in the future. An example of a pure discount loan is when you borrow a sum of money from your parents, say to buy a car, and agree to repay the amount borrowed in five years' time without paying any interest over the life of the loan. Because your parents are not charging you for the time-value of the money (interest), your parents are giving you a discount on the amount they have lent you.

Interest-only loan

An *Interest-only loan* requires the borrower to pay interest each period and then to repay the entire principal at some point in the future. An example of an interest-only loan is a bond or debenture, whereby the issuer borrows the face-value of the instrument and agrees to pay to the holder of the certificate an agreed amount of interest each period, then repays to the holder the full amount borrowed (face value) on the maturity date.

Amortised loan

The third main type of loan is an *amortised loan*, which is the most common type of loan, and which requires the borrower to repay both the principal and interest over the life of the loan. If you have a loan from a bank it is most likely an amortised loan, meaning that with each payment you make on the loan you are paying-off part of the outstanding balance on the loan and also paying interest on the loan. In Business Finance it is amortised loans that we are really interested in, although you could be asked to define a pure discount loan and an interest-only loan.

3.8.1 Amortised Loan – Calculations

We now know that with an amortised loan each payment made in part pays-off a portion of the outstanding balance on the loan and also pays interest on the loan. With an amortised loan, the amount borrowed is the present value of all the repayments at the time the loan is taken out, i.e. at time-period zero. Therefore, if one knows the amount being borrowed (PV_0), the number of repayments (n) on the loan, and the interest rate per compounding/repayment period (r), then you can calculate the regular periodic payment (PMT) using the present value of an ordinary annuity formula.

The following numerical example will help to explain how to calculate the repayment on a loan.

■ **Example 3.14** Assume that you are an investment adviser and that one of your clients, Marlon, comes to see you and says that he plans to borrow \$7,500 from CBA Bank to buy a second-hand car. Marlon agrees to repay the loan by way of equal monthly repayments over five years, with an interest rate on the loan of 12% per annum, compounded monthly. What will be the amount of each monthly repayment on Marlon's loan.

The repayments that Marlon will make on the loan are a fixed annuity stream, with each payment to be made at the end of each monthly period. Therefore, you will use the present value of an ordinary annuity formula to work out his regular monthly repayment as follows

$$PV_0 = \$7,500 \quad r = 0.12/12 = 0.01, \quad n = 5 \times 12 = 60$$

Then

$$\$7,500 = PMT \left[\frac{1 - (1.01)^{-60}}{0.01} \right]$$

$$\$7,500 = PMT \times 44.955$$

$$\begin{aligned} PMT &= \frac{\$7,500}{44.955} \\ &= \$166.83 \end{aligned}$$

meaning that to pay the loan off in five years, Marlon will have to make 60 payments of \$166.83 each. ■

■ **Example 3.15** Stacey is considering taking out a housing mortgage loan of \$350,000 to be repaid by equal instalments of principal and interest over 25 years at 7.8% p.a. compounding monthly. As Stacey's investment adviser, advise her on the following:

1. The monthly instalment;
2. The interest and principal portions of the 17th payment; and
3. The amount of the loan outstanding after the 100th payment.

$$PV_0 = \$350,000 \quad r = 0.078/12 = 0.0065, \quad n = 25 \times 12 = 300$$

To find the monthly instalment we will use the PV of an ordinary annuity formula as follows

$$\$350,000 = PMT \left[\frac{1 - (1.0065)^{-300}}{0.0065} \right]$$

$$\$350,000 = PMT \times 131.82$$

$$\begin{aligned} PMT &= \frac{\$350,000}{131.82} \\ &= \$2,655.15. \end{aligned}$$

In regards to finding the interest and principal portions of the 17th payment, we must be aware that the interest component of any payment is the interest owing on the amount outstanding at the start of the payment period. In this case, interest paid in the 17th period depends on the balance of the loan outstanding at the beginning of the 17th period, i.e. just after the 16th payment has been made, and there are still 284 payments remaining, i.e. $n = 300 - 16 = 284$. Thus, as a first

step we must find the *PV* of the last 284 *PMT*s, using the *PV* of an ordinary annuity formula, where *PMT* is \$2,655.15, *r* is 0.0065, and $-n$ is 284, as follows

$$\begin{aligned} PV_{16} &= \$2,655.15 \left[\frac{1 - (1.0065)^{-284}}{0.0065} \right] \\ &= \$2,655.15 \times 129.413 \\ &= \$343,612.00 \end{aligned}$$

Interest paid in 17th payment = $0.0065 \times \$343,612 = \$2,233.48$,

which means that the principal portion of the 17th payment (the amount of the loan actually paid-off in the 17th period) is \$421.67. That is

$$\begin{aligned} \text{Principal payment} &= PMT - \text{interest payment} \\ &= \$2,655.15 - \$2,233.48 \\ &= \$421.67. \end{aligned}$$

Finally, to find the amount of the loan outstanding after the 100th payment we simply find the *PV* of the 200 payments remaining after the 100th payment, which we find as follows:

$$\begin{aligned} PV_{100} &= \$2,655.15 \left[\frac{1 - (1.0065)^{-200}}{0.0065} \right] \\ &= \$2,655.15 \times 111.7414 \\ &= \$296,690.18. \end{aligned}$$

■

In practice, most of the calculations for annuity payments and amortised loans are done by financial calculators. In the following example, we consider a comprehensive scenario in which we employ a financial calculator to perform our numerical calculations. In the appendix, the steps and keystrokes involved in using a financial calculator to find an unknown variable for an ordinary annuity, such as an amortised loan, are explained.

■ **Example 3.16 — Using financial calculator.** John is going to borrow \$200,000 to buy an investment apartment and will have monthly repayments on the loan, with the interest rate charged being 6.6% p.a., compounded monthly, for which John wishes to pay off over 25 years. As John's investment adviser, calculate what his monthly repayments will be.

In working out the answer, note the following

$$PV_0 = \$200,000 \quad r = 0.066/12 = 0.0055, \quad n = 25 \times 12 = 300$$

Then,

$$\begin{aligned} \$200,000 &= PMT \left[\frac{1 - (1.0055)^{-300}}{0.0055} \right] \\ \$200,000 &= PMT \times 146.74 \\ &= \$1,363.00. \end{aligned}$$

The repayment that John will have to make every month to get the loan paid-off in 25 years is \$1,363.00.

Now assume that John asks you to calculate how much interest he would save over the life of the loan if he pays an extra \$200 per month off the loan, i.e. if he pays \$1,563.00 per month.

We use a financial calculator to find the number of repayments on the loan, which turns out to be 222 months, i.e. 18 years and six months. Compare that to the original 300 months (or 25 years) repayment, it saves John 78 months, i.e. six years and six months, of repayments. The amount of interest that John would save by paying the extra \$200 a month is

$$\text{Total payment in 25 years} = \$1,363 \times 300 \text{ months} = \$408,900$$

$$\text{Interest paid in 25 years} = \$408,900 - \$200,000 = \$208,900$$

$$\text{Total payment in 18.5 years} = \$1,563 \times 222 \text{ months} = \$346,986$$

$$\text{Interest paid in 18.5 years} = \$346,986 - \$200,000 = \$146,986$$

Then

$$\text{Saving in interest} = \$208,900 - \$146,986 = \$61,914$$

Continuing with John, assume that he now states that he anticipates that the interest rate on the loan may rise from 6.6% p.a to 8.0% p.a. and he would like to know what his monthly payments would have to increase to in order to still get the loan paid off in the original 25 years.

We could again use a financial calculator to find that *PMT* increases to \$1,544 per month. With a repayment of \$1,544 per month the total interest that John would pay over 25 years would be

$$\text{Total payment in 25 years} = \$1,544 \times 300 \text{ months} = \$463,200$$

$$\text{Interest paid in 25 years} = \$463,200 - \$200,000 = \$263,200$$

which is \$54,300 more than the total interest paid of \$208,900, when the repayment was \$1,363 per month for 300 months.

If interest rates do rise to 8.0% p.a. and John keeps his payments at the original amount of \$1,363 per month, using a financial calculator, we would find that the time taken to pay off the loan would increase from 25 years (300 months) to 48 years (576 months). How much extra interest would John pay in this case?

$$\text{Total payment in 25 years} = \$1,544 \times 300 \text{ months} = \$463,200$$

$$\text{Interest paid in 25 years} = \$463,200 - \$200,000 = \$263,200$$

Total interest paid over 25 years at a monthly repayment of \$1,363 is \$208,900, while total interest paid over 48 years at a monthly repayment of \$1,363 is

$$\text{Total payment in 48 years} = \$1,363 \times 576 \text{ months} = \$785,088$$

$$\text{Interest paid in 48 years} = \$785,088 - \$200,000 = \$585,088$$

an increase of \$376,188 in interest paid. ■

The above result, where John is faced with larger interest bills after the rise in interest rates, highlights the fact that if a loan is taken out and the interest rate charged on the loan increases, in order for the borrower to avoid paying more interest and for the loan repayment period to increase, the periodic repayment must be increased, assuming the borrower has the capacity to increase his/her repayments.

Increasing the Frequency of Repayments on a Loan

If an individual or firm has borrowed money one way in which interest can be saved is by increasing the frequency of the repayments on the loan. Examples of this could be where a monthly repayment is divided in half and, instead, payments are made fortnightly (every two weeks), or where a monthly payment is divided by four and payments are made weekly.

In order to prove this, we will continue with the example of John, above, and show that if he were to divide his monthly repayments by two and make payments fortnightly on his loan he would save money by paying less interest on the loan.

■ **Example 3.17 — Continuing with Example 3.16.** We will begin by assuming the original situation for John where he borrowed \$200,000 over 25 years at 6.6% p.a. compounded monthly and his monthly payments are \$1,363. However, to save John money we could advise him to divide his monthly payments in half and make fortnightly payments on the loan instead, meaning that the fortnightly payments on the loan would be $\$1,363/2 = \681.50 , and the total number of payments per year would be $52\text{weeks}/2 = 26$. If this was the case, using a financial calculator, we would find that the total number of fortnightly payments of \$681.50 needed to pay off the loan would be $n = 539$. This implies it would now take 20 years and nine months for John to pay off the loan, which would save him four years and three months to pay off the loan.

In terms of the money John would save, by paying fortnightly John is paying $26 \times \$681.50 = \$17,719$ a year, which is greater than the amount he would pay if paying monthly $12 \times \$1,363 = \$16,356$. This means that John would pay \$1,363 more a year on the loan, which on its own will save John $\$1,363 \times 4.25 \text{ years} = \$5,792.75$. However, the majority of the money John will save would come from the increased frequency of the payments, and not from the fact that his total repayments each year would be higher.

How much interest will John save in total by paying fortnightly instead of monthly? For the monthly payments total interest paid is \$208,900, while for fortnightly payments total interest paid is

$$\text{Total payment} = \$681.50 \times 539 \text{ months} = \$367,328.5$$

$$\text{Interest paid} = \$367,328.5 - \$200,000 = \$167,329$$

This means by paying fortnightly John saves \$41,571 in interest. ■

Highlight 3.4 — Is It Better to Increase the Frequency of Payments?. In the above example, by increasing the frequency of his payments John saves for two reasons. Firstly, he will pay more in total each year. Secondly, and more importantly, when John makes a half-payment mid-way through the month this means the balance of the loan outstanding at the end of the month is slightly less than what it otherwise would have been, meaning slightly more of the second payment per month goes towards paying off the principal of the loan and slightly less goes towards paying off interest. There is only a small difference each month but over 20 or 30 years it makes a big difference.

3.9 Revision Problems

Problem 3.1 An interest rate of 10% p.a. compounding quarterly represents and effective interest rate of:

- (a) 10.18% p.a.
- (b) 10.25% p.a.
- (c) 10.38% p.a.
- (d) 10.50% p.a.

Problem 3.2 Five years ago Chris entered into a loan agreement with Bangkok Bank to borrow \$200,000 and repay the loan over 20 years through equal monthly instalments. If the interest rate was fixed at 8% p.a. for the entire term of the loan, what is the amount of each monthly instalment?

- (a) \$1,333
- (b) \$16,667
- (c) \$833
- (d) \$1,673

Problem 3.3 Frank Lewis has a 30-year, \$500,000 mortgage with Westpac Bank at a nominal interest rate of 8% p.a. and monthly compounding. Which of the following statements regarding his mortgage is most correct?

- (a) The monthly payments will decline over time.
- (b) The proportion of the monthly payment which represents interest will be lower for the last payment than for the first payment on the loan.
- (c) The total dollar amount of principal being paid off each month gets larger as the loan approaches maturity.
- (d) Statements (b) and (c) are correct.

Problem 3.4 You bought a painting 10 years ago as an investment and you paid \$85,000 for it. If you sold it for \$484,050 what was your annual (compounded) return on investment?

- (a) 47%
- (b) 4.7%
- (c) 12.8%
- (d) 19%

Problem 3.5 Assuming an interest rate of 10% p.a., what is the amount of money a person must be given now to make them indifferent to receiving a stream of five annual payments of \$50 each, where the first payment is received immediately:

- (a) \$208
- (b) \$190
- (c) \$305
- (d) \$232

Problem 3.6 A bank offers an interest rate of 6% p.a. compounding semi-annually. How much would the future sum be if you made six semi-annual deposits of \$50 each and the last deposit earned interest for one period?

- (a) \$185
- (b) \$333
- (c) \$271
- (d) \$365

Problem 3.7 Joanne borrows \$20,000 from HSBC Bank to buy a car. The loan involves equal annual repayments over 20 years and $r = 10\%$ p.a.

- (a) How much are Joanne's annual repayments?
- (b) What is the value of the interest element of the 14th repayment?
- (c) How much is needed to pay off the loan at the time of the 16th repayment assuming the 16th repayment has not been made?

Appendix: Hewlett Packard 20b

3.10 Financial Calculator Steps and Keystrokes

The Hewlett Packard 20b Business Consultant is a very easy to use financial calculator that will serve you well in all finance courses. This tutorial will demonstrate how to use the financial functions to handle time value of money problems and make financial math easy. Examples provided are rather elementary, but understanding the basics is all that is necessary to learn the calculator. Please note that in the following text the blue key (above the **ON/CE** key) is referred to as **Shift** because it is used to shift to the blue-colored function below the key that is pressed next.

Initial Steps

Before we get started, we need to correctly set up the calculator. The 20b comes from the factory set to assume monthly compounding. In order to change this, for example to annual compounding, use the following steps:

Step 1: press 1 (once per year)

Step 2: press **Shift**

Step 3: and finally press **PMT**

To check that it has taken.

Step 1: press **RCL** (recall)

Step 2: press **Shift** **PMT**

You should see $P/YR = 1$ on the screen. Now, just make sure that you always enter the total number of periods (not necessarily years) into **N**, the per period interest rate into **I/YR**, and the per period payment into **PMT**.



Present Value of Annuities

Suppose that you are offered an investment that will pay you \$1,000 per year for 10 years. If you can earn a rate of 9% per year on similar investments, how much should you be willing to pay for this annuity?

In this case we need to solve for the present value of this annuity since that is the amount that you would be willing to pay today.

Step 1: enter 10 → **N**

Step 2: enter 9 → **I/YR**

Step 3: enter 1000 (a cash inflow) → **PMT**

Step 4: enter 0 → **FV**

Step 5: press **PV** to solve for the present value

The answer is -6,417.6577. Note that this is negative because it represents the amount that you would have to pay (a cash outflow) to purchase this annuity.

If you got the wrong answer it is almost certainly because you have P/YR set to something other than 1, or you are in BEGIN mode (press **Shift** **PV** to return to END mode).

Solving for the Payment Amount

We often need to solve for annuity payments. For example, you might want to know how much a mortgage or auto loan payment will be. Maybe you want to know how much you will need to save each year in order to reach a particular goal (saving for college or retirement).

Suppose that you are planning to send your daughter to college in 18 years. Furthermore, assume that you have determined that you will need \$100,000 at that time in order to pay for expenses. If you believe that you can earn an average annual rate of return of 8% per year, how much money would you need to invest at the end of each year to achieve your goal?

Step 1: enter 18 → **N**

Step 2: enter 8 → **I/YR**

Step 3: enter 100,000 → **FV**

Step 4: enter 0 → **PV**

Step 5: press **PMT** to solve for the payments amount

You need to invest \$2,670.21 per year for the next 18 years to meet your goal of having \$100,000.

Solving for the Number of Periods

Solving for N answers the question, “How long will it take...”:

Imagine that you have just retired, and that you have a nest egg of \$1,000,000. This is the amount that you will be drawing down for the rest of your life. If you expect to earn 6% per year on average and withdraw \$70,000 per year, how long will it take to burn through your nest egg (in other words, for how long can you afford to live)? Assume that your first withdrawal will occur one year from today (End Mode).

Step 1: enter 6 → **I/YR**

Step 2: enter -1,000,000 → **PV** (negative because you are investing this amount)

Step 3: enter 0 → **FV**

Step 4: enter 70,000 → **PMT**

Step 5: press **N** to solve for the number of payments.

You will see that you can make 33.40 withdrawals. Assuming that you can live for about a year on the last withdrawal, then you can afford to live without working for about another 34.40 years.

Solving for the Interest Rate

Solving for I/YR works just like solving for any of the other variables. Be sure to pay attention to the signs of the numbers that you enter into the TVM keys. Any time you are solving for N, I/YR, or PMT there is the potential for a wrong answer or error message if you do not get the signs right. Let us look at an example of solving for the interest rate:

Suppose that you are offered an investment that will cost \$925 and will pay you interest of \$80 per year for the next 20 years. Furthermore, at the end of the 20 years, the investment will pay \$1,000. If you purchase this investment, what is your compound annual rate of return?

Step 1: enter 20 → **N**

Step 2: enter 80 → **PMT**

Step 3: enter 1000 → **FV**

Step 4: enter -925 → **PV**

Step 5: press **I/YR** to solve for the annual rate of return

You will find that the investment will return an average of 8.81% per year.

Annuities Due

In the examples above, we assumed that the first payment would be made at the end of the year, which is typical. However, what if you plan to make (or receive) the first payment today? This changes the cash flow from a regular annuity to an annuity due.

Normally, the calculator should be working in "End Mode". It assumes that cash flows occur at the end of the period. In this case, though, the payments occur at the beginning of the period. Therefore, we need to put the calculator into "Begin Mode". To change to "Begin Mode", press **Shift** **PV**. The screen will now show BEG in the upper-right corner. Note that nothing will change about how you enter the numbers. The calculator will simply shift the cash flows for you. Obviously, though, you will get a different answer than if you were in "End mode".

Be sure to switch back to "End Mode" after solving the problem. Since you almost always want to be in "End Mode", it is a good idea to get in the habit of switching back. Press **Shift** **FV**. When in End Mode, you will not see any indication on the screen. Only when you are in "Begin Mode" do you see BEG on the screen.

Debt Financing

- Government Financing
- Cost of Debt

Bond

- Bond Pricing Model
- Cost of Debt and Yield To Maturity
- Bond Price Sensitivity to YTM

Interest Rate Risk

- Yield Curve
- Explaining the Shape of the Yield Curve

Revision Problems

4. Bond Valuation

This chapter is designed to give you an understanding of valuation methods for debt financing, particularly bonds. Amongst other things, we will look at bond valuation issues and concepts such as the bond pricing model, yields to maturity, and the term structure of interest rate. Additionally, We will cover concepts and issues such as the different sources of finance available to a firm; bonds terminologies; the bond pricing model; the cost of debt and yield to maturity; interest rate risk; yield curves; expectations theory and the term structure of interest rates; and how to derive an unknown interest rate.

4.1 Debt Financing

Debt financing refers to where a company raises capital, or finance, by borrowing money. The firm may borrow the money from a traditional lending institution, such as a bank or building society, in the form of a business loan, mortgage, or overdraft, for example. It may issue debt securities to institutional and/or private investors to raise finance, for example, issue bonds or bond-like instruments, such as debentures or term loans, as well as short-term instruments such as bills or notes. In return for lending the money, the individuals or institutions become creditors and receive a promise that the principal and interest on the debt will be repaid. In other words, if the debt payments are not received by the creditors, when they fall due, they can seek legal redress (i.e. undertake legal or court action) to receive their payments, and ultimately take control of the assets of the company to which they have lent money.

Business Snapshot 4.1 — Virgin raises \$344 million in its first corporate bond issue.

In November 2014 Virgin Australia raised \$US300 million (\$344 million) from its first corporate bond issue to global investors to shore-up its balance sheet as it takes full control of loss-making budget airline Tigerair. The deal involved the sale of unsecured notes to investors primarily in the US and was priced at a coupon rate of 8.5 per cent, equivalent to 6.86 per cent above US Treasury bonds. Similarly rated American Airlines' bonds sold in September 2014 are trading at about 5.5 per cent, signifying the premium Virgin has paid to debut in the market. Australia's second-largest airline tapped capital markets in the US in October 2013 in a \$770 million financing deal whereby it used 24 aircraft as security. In contrast, Virgin's first public issue was unsecured.

4.1.1 Government Financing

Governments have a number of means of raising finance to support their spending. Firstly, and most obviously, governments levy taxes on individuals and businesses within their own country in order to raise finance. Additionally, governments raise funds via issuing of fines and penalties (such as speeding-fines) and the charging of fees for services they provide. For example, the fee you pay when applying for a passport. Also, in many countries governments are involved in productive economic activities, such as through state-owned enterprises in China, which generate revenues for governments. Another way that governments are able to raise funds is by simply printing money, although this has been shown in the past (for example, Germany in the 1920s and 1930s) to have the potential to cause economic and social problems, such as hyper-inflation. Governments can also raise capital by issuing bonds in IPOs in primary markets. For example, the Australian government often issues Commonwealth Government Bonds when it needs to raise finance, while the US government issues Treasuries, the German government issues *Bunds*, the British governments issues *Gilts*, and the Japanese government issues *JGBs* (Japanese Government Bonds).

Business Snapshot 4.2 — Strong demand for Australian Government Bond issue. The Australian Federal Government has issued a record \$7 billion, 12-year bond – the government’s largest ever syndicated debt raising^a. The April 2026 bond issue surpassed November 2013’s record \$5.9 billion, 20-year bond issue, putting the government ahead of its debt funding run rate while extending the maturity of its debt. The yield on the bond was 4.375%.

The Australian Office of Financial Management (AOFM), which manages the government’s debt issuance, has now raised \$61.5 billion in the current financial year, just \$13.5 billion shy of its stated \$75 billion fund raising target. Rob Nicholl, the Chief Executive of the AOFM said that the bond offer attracted an order book of \$10 billion with half of the bids coming from fund managers and central banks. Around a third of the buyers were foreign while domestic investors accounted for two thirds, a reversal of demand for November’s 20 year bond. About a third of the total deal was sold to Asian investors. Central banks accounted for 16% of the \$7 billion, while hedge funds were allotted 18% of the deal.

^aThe Age, 12/03/2014, Jonathan Shapiro.

4.1.2 Cost of Debt

The cost of debt finance for the borrowing firm is the interest expense that must be paid at agreed points in time on the money borrowed, for example, monthly, half-yearly, or yearly. These periodic cash-flows are fixed in terms of their size or value, assuming that the interest rate or term of the loan does not change. Generally, with debt finance, the loan is of a fixed term with a maturity date specified for some time in the future. For example, with a bond it may be a 10-year bond, meaning that the original amount borrowed must be repaid in 10 years’ time. In the case of a mortgage loan, the term may be 25 years, meaning that the loan must be paid-off within 25 years, and because the full amount borrowed must be repaid we say the redemption on debt finance is in full at the face-value of the debt, i.e. at the value of the original amount borrowed. As previously stated, debt-holders have a legal right to receive payment on the money they have lent, so the relative riskiness of debt investment for the lender is lower than for equity/share investment.

Business Snapshot 4.3 — Euro-zone sovereign debt crisis.

Since the onset of the global financial crisis (GFC) in 2007/08, Europe has been subject to a sovereign debt crisis, with a number of European governments in financial difficulty. For the governments in trouble, particularly the governments of Portugal, Italy, Ireland, Greece, and Spain (who are collectively referred to as the 'PIIGS'), this has meant that they have had difficulty in raising finances through issuing bonds in bond markets.

The reason for this is that investors have become very wary of the ability of these governments to pay their debts, which has meant that either investors will not buy the bonds of the governments of some countries (particularly Greece) or are demanding that very high interest rates (coupon rates) be paid on the bonds issued by some of the other countries (particularly Italy, Spain, and Portugal). Consequently, to partly solve the problem, organisations such as the ECB (European Central Bank) and the World Bank have had to either provide temporary funding directly to governments (in the case of Greece) or purchase the bonds of other governments in bond markets (particularly those of Italy, Spain, and Portugal). Below is a very brief history of the European sovereign debt crisis taken from 'The Economist Magazine' that helps explain its causes and consequences.

The European Sovereign Debt Crisis: A Very Short History^a

In German eyes this crisis is all about profligacy. Greece set the tone when it lied about its circumstances and lived beyond its means. There is no disputing Greek dissipation, nor the fact that the euro zone's troubled members, which also include the PIIGS, must now pay a heavy price. But those other troubled countries were not exactly profligate. Before the crisis the governments of both Ireland and Spain ran budget surpluses. Both meticulously kept within the limits for deficits and debts set down by the stability and growth pact. Unlike Germany, which flouted the rules for four years from 2003, and avoided punishment. Nor did Italy lurch into extravagance.

Even where troubled euro-zone countries had not been profligate, they have been running unsustainable current-account deficits. Low interest rates fuelled domestic spending and spurred inflation in wages and goods, which in turn made their exports more expensive and left imports relatively cheaper. But it was also because Germany was recycling the surpluses produced by its export machine, financing their consumption.

Germany's economy is remarkable in many ways, but it was as unbalanced as the euro zone's peripheral economies. In their determination to save, Germans seemed to forget that in the long run the point of exports is to pay for imports. They must now regret having invested their savings abroad in American sub-prime mortgages and Greek government debt.

Your debt, your fault

To end the crisis, the euro zone members agreed last month (October 2011) to write down half of the Greek debt owned by the private sector, recapitalise Europe's banks and boost the fund created as a firewall to protect solvent euro-zone governments. It is an ambitious plan, but Greece may need even more help and the firewall does not look strong enough to withstand a bout of contagion.

Even when the crisis has abated, restoring Europe to health will take many years. That is because the troubled countries need to control their governments' deficits and to re-establish sound current accounts by improving their competitiveness. Germans feel that

the responsibility for this lengthy adjustment lies exclusively with borrowers, which must urgently restore budget discipline. Significantly, the German word for debt, *Schulden*, is the plural of *Schuld*, meaning guilt or fault.

However, this strategy risks being self-defeating. By pushing for immediate austerity the euro zone is deepening recession in the troubled economies, which will only make their debt harder to service. Germany's approach suffers from a fallacy of composition. It is not possible for everyone to save their way to prosperity. As Keynes argued after the Depression, someone, somewhere must be consuming. In Europe that should be countries such as Germany and the Netherlands that were running vast current-account surpluses during the boom. But the creditors are loath to accept that they are part of the problem.

Creditor governments, most of all Germany, face a dilemma. They need to save troubled governments in order to prevent contagion. On the other hand, they also want to keep up market pressure for reforms and to establish the principle that governments are on their own, so that German taxpayers will not be landed with the bill every time some EU country goes on a spending spree. So far, Germany is trying to have it both ways and succeeding only in getting everyone deeper into the mire.

^aThe Economist, November 12th 2011

4.2 Bond

A *bond* is a debt investment in which an investor loans money to an entity (corporation or governmental) that borrows the funds for a defined period of time at a fixed interest rate. Bonds are used by companies, as well as state and federal governments to raise finance for their projects and activities. Bonds are commonly referred to as fixed-income securities or assets, because the interest payment that is made by the borrower is of a fixed dollar amount, and are one of the three main asset or investment classes, along with shares and cash (or cash-equivalents).

With bonds, the bond issuer, who is the borrower, issues the bond which will state the interest rate (coupon rate), and the frequency of coupons that will be paid. It also states when the loaned funds (bond principal or face value) are to be returned (maturity date). Interest on bonds is usually paid every six months (semi-annually).

The two main features that determine the interest rate to be paid on a bond are

- The credit worthiness of the bond issuer. The higher the credit worthiness then the lower the interest rate that will be paid.
- The duration of the bond, or how far in to the future is the bond's maturity date. Generally, the longer the maturity on the bond the higher the interest rate paid.

Bond maturities range from 90-days up to 30-years on some government bonds, while corporate bonds are usually in the three to 10-year maturity range.

Another type of bond is called a *debenture*, which has very similar characteristics to a bond except that debentures are unsecured, meaning that the payment of interest and principle on the debenture is guaranteed only by the creditworthiness of the borrower. This means that if the borrower fails the lender may not get their money back as easily as with a bond, which is secured. When a loan is secured, specific assets of the borrower have been pledged as security for the bondholders, in the event that the borrower is in financial difficulty and cannot meet the payments on the bonds.

Coupon and Maturity	Date of first issue	Face Value (AU\$m)	Next Coupon Payment Date
6.25% 15 April 2015	28 April 2002	14797	15 May 2015
4.75% 21 October 2015	01 July 2011	13899	21 May 2015
4.75% 15 June 2016	07 July 2010	21900	15 December 2014
6.00% 15 February 2017	08 June 2004	21096	15 February 2015
4.25% 21 July 2017	02 September 2011	18900	21 January 2015
5.50% 21 January 2018	24 November 2010	20500	21 January 2015
3.25% 21 October 2018	06 December 2013	10900	21 May 2015
5.25% 15 March 2019	17 January 2006	20847	15 March 2015
2.75% 21 October 2019	18 July 2014	7600	21 May 2015
4.50% 15 April 2020	29 May 2009	20397	15 May 2015
5.75% 15 May 2021	11 September 2007	21599	15 April 2015
5.75% 15 July 2022	07 May 2010	17500	15 January 2015
5.50% 21 April 2023	18 April 2011	21300	21 May 2015
2.75% 21 April 2024	20 June 2012	18700	21 May 2015
3.25% 21 April 2025	22 April 2013	13800	21 May 2015
4.25% 21 April 2026	12 March 2014	12800	21 May 2015
4.75% 21 April 2027	20 October 2011	13000	21 May 2015
3.25% 21 April 2029	10 October 2012	9000	21 May 2015
4.50% 21 April 2033	19 November 2013	8500	21 May 2015
3.75% 21 April 2037	15 October 2014	7000	21 May 2015

Table 4.1: Australian Commonwealth Government Treasury Bonds, as quoted by the Australian Office of Financial Management (AOFM)

Highlight 4.1 — Bond terminologies. Following is the list of terms used for Bonds.

Face value, which is the original price of the bond and the amount borrowed by the bond issuer. The face value will be printed on the front or “face” of the bond certificate.

Par value of a bond is the amount of finance or capital that the issuer receives on the sale of each bond in an IPO in the bond market.

Maturity date, which is the date the loan or amount borrowed (face value) must be repaid in full to the bond-holder.

Coupon rate, which is the rate of interest to be paid to the bond holder periodically by the bond issuer, as well as the periodic intervals at which the coupon rate will be paid (for example, yearly, half-yearly, or quarterly)

Once the coupon rate on the bond is known, it is then possible to work out the *coupon payment* that will be paid periodically on the bond. To do this you would simply multiply the coupon rate of the bond by the bond’s face value. For example, if a bond has a face value of \$1,000 and coupon rate of 10% with coupon payments once per year, then the annual coupon payment on the bond would be 10% of \$1,000, which works out to be \$100.

4.2.1 Bond Pricing Model

In order to price a bond, that is, find its current market value or the present value of cash-flows associated with the debt security, you must use the bond price formula, which is as follows

$$\text{Bond price} = PMT \left[\frac{1 - (1 + r_d)^{-n}}{r_d} \right] + \text{Face Value}(1 + r_d)^{-n},$$

where

PMT is the regular fixed coupon (interest) payment to be received at the end of each investment period;

“Face Value” is the original amount borrowed per bond by the bond issuer which will be repaid at the end of period n , i.e. on the maturity date;

r_d is the bond investor’s required rate of return, i.e. the current cost of debt; and

n is the number of coupon payments/investment periods left until maturity of the bond

4.2.2 Cost of Debt and Yield To Maturity

Looking at the bond price formula, one may ask how the cost of debt, i.e. r_d , is determined. This is an important question because the r_d used in the formula will have a significant impact on the current market price of the bond, i.e. the price that an investor is willing to pay for a bond now.

What we must remember here initially is that the r_d used to price a bond will be related to current interest rates in the market for similar risk bonds, as well as being related to the bond investor’s required rate of return, and is referred to as the current yield, or yield to maturity (YTM) on the bond.

In order to calculate r_d one must begin with at least the risk free rate of interest¹ (R_f) as the opportunity cost, then add compensation for anticipated inflation (Inf), a risk premium (R_p), and a term to maturity premium (R_{tm}). The risk premium for time value of money is related to the risk of the cash-flows to cover default risk, or is related to the term to maturity (R_{tm}) of the bond. Generally, as previously stated, the longer the term to maturity on the bond the higher will be the interest rate charged on the bond. This is because, the longer out in time we go, the harder it is to predict with accuracy what will happen. Therefore, the longer the maturity on the bond, the greater the uncertainty and, therefore, the greater the risk. Hence,

$$r_d = R_f + Inf + R_p + R_{tm}.$$

The current cost of debt (r_d) on any debt instrument in the market is referred to as its yield to maturity (YTM). It is the required rate of return that investors expect to receive from the lending and is the prevailing market interest rate that equates the instrument’s present value of interest and principal payments to its current price. Due to the inverse relationship between bond price and YTM, if you pay a higher price for a bond, your rate of return will be lower, while if you require a higher rate of return on the bond, then you must pay a lower price for it, *ceteris paribus*. The following simple example will help explain this concept:

■ **Example 4.1** Suppose that you purchase an investment asset in 2015 for \$1 million and your annual dollar return on the asset (rental income) is \$50,000. This means that your annual rate of return will be

$$\frac{\$50,000}{\$1 \text{ million}} \times 100 = 5\%$$

¹ A full discussion about risk free rate is provided in Section 7.2.1

However, if instead of paying \$1 million for the asset you had paid \$2 million, and your dollar return stays the same, then your annual rate of return is only

$$\frac{\$50,000}{\$2 \text{ million}} \times 100 = 2.5\%$$

Thus, if you pay a higher price and your dollar return stays the same, your YTM falls. But what if, instead of paying more than the original \$1 million, you had paid less, say only \$500,000 for the same annual dollar return of \$50,000. Then the YTM will be

$$\frac{\$50,000}{\$500,000} \times 100 = 10\%$$

Here, you have paid a lower price and your return (YTM) has gone up. ■

4.2.3 Bond Price Sensitivity to YTM

We now know that a bond's price is inversely related to its yield to maturity. We can see how this works graphically in Figure 4.1, below, where we are assuming that we have a bond with a face value of \$1,000, a 10% p.a. coupon rate (meaning that the annual coupon payment will be \$100), and 20 years until maturity.

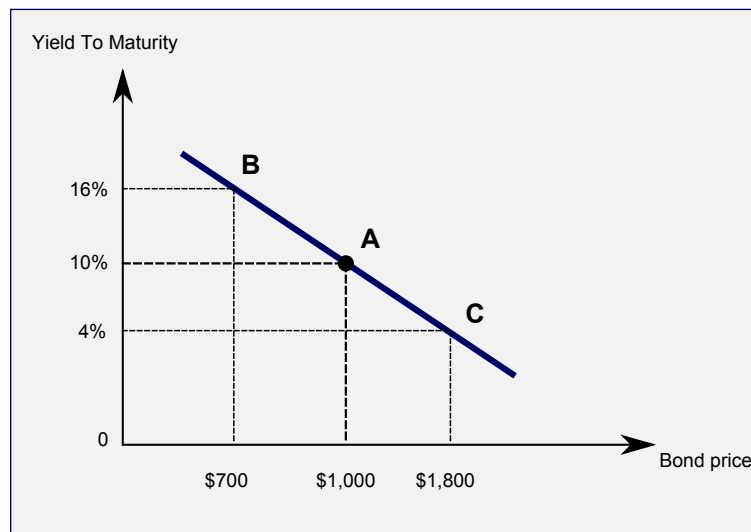


Figure 4.1: Bond price versus different yields

So, when YTM is 10% p.a the bond's price is \$1,000, equal to its face value, because the YTM equals the annual coupon rate on the bond. However, when the YTM on the bond is higher at 16%, the price of the bond falls to \$700, because for the investor to get a higher return they must pay a lower price given that the coupon payment and maturity on the bond are unchanged. Conversely, if the price of the bond in the market rises to \$1,800, the YTM falls to 4% p.a., as the investor is being forced to pay a higher price, but, again, the coupon payment and maturity on the bond are unchanged.

Calculating Bond Price

In order to calculate the current market price of a bond, use the bond price formula specified above and note the face value of the bond, the coupon rate and coupon payment, the term to maturity, and the current yield in the market for such a bond.

■ **Example 4.2** Suppose our company decides to issue 20-year bonds to raise finance and these bonds have a face value of \$1,000 with annual coupon payments. The return on other bonds of similar risk is 12% p.a., so we have decided to offer a 12% coupon interest rate on these bonds, meaning that the annual coupon payment we will pay on each bond is $0.12 \times \$1,000 = \120

Suppose, also, that an investor in the market is keen to purchase one of these bonds and the investor's required rate of return, YTM, is 12% p.a. What price will the investor pay for the bond?

To find the price the investor will pay we will use the bond price formula as follows (and remember, we use YTM as r_d in the formula):

$$\begin{aligned} \text{Bond price} &= PMT \left[\frac{1 - (1 + r_d)^{-n}}{r_d} \right] + \text{Face Value}(1 + r_d)^{-n} \\ &= \$120 \left[\frac{1 - (1.12)^{-20}}{0.12} \right] + \$1,000(1.12)^{-20} \\ &= \$120[7.4694] + \$1,000(0.1037) \\ &= \$1,000. \end{aligned}$$

We see that the investor would pay \$1,000 for this bond, which makes sense as the investor's required rate of return is exactly the same as the coupon rate of 12% p.a., so the current market price of the bond equals the face value.

Suppose, now, that there is a second investor who's required rate of return on the bond is 10% p.a. What would this investor pay for the bond?

$$\begin{aligned} \text{Bond price} &= PMT \left[\frac{1 - (1 + r_d)^{-n}}{r_d} \right] + \text{Face Value}(1 + r_d)^{-n} \\ &= \$120 \left[\frac{1 - (1.10)^{-20}}{0.10} \right] + \$1,000(1.10)^{-20} \\ &= \$120[8.5136] + \$1,000(0.1486) \\ &= \$1,170.23. \end{aligned}$$

Using 10% as the r_d we see that the second investor would pay a higher price for the bond, this being \$1,170.23, which makes sense as this investor is prepared to accept a lower rate of return on the bond (of only 10% p.a.), so is willing to pay a higher price. Remember, we are assuming that the coupon rate and coupon payments on the bond, as well as the term to maturity, are being held constant.

Now suppose a third investor comes along who would like an annual return on these bonds of 14%, in this case YTM and $r_d = 14\%$. What would this investor pay for the bond?

$$\begin{aligned} \text{Bond price} &= PMT \left[\frac{1 - (1 + r_d)^{-n}}{r_d} \right] + \text{Face Value}(1 + r_d)^{-n} \\ &= \$120 \left[\frac{1 - (1.14)^{-20}}{0.14} \right] + \$1,000(1.14)^{-20} \\ &= \$120[6.6231] + \$1,000(0.0728) \\ &= \$867.57. \end{aligned}$$

Using 14% as the r_d , we see that the third investor would pay a lower price for the bond, this being \$867.57, which makes sense as this investor is after a higher rate of return on the bond

(of 14% p.a.), so will only pay a lower price for the bond. Recall the coupon rate and coupon payments on the bond, as well as the term to maturity, are being held constant.

Highlight 4.2 — Premium vs discount. The general result, in terms of bond price, is that if YTM goes down the price of the bond will go up and if YTM goes up the price of the bond will go down. Further, if the YTM on the bond equals the coupon rate on the bond then the bond will sell for its face value in the market, while if the YTM on the bond is less than the coupon rate on the bond then the bond will sell for a *premium*. Further, if the YTM on the bond is greater than the coupon rate then the bond will sell for a *discount* in the market

4.3 Interest Rate Risk

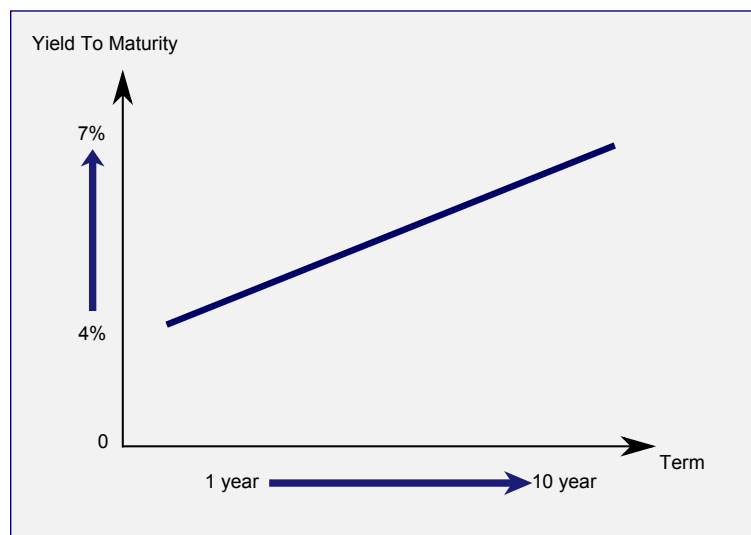


Figure 4.2: Bond maturities and yields

Interest rate risk is the risk that an investment or asset's value will change due to a change in the level of interest rates, and such changes usually effect securities inversely. Interest rate risk effects the value of bonds more directly than shares and it is a major risk to all bondholders. As interest rates rise, bond prices fall and vice versa. The rationale is that as interest rates increase, the opportunity cost of holding a bond decreases since investors are able to realize greater yields by switching to other investments that reflect the higher interest rate. For example, a 5% bond is worth more if interest rates decrease since the bondholder receives a fixed rate of return relative to the market, which is offering a lower rate of return as a result of the decrease in rates. Hence, demand for the bond will rise and so will its price, meaning that the bondholder will achieve a capital gain on sale of the bond.

There are two main types of interest rate risk that bondholders face, namely, *term-to-maturity risk* and *coupon rate risk*.

Term to maturity risk relates to the fact that, all other things being equal, the longer the term to maturity on a bond, the greater the interest rate risk. This is because, the further out in time we go the harder it becomes to predict what will happen with accuracy, and therefore, the greater the uncertainty and the greater the risk. Thus, in a normal market, with a normal upward-sloping yield curve, the yield on a 10 year bond will be higher than the yield on a 1 year bond, *ceteris*

paribus. Notice from Figure 4.2, that in a normal market the bond yield curve is upward sloping and the rate of return on a 10-year bond will be higher than the rate of return on an equivalent one-year bond.

Coupon rate risk relates to the fact that, all other things being equal, the lower the coupon rate on a bond the greater the interest rate risk. This is because the lower is the coupon rate the more likely it is that in the future interest rates in the market will be above the coupon rate. Remember, if current market interest rate goes above the coupon rate on a bond then the price of the bond will fall below the face value and the bond holder will suffer a capital loss.

As in Figure 4.3, if the coupon rate on a bond is relatively low then there is a greater chance that the bondholder will suffer a loss in the future, whereas if the coupon rate on the bond is relatively high there is a greater chance that the bondholder will achieve a profit in the future.

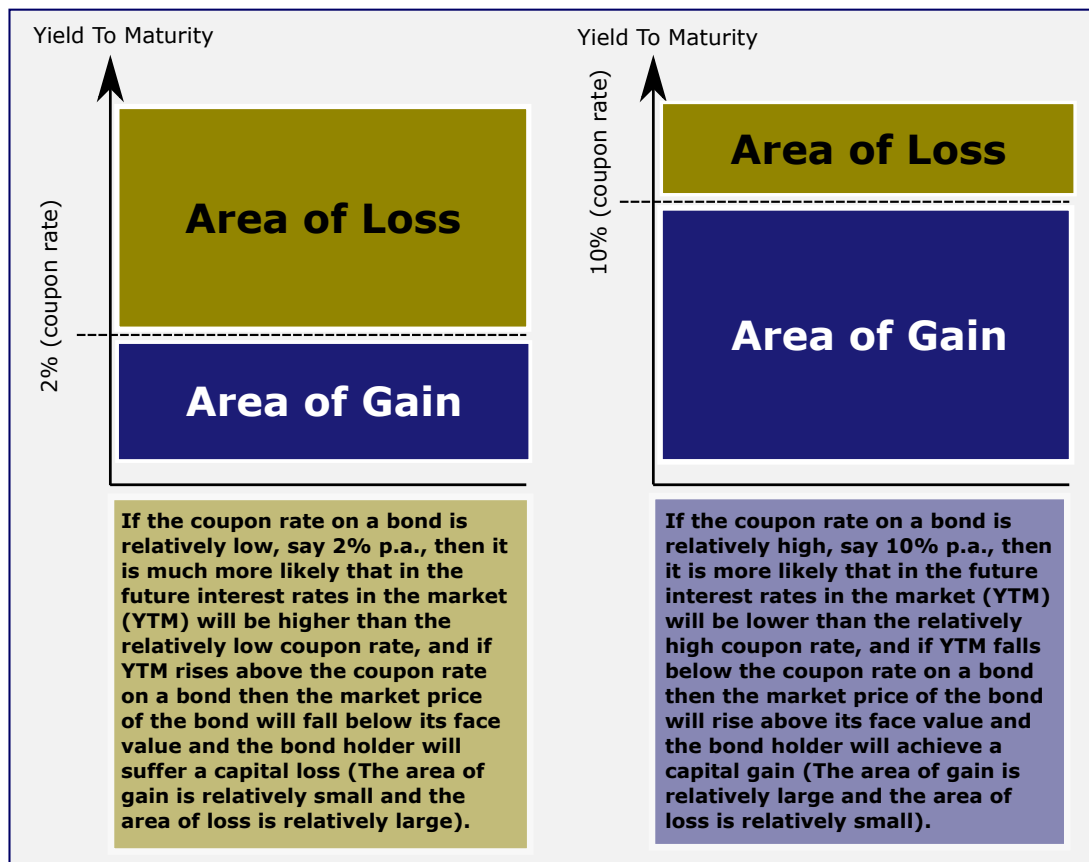


Figure 4.3: Coupon rate risk

4.3.1 Yield Curve

A yield curve is a line that plots interest rates, at a set point in time, of bonds that have the same credit quality, but differ in terms of their maturity dates. Yield curves are used to predict changes in economic output and growth. The shape of the yield curve helps to give an idea of what is expected to happen to interest rates in the future and to economic activity in an economy or market. There are three main types of yield curve shapes, which are as follows

- **The normal (upward sloping) yield curve:** A normal yield curve is one that is upward sloping over time and in which longer maturity bonds have a higher yield compared to shorter-term bonds due to the risks associated with time.

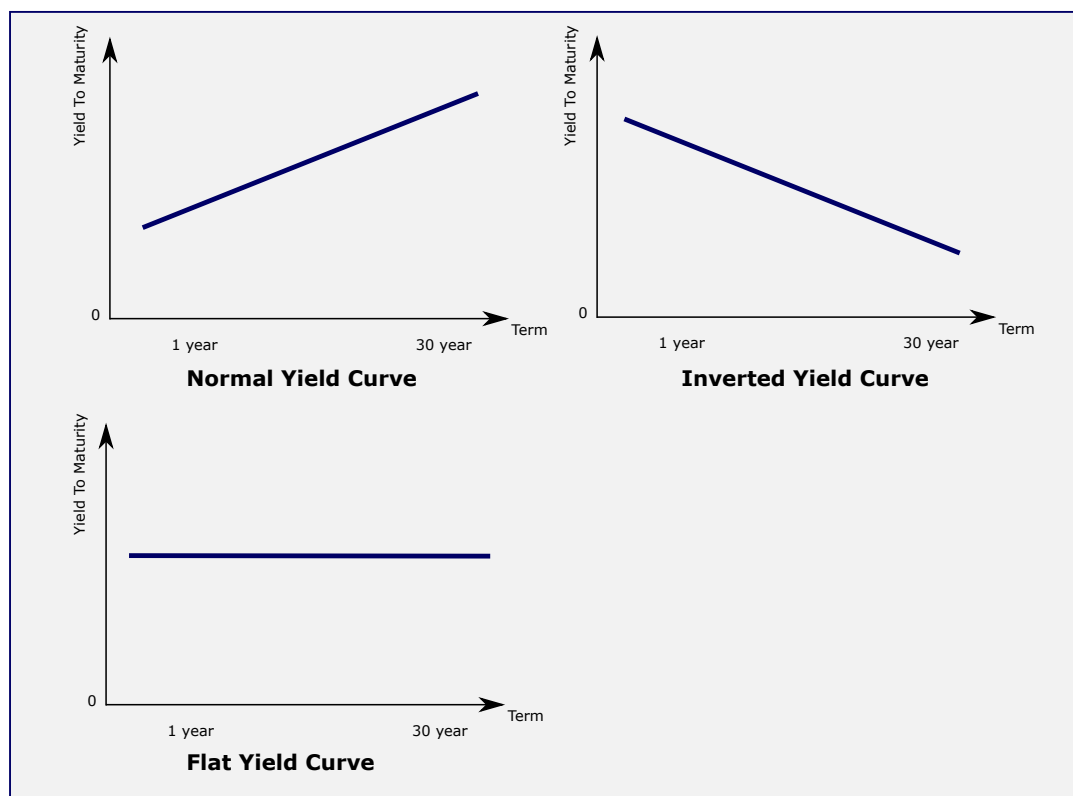


Figure 4.4: Different shapes of the yield curve

- **The inverted (downward sloping) yield curve:** An inverted yield curve is downward sloping over time and is one in which the shorter-term yields are higher than the longer-term yields. This is often taken as a sign of trouble ahead for an economy, such as a recession.
- **The flat yield curve:** A flat yield curve is one in which shorter and longer-term yields are very close to or the same as each other.

Figure 4.4 presents a plot of the various possible shapes of the yield curve.

4.3.2 Explaining the Shape of the Yield Curve

There are a number of theories that are used to explain the shape of the investment yield curve. The three most common theories are *Expectations Theory*, the *Liquidity Premium Theory*, and the *Market Segmentation Theory*. While all three theories are important in finance, in the subject Business Finance, we will concentrate only on Expectations Theory.

Expectations Theory in finance is the hypothesis that long-term interest rates contain a prediction of future short-term interest rates. Additionally, it states that if the market is operating efficiently and correctly then you would earn the same amount of interest by investing, for example, in a one-year bond today and then rolling that investment into a new one-year bond a year later, compared to buying a two-year bond today.

However, it should be noted that Expectations Theory is not universally accepted in the world of finance. While this theory is often used to explain the yield curve, it has sometimes proven inaccurate in practice as interest rates tend to remain flat when the yield curve is normal. In other words, Expectations Theory often overstates future short-term interest rates. An alternative theory that expands on Expectations Theory is preferred habitat theory, which is used to explain

why, in reality, longer-term bonds tend to pay higher interest than two shorter-term bonds that add up to the same maturity. Preferred Habitat Theory says that investors prefer short-term bonds and are only interested in longer-term bonds if they pay a risk premium. So, while Expectations Theory assumes that investors only care about yield, Preferred Habitat Theory assumes they care about maturity as well as yield.

The following example will explain how Expectations Theory works in predicting future interest rates.

Suppose the interest rate prevailing over Year 1 is 10% and the one year interest rate expected to prevail in Year 2 is 15%. If an investor invests \$100 for one year in Year 1, when the interest rate is 10%, then at the end of Year 1 the investor will have

$$\$100 \times 0.10 + \$100 = \$110.$$

If in Year 2 the investor takes their \$110 and invest it for another year at the interest rate prevailing in Year 2 of 15%, then at the end of Year 2 the investor would have

$$\$110 + \$110 \times 0.15 = \$126.50.$$

This means that over the two-year period the investor has earned total interest of \$26.50, meaning their total rate of return over the two years is

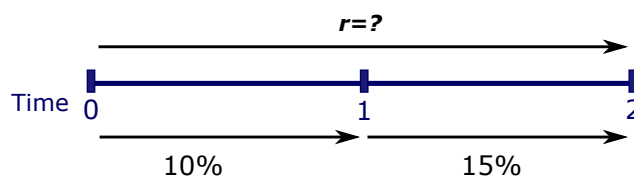
$$\frac{\$26.50}{\$100} \times 100 = 26.50\%.$$

Based on the above information, we could work out that the equivalent annual rate of return that the investor would have had to have received on a two-year investment purchased at the beginning of Year 1 would be 12.47%. Earning an annual return of 12.47% a year for two years on a \$100 investment will mean that at the end of Year 2 the investor will have \$126.50. This is the same amount as the investor would have if they invested in the two consecutive one-year securities as above.

In an efficient market, rational investors will only hold a two-year security if it gives at least the same return as holding two consecutive one-year securities. In the example used above, the first one-year security yields 10% and the second one-year security yields 15%. Therefore, the two-year rate will have to be 12.47% p.a.

In order to work out the two-year rate of 12.47% p.a. in the example above, we would use the two consecutive one-year rates. In working out the solution to such a problem, it is helpful to draw a timeline. For the above example the timeline would be as follows,

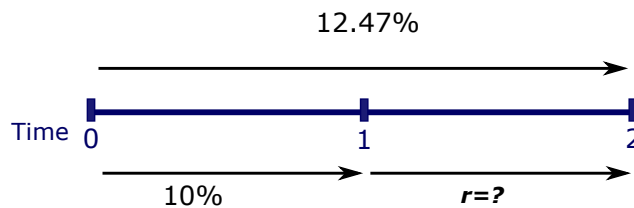
$$\begin{aligned}(1+r)^2 &= (1+r)^1(1+r)^1 \\ (1+r)^2 &= (1+0.10)^1(1+0.15)^1 \\ (1+r)^2 &= (1+0.10)(1+0.15) \\ (1+r)^2 &= 1.2650 \\ 1+r &= 1.2650^{1/2} \\ 1+r &= 1.1247 \\ r &= 0.1247 \\ &= 12.47\%\end{aligned}$$



According to Expectations Theory long-term per annum rates are the “average” of the short term rates expected to prevail during the investment period and the yield curve reflects investors’ expectations about the level of interest rates prevailing during the life of a security.

However, in reality, the situation at any given present time would be that we would know the one-year rate for the first year (10% in the example) and we would also know at the present time the rate per annum for longer term securities issued now. Therefore, in the example above we would know that the annual rate of return on a two-year security issued now is 12.47% p.a. Therefore, the market would then use this information to predict interest rates in the future, for example, the one-year rate in one years time, i.e. the one-year rate in Year 2. Then,

$$\begin{aligned}
 (1+r)^2 &= (1+r)^1(1+r)^1 \\
 (1+0.1247)^2 &= (1.10)^1(1+r)^1 \\
 \frac{1.2650}{1.10} &= 1+r \\
 1.15 &= 1+r \\
 r &= 0.15 \\
 &= 15\%
 \end{aligned}$$



At T_0 , we know that the rate of return on a one-year security issued now is 10% and that the rate of return on a two-year security issued now is 12.47% p.a. We can use this information to work out what will be the interest rate on a one-year security issued at the beginning of Year 2, i.e. just after time 1. The answer is 15%.

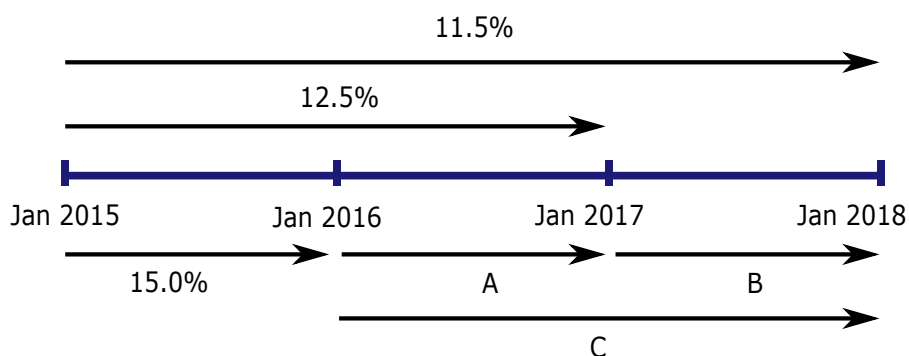
■ **Example 4.3 — Expectation theory.** Here is another example of how expectation theory (term structure of interest rates) is used to work out unknown future interest rates in the market. Suppose that on 1 January 2015 the following term structure of interest rates for bond securities was observed

- One-year bond rate (maturity on 31/12/2015) 15.0% p.a.
- Two-year bond rate (maturity on 31/12/2016) 12.5% p.a.
- Three-year bond rate (maturity on 31/12/2017) 11.5% p.a.

Now suppose that you are required to work out interest rates prevailing on the following bond securities:

- A one-year bond issued on 01/01/2016 (A)
- A one-year bond issued on 01/01/2017 (B); and,
- A two-year bond issued on 01/01/2016 (C).

The following timeline, pictorially, explains the problem.



To work out the rate of interest on a one-year bond issued at the beginning of January 2016 (unknown A) with Expectations Theory, remember that investing in a one-period security at the beginning of 2015 followed by another one-period security at the beginning of 2016 should give the same total return as investing in a two-period security at the beginning of 2015. To find the solution, we will use the rate of return on a one-year bond in 2015 and the rate of return per annum on a two-year bond spanning 2015 and 2016 to find the gap, A, which is the rate of return on a one-year bond issued at the beginning of 2016, as follows.

$$\begin{aligned}
 (1 + 125)^2 &= (1.15)^1 (1 + A)^1 \\
 1.2656 &= 1.15(1 + A) \\
 \frac{1.2656}{1.15} &= 1 + A \\
 1.1005 &= 1 + A \\
 A &= 0.1005 \\
 &= 10.05\%
 \end{aligned}$$

To work out the rate of interest on a one-year bond issued at the beginning of January 2017 (unknown B) with Expectations Theory, we must remember that investing in a two-period security at the beginning of 2015 followed by another one-period security at the beginning of 2017 should give the same total return as investing in a three-period security at the beginning of 2015. To find the solution, we will use the rate of return per annum on a two-year bond spanning 2015 and 2016 and the rate of return per annum on a three-year bond spanning 2015 to the end of 2017 to find the gap, B, which is the rate of return on a one-year bond issued at the beginning of 2017, as follows.

$$\begin{aligned}
 (1 + 0.115)^3 &= (1.125)^2 (1 + B)^1 \\
 1.3862 &= 1.265(1 + B) \\
 \frac{1.3862}{1.2656} &= 1 + B \\
 1.0953 &= 1 + B \\
 B &= 0.0953 \\
 &= 9.53\%.
 \end{aligned}$$

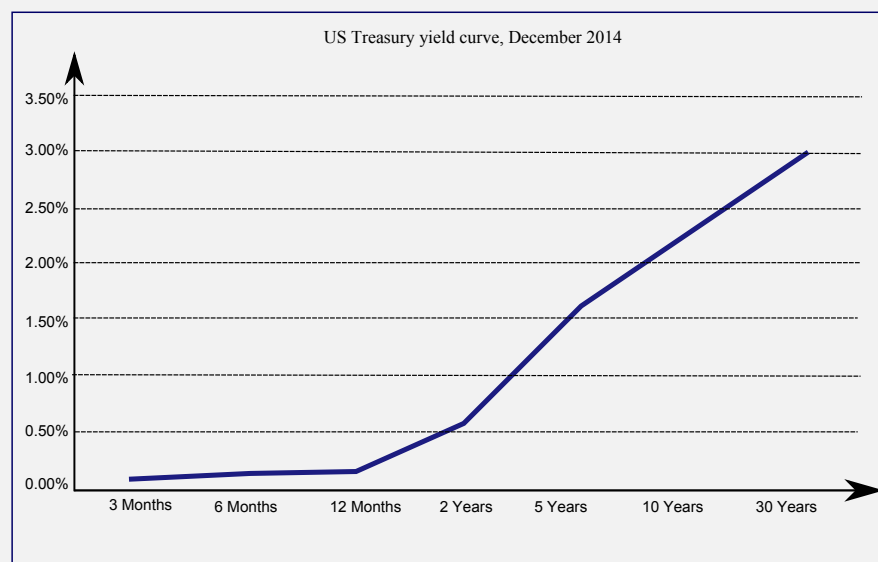
To work out the rate of interest on a two-year bond issued at the beginning of January 2016 (unknown C) with Expectations Theory, remember that investing in a one-period security at the beginning of 2016 followed by another one-period security at the beginning of 2017 should give the same total return as investing in a two-period security at the beginning of 2016. To find the solution, we will use the rate of return per annum on a one-year bond issued at the beginning of 2016 (which we have worked out above as unknown A of 10.05%) and the rate of return on a one-year bond issued at the beginning of 2017 (which we have worked out above as unknown B of 9.53%) to find the gap, C, which is the rate of return per annum on a two-year bond issued at

the beginning of 2016, as follows:

$$\begin{aligned}
 (1 + C)^2 &= (1.1005)(1.0953) \\
 (1 + C)^2 &= 1.2054 \\
 (1 + C)^{2/2} &= 1.2054^{1/2} \\
 1 + C &= 1.0979 \\
 C &= 0.0979 \\
 &= 9.79\%.
 \end{aligned}$$

It should be noted here that in this market the yield curve is inverted, or downward sloping, as longer-term rates are lower than short-term rates and consecutive one-period rates are decreasing (from 15.0% to 10.05%, to 9.53%). ■

Business Snapshot 4.4 — Global Financial Crisis and Yield Curve. As discussed, an inverted yield curve refers to a situation where short-term interest rates exceed long-term rates. From an economic perspective, an inverted yield curve is a noteworthy event as, generally, an inverted yield curve indicates an economic recession is expected sometime in the future. When short-term interest rates exceed long-term rates, market sentiment suggests that the long-term outlook is poor and that the yields offered on long-term fixed income assets (such as bonds and debentures) will fall. This fall in interest rates, that is expected in the future as the economy goes into recession, is consistent with standard economic theory, as a central bank's monetary policy means of dealing with a recession is to lower interest rates in order to boost economic activity and thus move the economy back out of recession.



Interestingly, in late-2005 the US Treasury yield curve inverted and US long-term interest rates were predicted to fall below short-term rates, indicating that the market expected that a recession was ahead for the US economy, which at the time was performing relatively well (with good GDP growth and relatively low unemployment). Given that the market yield curve had inverted, this was suggesting that at some stage in the future the US Federal Reserve (the US central bank, equivalent to Australia's Reserve Bank) would have to lower official interest rates. What was occurring in the US market at that time was a combination of higher energy

prices, higher interest rates, falling house prices, and weakening US consumer confidence and spending, a classic formula for a recession. Shortly following that, the global financial crisis began for the US and world economies in mid-2008, correlated with a surge in home loan defaults in the US. This caused bank failures (Lehman Brothers and Bear Sterns), negative GDP growth rate, much higher unemployment, and a significant stock market correction. For example, from its then record level of 14,164.53 points at October 9, 2007, the Dow-Jones Industrial Average, fell to 6,547.05 points by March 9, 2009 (a fall of 54%) and its lowest level since the Asian Financial Crisis hit in April 1997. As at December 2014, the US Treasury yield curve was of a normal shape, being upward sloping, as can be seen in the figure above.

4.4 Revision Problems

Problem 4.1 Wesfarmers P/L has issued bonds earning a 7% p.a. coupon rate. The interest is paid semi-annually and the bonds mature in eight years. Each bond's face value is \$1,000. If your required rate of return is 8% p.a. what is the price you should pay for each of the bonds?

Problem 4.2 You have just purchased 10 newly-issued \$100 five-year Avco Ltd. debentures at par. These debentures pay \$6 (per debenture) in interest semi-annually. You are also negotiating the purchase of 10 \$100 debentures issued by Amcos Ltd. four years ago that return \$3 per debenture in semi-annual interest payments and have six years remaining to maturity. What is the maximum price you should offer for the Amcos Ltd. debentures assuming Amcos Ltd. is now in the same risk class as the Avco Ltd.?

Problem 4.3 If at T_0 the one-period rate of interest is 6%, the two-year rate is 8% p.a., and the three-year rate is 10% p.a., what is the two-period per annum interest rate expected to prevail at T_1 ?

Problem 4.4 Using semi-annual compounding a 15 year zero-coupon bond that has a face value of \$1,000 and a required return of 8% p.a. would be priced at:

- (a) \$308
- (b) \$315
- (c) \$464
- (d) \$555

Problem 4.5 You buy a two-year bond which pays interest of 12% p.a. At the end of year two you purchase a one-year bond that pays interest of 10%. According to the expectations theory of interest rates you could have purchased a three-year bond today that pays interest each year of:

- (a) 11.0%
- (b) 11.3%
- (c) 11.5%
- (d) cannot be determined

Problem 4.6 Which of the following statements is most correct?

- (a) All else equal, if a bond's yield to maturity increases its current yield will fall.
- (b) All else equal, if a bond's yield to maturity increases its price will fall.
- (c) If a bond's yield to maturity exceeds the coupon rate the bond will sell at a premium.
- (d) All of the answers above are correct.

Problem 4.7 Two years ago you bought a Commonwealth Government bond for \$1,000 because you liked the 10% p.a. coupon interest payment that you would receive for 10 years. Interest on the bond is paid annually. Two years later when the market interest rate has fallen to 8% p.a. what is the value of your Commonwealth Government bond?

Equity Financing

Cost of Equity

Share Valuation Models**Dividend Valuation Model**

Constant Growth Stock

Dividend Yield, Capital Gains Yield and Total Return

Zero Growth in Dividends

Required Rate of Return

Negative Dividend Growth

Modified Constant Growth

Short-Term vs. Long-Term Performance

Sensitivity to the Dividend Growth Rate

Sensitivity to Required Rate of Return

Price/Earnings Model of Share Valuation**Revision Problems**

5. Share Valuation

In this chapter we will focus on valuation of equity or shares. You will learn about the dividend valuation model; constant growth stocks and the constant growth in dividends pricing model; dividend yields; capital gains yields; and total return; the dividend valuation model with zero growth in dividends and with negative growth in dividends; preference shares; the non-constant growth in dividends pricing model; the effects of a company's short-term versus long-term performance on its share price; share price sensitivity to dividend growth rates and to the required rate of return; and the price-earnings model of share valuation.

5.1 Equity Financing

Equity finance refers to the process of raising capital through the sale of shares in a corporation, and essentially involves the sale of an ownership interest in a company to raise funds for business purposes. Equity financing generally involves an initial public offering (IPO) of shares on a stock exchange where shares in the company are sold for the first time to raise funds for the issuing company, with the purchases of the shares (the shareholders/investors) hoping to receive a return on the shares in the form of a periodic dividend payment and a capital gain if and when they sell the shares (if the market price of the shares has risen since they purchased them). Most commonly, ordinary shares are issued in an IPO, although preference shares are sometimes issued and these give the shareholders additional ownership benefits over ordinary shareholders. For example, preference shareholders are paid a dividend before ordinary shareholders but, generally, this comes at a higher cost. Equity financing is distinct from debt financing, which refers to funds borrowed by a business, whereas with equity financing ownership shares in the business are sold and there is no borrowing of money. Unlike debt financing, where the debt holders have a contractual right to receive payment, share owners have a residual claim to the cash-flows generated by the real assets of the business. This means that if they do not receive payment, or a return, on their investment they cannot seek legal redress to force the company to make any payments.

5.1.1 Cost of Equity

The cost of equity financing for the issuing firm is the dividend payments that are made, although the timing of these payments can be variable. For example, a company may pay no dividends in some periods, while in another period it could, for example, pay a dividend of 25 cents per share, followed in later periods by a dividend of, say, 10 cents per share. So the size of the dividend payment and whether a dividend payment is actually made is variable. Also, generally, we assume no maturity date on shares. This means, as long as the company continues to operate and be listed on the stock exchange, then the shares will not mature, meaning that the shareholders do not have the right to go to the issuing company and demand that the company pay the shareholder the value of the shares. Nevertheless, sometimes an issuing company will buy its share back from shareholders in what is called a *share buy-back*. Additionally, because shareholders have a residual right to receive payments on their share investment, the relative riskiness of share investing is higher than for debt investing.

5.2 Share Valuation Models

There are two main categories of equity valuation for ordinary shares, sometimes referred to as 'common stock'. The first of these two main methods are the *absolute valuation models* that attempt to find the intrinsic or "true" value of an investment based on fundamentals. In this method, the focus is on dividends, cash—flows, and growth rates for a single company, without comparison to other companies. The main type of absolute valuation model that we will learn about is the dividend valuation model, also referred to as the dividend discount model.

The second main category of ordinary share valuation methods are relative valuation models that involve comparing the company in question to other similar companies. These methods generally involve calculating multiples or ratios. The ratio that we will focus on is the P/E or price-to-earnings ratio, and, usually, these ratios are compared to the ratios of other companies. For instance, if the P/E of Firm A is lower than the P/E of Firm B, then Firm A may be said to be relatively undervalued. We should note at this stage that, generally, equity valuation is more difficult than debt valuation because of the uncertainty of the future cash-flows associated with equity. Remember, the payment of dividends on shares is variable, unlike repayments on debt, and, generally, shares do not have a maturity date.

5.3 Dividend Valuation Model

The *dividend valuation model* is one of the most basic absolute valuation models. The dividend model calculates the "true" value of a firm based on the dividends the company pays its shareholders. With this model the market value of a share is the present value of all the expected future dividends to be received from the share discounted at a rate of return that reflects the riskiness of those dividends.

The justification for using dividends to value a company is that dividends represent the actual cash flows going to the shareholder, thus valuing the present value of these cash flows should give you a value for how much the shares should be worth. The first thing that needs to be checked, if this model is going to be used, is to see whether the company in question is actually paying a dividend.

However, for the dividend valuation model to work, it is not enough for the company to just pay a dividend, the dividend should also be stable and predictable. Companies that pay stable and predictable dividends are usually mature "blue chip" companies, such as BHP-Billiton, Rio Tinto, and the Commonwealth Bank, in mature and well-developed industries. These type of

companies are often best suited to this valuation method.

With the dividend valuation model the current price of a share is the present value (*PV*) of all expected future dividends to be earned on that share. An implication of this is that in finding the current price of the share, the future selling price of the share is ignored, as the future selling price at Time-period t is the *PV* to the buyer at that time of all expected future dividends to be earned on the share. Further, the (present) value of a share simply comes from the *PV* of all the expected future dividends on that share. The formula used to find the current price of the share (P_0) or PV_0 of all the expected future dividends on the share is as follows.

$$P_0 = D_1(1 + r_e)^{-1} + D_2(1 + r_e)^{-2} + D_3(1 + r_e)^{-3} + \dots + D_\infty(1 + r_e)^{-\infty}.$$

where r_e is the required return of shareholders in the company, i.e. it is the cost of equity for the company. The above formula is treating each of the expected future dividends to be earned on the share as a single sum, and discounting all these expected future dividends up to infinity (∞) back to T_0 as single sums. Note that it is only expected future dividends that are taken into account with this model, not the current dividend (D_0) or any previous dividends, e.g. D_{t-1} , the dividend in the previous period.

5.3.1 Constant Growth Stock

Looking at the dividend valuation model formula above, it can be seen that dividends need to be calculated up to and discounted back from infinity, which, in reality, is impossible. For the dividend valuation model to work either the dividends paid by a company on its shares must be constant, i.e. unchanging, or be growing at a constant rate.

A constant growth stock is one whose dividends are expected to grow at a constant rate, g , forever, with dividends growing by the same percent each time period,. This creates the Constant Growth In Dividends Pricing Model. When we have constant growth in dividends we use the constant growth in dividends pricing model to find the current price of the share as follows

$$P_0 = \frac{D_0(1 + g)}{r_e - g} = \frac{D_1}{r_e - g}$$

where D_0 is the most recently paid dividend (dividend paid at T_0), g is the long-term constant growth rate in dividends, and D_1 is the next dividend to be paid at T_1 (at the end of Period 1). Note, this is basically the same formula as that we used for the *PV* of a growing perpetuity annuity we covered in Chapter 2.

■ **Example 5.1** Let us assume that Sumitomo Bank has just paid a dividend of 15 cents per share and this is expected to grow at 5% per annum indefinitely. What price should an investor pay for the Sumitomo Bank shares if the investor's required rate of return on the investment is 10% p.a.?

From this information, we know that the most recently paid dividend on the Sumitomo Bank shares is \$0.15, that the constant growth rate of these dividends, $g = 5\%$ p.a., and that the investor's required rate of return on the equity, $r_e = 10\%$ p.a. Since we have a constant growth in dividends stock we will use the constant growth in dividends pricing formula to find the current

price of these shares. Remember, that formula is as follows.

$$\begin{aligned} P_0 &= \frac{D_0(1+g)}{r_e - g} \\ &= \frac{\$0.15(1.05)}{0.10 - 0.05} \\ &= \$3.15. \end{aligned}$$

■

■ **Example 5.2** Assume that D_0 is \$2.00, g is 6% p.a., and r_e is 13% p.a. Since we have constant growth in dividends we will use the constant growth in dividends pricing model to find the current price of these shares as follows

$$\begin{aligned} P_0 &= \frac{D_0(1+g)}{r_e - g} \\ &= \frac{\$2.00(1.06)}{0.13 - 0.06} \\ &= \$30.29. \end{aligned}$$

The price of \$30.29 is the PV of the dividends to be received in Year 1, Year 2, ..., Year 20, ..., ∞ .

Continuing with this example, what will be the share's price one year from now, i.e. what will be P_1 ? D_1 will have been paid, so expected future dividends are $D_2, D_3, D_4, \dots, D_\infty$. Therefore,

$$\begin{aligned} P_1 &= \frac{D_1(1+g)}{r_e - g} \\ &= \frac{D_2}{r_e - g} \\ &= \frac{\$2.12(1.06)}{0.13 - 0.06} \\ &= \$32.10. \end{aligned}$$

So, P_1 is the PV at the end of the first period of $D_2, D_3, D_4, \dots, D_\infty$.

■

5.3.2 Dividend Yield, Capital Gains Yield and Total Return

The total return on a share over a given period is the dividend yield plus the capital gains yield, therefore,

$$R_i = DY + CGY,$$

where R_i is the total return on the individual asset i , DY is dividend yield (the return on the share received from the periodic income stream) and CGY is the capital gain yield.

■ **Example 5.3 — Continuing with Example 5.2.** We can find the dividend yield (DY) and capital gains yield (CGY) on this share for Year 1, as follows.

$$DY = \frac{D_1}{P_0} = \frac{\$2.12}{\$30.29} = 0.07 = 7\%.$$

while the *CGY* is the return on the share received from the change in its price over the period, i.e. the change in its price from T_0 to T_1 , as follows

$$CGY = \frac{P_1 - P_0}{P_0} = \frac{\$32.10 - \$30.29}{\$30.29} = 0.06 = 6\%.$$

Then, the total return on a share over a given period is

$$R_i = DY + CGY = 7\% + 6\% = 13\%.$$

It should be noted that this total return of 13% is equal to the required rate of return on the share specified above of 13% p.a. ■

For a constant growth in dividends share the capital gains yield will equal the constant growth rate in dividends, g , so for such a share the total return R_i will equal $DY + g$. In the example above, where we have a constant growth rate in dividends of 6% p.a., the total return is $R_i = 7\% + 6\%$.

Highlight 5.1 In order for the constant growth in dividends pricing model to work it is important to realise that the constant growth rate in dividends, g , must be less than the required rate of return on the share for the investor, r_e . This means that the denominator in the pricing formula must be positive. If g is greater than r_e the denominator in the formula will be negative, and this will lead to a negatively-priced stock, which is not sensible. A negatively-priced stock means that the person buying the share would be paid to buy it by the seller. Therefore, in order to use the constant growth in dividends pricing model to value a share $g < r_e$.

Business Snapshot 5.1 — Constant Growth In Dividends Pricing Model.

Why I Am Buying BHP Billiton PCL (BBL)?^a

David Mantra

Date: 25/11/2014

My opinion on the valuation of shares in BBL (BHP-Billiton Limited) hasn't changed materially in the last month as nothing has fundamentally changed with the company over that time frame. I valued the shares using a dividend discount model analysis with the same input as last month, a 10% discount rate and 5.5% long-term growth rate. This growth rate seems reasonable considering it's about half of BBL's 10-year EPS growth rate. I purposely use fairly conservative numbers with more uncertain investments like BBL due to the volatile nature of commodities. However, even using this conservative valuation model I still get a fair value of \$58.14. That means the shares potentially offer a 15% margin of safety even after factoring in somewhat low growth, which indicates to me that the shares are a value here. [And] Morningstar rates BBL as a 4/5 star value, with a fair value estimate of \$70.00.

^aThis article is referencing to BBL shares that trade on the London Stock Exchange (LSE) and are offered as ADR (American Depositary Receipt) shares on the New York Stock Exchange (NYSE) for US investors.

5.3.3 Zero Growth in Dividends

The constant growth in dividends pricing model is based on the assumption that the dividends expected to be paid on a share will grow at a constant rate. However, it is possible that there will be no growth in the dividends paid on a share, which means that the dividend paid is constant or fixed. An example of a share that has a fixed constant dividend is a preference share. In this situation, since $g = 0\%$, we would have a constant dividend that we would treat as a perpetuity. In order to find the price of such a share the *PV* of a perpetuity formula (see Chapter 2) would be used, with the formula being of the following form

$$PV_0 = P_0 = \frac{PMT}{r_e} = \frac{D}{r_e}.$$

■ **Example 5.4** If we had a share, such as a preference share, which has a fixed dividend of \$5 per share and the investor's required rate of return is assumed to be 10% p.a., the price of this share using the dividend valuation model would be as follows

$$P_0 = \frac{\$5}{0.10} = \$50.$$

■

5.3.4 Required Rate of Return

If the price of a share is known and the dividend is also known, then it is possible to work out the unknown required rate of return on that share by re-arranging the pricing formula of $P_0 = \frac{D}{r_e}$ to become $r_e = \frac{D}{P_0}$.

This is simply the dividend yield (*DY*) formula we saw earlier. This required rate of return formula is dividing the price of the share at T_0 into the dividend paid on the share at T_1 .

■ **Example 5.5** Work out the required rate of return, r_e , on a share that had a price in the market at the beginning of the period (P_0) of \$50 and has paid a dividend at the end of the period (D_1) of \$5.

$$r_e = \frac{D}{P_0} = \frac{\$5}{\$50} = 0.10 = 10\%.$$

■

5.3.5 Negative Dividend Growth

Up to this point, when using the constant growth in dividends pricing model, we have assumed that dividends are growing at a constant rate each period. However, it is possible to have a situation where the fixed percentage change in dividends each period, g , is negative, meaning that the dividends on the share are decreasing by a fixed percentage each period. In this situation of negative dividend growth, we would like to know whether anyone would still buy the share and would the share still have any value. The answer to those questions is yes, and we can still use the constant growth in dividends pricing model to find the price that investors would pay for the share.

The following example will explain how we can price a share when the dividends on the share are decreasing by a fixed percentage each period.

■ **Example 5.6** Assume that a firm has just paid a dividend of $D_0 = \$2.00$ and g is equal to -6% , i.e. dividends are decreasing by 6% each period. We can find the current price of this share using the constant growth in dividends pricing formula as follows:

$$\begin{aligned}
 P_0 &= \frac{D_0(1+g)}{r_e - g} = \frac{D_1}{r_e - g} \\
 &= \frac{\$2.00(1 + (-0.06))}{0.13 - (-0.06)} \\
 &= \frac{\$2.00(1 - 0.06)}{0.13 + 0.06} \\
 &= \frac{\$1.88}{0.19} = \$9.89.
 \end{aligned}$$

Note 1: When a negative is multiplied by a positive the result is a negative. Thus the numerator here becomes $\$2.00(1 - 0.06)$.

Note 2: When a negative is multiplied by a negative the result is a positive. Thus the denominator here becomes $0.13 + 0.06$.

Even though dividends on these shares are decreasing (by a fixed percentage each period) the shares still have value, at $\$9.89$ in this example.

In this situation of constant negative dividends growth, we are still able to work out the dividend yield (DY) and capital gains yield (CGY) on the shares, where the CGY is simply the negative growth rate $-g = -6\%$ p.a. The DY is found as $r_e - (-g)$, which in this example is $13\% - (-6\%) = 19\%$. Because we have a constant percentage change in the dividends on these shares both yields are constant over time, with the high DY (19% p.a.) offsetting the negative CGY (-6% p.a.). ■

5.3.6 Modified Constant Growth

While the constant growth in dividends pricing model assumes that the growth rate in dividends on a share is expected to be constant into perpetuity, it is possible that there may be varying growth rates in the dividends expected to be paid over the short-to-medium term, before a constant long-term growth rate into perpetuity comes into effect. In such a situation, in order to value a share, the model that is used is an extension of the constant growth in dividends pricing model, which is termed the *Modified Constant Growth* model.

The modified constant growth in dividends pricing model assumes that there will be non-constant growth rates in the dividends paid on a share over a fixed length of time, which may exceed r_e in the short-to-medium term. However, in the long-run the growth rate in the dividends, g , will become fixed and will be less than r_e as required by the constant growth in dividends pricing model.

■ **Example 5.7** We will assume that the growth rate in dividends on a share is expected to be 30% for the first three years before a long-run constant growth, g , of 6% commences in Year 4. We will also assume that the most recently dividend paid on the share, D_0 , is $\$2.00$, and that the required rate of return on the share, r_e , is 13% p.a.

Note, this required rate of return of 13% p.a. is less than the growth rate in dividends of 30% in Years 1, 2, and 3. In those three years we cannot use the constant growth in dividends model to price the shares as, firstly, the growth rate in dividends is not a long-term constant rate in Years 1, 2, and 3. Secondly, the growth rate in dividends in those three years is greater than r_e .

To find P_0 , we will use the modified constant growth in dividends pricing model. It is important to note that when using the modified constant growth in dividends pricing model dividends only have to be calculated up to the first period in which the long-term constant growth rate commences, so in this example that means that we only have to calculate dividend up to D_4 , as Year 4 is the period in which the long-term constant growth rate of 6% commences.

To calculate dividends up to D_4 we commence with the most recent dividend paid, $D_0 = \$2.00$ and multiply it by the growth rate in dividends in Year 1 of 30% to get

$$D_1 = \$2.00(1.30) = \$2.60.$$

Then multiply D_1 by the growth rate in dividends in Year 2 of 30%, to get

$$D_2 = \$2.60(1.30) = \$3.38.$$

Similarly,

$$D_3 = \$3.38(1.30) = \$4.394, \text{ and}$$

$$D_4 = \$4.394(1.06) = \$4.6576.$$

For the next step in the process, we apply the constant growth in dividends pricing formula to the first of the long-term constantly growing dividends, which in this example is D_4 , which gives us the present value of all the long-term constantly growing dividends at the end of the period before the long-term constant growth rate commences.

In this example it gives us P_3 , which is the present value of all the long-term constantly growing dividends from Year 4 onwards at the end of Year 3. Thus,

$$\begin{aligned} P_n &= \frac{D_{n+1}}{r_e - g} \\ P_3 &= \frac{D_4}{r_e - g} \\ &= \frac{\$4.6576}{0.13 - 0.06} = \$66.54. \end{aligned}$$

Note that D_{n+1} is the dividend to be paid at the end of the first period of a long-term constant growth rate in dividends into perpetuity. This price, P_3 , of \$66.54 is the present value at the end of Year 3 of Dividends 4 through to infinity (i.e. D_4 to D_∞).

The next step in the process is to find the present value of all the future dividends expected to be paid on this share and then find their sum to give P_0 . Because each of the dividends is of a different value, to find their PV_0 we must treat each dividend as a single sum and use the PV of a single sum formula, which is $PV_0 = D_n(1 + r)^{-n}$.

So, in this example we have the following.

$$P_0 = D_1(1 + r)^{-1} + D_2(1 + r)^{-2} + D_3(1 + r)^{-3} + P_3(1 + r)^{-3},$$

which means that we are discounting all the dividend payments back to T_0 . Then

$$\begin{aligned} P_0 &= \$2.60(1.13)^{-1} + \$3.38(1.13)^{-2} + \$4.394(1.13)^{-3} + \$66.5371(1.13)^{-3} \\ &= \$2.301 + \$2.6469 + \$3.0455 + \$46.1169 \\ &= \$54.11 \end{aligned}$$

The price of \$54.11 is the present value at T_0 of Dividends 1 through to infinity (PV_0 of $D_1 \dots D_\infty$) on this share.

Highlight 5.2 Whenever you apply the constant growth in dividends pricing formula to the long-term constantly growing dividends, it will give you the *PV* or price of those long-term constantly growing dividends at the end of the period before the long-term constant growth rate commences.

■ **Example 5.8** Assume that you are a financial adviser and that your client, Samantha, would like to know the price she should pay for AMP Shares. AMP has just paid a dividend of \$0.15 (i.e. 15 cents) per share and the dividend is expected to grow at a rate of 20% p.a. for the next five years, and then at a rate of 5% p.a. after that. Samantha's required rate of return on the AMP Shares is 10% p.a.

You will use the modified constant growth rate in dividends pricing model to work out the correct price for Samantha as follows

Step 1: Calculate dividends up to the first period in which the long-term constant growth rate commences (Year 6 in this example):

$$\begin{aligned} D_0 &= \$0.15, \\ D_1 &= \$0.15(1.20) = \$0.18, \\ D_2 &= \$0.18(1.20) = \$0.216, \\ D_3 &= \$0.216(1.20) = \$0.259, \\ D_4 &= \$0.259(1.20) = \$0.311, \\ D_5 &= \$0.311(1.20) = \$0.373, \\ D_6 &= \$0.373(1.20) = \$0.392. \end{aligned}$$

Step 2: Use the constant growth in dividends pricing formula to find the present value of all the long-term constantly growing dividends at the end of the period before the long-term constant growth rate commences (end of Year 5 in this example):

$$P_5 = \frac{\$0.392}{0.10 - 0.05} = \$7.84.$$

Step 3: Find P_0 by using the *PV* of a single sum formula to discount back to T_0 Dividends 1 to 5 and P_5 at Samantha's required rate of return of 10% p.a.:

$$\begin{aligned} P_0 &= \$0.18(1.10)^{-1} + \$0.216(1.10)^{-2} + \$0.259(1.10)^{-3} + \$0.311(1.10)^{-4} \\ &\quad + \$0.373(1.10)^{-5} + \$7.84(1.10)^{-5} \\ &= \$0.164 + \$0.179 + \$0.195 + \$0.212 + \$0.232 + \$4.868 \\ &= \$5.85 \end{aligned}$$

5.3.7 Short-Term vs. Long-Term Performance

In most equity markets around the globe it is the performance of a company in the short-term that is seen as the main determinant of the current market price of the share because it is the quarterly earnings and performance forecasts of a company that are focused on most by traders. An important question to be asked is how much of a share's current market price is derived from its short-term performance as compared to its long-term performance. Using the constant growth in dividends pricing model it is possible to answer this question. With this model, we can find the present value at T_0 of all the long-term constantly growing dividends on a share and express this as a percentage of the current market price of the share. The following example will explain how this can be done.

■ **Example 5.9 — Continuing with Example 5.7.** The PV_0 of dividends beyond Year 3 (i.e. the PV_0 of all the long-term constantly growing dividends) was found to be \$46.12, while the current market price of the share, P_0 , is \$54.11. Thus, the percentage of the share's price or value due to "long-term" dividend growth (or long-term performance) is 85.23% (i.e. $\$46.12/\$54.11 = 0.8523$). ■

Thus, at least based on the constant growth in dividend pricing model, it is the long-term performance of a share that is most important in determining its current market price. However, as stated above, the market focuses much more on the short-term (quarterly) performance of a company in determining demand for a share. The reasons for this are that changes in quarterly earnings are sometimes taken as a signal of changes in future cash-flows, thus affecting the current share price. Also, corporate managers usually have their pay bonuses tied to quarterly earnings, so they are more motivated to focus their attention and resources on ensuring the company performs best in the short-term rather than long-term.

5.3.8 Sensitivity to the Dividend Growth Rate

Generally, there is a positive relationship between the dividend growth rate, g , and a share's price as the higher is a share's dividend growth rate, g , the higher will be the share price, and *vice versa*. The reason for this are that, firstly, in terms of the constant growth in dividends pricing formula, the higher is g the lower will be the denominator ($r_e - g$). Therefore, with a fixed numerator (D_1) the result (P_0) must be higher:

$$P_0 = \frac{D_1}{r_e - g}.$$

Secondly, an intuitive explanation is that if g is higher, then future dividends will be higher. Therefore, more investors will want to own the share in order to obtain the higher future dividends. Hence, demand for the share in the market will go up, and with a fixed supply of the share and a free market, the price of the share must rise.

5.3.9 Sensitivity to Required Rate of Return

Generally, there is a negative relationship between the required rate of return on a share, r_e , and a share's price as the higher is r_e , the lower will be the share price and *vice versa*. The reason for this are that, firstly, in terms of the constant growth in dividends pricing formula, the higher is r_e the higher will be the denominator ($r_e - g$). Therefore, with a fixed numerator (D_1) the result (P_0) must be lower. Secondly, an intuitive explanation is that if r_e is higher then investors require a higher rate of return on their investment. Therefore, holding other factors constant, to obtain a higher return they must pay a lower price.

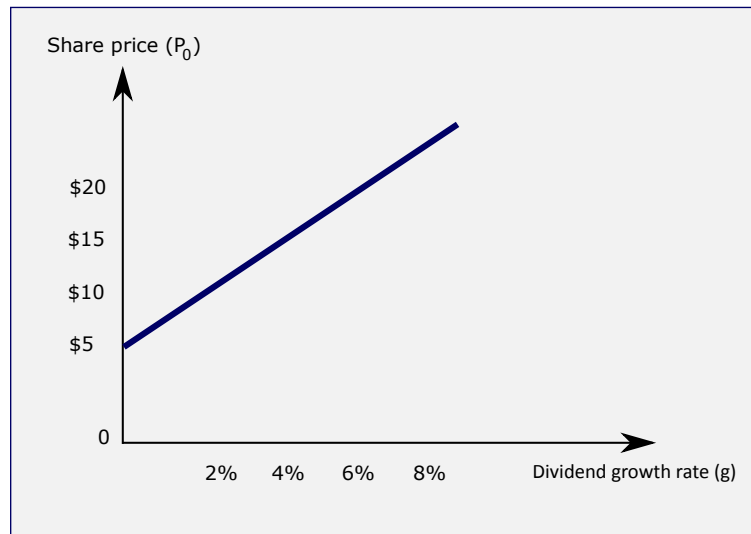


Figure 5.1: Relationship Between Dividend Growth Rate And Market Price: There is a positive relationship between the dividend growth rate, g , and share price, P_0 .

5.4 Price/Earnings Model of Share Valuation

The Price/Earnings Model (P/E Model) is another method used to estimate a firm's share price and is based on a company's P/E ratio. When it comes to valuing stocks, the price/earnings ratio is one of the oldest and most frequently used metrics. The P/E is the ratio of a company's share price to its per-share earnings. To calculate the P/E , divide the current share price of a company by its earnings per share (EPS) as follows.

$$\frac{P}{E} = \frac{\text{Market price of share}(P_0)}{EPS},$$

Historically, the average P/E ratio in the Australian market has been around 14 to 15, while in the US market it has been higher at around 20 to 25 .

A share's P/E tells us how much investors are willing to pay per dollar of earnings on a share. For this reason it is also called the "multiple" of a share or stock. For example, a P/E of 20 suggests that investors in the share are willing to pay \$20 for every \$1 of earnings that the company generates.

The P/E is a reflection of the market's optimism concerning a company's growth prospects. If a company has a P/E higher than the market or industry average, this means that the market is expecting big things over the next few months or years. The P/E is a much better indicator of the value of a share than the market price alone. For example, all things being equal, a \$10 stock with a P/E of 75 is much more "expensive" than a \$100 stock with a P/E of 20. However, generally, it is difficult to say whether a particular P/E is high or low without taking into account factors such as growth rates, risk and the industry in which the company operates. P/E can vary widely between different companies and industries. It is useful to compare the P/E of one company to another in the same industry, to the market in general, or to the company's own historical P/E ratios.

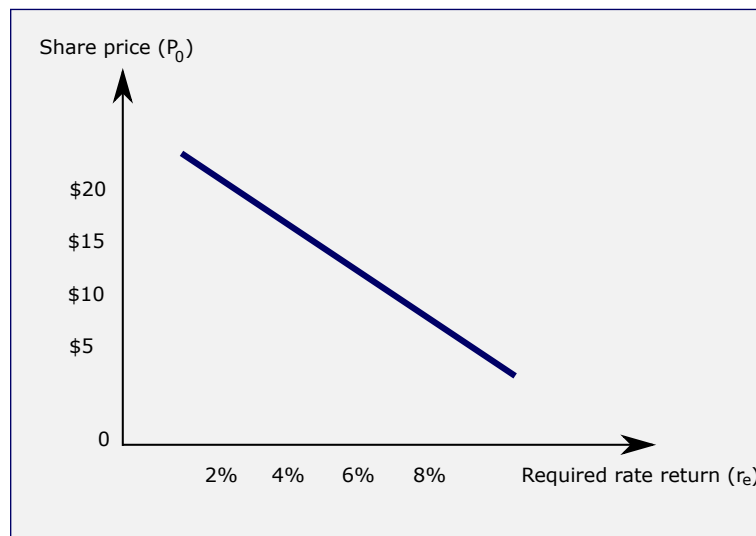


Figure 5.2: Relationship Between Required Rate of Return and Market Price: There is a negative relationship between the required rate of return, r_e , and share price, P_0 .

Business Snapshot 5.2 — Price/Earning Model of Share Valuation.

Analysts Agree: The Big Four Banks Are Expensive

The Sydney Morning Herald

Date: 28/05/2013

Analysts are again questioning whether Australian banks are overvalued, after the financial sector led last week's share-market sell-off.

The big four banks have been trading near their historic valuation highs, with Commonwealth Bank's price-to-earnings ratio recently lifting to about 15.5 times its annual earnings. Before last week's falls, CBA's share price had risen more than 50% in 12 months. The share price closed at \$68.19 on Monday, 38% higher than last May.

CIMB analysts John Buonaccorsi and Ashley Dalziell said local banks were about 20% overvalued on most fundamental ratios, and Platypus Asset Management's chief investment officer, Don Williams, said CBA's highest P/E ratio was in 1999 at about 17 times earnings. "We would argue that 14 or 15 times is at the high end of its valuation range and the low end is around 10", he said. Last week, UBS analyst Jonathan Mott described CBA as the "most expensive large bank in the world by nearly every measure".

Two of the main factors influencing a company's P/E ratio are, firstly, the perceived riskiness of the company, as the riskier the investment the lower will the P/E ratio. The reason for this is because the riskier the investment, the higher the investors' required rate of return. Further, to receive a higher rate of return investors will only be willing to pay a lower price, and the lower the price the lower must be the P/E ratio.

The second main influencing factor is the expected growth rate of earnings for the company, as the greater the growth rate in the company's earnings, the higher the P/E ratio. The reason for this is that the higher future earnings, the higher the demand for the company's shares, and with a fixed supply, the price of the share will rise. Also, the higher the price of the share the higher the P/E ratio, because, in the P/E the numerator, price (P), will be higher. The formula

that is used to calculate a firm's P/E is as follows

$$\frac{P}{E} = \frac{(1-b)}{r_e - g},$$

where

b = percentage of earnings per share retained by the firm;

r_e = the required rate of return of the firm's shareholders; and

g = the long-term constant growth rate of earnings per share (and not of dividends).

With the P/E model of share valuation, share price is calculated by multiplying the firm's expected earnings per share (EPS) by the firm's P/E ratio. Thus,

$$P = EPS \times P/E.$$

Although, sometimes the average P/E ratio for the particular industry is used.

■ **Example 5.10** We will assume that Argyle Mines Pty. Ltd. is to retain 40% of its expected earnings per share of \$0.15, and that its earnings are expected to grow at 2% p.a. into perpetuity. We will also assume that the required rate of return of its equity holders is 7% p.a. Using the P/E model of share valuation, firstly we calculate Argyle's P/E ratio as follows.

$$\begin{aligned} \frac{P}{E} &= \frac{(1-b)}{r_e - g}, \\ &= \frac{1-0.4}{0.07-0.02} = 12. \end{aligned}$$

This means investors in Argyle are prepared to pay \$12 per \$1 of earnings on Argyle shares. Then the share price is

$$\begin{aligned} P_0 &= P/E \times EPS \\ &= 12 \times \$0.15 = \$1.80 \end{aligned}$$

■

■ **Example 5.11** The directors of Parmalat Pty. Ltd. have provided you, the Corporate Finance Manager, with the following information and asked you to calculate the company's P/E ratio. Explain what it means and, also, calculate the current market price of a Parmalat share based on the P/E share valuation model:

Expected $EPS = \$1.50$

$r_e = 11\% p.a.$

EPS to be retained by the firm = 60%

$g = 5\% p.a.$

Therefore, Parmalat's P/E ratio is found as follows.

$$P/E = \frac{1-0.6}{0.11-0.05} = 6.67$$

This means that equity investors in Parmalat are prepared to pay \$6.67 per \$1 of earnings on each Parmalat P/L share, while Parmalat's share price is found as follows.

$$\begin{aligned} P_0 &= P/E \times EPS \\ &= 6.67 \times \$1.50 = \$10.00 \end{aligned}$$

■

Business Snapshot 5.3 As at the 11th of December 2014, the following information applied to Australian wealth management company AMP Limited.

(Forecast) P/E Ratio 2014 = 15.25

(Forecast) EPS 2014 = \$0.34

$EPS \times P/E = \$5.19$

(Actual) Share Price = \$5.42

Based on the information in the table, we have AMP's price as per the P/E model at \$5.19, whilst the actual price of an AMP Limited share was higher at \$5.42, indicating that AMP Limited shares are overvalued and headed for a downward price correction.

5.5 Revision Problems

Problem 5.1 Why does it not matter to the price of a share if the intended buyers holds it indefinitely or intends to sell it in some future period?

Problem 5.2 AMP Pty Ltd last paid a dividend of \$0.30 per share. This dividend is expected to grow at 12% per annum for three years, then at 10% per annum for the next three years, after which it is expected to grow at a 4% p.a. rate forever. Required:

- (a) What is the price you would pay for an AMP share if your required rate of return is 8% p.a?
- (b) Would the price change if you expected to hold the share for only three years? Explain.

Problem 5.3 Leighton Pty Ltd's last dividend, D_0 , was \$0.55 and the company expects to experience no growth in dividends for the next two years. However, Leighton's dividends will grow at an annual rate of 5% in the third and fourth years, and beginning with the fifth year, its dividends should attain a 10% p.a. growth rate which it will sustain thereafter. Leighton's shareholders have a required rate of return of 12% p.a. Calculate the value of a Leighton share today.

Problem 5.4 Calculate the dividend yield for Santos Pty Ltd, assuming that the required rate of return is 25% p.a. and that the expected growth rate in dividends is 17% p.a.

- (a) 8.5%
- (b) 8.0%
- (c) 6.9%
- (d) 9.0%

Problem 5.5 You are considering acquiring a share and selling it at the end of one year. You expect to receive both \$1.50 in dividends and \$26 from the sale of the share at the end of the year. What is the maximum price you would pay for the share today if you wanted to earn a 15% return?

- (a) \$23.91
- (b) \$24.11
- (c) \$27.30
- (d) \$27.50

Problem 5.6 Given the following information about Flinstone Ltd., what is the required rate of return of its ordinary shareholders? The expected growth rate of dividends is 4.5% p.a., the current market price of its ordinary shares is \$43.67 and the last dividend paid to the firm's ordinary shareholders was \$1.75.

- (a) 5.00%
- (b) 8.00%
- (c) 8.69%
- (d) 9.51%

Problem 5.7 The directors of QBE Insurance Pty. Ltd. have provided you, the Corporate Finance Manager, with the following information:

Expected earnings per share	\$1.30
Shareholders' required return	10% p.a.
Earnings per share to be retained by the firm	40%
Growth rate in earnings per share	6% p.a.

Required:

- (a) Calculate QBE's P/E ratio. In simple terms, what does this ratio tell you?
- (b) Calculate QBE's share price using the P/E ratio calculated in Part (a).

Capital Budgeting

Methods of Project Evaluation

- Non-Discounted Cash-Flow Analysis

- Discounted Cash-Flow Analysis

Detailed NPV Analysis

- NPV Investment Evaluation Process

- Incremental Cash-Flows Analysis

Evaluating Projects with Different Lifespans

Depreciation, Inflation & Effective Write-off

Capital Rationing

- Profitability Index (Benefit-Cost Ratio)

Revision Problems

6. Capital Budgeting

This chapter is designed to give you an understanding of capital budgeting and methods of project evaluation. We will touch on related issues and concepts such as non-discounted and discounted methods of project evaluation; incremental cash-flows; sunk costs; opportunity costs; and side-effects; financing costs and taxation; and book and capital gains and losses. Further, you will learn about how to account for book-gains and losses in project evaluation and the difference between these and capital gains and losses, under different project evaluation methods.

6.1 Capital Budgeting

Capital budgeting, also known as project evaluation, is the process in which a corporation analyses the cash-flows to be generated by investment in a potential new asset, product, or project. This is done to decide on whether the investment is beneficial for the company in terms of achieving the main corporate objective of maximising the market value of the company.

Put simply, capital budgeting involves analysis of potential additions to a company's real (productive) assets, usually of a long-term nature and involving large expenditures. Capital budgeting is usually undertaken because corporations have limited funds available for investment and, so, must decide on which projects are best in order to increase the wealth of the company's shareholders by maximising the market value of the company. In terms of an accounting balance sheet, the capital budgeting decisions of a company are reflected in the left-hand side of the statement. The assets of the firm reflect investment in real assets and its project evaluation decisions, and which real assets the company has invested in in order to maximise its market value.

6.2 Methods of Project Evaluation

There are two main methods of capital budgeting or project evaluation analysis, which are the non-discounted cash-flow and the discounted cash-flow methods.

Business Snapshot 6.1 BHP-Billiton's consolidated balance sheet shows, on the left-hand side, its real assets, current assets and non-current assets, which reflect its capital budgeting decisions and investment in real assets that are used to produce goods and services that are then sold to generate cash-inflows. The right-hand side of the balance sheet shows BHP-Billiton's financial assets, its current and non-current liabilities (debts), and equity (proprietorship), which reflect the company's capital structure decisions and sources of funds (capital) to invest in the real assets.

BHP-Billiton			
Consolidated Balance Sheet as at 30 June 2013			
Assets	US\$m	Liabilities	US\$m
Current Assets		Current Liabilities	
Cash and cash equivalents	6,060	Trade and other payables	10,881
Trade and other receivables	6,728	Interest bearing liabilities	5,303
Other financial assets	159	Liabilities held for sale	220
Inventories	5,822	Other financial liabilities	217
Assets held for sale	286	Current tax payable	1,148
Current tax assets	327	Provisions	2,395
Other	404	Deferred income	208
Total current assets	19,786	Total current liabilities	20,372
Non-Current Assets		Non-Current Liabilities	
Trade and other receivables	1,579	Trade and other payables	293
Other financial assets	1,698	Interest bearing liabilities	29,862
Inventories	622	Other financial liabilities	582
Property, plant and equipment	102,927	Deferred tax liabilities	6,469
Intangible assets	5,226	Provisions	8,237
Deferred tax assets	6,136	Deferred income	259
Other	135	Total non-current liabilities	45,702
Total non-current assets	118,323	Total liabilities	66,074
Total assets	138,109		
		Equity	
		Share capital	1,186
		(BHP Billiton Limited)	
		Share capital – BHP Billiton Plc	1,069
		(BHP Billiton Plc)	
		Treasury shares	-540
		Reserves	1,970
		Retained earnings	66,979
		Total equity attributable	
		to members of BHP Billiton Group	70,664
		Non-controlling interests	1,371
		Total equity	72,035
		Total liabilities and equity	138,109

6.2.1 Non-Discounted Cash-Flow Analysis

With non-discounted cash-flow methods of project evaluation, future cash-flows associated with an investment are not discounted back to time-period zero at an appropriate discount (interest) rate. As such, the non-discounted cash-flow capital budgeting methods do not take into account the time-value of money. The two main methods of non-discounted cash-flow capital budgeting

analysis are the payback period method and the accounting rate of return method.

Payback Period Method

The payback period method of project evaluation is a relatively simple technique which calculates the amount of time required for an investment to generate net cash-flows to cover the initial cost of the investment. With this approach, generally, an investment is acceptable if its payback period is less than some prescribed number of periods and, normally, projects with shorter payback periods are preferred.

■ **Example 6.1 — Payback Period Method – Calculating the payback period** . Assume the initial outlay or cost of a three-year project is \$100 million. Also assume that the net cash-inflows associated with the project are as follow.

Net Cash-Flows

Year 1	\$20 million
Year 2	\$40 million
Year 3	\$60 million

To work out the payback period on this project we need to calculate how long it takes for the net cash-inflows of the project to cover the initial cost of \$100 million. To do that we will need to work out the accumulated net cash-inflows at the end of each year for the project as follows.

Accumulated Net Cash-Inflows

Year 1	\$20 million
Year 2	\$60 million (i.e. \$20 million + \$40 million)
Year 3	\$120 million (i.e. \$60 million + \$60 million)

By the end of Year 2 the accumulated net cash-inflows of the project are \$60 million, meaning another \$40 million is required to cover the cost of the project of \$100 million. By the end of Year 3, the accumulated net cash-inflows are \$120 million, \$40 million greater than the cost of the project. Hence, payback on this project occurs some time between the end of Year 2 and the end of Year 3. Since at the end of Year 2 there is still \$40 million left to cover the cost of the project, and since the net cash-inflow in Year 3 is \$60 million, we can say that payback occurs sometime after two years. Specifically, the accumulated net cash-inflow is \$60 million, plus another 2/3 of a year, calculated as follows.

$$\frac{\text{Remaining cash-flow at the end of Year 2}}{\text{Net cash-flow in Year 3}} = \frac{\$40 \text{ million}}{\$60 \text{ million}} = \frac{2}{3}.$$

Hence, the payback period for this project is 2.67 years, i.e. 2 years and 8 months. ■

■ **Example 6.2 — Payback period: choosing amongst projects**. Assume a company has a \$50 million capital budget limit and has the opportunity to invest in any one of three possible projects, i.e. the projects are mutually exclusive. Each of the projects has a three-year lifespan. The annual net cash-inflows associated with each of the three projects, Projects A, B, and C, are listed here along with the payback periods.

	Cost	Net cash-inflow			Payback Period
	Year 0	Year 1	Year 2	Year 3	
Project A	-\$50 million	\$1	49	0	2 years
Project B	-\$50 million	\$49	0	1	3 years
Project C	-\$50 million	\$0	0	500	3 years

Each of the projects has the same cost, listed as a \$50 million cash-outflow at time-period zero (T_0), but different cash-inflows and different payback periods. Project A repays itself in two years, Project B repays itself in three years, and for Project C, the repayment period is three years. Hence, given that the projects are mutually exclusive, the payback period methods says that we should choose Project A as it pays itself off most quickly (i.e. Project A has the shortest payback period).

It is important to note in this example that the total cash-inflows associated with Project C are much larger than those associated with both Projects A and B; this point will be discussed further in the next section. ■

Advantages and Disadvantages of the Payback Period Method

The advantages of the payback period method of project evaluation are that it is a simple method that is easy to understand and does not require too much detailed analysis. Also, it adjusts for the uncertainty of later cash flows. In other words, we can be more certain about the cash-flows associated with shorter-term projects, because, the further out in time we go the harder it becomes to predict with accuracy. Finally, it is biased towards liquidity, meaning that projects that pay themselves off more quickly give the company access to cash more quickly.

The disadvantages of the payback period method of project evaluation are that the time-value of money and risk are ignored, since there is no discounting back to T_0 of future net cash-flows at an appropriate rate of interest. Furthermore, the determination of an acceptable payback period is ad-hoc, as there is no scientific method for determining the maximum time period for projects to pay themselves off.

Another disadvantage of the method is that it ignores cash-flows beyond the cut-off date. For example, Project A is preferred over Project C in Example 6.2, although Project C has much higher cash-inflows beyond the two-year cut-off payback period. Therefore, the payback period is biased against long-term projects that could increase the market value of the firm.

Accounting Rate of Return

The *accounting rate of return* (ARR) method of project evaluation is a relatively simple technique which is a measure of an investment's profitability. ARR divides the average profit of a project by the initial cost of the investment in order to get the ratio or return that can be expected, allowing investors or business owners to compare the profit potential for projects, products and investments. With this approach, an investment is acceptable if its ARR is greater than some target rate of return. The formula for the ARR is as follows.

$$\text{ARR} = \frac{\text{average net profit}}{\text{average book value}}.$$

■ **Example 6.3 — Calculating the ARR on a Project.** Assume that the cost of a project is \$240 million and that it is a three-year project with the following relevant information:

	Year 1 (\$ million)	Year 2 (\$ million)	Year 3 (\$ million)
Sales	440	240	160
Expense	220	120	80
Gross profit	220	120	80
Depreciation	80	80	80
Earnings before tax	140	40	0
Tax (25%)	35	10	0
Net profit	105	30	0

Therefore,

$$\begin{aligned}
 \text{Average net profit} &= \frac{\sum \text{Net profit}}{n} \\
 &= \frac{\$105 \text{ million} + \$30 \text{ million} + \$0 \text{ million}}{3} \\
 &= \frac{\$135 \text{ million}}{3} = \$45 \text{ million}
 \end{aligned}$$

and

$$\begin{aligned}
 \text{Average book value} &= \frac{\text{Initial investment} + \text{Book value at the end}}{2} \\
 &= \frac{\$240 \text{ million} + \$0 \text{ million}}{2} \\
 &= \$120 \text{ million}
 \end{aligned}$$

Note that in Business Finance we will always assume that book-value at the end of a project is \$0.

Hence,

$$\begin{aligned}
 \text{ARR} &= \frac{\text{Average net profit}}{\text{Average book value}} \\
 &= \frac{\$45 \text{ million}}{\$120 \text{ million}} \\
 &= 37.5\%
 \end{aligned}$$

Therefore, if in this example the target return is 20%, then because this project has an ARR of 37.5% (which is greater than 20%) we would proceed with the project. ■

Advantages and Disadvantages of the Accounting Rate of Return

The advantages of the ARR method of project evaluation are that, like the payback period method, it is a simple method that is easy to understand and does not require too much detailed analysis. Also, it considers all profits of a project.

The disadvantages of the ARR method of project evaluation are that, similar to the payback period method. In particular, the time-value of money and risk are ignored, as there is no discounting back to T_0 of future net cash-flows at an appropriate rate of interest and this is

because both these methods are non-discounted cash-flow methods. Furthermore, the measure is not a 'true' reflection of return, because it is not a compounded return. Additionally, the method involves the use of profits and book values instead of cash-flows and market values. Finally, akin to the payback period method, the target rate is determined in a subjective manner.

6.2.2 Discounted Cash-Flow Analysis

Discounted cash-flow analysis is used to determine whether an investment is worthwhile, and, generally, involves discounting the future net cash-inflows of a project back to time-period zero to find their present values, which are then compared to the cost of the project to make a decision as to whether or not the project is accepted.

The two main methods of discounted cash-flow capital budgeting analysis are the net present value method and the internal rate of return method. The great advantages of these discounted cash-flow methods is that, unlike with non-discounted cash-flow methods, the time-value of money and riskiness of the future cash-flows are taken into account and the methods are based on cash-flow analysis.

Net Present Value Method

The net present value (NPV) method of project evaluation gives the difference between an investment's market value (in today's dollars) and its cost (also in today's dollars). It takes the difference between the present-value of the cash-inflows and cash-outflows associated with a project, and measures how much value is created by undertaking an investment. With the NPV method the future net cash-inflows of a project are discounted back to time-period zero to find their present value, and the combined present-value of the future net cash-inflows is then compared to the cost of the project to make a decision as to whether or not the project is accepted. An investment should be accepted if its NPV is positive and rejected if it is negative.

The formula used to find the NPV of a project is follows.

$$NPV = \sum_{t=1}^n \frac{CF_t}{(1+r)^t} - CF_0$$

where CF_t are the future net cash-inflows of the project that are discounted back to time-period zero at an appropriate discount rate to find their present-values, which are then summed, and CF_0 is the initial cost of the project, which is a net cash-outflow at time-period zero. This initial cost is then subtracted from the sum of the present-values of the future net cash-inflows to find the net present-value of the project at time-period zero. Put simply, there are five steps involved in a net-present value analysis:

- Step 1:** Estimate the net cash-flows (inflows less outflows) of the project;
- Step 2:** Assess the riskiness of the NCFs;
- Step 3:** Determine r (discount rate for the project – based on the riskiness of the cash-flows);
- Step 4:** Find the NPV of the project; and
- Step 5:** Accept the project if $NPV > 0$ (and reject the project if $NPV < 0$).

■ **Example 6.4 — A Simple Net Present Value Analysis¹**. Assume the following for Project X, which is a two-year project that has a required rate of return of 10% p.a.:

Year	Revenues	Expenses	NCF	PVIF	PV
	(\$ million)	(\$ million)	(\$ million)	(\$ million)	(\$ million)
0			-\$1,100	$(1.10)^0 = 1$	-\$1,100.00
1	\$1,000	- \$500	= \$500	$(1.10)^{-1} = 0.9090$	+ \$454.50
2	\$2,000	- \$1,000	= \$1,000	$(1.10)^{-2} = 0.8265$	+ \$826.45
				NPV	= \$180.95

In the table above, note the following.

- Net cash flows (NCFs) are the net cash-inflows at the end of each year of the project. At time zero (T_0) the initial NCF is considered the cost of the project;
- The net present-value of the project is the difference between the initial cost of the project and the present-value of the future net cash-inflows of the project; and
- Since, in this example, the NPV is positive, we would proceed with the project as the market-value of the company at T_0 would increase by \$180.95 million if the project is undertaken.

■

Internal Rate of Return Method

The *internal rate of return (IRR) method* of capital budgeting analysis is the second of the discounted cash-flow methods of project evaluation. Put simply, IRR is the actual rate of return on a project that makes the present value of the project's future net cash-inflows equal to the cost of the project. Generally speaking, for individual projects, the project is accepted if its IRR is greater than the required rate of return on the project, and the higher an investment's IRR, the better the investment. Consequently, IRR is often used to choose among alternative projects, with projects having higher IRRs chosen.

There is no “magic formula” for finding the IRR, and unless you are using a financial calculator, you have to find a project's IRR through trial and error, i.e. you have to try different discount rates/interest rates until you find the rate that makes the net present value (NPV) of the project equal to \$0.

Highlight 6.1 — NPV and IRR. The rationale for the IRR method is that if a project's IRR is greater than its required rate of return, r , then the project's actual rate of return is greater than its cost. Therefore, NPV of the project will be positive, and some return is left over to increase shareholders' wealth. Hence, the criteria for this method is that a project is accepted if its IRR is greater than its r as the NPV of the project will be positive, and rejected if its IRR is less than its r as NPV will be negative.

Below is a numerical example which shows how it is possible to work out the IRR for a particular project.

■ **Example 6.5** Assume that the initial cost of a project, i.e. net cash-outflow at T_0 , is \$200 million and the following are the net cash-inflows for the project over its three-year lifespan:

¹This is referred to as a simple net present-value analysis because the revenues, expenses, and NCFs have already been calculated. In a detailed NPV analysis you would be required to calculate revenues, expenses, and NCFs yourself.

Year	Net Cash Flow
1	\$50 million
2	\$100 million
3	\$150 million

In working out this project's IRR we must find such that at IRR the NPV of the project = \$0.

$$\$0 = -\$200 + \$50(1 + IRR)^{-1} + \$100(1 + IRR)^{-2} + \$150(1 + IRR)^{-3},$$

which can be re-arranged to read as follows.

$$\$200 = \$50(1 + IRR)^{-1} + \$100(1 + IRR)^{-2} + \$150(1 + IRR)^{-3}.$$

To calculate IRR we use trial-and-error. In other words, we will try different interest rates until we find the one that makes the NPV of the project \$0. For example, if you used the interest rates listed below, you would get the corresponding NPVs.

Discount Rate	NPV
0%	\$100 million
5%	\$68 million
10%	\$41 million
15%	\$18 million
20%	-\$2million

Let us try some different interest rates and examine the NPV of the project. Remember, at the IRR the project's NPV will equal \$0.

Trial 1: Interest rate (r) = 0.00%. Therefore,

$$\begin{aligned} NPV &= -\$200 + \$50(1.00)^{-1} + \$100(1.00)^{-2} + \$150(1.00)^{-3} \\ &= -\$200 + \$50(1.00) + \$100(1.00) + \$150(1.00) \\ &= -\$200 + \$50 + \$100 + \$150 \\ &= \$100. \end{aligned}$$

At 0% the NPV is positive \$100. Therefore, 0% must be too low as a discount rate. So, we must now try a higher interest rate.

Trial 2: Interest rate (r) = 15.00%. Therefore,

$$\begin{aligned} NPV &= -\$200 + \$50(1.15)^{-1} + \$100(1.15)^{-2} + \$150(1.15)^{-3} \\ &= -\$200 + \$50(0.8696) + \$100(0.7561) + \$150(0.6575) \\ &= -\$200 + \$43.48 + \$75.61 + \$98.63 \\ &= \$18. \end{aligned}$$

At 15.00% the NPV is positive \$18, therefore, 15.00% must be too low as a discount rate. So again, we must now try a higher interest rate.

Trial 3: Interest rate (r) = 20.00%. Therefore,

$$\begin{aligned} NPV &= -\$200 + \$50(1.20)^{-1} + \$100(1.20)^{-2} + \$150(1.20)^{-3} \\ &= -\$200 + \$50(0.8333) + \$100(0.6944) + \$150(0.5787) \\ &= -\$200 + \$41.67 + \$69.44 + \$86.81 \\ &= -\$2. \end{aligned}$$

At 20% the NPV is negative \$2, therefore, 20.00% must be too high as a discount rate. So, we would now have to try a lower discount rate to get the NPV of the project equal to \$0.

Eventually, after going back and forward with different interest rates, we would find that the interest rate that makes the NPV of this project equal to \$0 is 19.44%. This rate will make the present value of the future net cash-inflows of this project equal to its cost (net cash outflow at T_0) and is the actual rate of return on the project.

Trial 4: Interest rate (r) = 19.44%. Therefore,

$$\begin{aligned} NPV &= -\$200 + \$50(1.1944)^{-1} + \$100(1.1944)^{-2} + \$150(1.1944)^{-3} \\ &= -\$200 + \$50(0.8372) + \$100(0.7010) + \$150(0.5869) \\ &= -\$200 + \$41.86 + \$70.10 + \$88.04 \\ &= \$0 \end{aligned}$$

At 19.44%, the NPV of the project will be \$0. Therefore, $IRR = 19.44\%$. ■

Problems with the IRR Method

There are two major problems with the IRR method of project evaluation which can cause problems if this method is used in project evaluation. The first of these problems is that if there is more than one negative net cash-flow for a project this will lead to the problem of multiple internal rates of return. For example, assume that you are considering a project which has the following net cash-flows:

Year	Net Cash Flow
0	-\$252 million
1	\$1,431 million
2	-\$3,035 million
3	\$2,850 million
4	-\$1,000 million

Using the IRR method we would then use trial-and-error until we find the rate that makes the NPV of the project \$0. The problem here is that, because there is more than one negative net cash-flow associated with this project, in periods 0, 2, and 4, we would find that the NPV of the project would be \$0 at four different rates: 25.00%, 33.33%, 42.86%, and 66.67%. This is problematic as there can only ever be one IRR for a project, meaning that to have multiple IRRs is erroneous. Thus, if a project has more than one negative net cash-flow you cannot use the IRR method to evaluate the project.

The second problem with the IRR method arises if multiple projects are being compared and the projects are not independent, but are mutually exclusive. For independent projects the NPV and IRR methods are complementary to one another and will both lead to the same accept or reject decision for the projects. This means that if one is analysing independent projects the

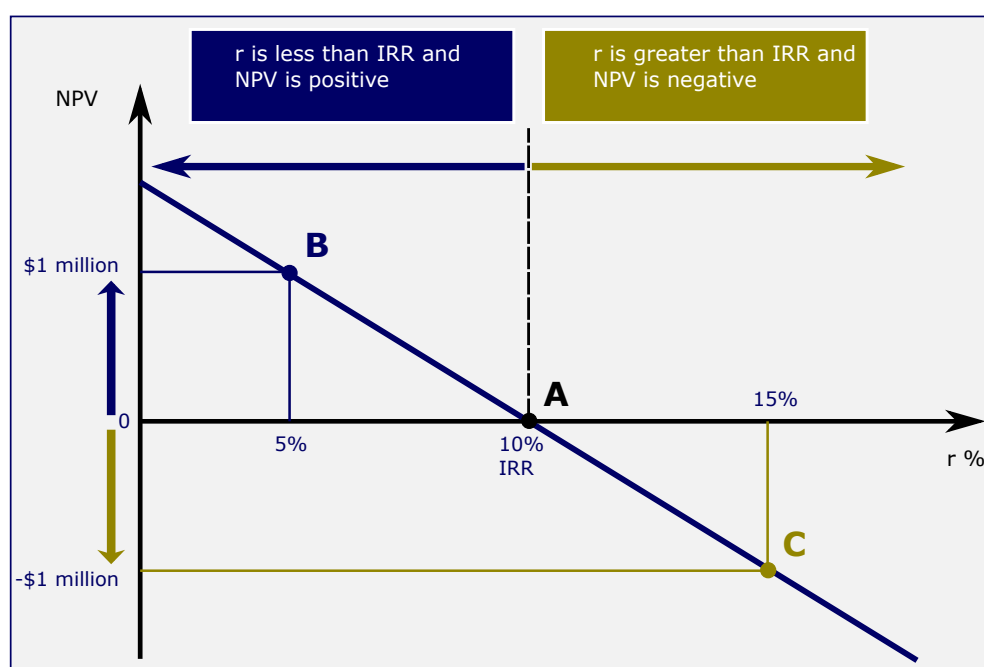


Figure 6.1: The IRR method for the evaluation of independent projects

NPV method and/or the IRR method can be used for project evaluation. For mutually exclusive projects, the IRR cannot be used as the cash-flows of one project are affected by the acceptance of another project.

Highlight 6.2 — Independent vs mutually exclusive projects. An independent project is one whose acceptance or rejection is independent of the acceptance or rejection of other projects. Thus, the funding of an independent project does not depend on another project receiving funding first. Mutually exclusive projects are those where acceptance of one project rules out acceptance of another project.

Applying The IRR Method to Evaluation of an Independent Project

Assume we have a project that is an independent project and we have worked out that the IRR on the project is 10%, as shown on the horizontal axis at Point A in Figure 4.1 below, where the horizontal axis measures interest rates and the vertical axis measures NPV in dollar terms. If we assume that the actual required rate of return on this project is, say, 5%, then on the horizontal axis of Figure 6.1 we would be to the left of the IRR of 10% and based on IRR analysis we would choose the project, as its IRR (actual rate of return) is greater than its r . Additionally, based on the NPV method of project evaluation, if we assumed that the r of the project is 5% we would also proceed with the project as on the downward sloping NPV function in Figure 6.1 (blue line), we would be above the horizontal and at Point B and NPV would be positive (\$1 million), so we would proceed with the project. Hence, both IRR and NPV analysis have told us to accept this project.

Similarly, if we were to assume that the actual required rate of return on this project is, say, 15%, then on the horizontal axis of Figure 6.1 we would be to the right of the IRR of 10% and based on IRR analysis we would not choose the project as its IRR (actual rate of return) is less than its r . Additionally, based on the NPV method of project evaluation, if we assumed that the r of the project is 15%, we would also not proceed with the project, as on the downward

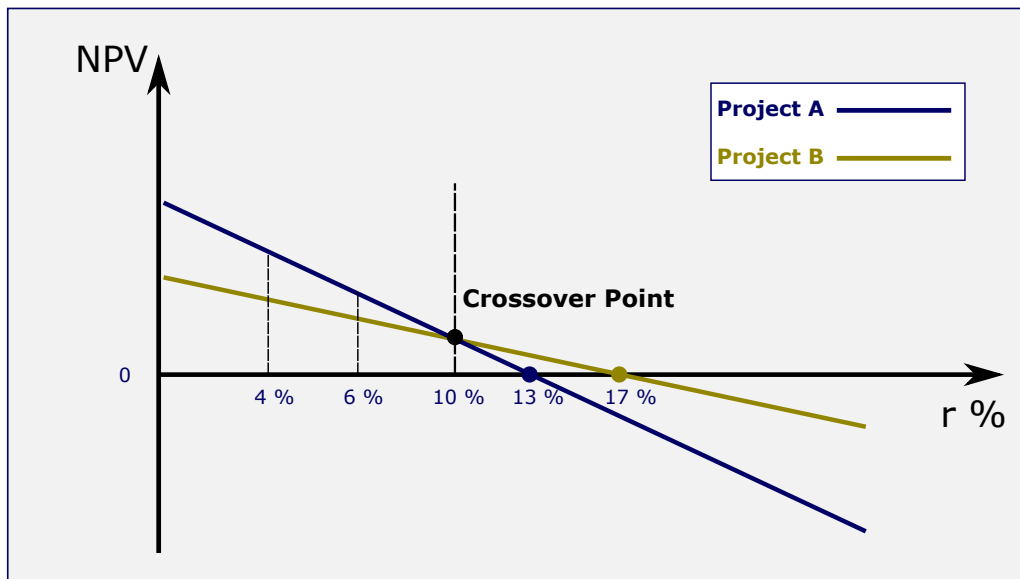


Figure 6.2: The IRR method for the evaluation of mutually exclusive projects

sloping NPV function in Figure 6.1 (blue line) we would be below the horizontal. At Point C, NPV would be negative (-\$1 million), so we would not proceed with the project. Hence, both IRR and NPV analysis have told use not to accept this project.

For an independent project, when NPV is positive IRR will be greater than r , so, using either the NPV or IRR approach we will accept the project. With the same token, if the NPV is negative the IRR will be less than r and using either the NPV or IRR approach we will reject the project. Thus, when dealing with independent projects we can use either the NPV approach or the IRR approach to evaluate the projects.

The IRR Method Applied to Mutually Exclusive Projects

For mutually exclusive projects, using the IRR method may lead to an incorrect accept or reject decision being made. Assume we have two projects, Projects A and B, which are mutually exclusive of one-another, and where the IRR of Project A is 13% and the IRR of Project B is 17%. Hence, based on IRR analysis, since the IRR of Project B is higher than the IRR of Project A, and we can only choose one of the projects, we would choose Project B. However, if we were to base our project evaluation analysis on the NPV method, our decision may be different and we may not choose Project B over Project A. Figure 6.2 illustrates this concept.

It can be seen that at required rates of return greater than 10% (the cross-over point for the NPV functions of the two projects) the NPV of project B is greater than the NPV of Project A. On the right-hand-side of the cross-over point the NPV function of Project B is above the NPV function of project A, so at any required rate of return greater than 10% both the IRR and NPV methods of project evaluation would lead to Project B being chosen. However, at discount rates lower than 10%, on the left-hand-side of the cross-over point, the NPV function of Project A is above that of Project B. This means that we should choose Project A, rather than Project B, as project A will increase the market value of the firm by more than would Project B. However, regardless of what the required rate of return is, and regardless of which project has the higher NPV, the IRR method would always lead to Project B being chosen as Project B has the higher IRR. This is the problem with the IRR approach when dealing with mutually exclusive projects. Consequently, because of the possible errors, you cannot use IRR approach to evaluate mutually

exclusive projects, rather only use the NPV approach for this purpose.

Incremental Cash-Flows

In NPV analysis the focus is on incremental cash-flows, which are the additional operating cash-flows that occur as a result of taking on a new project. In undertaking an NPV project evaluation we need to consider what difference it makes to the total cash-flows of the firm whether the firm does or does not undertake the project under consideration. In addressing the issue of the incremental cash-flows associated with a potential new project we must address three related issues, these being sunk costs, opportunity costs, and side effects.

Sunk Costs

A *sunk costs* is a cost that has already been incurred and, thus, cannot be recovered because it has already happened. Therefore, it is independent of any event that may occur in the future. Sunk costs are unavoidable (incurred in the past) cash-outflows no longer relevant to influencing whether a project should be undertaken. In NPV project evaluation we ignore sunk costs. Examples, of sunk costs include the costs of marketing or feasibility studies for a potential new product as these costs are incurred before the product is developed and, also, must be paid regardless of whether the company proceeds with development of the product.

■ **Example 6.6 — Sunk costs in NPV analysis.** Suppose a \$10,000 payment is to be made to a marketing company for assessing the market for a potential new product, which costs \$125,000 (net cash-outflow at T_0) and yields net cash-inflows of \$75,000 a year for two years, and the discount rate is 10% p.a. Should this project be taken on?

If the NPV analysis is done *incorrectly* and the sunk cost of the marketing study is included in the analysis as a cost of the potential new product, meaning that the net cash-outflow at T_0 will be $\$125,000 + \$10,000 = \$135,000$, then the NPV analysis will yield the following result.

$$\begin{aligned}\text{NPV} &= -\$135,000 + \$75,000(1.10)^{-1} + \$75,000(1.10)^{-2} \\ &= -\$4,800\end{aligned}$$

Therefore, since the NPV is negative we would reject the project, which is an *incorrect* decision.

If the NPV analysis is done correctly the \$10,000 sunk cost of the marketing study would not be included in the analysis and the following would be the result:

$$\begin{aligned}\text{NPV} &= -\$125,000 + \$75,000(1.10)^{-1} + \$75,000(1.10)^{-2} \\ &= \$5,200\end{aligned}$$

Therefore, since the NPV is positive we would accept the project. This is the *correct* decision, which may lead to increasing the market value of the company by \$5,200. ■

Opportunity Costs

An *opportunity cost* is the the cost of an alternative that must be forgone in order to pursue a certain action or, stated alternatively, it is the benefit that could have been received by taking an alternative action. In NPV analysis an opportunity cost refers to a situation where factors of production (FoP) or resources of a company must be used in a new project, but previously they were being used for another purpose. In NPV analysis, if a project uses resources which could be put to some other use then the dollar value of the alternative use must be included as an expense and a cash-outflow in the project evaluation.

■ **Example 6.7 — Opportunity Costs in NPV Analysis.** Suppose that a company that manufactures aluminium cans leases out canning machinery that generates a rental income for the company of \$3 million per year. Now assume that the canning machinery will be used in a project for the company that will yield the company \$20 million a year for two years and will cost \$30 million (net cash-outflow at T_0). Also assume that the required rate of return for the company is 10% p.a. Should this project be taken on?

If the NPV analysis is done incorrectly and the opportunity cost of the lost rental income on the canning machinery of \$3 million per year is ignored, the NPV analysis will yield the following result.

$$\begin{aligned}\text{NPV} &= -\$30 \text{ million} + \$20 \text{ million}(1.10)^{-1} + \$20 \text{ million}(1.10)^{-2} \\ &= \$4.7 \text{ million}\end{aligned}$$

Therefore, since the NPV is positive, we would accept the project. This is an *incorrect* decision.

If the NPV analysis is done correctly the opportunity cost of the \$3 million in lost rental income would be subtracted from the \$20 million net cash-inflow per year that the new project would yield the company, meaning that the net cash-inflow each year for the two years would fall to \$20 million – \$3 million = \$17 million and the following would be the result.

$$\begin{aligned}\text{NPV} &= -\$30 \text{ million} + \$17 \text{ million}(1.10)^{-1} + \$17 \text{ million}(1.10)^{-2} \\ &= -\$0.5 \text{ million}\end{aligned}$$

Therefore, since the NPV is negative we would reject the project – this is the *correct* decision as the new project would, in reality, decrease the market value of the company and the wealth of the owners (shareholders) of the company by \$500,000. ■

Side Effects

In NPV analysis a *side effect* refers to a situation where the sale of a new product by a company affects the sales, either positively or negatively, of other products sold by the company. In a NPV analysis we must include any positive or negative cash-flows that occur in other aspects of the business as a result of taking on the new activity. Side effects are similar to opportunity costs in the sense that we are taking into account effects on the cash-flows of the business as a result of a new project or investment. The difference between a side effect and an opportunity costs is that, while opportunity costs relate to use of factors of production, side effects relate to the products or services produced by the business.

■ **Example 6.8 — Side Effects in NPV Analysis.** Suppose that a car manufacturer has decided to switch production from sedan cars to sports cars, with 2,000 sports cars that net the company \$40,000 each to be manufactured and sold at the cost of 2,000 sedan cars that were netting the company \$30,000 each. The cost of switching production at the plant from sedan to sports cars is estimated to be \$100 million (net cash-outflow at T_0). Also assume that the required rate of return for the company is 10% p.a. Should this project be taken on?

If the NPV analysis is done incorrectly and the side effect of the lost revenue from no longer selling the 2,000 sedan cars is ignored, the NPV analysis will yield the following result:

$$\begin{aligned}\text{Cash-flow from sales} &= 2,000 \text{ sports cars} \times \$40,000 \text{ each} \\ &= \$80 \text{ million}\end{aligned}$$

$$\begin{aligned}\text{NPV} &= -\$100m + \$80 \text{ million}(1.10)^{-1} + \$80 \text{ million}(1.10)^{-2} \\ &= \$38.84 \text{ million}\end{aligned}$$

Therefore, since the NPV is positive we would accept the project, which is an *incorrect* decision.

If the NPV analysis is done correctly, the side effect of the foregone revenue from the company no longer selling 2,000 sedans would need to be subtracted from the \$80 million net cash-inflow per year that the new project would yield with the sale of the 2,000 sports cars and the following would be the result.

$$\begin{aligned}\text{Cash-flow from sales} &= \$80 \text{ million} (2,000 \text{ sports cars} \times \$40,000 \text{ each}) \\ &\quad - \$60 \text{ million} (2,000 \text{ sedan cars} \times \$30,000 \text{ each}) \\ &= \$20 \text{ million}\end{aligned}$$

$$\begin{aligned}\text{NPV} &= -\$100m + \$20 \text{ million}(1.10)^{-1} + \$20 \text{ million}(1.10)^{-2} \\ &= -\$68.29 \text{ million}\end{aligned}$$

Therefore, since the NPV is negative we would reject the project. This is the *correct* decision. The new project would, in reality, decrease the market value of the company and the wealth of the owners (shareholders) of the company by \$65.29 million. ■

Financing Costs

Financing costs are interest expenses for debt finance and dividend payments for equity finance. In project evaluation analysis the costs of financing can be taken account of either in the cash-flows or via the required rate of return r .

Interest expense represents the cost incurred by an entity for borrowed funds and can arise from borrowings through bonds or debentures, as well as from banks. It is calculated as the interest rate on the borrowings times the outstanding principal amount of the debt. Generally, for a corporation interest expense is tax-deductible, meaning that it can be included as an expense in the profit and loss statement.

Dividends, on the other hand, represent distribution of a portion of a company's earnings, as decided by the senior management of the company, to the shareholders and represent the cost of the company's equity finance. Dividends are not tax-deductible, so cannot be included as an expense in the P&L statement.

In *capital budgeting analysis* we take account of financing costs implicitly through the required rate of return or discount rate. Therefore, it is important financing costs are not also taken into account in the cash-flows or there will be double counting. The required rate of return is the return that the firm has to earn on the project in order to satisfy the providers of financial capital for any project or investment, so we assume that r covers the costs of finance.

Taxation

Taxes are usually levied by federal, state, and local government and means by which governments finance their expenditures by imposing charges on corporations and citizens. In terms of companies, corporate taxation represents a levy placed on the profit of a firm earned during a

given taxable period and applied to the company's operating earnings, after expenses such as cost of good sold, depreciation, and interest paid have been deducted from revenues.

The major impact of taxation on a corporation is that the payment of any tax liability represents a cash-outlay or cash-outflow, while tax-deductible expenses, such as depreciation and interest charges, provide the company with what is called a 'tax-shield'.

Highlight 6.3 — Book Gain or Loss on Sale versus Capital Gains or Losses. In terms of tax treatment, a book gain or loss on the sale of a productive asset, i.e. on the sale of a factor of production, is different to a capital gain or loss on the sale of an investment asset.

Gain or loss on sale of factor of production: If the salvage value (sale price) of a productive asset is greater than the book value of the asset we have a gain on sale. So tax must be paid on the gain. Similarly, if the salvage value (sale price) of a productive asset is less than the book value we have a loss on sale, and the loss provides a tax rebate (tax shield).

Example: Book Gain on Sale of Productive Asset

- Purchase price of asset (machine) \$1 million, with useful life of machine 10 years
- Depreciation (Straight-line) per annum = \$1 million/10 years = \$100,000
- After seven years company decides to sell the asset.
- Book value of asset after seven years is
purchase price (\$1 million) - accumulated depreciation ($7 \times \$100,000$) = \$300,000.
- Salvage value of asset at seven years = \$350,000
- Gain on sale = salvage value (\$350,000) - book value (\$300,000) = \$50,000.
- Tax rate = 30%; therefore
Tax on book gain = book gain (\$50,000) $\times$ tax rate (0.30) = \$15,000.

Note: we do take depreciation into account in calculating the book gain on a productive asset.

Disposal of Investment Assets: Capital Gains or Capital Losses: Capital gains (losses) on investment assets increase (decrease) the amount of tax paid. If the sale price of an investment asset is greater than the purchase price of the asset, we have a capital gain, and tax must be paid on the capital gain. If the sale price of an investment asset is less than the purchase price of the asset we have a capital loss. However, capital losses do not provide a tax rebate, but can be used to offset capital gains (in the current period and/or in the future).

Example: Capital Gain on Sale of an Investment Asset

- An investment property was purchased in 2011, with the purchase price of \$1 million, and sold in 2015 for \$1.3 million.
- Capital gain on sale = \$1.3 million - \$1 million = \$300,000.
- Tax rate = 40%; therefore,
Tax liability on capital gain = capital gain $\times$ tax rate = \$300,000 $\times$ 0.40 = \$120,000.

Note: we do not take depreciation into account when working out the capital gain (or loss) on an investment asset.

Real-World use of Capital Budgeting Techniques

Truong, Partington, and Peat (2008) in their paper entitled Cost-of-Capital Estimation and Capital-Budgeting Practice in Australia appearing in the Australian Journal of Management (Vol. 33, No. 1, pp. 95-121) employed a sample survey to analyse the capital-budgeting practices of Australian publicly-listed companies and reported the popularity of the use of various capital budgeting techniques. Figure 6.3, presents the results.

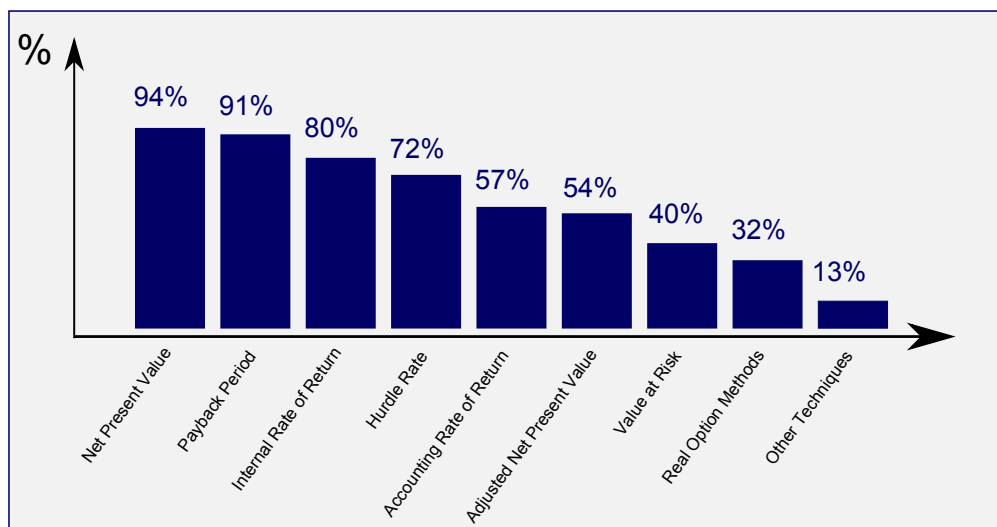


Figure 6.3: Capital budgeting techniques used by Australian companies, ranked based on popularity (Source: Truong, Partington, and Peat (2008))

6.3 Detailed NPV Analysis

6.3.1 NPV Investment Evaluation Process

Here and below we will discuss and see an example of how to undertake a detailed net present value analysis. A detailed NPV analysis is one where, generally, an analyst must calculate the net cash-flows for an investment/project/product, as distinct from where the net cash-flows have already been calculated. With a detailed NPV analysis there are, generally, seven steps involved:

- Step 1:** Calculate depreciation per annum;
- Step 2:** Calculate any gain or loss on sale of productive assets;
- Step 3:** Calculate taxable income, i.e. profit or loss;
- Step 4:** Calculate the tax item, i.e. tax liability or tax rebate;
- Step 5:** Calculate net cash-flows;
- Step 6:** Discount net cash-flows;
- Step 7:** Find NPV and reach conclusion
 - NPV positive – accept the project,
 - NPV negative – reject the project.

■ **Example 6.9 — A Detailed NPV Analysis.** Here we will see an example of a detailed NPV analysis. This is regarded as a ‘two-step’ analysis as both a profit and loss and a cash-flow statement are prepared.

Notes For Analysis

1. Purchase price of new machine: \$42 000;
2. Useful life of new asset: three years;
3. Depreciation: straight- line;
4. Salvage value of new asset at end of Year 3: \$1,000;
5. Operating net cash-inflows p.a.: Year 1: \$31 000; Year 2: \$29 000; Year 3: \$27 000;
6. Feasibility study cost: \$4,000;
7. Warehouse previously rented out for \$8,000 p.a. to be used in new project;
8. New technician to replace existing technician - existing technician’s salary: \$65 000 p.a.,

- new technician's salary: \$70 000 p.a.;
9. Old machine to be sold for \$2,500; book value: \$3,000;
10. Tax rate: 30%;
11. Required rate of return: 12% p.a.

Step 1: Calculate depreciation per annum.

At Step 1, we must work out the annual depreciation charge on the new asset, and depreciation p.a. is calculated as follows

$$\text{Depreciation p.a.} = \frac{\text{cost of asset}}{n}$$

where n = number of years. Therefore, in this example depreciation per annum will be:

$$\text{Depreciation p.a.} = \frac{\$42,000}{3} = \$14,000 \text{ p.a.}$$

Step 2: Calculate any gain or loss on sale of productive assets.

At Step 2, we must calculate any gain or loss on the sale of productive assets or factors of production, and this is for both any old assets sold and for the new asset when it is sold at the end of its useful life. We calculate any gain or loss on the sale of a productive asset as the salvage value (SV) of the asset less the book value of the asset at the point in time when it is sold (BV_t). The book value is calculated as the purchase price of the asset less the accumulated depreciation at that point in time. In this example book gain (loss) will have to be calculated for the old machine and for the new machine, as follows.

$$\begin{aligned} \text{Old Machine:} \quad \text{loss on sale} &= SV_0 - BV_0 \\ &= \$2,500 - \$3,000 = -\$500 \end{aligned}$$

Note that for the old machine we calculate gain (loss) on sale at T_0 as this is when the new asset is purchased and replaces the old asset. Also, for the old machine we do not need to calculate book value₀ as it has been given to us.

$$\begin{aligned} \text{New Machine:} \quad \text{Gain on sale} &= SV_3 - BV_3 \\ &= \$1,000 - \$0 = \$1,000 \end{aligned}$$

$$\text{Note:} \quad BV_3 = \$42,000 - (3 \times \$14,000) = \$0$$

For the new machine we calculate gain (loss) on sale at T_3 , as this is the end of its useful life and when it is sold. Also, for the new machine, book value at the end of its useful life will be \$0.

Step 3: Calculate taxable income, i.e. profit or loss, and

Step 4: Calculate the tax item, i.e. tax liability or tax rebate.

At Step 3 we must calculate the profit or loss for each time period over the life of the asset via a profit and loss (P&L) statement in order to then undertake Step 4, which is to work out the

tax liability (on a profit) that must be paid to the tax department or tax-rebate (on a loss) that will be received from the tax department.

In this example, profit or loss is calculated for each time period, more specifically, for the end of each time-period, with the items relevant to the P&L statement being included as follows:

Profit and Loss Statement	Year 0	Year 1	Year 2	Year 3
Operating Cash-flows		\$31,000	\$29,000	\$27,000
Depreciation		(\$14,000)	(\$14,000)	(\$14,000)
Rent Foregone		(\$8,000)	(\$8,000)	(\$8,000)
Gain & Loss on Sale	(\$500)			\$1,000
Incremental Salary		(\$5,000)	(\$5,000)	(\$5,000)
Taxable Income (P&L)	(\$500)	\$4,000	\$2,000	\$1,000
Tax Paid (30%)	\$150	(\$1,200)	(\$600)	(\$300)

We can see in the above table the items that are relevant in calculating profit or loss in this example. These relevant items are

- The operating cash-flows (inflows) of \$31 000, \$29 000, and \$27 000 for Years 1, 2, and 3, respectively, which are treated as revenues in the P&L statement.
- The annual depreciation charge of \$14,000, which is treated as an expense in Years 1, 2, and 3.
- The rent revenue foregone on the warehouse, which is an opportunity cost as the warehouse, which is a resource or factor of production of the company, was previously rented out for \$8,000 p.a. and is now to be used in the new project, so is entered as an \$8,000 p.a. expense in Years 1, 2, and 3 in the P&L statement.
- Any gains or losses on sale of productive assets, these being a loss on sale of the old machine of \$500 at Year 0 (or Time-period 0, T_0), and a gain on sale of \$1,000 on the new machine at the end of its useful life at Year 3.
- The incremental salary of \$5,000 p.a. as an expense for each of Years 1, 2, and 3, this being the difference between the new technician's annual salary and the existing technician's salary of \$65 000 p.a., as the purchase of the new machine requires the company to hire a new technician who will replace the existing technician.

Following entry of all the relevant items in the P&L statement the annual profit or loss is calculated by summing the entries in each column, then

- The tax item is worked out as 30% of the annual profit or loss, with the profit or loss and tax item for each time-period being a \$500 loss at Year 0, giving a \$150 tax-rebate (i.e. $\$500 \times 0.30 = \150).
- A profit of \$4,000 for Year 1, giving a tax liability of $\$4,000 \times 0.30 = \$1,200$, for that year.
- A profit of \$2,000 for Year 2, giving a tax liability of $\$2,000 \times 0.30 = \600 , for that year.
- A profit of \$1,000 for Year 3, giving a tax liability of $\$1,000 \times 0.30 = \300 .

It should be noted that, as it is not stated otherwise, we are assuming that all profits and losses (and cash-flows and net cash-flows) are occurring at the end of each period.

In this example each time-period has been allocated a separate column in the P&L statement. However the time-periods and columns are set-out in the P&L statement, the same set-out must be adopted in the cash-flow statement, which is discussed further below. Here the reason why each time-period has been allocated a separate column is because it must be the case that for

each time-period there is at-least one unique entry for that time-period in the P&L or cash-flow statement, meaning that the profit or loss and/or net cash-flow for each period will be different to all the other periods.

However, while in this example, each time-period has been allocated its own column, it may be the case that in some detailed net present-value analyses multiple time periods can be combined into a single column. For this to be able to happen the time-periods must be consecutive time-periods, and the entries for these consecutive time-periods must be exactly the same in both the P&L and cash-flow statements, meaning that the profit or loss and net cash-flow for these consecutive periods will be the same. To save time in the analysis, we can combine the columns in both the P&L and cash-flow statements for these consecutive time-periods.

Step 5: Calculate net cash-flows

In a detailed two-step NPV analysis the main reason why a P&L statement is prepared is to work out the tax item for each time-period, and this tax item is then to be transferred to the cash-flow statement, preparation of which is Step 5 in the process. In this example, net cash-flow is calculated for each time period, more specifically, for the end of each time-period, with the items relevant to the cash-flow statement being included, as follows:

Cash-Flow Statement	Year 0	Year 1	Year 2	Year 3
Tax Paid (30%)	\$150	(\$1,200)	(\$600)	(\$300)
Operating Cash-Flows		\$31,000	\$29,000	\$27,000
Rent foregone		(\$8,000)	(\$8,000)	(\$8,000)
Salvage Value	\$2,500			\$1,000
Incremental Salary		(\$5,000)	(\$5,000)	(\$5,000)
Initial Outlay	(\$42,000)			
Net Cash-Flows	(\$39,350)	\$16,800	\$15,400	\$14,700

We can see in the above table the items that are relevant in calculating net cash-flows in this example. These relevant items are,

1. The tax items that are transferred down from the P&L statement, these being a tax-rebate/cash-inflow at Year 0 of \$150, i.e. the tax department will pay the company a rebate based on a loss in the P&L statement, so cash will flow into the business.
2. Tax liabilities and cash-outflows of \$1,200, \$600, and \$300 at the end of Years 1, 2, and 3, respectively, i.e. the company will have to pay the tax department at the end of each of these three years as the company has recorded profits at the end of each of these three years in the P&L statement, so cash will flow out of the business.
3. The operating cash flows (inflows) of \$31 000, \$29 000, and \$27 000 for Years 1, 2, and 3, respectively, which are treated as cash-inflows in the cash-flow statement.
4. The rent revenue foregone on the warehouse (an opportunity cost), which is entered as an \$8,000 p.a. cash-outflow in Years 1, 2, and 3 in the cash-flow statement.
5. Any salvage values from the sale of productive assets, which are entered as cash-inflows in the cash-flow statement, so in this example we have a salvage value at Year 0 of \$2,500 on the old machine, and a salvage value of \$1,000 in Year 3 for the old machine. It should be remembered that in the P&L statement it was the gain (as a revenue) or loss (as an expense) on sale of these productive assets that was recorded, while in the cash-flow statement it is the actual amount of money flowing into the business from the sale of these assets that is

recorded as a cash-inflow.

6. The incremental salary of \$5,000 p.a. as a cash-outflow for each of Years 1, 2, and 3 (which was recorded as an expense in the P&L statement).
7. The initial outlay (purchase price) of the new machine, which is recorded as a cash-outflow at Year 0 as this is when the new machine is purchased and paid for.

To be noted here is that, firstly, the annual depreciation expense is not recorded in the cash-flow statement, and this is because depreciation is a non-cash expense. This means while in the P&L statement for each of Years 1, 2, and 3, a depreciation charge of \$14,000 is recorded as an expense, the company is not actually paying this expense of \$14,000 in each of those years. Therefore, in the cash-flow statement, which records actual flows of cash into and out of the business in each time-period, no depreciation expense is recorded. However, also to be noted is that the initial purchase price of the new machine (\$42,000) is recorded as a cash-outflow in the cash-flow statement at Year 0, when the payment for the new machine occurs; but, neither at Year 0 nor in any other year is a \$42,000 purchase or expense is recorded in the P&L statement. However, the purchase price for the new machine of \$42,000 is actually taken into account in the P&L statement via the annual depreciation charge that is recorded as an expense, that is

$$\$14,000 \times 3(\text{years}) = \$42,000$$

This is equal to the initial purchase price of the new asset. Thus, the payment for the purchase of the new asset is recorded as a cash-outflow when it occurs (usually at Year 0) in the cash-flow statement, but the cost of the asset is usually expensed, or claimed, over its useful life as an annual depreciation expense in the P&L statement.

Following the entry of all the relevant items in the cash-flow statement, the annual net cash-flow is calculated by summing the entries in each column. It should be noted that, as it is not stated otherwise, we are assuming that all cash-flows and net cash-flows are occurring at the end of each period.

In any NPV analysis, regardless of whether it is a simple or a detailed analysis, at Year 0 the net cash-flow occurring will be a net cash-outflow, and this net cash-outflow we regard as the cost of the new asset. In this example, the net cash-outflow at Year 0 is \$39,350, while for the remaining time periods we will have net cash-inflows, which are \$16,800, \$15,400, and \$14,700 at the end of Years 1, 2, and 3, respectively. This is, typically, the case when working out detailed NPV questions.

Step 6: Discount net cash-flows, and

Step 7: Find NPV and reach conclusion:

NPV positive – accept the project,

NPV negative – reject the project.

As Step 6 we must now discount back to T_0 the net cash-inflows associated with the new asset to find their present values, and then in Step 7 we subtract from these PVs the initial cost of the asset, which is the net cash-outflow at T_0 , to find the net present value of the project and we then make our accept or reject decision.

In this example the assumed required rate of return for the company/new asset is 12% p.a., so we must discount back to T_0 the net cash-inflows of the project at that discount rate to find their PVs at T_0 , and then must subtract the initial cost of the project/net cash-outflow at T_0 of \$39,350 to find the net present value of the project. And important to note here is that as each of the net cash-inflows is of a different value, to find their respective PVs at T_0 we must treat each of the net cash-inflows as single sums and use the present value of a single sum formula to find

their PV_0 (and this formula was discussed in more detail in Chapter 2 Financial Mathematics And The Time Value of Money) with the formula being $PV_0 = FV_n(1+r)^{-n}$. So, the NPV of the new asset is found as follows:

$$NPV = -\$39,350 + \$16,800(1.12)^{-1} + \$15,400(1.12)^{-2} + \$14,700(1.12)^{-3}$$

$$NPV = -\$39,350 + \$16,800(0.8929) + \$15,400(0.7972) + \$14,700(0.7118)$$

$$NPV = -\$39,350 + \$15,001. + \$12,277 + \$10,463$$

$$NPV = -\$39,350 + \$37,741$$

$$NPV = -\$1,609 \text{ (approximately)}$$

Since the NPV of the new asset is found to be negative (-\$1,609), we conclude that if the new asset is purchased then the market value of the company at T_0 will decrease by \$1,609. Therefore, we must reject the purchase of the new machine. ■

6.3.2 Incremental Cash-Flows Analysis

Remember that in NPV capital budgeting analysis we are interested in incremental cash-flows, i.e. the additional cash-flows that occur as a result of taking on a new project. Here again we have an example that helps to further explain the treatment of incremental cash-flows in NPV analysis:

■ **Example 6.10 — Incremental Cash-Flows Analysis.** We will assume that we have a firm that is considering replacing a machine purchased two years ago with a useful life of five years with a new machine that has a useful life of three years and the relevant information for this analysis is as follows.

	Old Machine	New Machine
Initial Cost	\$240,000	\$360,000
Annual Revenues	\$100,000	\$150,000
Annual Cost	\$60,000)	\$70,000
Annual Depreciation	\$48,000	\$120,000
Salvage Value	\$80,000 (Now)	\$100,000 (End of Year 3)
Tax Rate	30%	
Required Rate of Return	10% p.a.	

Step 1: Calculate depreciation per annum.

As per the previous detailed NPV analysis, at Step 1 we must work out the annual depreciation charge on the new asset. However, the difference here is that we also have the old machine, which is yet to reach the end of its useful life. Thus, as the annual depreciation charge in this example, we must calculate the annual incremental depreciation charge, which is the difference in annual depreciation between the new and old machines.

For the new machine the annual depreciation expense is given as \$120,000, while for the old machine it is given as \$48,000, so the incremental annual depreciation charge we will need to record as an expense in the P&L statement will be \$72,000, i.e. $\$120,000 - \$48,000 = \$72,000$.

Step 2: Calculate any gain or loss on sale of productive assets.

At Step 2 we must calculate any gain or loss on the sale of productive assets or factors of production, and this is for both any old assets sold and for the new asset when it is sold at the end of its useful life. Remember, we calculate any gain or loss on the sale of a productive asset as the salvage value of the asset less the book value of the asset at the point in time when it is sold. In this example, book gain (loss) will have to be calculated for the old machine and for the new machine, as follows.

Old Machine:

$$\begin{aligned}
 BV_0 &= \text{purchase price} - \text{accumulated depreciation} \\
 &= \$240,000 - (2 \times \$48,000) \\
 &= \$240,000 - \$96,000 = \$144,000 \\
 SV_0 &= \$80,000 \\
 \text{Loss on sale at time 0} &= \$80,000 - \$144,000 = -\$64,000
 \end{aligned}$$

For the old machine we calculate gain (loss) on sale and its book value at T_0 , as this is when the asset is sold, two years into its useful life of five years.

New Machine:

$$\begin{aligned}
 BV_3 &= \text{purchase price} - \text{accumulated depreciation} \\
 &= \$360,000 - (3 \times \$120,000) \\
 &= \$360,000 - \$360,000 = \$0 \\
 SV_0 &= \$100,000 \\
 \text{Gain on sale at time 3} &= \$100,000 - \$0 = \$100,000
 \end{aligned}$$

For the new machine we calculate gain (loss) on sale at T_3 , as this is the end of its useful life and when it is sold. Also, for the new machine we must calculate its book value at the end of its useful life.

Step 3: Calculate taxable income, i.e. profit or loss.

Step 4: Calculate the tax item, i.e. tax liability or tax rebate.

At Step 3 we must calculate the profit or loss for each time period over the life of the asset via a P&L statement in order to then undertake the next step. In this example, profit or loss is calculated for each time period, more specifically, for the end of each time-period, with the items relevant to the P&L statement being included as follows:

Profit and Loss Statement	Year 0	Year 1	Year 2	Year 3
Incremental Revenues		\$50,000	\$50,000	\$50,000
Incremental Costs		(\$10,000)	(\$10,000)	(\$10,000)
Incremental Depreciation		(\$72,000)	(\$72,000)	(\$72,000)
Gain & Loss on Sale	(\$64,000)			\$100,000
Taxable Income (P&L)	(\$64,000)	(\$32,000)	(\$32,000)	\$68,000
Tax Paid (30%)	\$19,200	\$9,600	\$9,600	(\$20,400)

We must remember that in NPV capital budgeting analysis we are interested in including incremental cash-flows that occur as a result of taking on a new project. Therefore, in preparing the above P&L statement it is the incremental cash-flows that have been included as the revenues and expenses associated with the purchase of the new machine. Hence, the relevant incremental items for the P&L statement for Years 1, 2, and 3 are

- (a) the incremental cash revenues, calculate as

$$\begin{array}{r} \$150,000 \text{ (the annual revenue earned with the new machine)} \\ - \$100,000 \text{ (the annual revenue earned with the old machine)} \\ \hline \$50,000 \end{array}$$

- (b) the incremental cost, calculate as

$$\begin{array}{r} \$70,000 \text{ (the annual cost with the new machine)} \\ - \$60,000 \text{ (the annual cost with the old machine)} \\ \hline \$10,000 \end{array}$$

- (c) the incremental depreciation expense p.a. for Years 1, 2, and 3 of \$72,000 (Step 1).
 (d) the loss on the sale of the old machine of \$64,000 at Year 0 (Step 2) and the gain on sale of the new machine at Year 3 of \$100,000.

Following the entry of all the relevant items in the P&L statement the annual profit or loss is calculated by summing the entries in each column. Then the tax item is worked out as 30% of the annual profit or loss. With the profit or loss and tax item for each time-period being a \$64,000 loss at Year 0, giving a tax rebate of

$$\text{tax-rebate at Year 0} = \$64,000 \times 0.30 = \$19,200,$$

a loss of \$32,000 for both Year 1 and Year 2,

giving a tax-rebate for Years 1 and 2 = $\$32,000 \times 0.30 = \$9,600 = \$9,600$, and

a profit of \$68,000 for Year 3,

giving a tax liability at Year 3 = $\$68,000 \times 0.30 = \$20,400$.

Here the time-periods have been allocated to columns in the P&L statement, which must also be done in the same way in the cash-flow statement. In this example, the reason that Year 0 and Year 3 have been allocated a separate column each is because, for each of these two time-periods, there is at-least one unique entry for that time-period in the P&L or cash-flow statement, meaning that the profit or loss and/or net cash-flow for each of these periods will be different to all the other periods.

However, it can be seen in the above P&L statement and will be seen again in the cash-flow statement below, that Year 1 and Year 2 could have been combined into one column. This is because they are consecutive time-periods for which the entries are exactly the same in both the P&L and cash-flow statements, meaning that the profit and loss figure (and net cash-flow) for these consecutive periods is (will be) the same. To save time in the analysis, we could have combined Years 1 and 2 into a single column in both the P&L and cash-flow statements.

Step 5: Calculate net cash-flows.

As we learned in the previous detailed NPV analysis example, the main reason a P&L statement is prepared is to work out the tax item for each time-period, which is then to be transferred to the cash-flow statement, preparation of which is Step 5 in the process.

In this example net cash-flow is calculated for each time period, more specifically, for the end of each time-period, with the items relevant to the cash-flow statement being included, as follows.

Cash-Flow Statement	Year 0	Year 1	Year 2	Year 3
Tax Paid (30%)	\$19,200	\$9,600	\$9,600	(\$20,400)
Incremental Revenues		\$50,000	\$50,000	\$50,000
Incremental Costs		(\$10,000)	(\$10,000)	(\$10,000)
Salvage Value	\$80,000			\$100,000
Initial Outlay	(\$360,000)			
Net Cash-Flows	(\$260,800)	\$49,600	\$49,600	\$119,600

We can see in the above table the items that are relevant in calculating net cash-flows in this example. These relevant items are as follows.

- The tax items that are transferred down from the P&L statement, these being a tax-rebates (cash-inflows) of \$19,200 at Year 0, \$9,600 of at the end of both Year 1 and Year 2, and a tax liability (cash-outflows) of \$20,400 at the end of Year 3.
- The incremental revenues of \$50,000 a year for Years 1, 2, and 3, which are treated as cash-inflows in the cash-flow statement (also treated as revenues in the P&L statement).
- The incremental costs of \$10,000 a year for Years 1, 2, and 3, which are entered as cash-outflows in the cash-flow statement (also entered as an expense in the P&L statement).
- The salvage values from the sale of productive assets, which are entered as cash-inflows in the cash-flow statement. Here, we have the salvage value at Year 0 of \$80,000 on the old machine, and the salvage value of the new machine of \$100,000 at Year 3. Recall that in the P&L statement it was the gain (as a revenue) or loss (as an expense) on sale of these productive assets that was recorded, while in the cash-flow statement it is the actual amount of money flowing into the business from the sale of these assets that is recorded as a cash-inflow.
- The initial outlay (purchase price) of the new machine of \$360,000, which is recorded as a cash-outflow at Year 0 as this is when the new machine is purchased and paid for.

Following the entry of all the relevant items in the cash-flow statement, the annual net cash-flow is calculated by summing the entries in each column, with a net cash-outflow at Year 0 of \$260,800, while for the remaining time periods we have net cash-inflows of \$49,600 for both of Year 1 and Year 2, and \$119,600 for Year 3.

Step 6: Discount net cash-flows.

Step 7: Find NPV and reach conclusion:

NPV positive – accept the project,

NPV negative – reject the project.

At Step 6 we must now discount back to T_0 the net cash-inflows associated with the new machine to find their present values. Then in Step 7, in order to find the net present value of the project, we subtract from these PVs the initial cost of the new machine at T_0 . Therefore, we discount the net cash-inflows of the project at the discount rate of 10%p.a. back to T_0 , and then, subtract the initial cost of the project (\$260,800) to find the net present value of the project.

It is important to note here that while with the previous detailed NPV analysis question each of the net cash-inflows was of a different value. As a result, in order to find their respective PVs at T_0 , we treated each of the net cash-inflows as single sums and used the present value of a single sum formula to find their PV_0 . However, in this example the net cash-inflows for two of the consecutive years, Year 1 and Year 2, are the same at \$49,600. Therefore, to save time in our

analysis, we can treat these net cash-inflows as an ordinary annuity stream. Then, we have

$$\text{NPV} = -\$260,800 + \$49,600 \left[\frac{1 - (1.10)^{-2}}{0.10} \right] + \$119,600(1.10)^{-3}$$

$$\text{NPV} = -\$260,800 + \$49,600 \times 1.7355 + \$119,600 \times 0.7513$$

$$\text{NPV} = -\$260,800 + \$86,080.80 + \$89,855.48$$

$$\text{NPV} = -\$260,800 + \$175,936.28$$

$$\text{NPV} = -\$84,863.72.$$

Since the NPV of the new asset is found to be negative (-\$84,863.72), we conclude that if the new asset is purchased the market value of the company and the wealth of the shareholders at T_0 will decrease by \$84,863.72. Thus, we must reject the purchase of the new machine. ■

The table below summarises the treatment of items in the P&L and Cash-Flow Statements in NPV analysis.

	When	In Profit and Loss	In Cash-flow
Initial investment or cost	Year 0	No	Yes
Depreciation	Year 1 to Year n	Yes	No
Salvage value	Year 0 and/or Year n	No	Yes
Gain or Loss on Disposal	Year 0 and/or Year n	Yes	No
Incremental revenues	Year 1 to Year n	Yes	Yes
Incremental expenses	Year 1 to Year n	Yes	Yes
Opportunity costs	Year 1 to Year n	Yes	Yes
Side effects	Year 1 to Year n	Yes	Yes
Working capital	Year 1 and Year n	No	Yes

Working Capital

Working capital refers to smaller assets that are used in conjunction with a larger asset. For example, if a large piece of machinery is purchased and there are smaller assets that have to be used in conjunction with the larger piece of machinery, then these smaller assets are referred to as working capital.

In capital budgeting analysis in the subject Business Finance, working capital is treated as a cash-outflow in the cash-flow statement at T_0 equal to the purchase price of the working capital. Then this working capital is sold at the end of the useful life of the new asset for the same price at which it was purchased, with the sale price representing a cash-inflow in the cash-flow statement at the end of the useful life of the new asset. This means that we do not depreciate the working capital, and, because the salvage value of the working capital will be the same as the purchase price and we will not have any gain or loss on the sale of the working capital.

In the real world, treatment of working capital may need to be different. In some cases the working capital may be depreciated, requiring recognition of the depreciation in the P&L statement, and the salvage value of the working capital may be different to its book-value at the time of sale. Therefore, there may be a gain or loss on the sale of the working capital.

One-Step Process

Up to this point the detailed NPV analysis that have been conducted have involved a two-step process, requiring preparation of both a P&L and a cash-flow statement. The preparation of the P&L statement has been necessary in order to work out the profit or loss for each time-period over the life of any new asset in order to calculate the tax item (liability or rebate) that needs to be transferred to the cash-flow statement, as either a cash-inflow (for a tax rebate) or cash-outflow (for a tax liability).

Alternatively, rather than adopt a two-step process to detailed NPV project evaluation, a one-step process can be undertaken, which involves preparation of the cash-flow statement only (without preparing the P&L statement). However, when a one-step process is adopted it is necessary to account for all tax items in the cash-flow statement. Instead of working out taxable income (profit or loss) and the tax item via the P&L statement and then net cash-flows via the cash-flow statement, we must directly calculate after-tax net cash-flows for each relevant item in the cash-flow statement. To explain the one-step process we will do the example just undertaken above in one step, but before getting to the example there are a number of points that must be explained further.

Revenue

Incremental cash *revenue* is a cash-inflow that occurs due to sale of goods and services and it is taxable. If we are undertaking a detailed NPV analysis as a one-step process we must include in the cash-flow statement the net cash-inflow after tax for the incremental cash revenue, as the taxable component of incremental cash revenue has not been accounted for in a P&L statement. In the cash-flow statement for incremental cash revenue we will list the net cash-inflow after-tax, calculated as.

$$\begin{aligned}\text{Net cash-inflow after tax} &= \text{Incremental cash revenue} - (\text{Incremental revenue} \times t_c) \\ &= \text{Incremental cash revenue} \times (1 - t_c)\end{aligned}$$

where t_c is the corporate tax-rate.

Expenses

Incremental *cash expenses* are cash-outflows that occur due to the expenditures of production and they are usually tax-deductible. They reduce tax payable, and provide a tax-shield. If we are undertaking a detailed NPV analysis as a one-step process we must include the net cash-outflow after tax for the incremental cash expenses in the cash-flow statement, because the tax-shield component of incremental cash expenses has not been accounted for in a P&L statement. In the cash-flow statement for incremental cash expenses we will list the net cash-outflow after-tax, calculated as follows.

$$\begin{aligned}\text{Net cash-outflow after tax} &= \text{Incremental cash expenses} - (\text{Incremental expenses} \times t_c) \\ &= \text{Incremental cash expenses} \times (1 - t_c)\end{aligned}$$

Depreciation Tax-Shield

Depreciation is a non-cash expense and is not included as a cash-outflow in the cash-flow statement. However, depreciation is an expense that is an allowable tax-deduction, thus reducing tax payable. Thus, the amount by which it reduces tax-payable must be included as a tax-rebate cash-inflow in the cash-flow statement, calculated as follows.

Cash-inflow from depreciation = Depreciation $\times t_c$

Book Gain or Loss

In a two-step detailed NPV analysis, any book-gain (or book-loss) on disposal of productive assets would be recorded as a revenue (or expense) in the P&L statement and would increase (or decrease) profit or loss accordingly. See Highlight 6.3 for more details.

In order to calculate a book-gain or book-loss on disposal of productive assets, we take the salvage value (SV) of the asset and subtract the book-value (BV) of the asset at the point in time when it is sold. If $SV > BV$ we have a book-gain on disposal of the asset and a taxable profit arises, with the tax on the profit or gain needing to be accounted for in the cash-flow statement. The tax due on the gain is calculated as follows:

Tax due on gain = Book gain $\times t_c$

This tax liability is then subtracted from the salvage value of the productive asset as follows.

Net after-tax cash-inflow from disposal = $SV - (\text{Book gain} \times t_c)$.

However, if $SV < BV$ we have a book-loss on disposal of the asset and a taxable deductible loss arises, with the tax-rebate on the loss needing to be accounted for in the cash-flow statement, and with the tax-rebate on the loss calculated as follows.

Tax-rebate on loss = Book loss $\times t_c$

and this tax-rebate will be added to the salvage value/sale price of the productive asset as follows to give:

Net after-tax cash-inflow from disposal = $SV + (\text{Book loss} \times t_c)$

■ **Example 6.11 — Re-work Example 6.10 in a one-step process.** We can see the adjustments that are made in the cash-flow statement, which are as follows.

1. The depreciation tax-saving of $\$72,000 \times 0.30 = \$21,600$ is included as a cash-inflow for each of Years 1, 2, and 3,
2. Incremental after-tax cash revenues of $\$50,000 \times (1 - 0.30) = \$35,000$ are included as a cash-inflow for each of Years 1, 2, and 3,
3. Incremental after-tax cash expenses of $\$10,000 \times (1 - 0.30) = \$7,000$ as a cash-outflow for each of Years 1, 2, and 3.
4. The net after-tax cash-flows from asset disposal of old machine of $\$80,000 + (\$64,000 \times 0.30) = \$99,200$ as a cash-inflow for Year 0.
5. The net after-tax cash-flows from asset disposal of new machine of $\$100,000 - \$100,000 \times 0.30 = \$70,000$ as a cash-inflow for Year 3.
6. The initial outlay of the new machine of $\$360,000$ as a cash-outflow at Year 0, with the same net cash-flows occurring at the end of each period and the same NPV figure of $-\$84,863.72$.

Cash-Flow Statement	Year 0	Years 1	Years 2	Year 3
After Tax Cash-Flows				
Depreciation		\$21,600	\$21,600	\$21,600
Cash Saving		= \$72,000(0.3)	= \$72,000(0.3)	
Incremental		\$35,000	\$35,000	\$35,000
After Tax Revenue		= \$50,000(1 - 0.3)	= \$50,000(1 - 0.3)	
Incremental		(\$7,000)	(\$7,000)	(\$7,000)
After Tax Costs		= -\$10,000(1 - 0.3)	= -\$10,000(1 - 0.3)	
Initial Outlay	(\$360,000)			
Net After Tax	\$99,200			\$70,000
Cash-Flows	= \$80,000			= \$100,000
from Disposal	+\$64,000(0.3)			-\$100,000(0.3)
After Tax	(\$260,800)	\$49,600	\$49,600	\$119,600
Cash-Flow				
NPV	(\$84,863.72)			

■

6.4 Evaluating Projects with Different Lifespans

Normally, if two projects are mutually exclusive then they are competing projects and a choice between the two projects can be made by choosing the project with the higher NPV. However, project evaluation can become complicated if mutually exclusive projects being compared differ in lifespan. If the two projects have different lifespans we cannot base a NPV analysis of the two projects simply on which project has the higher NPV, and a common base is required for comparison.

When comparing two mutually exclusive projects with different lifespans, it is necessary to make comparisons over the same time period. To make such a comparison we must calculate and compare each project's equivalent annual annuity (EAA). In effect, with EAAs we are comparing each project's net cash-flows calculated on an annual basis, and once each project's EAA has been calculated we then select the project with the highest EAA.

Highlight 6.4 — Equivalent annual annuity. This is the constant annual cash-flow generated by a project over its lifespan, as if it was an annuity, with the present-value of the constant annual cash flows exactly equal to the project's NPV.

$$EAA = NPV \div \left[\frac{1 - (1 + r)^{-n}}{r} \right]$$

In order to calculate the EAA of a project we must calculate the NPV of the project over its life, as if it were “one-off”. Then we must convert the NPV of the project into an equivalent annuity for the life of each project. As such, we reduce a problem with different time-horizons to a choice between two annuities, with each EAA representing the constant annual cash-flow

generated by the respective project over its lifespan, as if it was an annuity, with the present-value of the constant annual cash-flow stream exactly equal to the project's NPV.

■ **Example 6.12 — Equivalent Annual Annuity.** Suppose a firm has to choose between two new machines that differ in terms of economic life and capacity. The firm wishes to undertake NPV analysis to determine which machine to purchase. The required rate of return for the firm is 14% p.a. and the after-tax net cash-flows for each machine are as follows.

Year	Machine 1 Net cash flows	Machine 2 Net cash flows
0	(\$45,000)	(\$45,000)
1	\$20,000	\$12,000
2	\$20,000	\$12,000
3	\$20,000	\$12,000
4		\$12,000
5		\$12,000
6		\$12,000

As the machines each have a different lifespan, to undertake an NPV analysis of the machines and decide which of the two the company should invest in it is necessary that we calculate the EAA of each machine. We calculate the EAA for each machine based on the cash-flow information in the table above:

Step 1: Calculate each project's NPV

Machine 1:

$$\begin{aligned}
 NPV_1 &= -\$45,000 + \$20,000 \left[\frac{1 - (1.14)^{-3}}{0.14} \right] \\
 &= -\$45,000 + \$20,000 \times 2.322 = -\$45,000 + \$46,432.64 \\
 &= \$1,432.64.
 \end{aligned}$$

Machine 2:

$$\begin{aligned}
 NPV_2 &= -\$45,000 + \$12,000 \left[\frac{1 - (1.14)^{-6}}{0.14} \right] \\
 &= -\$45,000 + \$12,000 \times 3.888 = -\$45,000 + \$46,664.01 \\
 &= \$1,664.01.
 \end{aligned}$$

The question to now be asked is that, since Machine 2 has a higher NPV, does this mean Machine 2 is better? At this stage we are unable to say whether or not Machine 2 is preferable to Machine 1, or vice versa. We must go one step further and calculate each machine's EAA based on the NPV of each machine. Recall, Highlight 6.4, then

Step 2: Calculate each project's EAA

Machine 1:

$$\$1,432.64 = EAA_1 \left[\frac{1 - (1.14)^{-3}}{0.14} \right]$$

$$\$1,432.64 = EAA_1 \times 2.322$$

$$EAA_1 = \frac{\$1,432.64}{2.322}$$

$$EAA_1 = \$617.08.$$

Machine 2:

$$\$1,664.01 = EAA_2 \left[\frac{1 - (1.14)^{-6}}{0.14} \right]$$

$$\$1,664.01 = EAA_2 \times 3.888$$

$$EAA_2 = \frac{\$1,664.01}{3.888}$$

$$EAA_2 = \$427.91.$$

Hence, we would choose Machine 1 as it has the higher EAA.

■

6.5 Depreciation, Inflation & Effective Write-off

Depreciation is a method of allocating the cost of an asset over its useful life. For accounting purposes, depreciation indicates how much of an asset's value has been used up, while for tax purposes, businesses can deduct the cost of the assets they purchase as business expenses.

Inflation, on the other hand, refers to the rate at which the general level of prices for goods and services in an economy is rising, and, subsequently, purchasing power is falling. Most countries' central banks aim to sustain a 'healthy' inflation rate of 2-3% per year.

Effective write-off of depreciation charges relates to the fact that, as inflation rises, every dollar will buy a smaller percentage of a good. For example, if the inflation rate is 2%, then a \$1 pack of gum will cost \$1.02 in a year's time. This means that in reality, in project evaluation if inflation is not adjusted for then errors may occur in capital budgeting decisions.

In what follows, we will see how rising prices adversely affect depreciation charges, because, generally, depreciation on a productive asset is calculated based on historical cost², and not on replacement cost³. As depreciation is based on historical cost, inflation erodes the real value of any depreciation tax deduction and therefore discourages capital investment in an inflationary period. If we include inflation in our analysis, then the farther into the future is the depreciation claim, the lower is its real present value. In Business Finance, we include inflation implicitly in the discount rate r of projects.

When there is inflation, we can also calculate the effective write-off of a productive asset. To do so, we work out the sum of the present value of the depreciation deductions at the inflation rate, and then divide this present value by the cost of the asset. To do this we use the following formula.

$$\text{Effective write-off} = \frac{\sum (\text{Depreciation p.a.})(1 + I)^{-n}}{\text{Purchase price of asset}}$$

²Based on the cost of the asset at the time it was bought

³The cost of the asset when it has to be replaced in the future

Where I is the inflation rate.

■ **Example 6.13 — Effective Write-Off of a Productive Asset with Inflation – A Real⁴ Analysis.** Acme Ltd., a scrap metal dealer, is considering the acquisition of a “Crusher” metal compactor at a cost of \$25,000.

- The compactor is estimated to have a five-year lifespan;
- Tax allowable depreciation is 20% prime cost (purchase price) p.a., i.e. we are using straight-line depreciation, so the annual depreciation charge is \$5,000;
- The company tax rate is 40% p.a.;
- The expected inflation rate for the next 5 years 8% p.a.

In order to work out the effective write-off of this new asset we must discount back to T_0 the depreciation charge each year for the life of the asset using the expected annual inflation rate as the discount rate. So, to find the present value at T_0 of each years' depreciation charge we can use the present value of a single sum formula with the inflation rate as the discount rate. The present value of a single sum formula is as follows:

$$PV_0 = FV_0(1 + I)^{-n}$$

In this example to find the ‘real’ value of each years' depreciation charge, we do the following.

Real value at T_0 of annual depreciation expense:

Year	Depreciation	PVIF	PV ₀
1	\$5,000	$(1.08)^{-1} = 0.9259$	\$4,629.50
2	\$5,000	$(1.08)^{-2} = 0.8573$	+ \$4,286.50
3	\$5,000	$(1.08)^{-3} = 0.7938$	+ \$3,969.00
4	\$5,000	$(1.08)^{-4} = 0.7350$	+ \$3,675.00
5	\$5,000	$(1.08)^{-5} = 0.6806$	+ \$3,403.00
Total			= \$19,963.00

The effective depreciation write-off:

$$\begin{aligned}
 \text{Effective write-off} &= \frac{\sum (\text{Depreciation p.a.})(1 + I)^{-n}}{\text{Purchase price of asset}} \\
 &= \frac{\$19,963.00}{\$25,000} = 0.7985 \\
 &= 79.85\%.
 \end{aligned}$$

Therefore, in this example, only 79.85% of the replacement cost of the asset is covered by the annual depreciation expense, and 20.15% (i.e. $1 - 0.7985 = 0.2015$) of the purchasing power of the sum of the annual depreciation expense is lost through inflation.

In this example the depreciation expense each year is of the same value, \$5000, meaning that it is a series of fixed cash-flows for a fixed period of time. Consequently, we could have calculated the sum of the PV_0 of the annual depreciation expenses as an ordinary annuity using

⁴In economics, finance, and accounting the term ‘real’ means that the effects of inflation, or rising prices, have been accounted for.

the *PV* of an ordinary annuity formula, with the discount rate being the inflation rate. Thus,

$$\begin{aligned}
 PV &= PMT \left[\frac{1 - (1 + I)^{-n}}{I} \right] \\
 &= \$5,000 \left[\frac{1 - (1.08)^{-5}}{0.08} \right] = \$19,963.00
 \end{aligned}$$

■

6.6 Capital Rationing

Capital rationing refers to the act of placing restrictions on the amount of new investments or projects undertaken by a company. In this case, a firm limits the total amount of funds to be invested in projects, so, even though certain projects may have a positive NPV, they could be rejected due to capital (financing) constraints.

There are two main types of capital rationing. The first is *hard capital rationing*, which is imposed by capital markets in the sense that markets will not provide sufficient financing for a project at an acceptable cost. In this situation, markets will not provide financing for the project at a rate of interest that will give the project a positive NPV. For example, management may base its project evaluation on a required rate of return of 7% p.a., which may give a positive NPV. However, the market (e.g. banks, shareholders) believe that the project is of higher risk and will require a rate of return of, say, 10% p.a., and using 10% p.a. may lead to the project having a negative NPV.

The second main type of capital rationing is *soft capital rationing*, which is imposed by upper-management of a company on lower-level management to ensure subsidiaries prioritise investments, which ensures discipline by lower-level management as subsidiaries only invest in the highest NPV projects.

6.6.1 Profitability Index (Benefit-Cost Ratio)

A project's *profitability index* (PI; or benefit-cost ratio) measures the return on the project relative to its cost, and is calculated, as follows.

$$PI = \frac{\text{PV of future cash-flows}}{\text{Cost}}$$

or as the net present value of the project plus its cost divided by the cost of the project, i.e. as:

$$PI = \frac{NPV + \text{Cost}}{\text{Cost}}$$

The general decision criteria for a project is that

- if its PI is greater than 1, it will be accepted, since, a PI of greater than 1 must mean that the NPV of the project is positive. Conversely,
- if a project's PI is less than 1 it will automatically be rejected, since a PI of less than 1 means that the NPV of the project must be negative.

When using the PI to select amongst competing projects, management should choose the project or projects that have the highest PIs and are within the capital (budget) constraint.

■ **Example 6.14 — Capital Rationing and the Profitability Index.** Assume ACME Ltd. has a \$15 million capital constraint (or limit) on the funds that it can invest in new projects, assets, products, or investments. Also assume that the management of ACME Ltd. is considering whether to invest in the following three projects, which each have a two-year duration, with their respective costs, net yearly cash-flows, NPVs, and PIs listed in the following table:

Project	Year 0 Net Cash-Flow	Year 1 Net Cash-Flow	Year 2 Net Cash-Flow	NPV	PI
A	-\$15 million	\$30 million	\$15 million	\$23.7 million	2.853
B	-\$8 million	\$4 million	\$25 million	\$15.5 million	2.938
C	-\$7 million	\$6 million	\$22 million	\$15.9 million	3.271

We can see from the information listed in the table, particularly the cost of each project (the respective net cash-outflow at Year 0) and the NPV of each of the projects, we are able to calculate the PI of each project as follows:

$$PI = \frac{NPV + \text{Cost}}{\text{Cost}}$$

Which for each project gives us the following:

$$PIA = \frac{(\$23.7\text{million} + \$15\text{million})}{\$15\text{million}} = 2.583$$

$$PIB = \frac{(\$15.5\text{million} + \$8\text{million})}{\$8\text{million}} = 2.938$$

$$PIC = \frac{(\$15.9\text{million} + \$7\text{million})}{\$7\text{million}} = 3.271$$

We can see that both Projects B and C have a higher PI than Project A and that the total cost of Projects B and C combined is \$8 million + \$7 million = \$15 million. Whereas the total cost of Project A alone is \$15 million. Hence, based on the capital constraint and the PI decision rule, and in order to maximise the total NPV for the company subject to the capital constraint, we would choose both Projects B and C as they both have a higher PI than Project A. Together, projects B and C are within the capital constraint of \$15million and their combined NPV is

$$\$15.5 \text{ million} + \$15.9 \text{ million} = \$31.4 \text{ million}$$

This is while the NPV of Project A is only \$23.7 million, meaning that the market value of Acme Ltd. will rise more by undertaking both Projects B and C, rather than just Project A alone.

■

6.7 Revision Problems

Problem 6.1 What is the payback period if the initial investment is \$60,000 and the net cash-flows are:

Year 1	\$20,000
Year 2	\$25,000
Year 3	\$30,000
Year 4	\$10,000
Year 5	\$ 5,000

- (a) 1.75 years
- (b) 2.25 Years
- (c) 2.45 Years
- (d) 2.50 Years

Problem 6.2 Other things being held constant, what happens to a project's net present value if the discount rate increases?

- (a) It rises
- (b) It falls
- (c) It must both rise and fall
- (d) It cannot change

Problem 6.3 When the Net Present Value of an investment is positive, then the Internal Rate of Return will be:

- (a) Equal to the opportunity cost of capital.
- (b) Greater than the opportunity cost of capital
- (c) Less than the opportunity cost of capital
- (d) Less than or equal to the opportunity cost of capital.

Problem 6.4 The internal rate of return is:

- (a) The discount rate that makes the NPV positive
- (b) The discount rate that equates the present value of the cash inflows with the present value of the cash outflows
- (c) The discount rate that makes NPV negative
- (d) The rate of return that makes the yield to maturity zero

Problem 6.5 You have determined the profitability of a planned project by finding the present value of all the cash flows from that project. Which of the following would cause the project to look more appealing in terms of the present value of those cash flows?

- (a) The discount rate decreases.
- (b) The cash flows are extended over a longer period of time, but the total amount of the cash flows remains the same.
- (c) The discount rate increases.
- (d) Answers (b) and (c) above.

Problem 6.6 The director of capital budgeting for KLM Ltd has identified a project with the following expected net cash flows (the project has a 10% p.a. cost of capital):

Year	Expected Net Cash Flows
0	(\$100m)
1	\$70m
2	\$50m
3	\$20m

What is the project' NPV?

- (a) \$19.98m
- (b) \$20.07m
- (c) \$21.46m
- (d) \$22.32m

Problem 6.7 Caterpillar Corporation is a US-based company whose core business is the production of machinery used in the heavy-industry sector. It has recently completed a \$400,000 two-year marketing study on whether to introduce a new machine to the market. Based on the results of the study, Caterpillar has estimated that 10,000 of its new machines can be sold annually over the next six years at a price of \$9,615 each. Variable costs per machine are \$7,400, and fixed costs total \$12 million a year. Working capital specifically for this project is estimated to be \$2 million and will be returned at the end of the project's life.

The cost of the machine includes \$40 million to build production facilities and \$2.4 million in land. The \$40 million investment in production facilities will be depreciated to zero over the life of the project. At the end of the project the facilities, including the land, will be sold for an estimated \$8.4 million. The market value of the land, which is not tax-deductable, is not expected to change.

Finally, start-up costs also entail fully tax-deductable expenses of \$1.4 million, which are deductible at the end of the first year of production. The tax rate applicable to Caterpillar is 30%. The after-tax discount rate is 10% p.a.

As the Corporate Financial Manager of Caterpillar, calculate the NPV of the project and advise management whether they should proceed

Problem 6.8 You have been observing the surge in health awareness in Australia for some time and realise that the time is right for you to start and run an aerobic exercise centre. Your family owns a warehouse, which will meet your needs, and until recently it has been renting at \$48,000 per year. You estimate that you will need to spend \$100,000 in total. This \$100,000 will be made up of an initial cost of \$50,000 to renovate the premises, \$45,000 for new equipment, and \$5,000 to install the equipment. You have done a market survey, at a cost of \$3,000, which leads you to believe that you will get 500 members each paying \$1,000 per year. You have also found 5 instructors you can hire at \$30,000 each per year. For tax reasons you will expense the renovation costs immediately and depreciate the equipment (including the installation cost) over ten years using the straight-line method. However you will expect the equipment to be full functional for 15 years, which is the life of the operation. Due to the nature of fitness equipment it will be unlikely to have a salvage value at the end of 15 years. Assume the initial investment is made today and all cash flows are received or paid at the end of each year. Your discount rate is 15% p.a. and your tax rate is 40%.

Should you invest in the project?

Problem 6.9 Which of the following is an incorrect statement regarding project evaluation analysis?

- (a) Depreciation results in a cash-inflow
- (b) The book gain on disposal of an asset results in a cash-inflow
- (c) Salvage value is treated as a cash-inflow
- (d) Dividend payments are not taken into account as part of the cash-flows

Problem 6.10 A company is considering a proposed expansion to its facilities. Which of the following statements is most correct?

- (a) In calculating the project's operating cash-flows, the firm should not subtract out financing costs such as interest expense, since these costs are already included in the cost of capital used to discount the project's net cash-flows.
- (b) Since depreciation is a non-cash expense, the firm does not need to know the depreciation rate when calculating the operating cash-flows.
- (c) When estimating the project's operating cash-flows, it is important to include any opportunity costs and sunk costs, but the firm should ignore cash-flows from externalities since they are accounted for elsewhere.
- (d) Statements (a) and (c) are correct.

Problem 6.11 Adams Audio is considering whether to make an investment in a new type of technology. Which of the following factors should the company consider when it decides whether to undertake the investment?

- (a) The company has already spent \$3 million researching the technology.
- (b) The new technology will affect the cash-flows produced by its other operations.
- (c) If the investment is not made, then the company will be able to sell one of its laboratories for \$2 million.
- (d) Factors b and c should be considered.

Problem 6.12 Greenberg Trading is considering two mutually exclusive projects, one with a four-year life and one with a nine-year life. The net cash-flows from the two projects are as follows:

Year	Project A	Project B
0	-\$160,000	-\$160,000
1	65,000	35,000
2	65,000	35,000
3	65,000	35,000
4	85,000	40,000
5		40,000
6		40,000
7		45,000
8		45,000
9		45,000

Assuming a 10% p.a. required rate of return on both projects, calculate each project's equivalent annual annuity (EAA). Which project should be selected?

Problem 6.13 Fine Fabrics Ltd is considering the purchase of a new printing machine that will enable its fabrics to be printed with more vibrant colours. The new machine costing \$84,000 is expected to have a useful life of 12 years and be able to be sold at that time for \$3,000. Fine Fabrics always uses straight-line depreciation for tax purposes. Fine Fabrics forecasts that the improved quality of its fabrics will generate an additional \$20,000 in revenue in each of the next 12 years. The new machine will also require additional costs in colour dyes each year. These

are expected to be only \$1,000 per year. Fine Fabric's current printing machine if replaced can be sold for \$6,000 today even though it has a book value of \$10,000 for tax purposes. The new printing machine will require an additional injection of \$5,000 in working capital, which will be recouped at the end of 12 years. In the first two years additional service costs of \$2,000 per year will be incurred. These service costs are not tax deductible. Tax rates are 30% and the required rate of return is 10% p.a.

Calculate the NPV and give your advice as to whether Fine Fabrics Ltd should proceed and purchase the new machine?

Problem 6.14 A company has the opportunity of buying a new high-tech metal cutter which will save the company \$14,000 each year in labour costs. This metal cutter will cost \$70,000 and will have a useful life of 7 years. It is expected to have a salvage value of \$16,000 and will be depreciated using a straight-line method of depreciation.

If the company goes ahead with the new metal cutter it can sell its old cutter for \$5,000, even though the machine has a book value for tax purposes of \$8,000.

The new machine will require a working capital injection of \$4,500 for the acquisition of additional scrap metal. The working capital would be recovered at the end of the 7 year period. The company's required rate of return is 10% p.a. and the tax rate is 30%.

Should the company acquire the new metal cutter?

Rate of Return

Historical Return
The Expected Return

Measuring Risk

The Risk-Free Asset
Company Risk
Distribution of Returns
Variance and Standard Deviation
Coefficient of Variation

Portfolio Expected Return**Portfolio Diversification**

Covariance and Correlation Coefficient

Portfolio Variance and Standard Deviation**Portfolio with more than two assets****Opportunity Set and Efficient Frontier**

Rational Investing
Risk-Return Preference

Revision Problems

7. Risk and Return

In this chapter you will learn about rates of return on an asset; calculating expected return on an individual asset; measuring risk for an individual asset; portfolio theory and diversification; calculating expected return on a portfolio of assets; correlation coefficients; measuring risk for a portfolio of assets; the opportunity set; rational investing; and risk-averse investing.

7.1 Rate of Return

The *rate of return* on an asset is the gain or loss on the investment over a specified period of time, expressed as a percentage change over the initial purchase price. Generally, financial securities are judged based on their rates of return, which are compared to other assets to determine the attractiveness of an investment.

There are three general types of return on an asset. The first of these is the historical rate of return, which is the return that the asset has already produced over a specified period of time. Then there is the expected rate of return on the asset, which, as the name suggests, is the return that the asset is expected to produce over some future period of time. Finally, there is the required rate of return, which is the return that an investor requires an asset to produce if he or she is to be a future investor in that asset.

7.1.1 Historical Return

The *historical rate of return* on an asset reflects the past performance of the investment and shows the return that has been earned on the individual asset, such as a share or a bond. It is generally made-up of two components, which are

1. the income stream that has been earned over the investment period, for example, the dividends received on a share or the coupon (interest) payments received on a bond.
2. the capital gain or loss, which comes from the change in the market price of the asset over the investment period.

So, the total historical return on an asset can be expressed as:

Total return = income yield + capital gains yield,

and is calculated as the percentage return on the asset over the investment period as follows:

$$R_i = \frac{\text{income paid at the end of period} + \text{change in market value over period}}{\text{market value at beginning of period}}$$

$$= \frac{I_1 + (P_1 - P_0)}{P_0}$$

where I_1 is the income paid at the end of the period, P_1 is the market price of the asset at the end of the period, and P_0 is the market price of the asset at the beginning of the period.

The following numerical example explains how to calculate the historical rate of return on an individual asset.

■ **Example 7.1** Suppose that the market price of a share purchased at the beginning of the year was \$37.00 (P_0) and that the share is sold at the end of the year for \$40.33 (P_1), while the dividend paid on the share at the end of the year is $D_1 = \$1.85$. Therefore, the rate of return earned on the share over the year can be found as follows:

$$R_i = \frac{D_1 + (P_1 - P_0)}{P_0}$$

$$= \frac{\$1.85 + (\$40.33 - \$37.00)}{\$37.00} = 0.1400 = 14.00\%.$$

The total return on the share over the year is 14%, of which five percentage points is coming from the income (dividend) stream ($\$1.85 \div \37.00) and nine percentage points is coming from the capital gain (increase in price, $\$3.33 \div \37.00). So, the total historical return is $R_i = 0.05 + 0.09 = 0.14 = 14\%$.

■

7.1.2 The Expected Return

The *expected return* on an asset is the return one anticipates receiving on the investment, given various known rates of return and their probabilities, usually based on historical data. To be able to calculate the expected rate of return on an individual asset you will need to be given a series of possible returns on the asset and the percentage probabilities of each of those returns occurring. The formula used to calculate the expected rate of return on an individual asset, as a weighted average return, is as follows.

$$\bar{R}_i = P(R_1) \times R_1 + P(R_2) \times R_2 + \dots + P(R_n) \times R_n$$

where

P is the probability of the return occurring;

R_1 is possible return 1;

R_2 is possible return 2; and

R_n is possible return n .

This can alternatively be written as follows.

$$\bar{R}_i = \sum_{i=1}^n P(R_i) \times R_i$$

where $\sum_{i=1}^n$ denotes the summation of all variables with the subscript i , counting from 1 to n .

■ **Example 7.2** Assume the following for two individual assets A and B:

		Asset A	Asset B
State of Economy	Probability	Return	Return
Recession	0.20	0.04	-0.10
Normal	0.50	0.08	0.18
Boom	0.30	0.14	0.30

Using the expected rate of return on an individual asset formula, we would find the expected rates of return for assets A and B as follows:

$$\bar{R}_A = 0.20 \times 0.04 + 0.50 \times 0.08 + 0.30 \times 0.14 = 9\%.$$

and

$$\bar{R}_B = 0.20 \times (-0.10) + 0.50 \times 0.18 + 0.30 \times 0.30 = 16\%.$$

■

7.2 Measuring Risk

The term risk can be defined as receiving an outcome different to what is expected. The greater the chance that the actual outcome could be different to the expected outcome, the greater the risk. For an individual asset risk refers to the probability or chance that the actual rate of return received on the asset will be different to the expected rate of return on the asset.

7.2.1 The Risk-Free Asset

Almost all assets in a market have at-least some risk associated with them. However, in every market there is one asset with, theoretically, no risk, which is referred to as the risk-free asset.

In most practitioners opinion, the risk-free asset in a market is a central government bond or treasury security. These assets are considered to be free of default risk as, theoretically, central governments do not go bankrupt. However, in reality, some governments do default on repayment of their debts (see Business Snapshot 7.1, for example). Hence, the rate of return on a treasury security for a given period will be the risk-free rate of return (R_f) in a country for that period. For example, if the rate of return on a one-year Australian Commonwealth Government bond is 2.0%, the one-year risk-free rate of return in Australia would be 2.0%.

7.2.2 Company Risk

While government debt is generally considered risk-free, for a company or corporation, no matter how big it is, there is a possibility that it will go bankrupt. Consequently, there is a chance that the company will default on its debt repayments and/or will not pay a dividend, and any investment in the company, whether debt or equity, could be lost. Hence, when determining the required rate of return on investment in a firm, or any other risky asset, a risk premium is added. The size of the risk premium will depend on the investment risk, which relates to the probability of earning a return different from that expected, and the greater the chance of receiving a return different from the expected return, the greater the investment risk.

Business Snapshot 7.1 — Sovereign risk. The table below documents cases of government (sovereign) debt defaults since the mid-1970s. A particularly bad period for sovereign default was the years 1981 to 1983, which was caused by sharply higher interest rates in the US in the mid-to-late 1970s (a result of oil-crisis-linked high inflation) and the resulting recession in the US that began in 1980 and ended in 1984.

Europe		Latin America		Africa		Asia	
Country	Year	Country	Year	Country	Year	Country	Year
Poland	1981	Argentina	1982; 2001	Cameroon	1989	Indonesia	1999
Romania	1981	Brazil	1983	Egypt	1984	Jordan	1989
Russia	1998	Chile	1983	Morocco	1983	Philippines	1983
Turkey	1978	Costa Rica	1983	Nigeria	1983	Pakistan	1981; 1999
Ukraine	1998	Dominican Republic	1982	South Africa	1985	Vietnam	1985
		Mexico	1982	Sudan	1979		
		Panama	1982	Uganda	1981		
		Peru	1978	Zaire	1976		
		Uruguay	2003				
		Venezuela	1982				

7.2.3 Distribution of Returns

One way to measure the risk associated with investment in an asset is to measure the *distribution of returns* on the asset by setting-up a histogram of the possible returns on the asset. The histograms in Figure 7.1 show the distribution of possible returns for the two assets, Asset A and Asset B from Example 7.2.

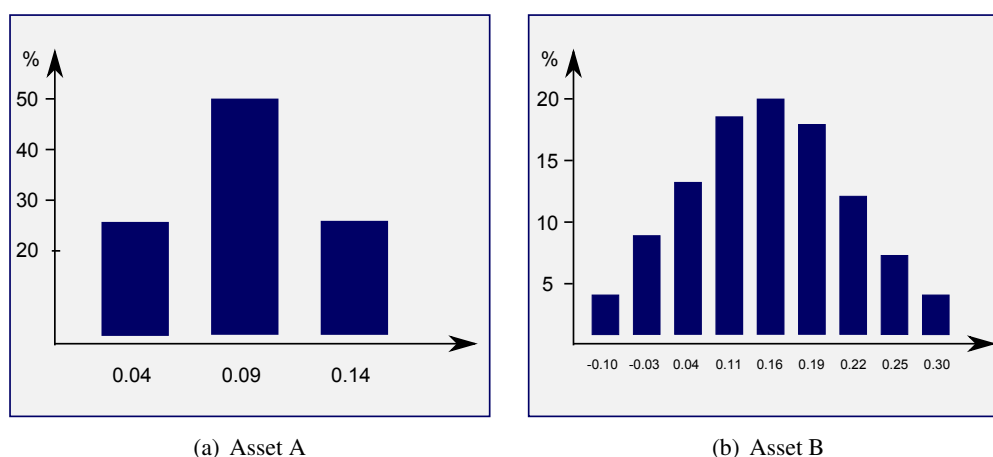


Figure 7.1: The histogram for the distribution of returns for Asset A and Asset B.

The histograms show the frequency of each possible return for assets A and B. For example, for Asset A the 9% return occurs most often (50% of the time) and the 4% and 14% returns each occur 25% of the time. For Asset B, the expected return of 16% occurs 20% of the time while, for example, the -10% and 30% returns both occur 2.50% of the time.

We know that investment risk relates to the probability of earning a return different from that expected, and the greater the chance of receiving a return different from the expected return, the greater the investment risk. Therefore, comparing Asset A to Asset B, the latter is riskier because there is a greater chance of getting a return other than the expected return (of 16%). For Asset B

the expected return of 16% will be received 20% of the time, while for Asset A the expected return of 9% will be received 50% of the time.

7.2.4 Variance and Standard Deviation

There are various ways to measure the risk associated with investment in an individual asset and we have seen above how the distribution of returns can be used as a measure of risk. Two measures of risk are of particular interest in Business Finance, namely *variance* and *standard deviation*.

An asset's variance and standard deviation of returns are both measures of dispersion of expected return. The greater the variance or standard deviation, the greater the uncertainty and, therefore, the greater the risk associated with the asset. To calculate the variance of returns on an individual asset the following formula is used.

$$\sigma_i^2 = \sum_{i=1}^n P(R_i)(R_i - \bar{R}_i)^2,$$

where $\bar{R}_i$ is the expected return, as discussed above. Further, the standard deviation of a random variable is defined as the square root of its variance. Then, the standard deviation of R_i is

$$\sigma_i = \sqrt{\sum_{i=1}^n P(R_i)(R_i - \bar{R}_i)^2}$$

■ **Example 7.3 — Continuing from Example 7.2.** For Asset A, the variance of returns is calculated as follows.

$$\begin{aligned}\sigma_A^2 &= (0.04 - 0.09)^2 \times (0.20) + (0.08 - 0.09)^2 \times (0.50) + (0.14 - 0.09)^2 \times (0.30) \\ &= (0.0005) + (0.00005) + (0.00075) \\ &= 0.0013 = 0.13\%,\end{aligned}$$

and then the standard deviation is

$$\sigma_A = \sqrt{0.0013} = 0.0361 = 3.61\%.$$

For Asset B the variance of returns is calculated as follows.

$$\begin{aligned}\sigma_B^2 &= (-0.10 - 0.16)^2 \times (0.20) + (0.18 - 0.16)^2 \times (0.50) + (0.30 - 0.16)^2 \times (0.30) \\ &= (0.01352) + (0.0002) + (0.00588) \\ &= 0.0196 = 1.96\%,\end{aligned}$$

and then the standard deviation is

$$\sigma_B = \sqrt{0.0196} = 0.14 = 14.00\%.$$

■

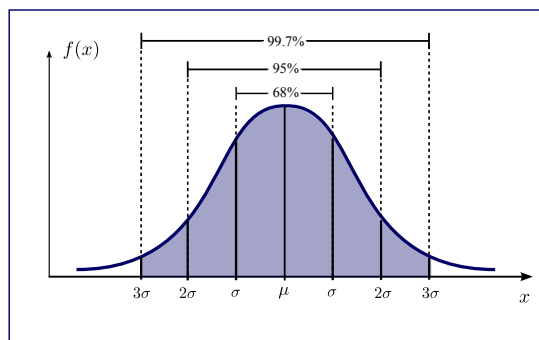


Figure 7.2: The Normal probability distribution function, commonly known as the bell curve. μ denotes the mean and σ denotes the standard deviation.

Highlight 7.1 — The Bell Curve. The term *bell curve* refers to the most common type of distribution function and is named after the graphical representation of a *normal distribution*, which looks like of a bell-shaped line. The bell curve is also known as a normal distribution and less commonly referred to as a *Gaussian distribution*, after German mathematician and physicist *Karl Gauss*. The highest point in the bell curve represents the most probable event, which is the mean, average or expected value.

In classical finance theory we often assume that the returns on an investment or asset are normally distributed, similar to Figure 7.2. The percentage numbers on the bell curve, e.g. 68% and 95%, represent the probability that the return at the end of the period will fall between that certain interval. For example, there is a 68% chance that the return from an asset will fall in the area one standard deviation away from the mean. With the same token, with a normal distribution there is only a 1% chance that the return will lie outside three standard deviations of the expected return.

■ **Example 7.4** The expected (mean or average) return of Asset A is 9% and the standard deviation of returns is 3.61%. This means that the range of one standard deviation of returns is $9\% \pm 3.61\%$, which gives 5.39% to 12.61%. This means since we are assuming normally distributed returns, there is a 68% chance that the return on Asset A will lie within the range 5.39% to 12.61%. For Asset A, two standard deviations of returns is 2×3.61 , which equals 7.22%. Therefore, the range of two standard deviations of returns is $9\% \pm 7.22\%$, which gives 1.78% to 16.22%. In other words, the probability that returns on Asset A will lie within the range 1.78% to 16.22% is 95%. Finally, three standard deviations of return is 3×3.61 , which equals 10.83%, meaning the range of three standard deviations of returns is $9\% \pm 10.83\%$, which gives -1.83% to 19.83%. Therefore, there is a 99% chance that the return on Asset A will lie within the range -1.83% to 19.83%, and only one time out of every hundred, i.e. 1% of the time, will the return for Asset A lie outside this range. ■

7.2.5 Coefficient of Variation

In probability theory and statistics, the *coefficient of variation* (CV) is a standardised measure of the dispersion of data points in a series around a mean or average. CV measures the dispersion of a probability distribution or frequency distribution and is defined as the ratio of the standard deviation to the mean.

In finance, the CV shows how much volatility (risk) is assumed in comparison to the amount of return expected from an investment. In other words, the CV shows risk, as measured by the

standard deviation, per unit of expected return. The lower the CV, the better, as this means that you have less risk per unit of return on your investment. The formula for CV is as follows:

$$CV_i = \frac{\sigma_i}{\bar{R}_i}.$$

■ **Example 7.5 — Continuing from Example 7.2.** Given that we know that for Asset A the expected return is 9% and the standard deviation of returns is 3.61%, the CV for Asset A can be calculated as follows:

$$CV_A = \frac{\sigma_A}{\bar{R}_A} = \frac{0.036}{0.09} = 0.4010.$$

which means that for Asset A, for every one unit (percent) of expected return the risk is 0.401 units (percent).

For Asset B we know that the expected return is 16% and the standard deviation of returns is 14%, meaning that we can calculate the CV for Asset B as follows:

$$CV_B = \frac{\sigma_B}{\bar{R}_B} = \frac{0.14}{0.16} = 0.8750.$$

which means that for Asset B, for every one unit (percent) of expected return the risk is 0.8750 units (percent). Therefore, since the CV of Asset B is higher than the CV of Asset A, Asset B has more risk for each unit of expected return. ■

Highlight 7.2 — Portfolio. A portfolio is a collection of assets, securities, or investments held by an investor, investment company, or financial institution. The types of assets that can make-up a portfolio include shares, fixed-interest assets (e.g. bonds and debentures), cash, property, and, possibly, other assets such as valuable artworks. In an Australian context an investment portfolio may comprise the shares of different Australian companies, Australian government bonds and Australian corporate debentures, Australian dollars held in an Australian bank account, and Australian property, as well as similar types of assets overseas, for example, Asian shares, US government treasury bills, European corporate debentures, British pounds held in an English bank account, and North American property.

7.3 Portfolio Expected Return

To calculate the *expected return on a portfolio of assets* we must find the weighted average return of the individual securities making up the portfolio, the weight being the fraction of total funds invested in each security. The formula used to calculate the expected return on a portfolio is as follows:

$$\begin{aligned}\bar{r}_p &= W_1 \times \bar{r}_1 + W_2 \times \bar{r}_2 + \dots + W_n \times \bar{r}_n \\ &= \sum_{i=1}^n W_i \times \bar{r}_i\end{aligned}$$

where

W_1 is the weighting of Asset 1 in the portfolio,

$\bar{r}_1$ is the expected return on Asset 1,

W_i is the weighting of Asset i in the portfolio,

$\bar{r}_i$ is the expected return on Asset i .

■ **Example 7.6** Assume that Eduardo has a two-asset investment portfolio, consisting of Asset 1 and Asset 2. The expected return on Asset 1 is 10% and the expected return on Asset 2 is 15%, while the weighting of each of the two assets in his portfolio, in terms of the percentage value of the portfolio that each asset comprises, are 55% and 45%, respectively. Calculate the expected return for Eduardo's portfolio:

$$\bar{r}_p = 0.55 \times 0.10 + 0.45 \times 0.15 = 0.1225 = 12.25\%$$

■

As with an individual security or asset, two measures of risk for a portfolio are the variance and the standard deviation of returns on the portfolio. However, for a portfolio, because it is made up of more than one asset, the overall riskiness of the portfolio depends on the measure of risk of each asset in the portfolio, measured by each individual asset's variance or standard deviation. Additionally, the measure of co-movement between the returns of the assets making-up the portfolio is a critical factor in calculating the overall risk of a portfolio. This is further discussed below.

7.4 Portfolio Diversification

Before we get to the formulae for the variance and standard deviation of returns on a portfolio, it is important that we discuss the concept of *portfolio diversification*. Portfolio diversification is a risk-management technique that involves having a wide variety of assets in a portfolio. The motive for portfolio diversification is that, by having different assets making-up the portfolio, overall risk will be reduced relative to the average return of the portfolio, compared to the individual constituents of the portfolio. This is because the positive performance of some assets will balance-out the negative performance of other assets. Ideally, an investor should construct a portfolio in accordance with their individual risk tolerance, the amount of risk they are prepared to accept, and their investment objectives.

In order to diversify a portfolio one could have the shares of different companies in the portfolio, as well as having other types of assets. For example, bonds, property, and cash in the domestic market or internationally.

We know that because a portfolio is made up of more than one asset that one of the factors influencing the overall risk of the portfolio is the the measure of co-movement between the prices or returns of the assets making-up the portfolio. The co-movement is defined as the covariance of the return of the assets making-up the portfolio. In a portfolio, pairs of assets whose prices or returns do not move exactly together provide diversification and risk-reduction benefits.

The aim of portfolio diversification is to eliminate the risk that comes from price changes and in an optimally-diversified portfolio the net price change in a given period will be zero. For example, in a well-diversified two asset portfolio of Asset A and Asset B, if the price of Asset A rises by 5%, the price of Asset B should fall by 5%, the net price change will be 0%.

Another example of portfolio diversification could be where we have a two asset portfolio, again made-up of Assets A and B and with each asset comprising 50% of the total value of the portfolio. In one year, the return for Asset A is 4.25%, which is a good positive return, while the return for Asset B is -1.75%, which is, obviously, not a good return. In this situation, the overall return on the portfolio is 1.25%, which is less than the return we would have received if we had only invested in Asset A, but better than the return we would have received if we had only invested in Asset B. Now suppose in the following year the return on Asset A is -4.5% (loss) while for Asset B the return is 2.5%. Therefore, the return on the portfolio is -1%, which

is less than the return we would have received if we had only invested in Asset B, but better than the return we would have received if we had only invested in Asset A.

In the two examples above, the return on the portfolio is less extreme than the returns on the two individual assets making-up the portfolio and this highlights the benefit of portfolio diversification, which is eliminating the risk that comes from changes in the prices of individual assets.

Given that the aim of portfolio diversification is to have a net price change in the portfolio in a given period of 0%, a reasonable question that could be asked is *how do we make any money out of the portfolio?* It would seem that if the net price change in a portfolio is zero, then there is no point of going to the effort of constructing a portfolio of assets?

In answering the above question and justifying portfolio diversification, we must remember that the total return on an asset (including a portfolio) is made up of two components, an income yield and a capital gains yield. The income yield on an asset comes from the periodic income earned on the asset, for example, the dividends paid on a share or the coupon payments received from owning on a bond or debenture. The capital gain yield on an asset (including a portfolio) comes from the change in the price of the asset in a given time period. Therefore, even if a portfolio is optimally diversified, a positive return can still be earned because of the periodic income. Therefore, the return on the portfolio will be positive and will be equal to the income yield.

7.4.1 Covariance and Correlation Coefficient

We know already that covariance refers to the co-movement between the prices or returns of assets making-up a portfolio. The covariance of two variables x and y is denoted by $\text{cov}(x, y)$, and it is a statistical measure of the degree to which the prices or returns on two assets *co-vary*, and it can be

- positive, meaning that the prices or returns on two assets move in the same direction;
- zero, meaning that the prices or returns on two assets are independent of each other; or
- negative, meaning that the prices or returns on two assets move in the opposite direction.

While covariance is a useful measure, it is somewhat limited, as it only indicates the direction of the relationship between the prices or returns on a pair of assets. Another measure of the co-movement between prices or returns on a pair of assets is the correlation coefficient, denoted by $\rho_{x,y}$. It is a standardised statistical measure of covariance that not only indicates the direction of the relationship, but also the strength of the relationship between the prices or returns on two assets.

A correlation coefficient between any two assets will always be within the range of -1 to +1. A correlation coefficient of +1 indicates perfectly positively correlated prices or returns between two assets. In other words, we would have exact proportional movements in the same direction for the prices or returns of two assets. For instance, if the price of Asset A goes up by 1%, the price of Asset B will also go up by 1%. A correlation coefficient of -1 indicates perfectly negatively correlated prices or returns between two assets, meaning that we would have exact proportional movements in the opposite direction for the prices or returns of the two assets. Therefore, in this example, if the price of Asset A goes up by 1%, the price of Asset B will go down by 1%. Finally, a correlation coefficient of 0 (zero) indicates that there is no relationship between the prices or returns of two assets. In this case, if the price of Asset A goes up by 1%, the price of Asset B will not change.

Generally, the lower the correlation coefficient between asset prices or returns, the greater the potential reduction of risk. The aim of portfolio diversification is to have the correlation

coefficient of each pair of assets in a portfolio being as close to -1 (perfectly negatively correlated) as possible. Alternatively, the closer the correlation coefficient between two assets' prices or returns is to +1 (perfectly positively correlated), the less the risk-reduction benefits of having those two assets in a portfolio. However, as long as the correlation coefficient between the returns of a pair of assets in a portfolio is less than perfectly positively correlated, i.e. less than 1, then there will be some risk reduction and diversification benefits of having the two assets in a portfolio.

7.5 Portfolio Variance and Standard Deviation

Returning to measuring risk for an investment portfolio, we know that two measures of such risk are the portfolio variance and standard deviation. For a two asset portfolio variance is measured as follows.

$$\sigma_p^2 = W_1^2 \sigma_1^2 + W_2^2 \sigma_2^2 + 2W_1 W_2 \rho_{1,2} \sigma_1 \sigma_2.$$

where

W_1 is the weighting of Asset 1;

W_2 is the weighting of Asset 2;

σ_1 is the standard deviation of Asset 1;

σ_2 is the standard deviation of Asset 2; and

$\rho_{1,2}$ is the correlation coefficient between Asset 1 and Asset 2.

Then, standard deviation is measured as follows

$$\sigma_p = \sqrt{W_1^2 \sigma_1^2 + W_2^2 \sigma_2^2 + 2W_1 W_2 \rho_{1,2} \sigma_1 \sigma_2}.$$

■ **Example 7.7** Assume that you are an investment adviser and a client, Manuela, comes to see you and says that she has a two-asset investment portfolio, namely Asset 1 and Asset 2. Manuela also tells you that the standard deviation of returns on Asset 1 is 20% and on Asset 2 is 28%. Additionally, the weighting of each of the two assets in her portfolio is 55% and 45%, respectively. Manuela also tells you that the correlation coefficient between Asset 1 and Asset 2 ($\rho_{x,y}$) is 0.30. Manuela would like you to calculate for her the variance and standard deviation of returns on her portfolio, which you do as follows.

$$\begin{aligned} \sigma_p^2 &= 0.55^2 \times 0.20^2 + 0.45^2 \times 0.28^2 + 2 \times 0.55 \times 0.45 \times 0.30 \times 0.20 \times 0.28 \\ &= 3.63\%. \end{aligned}$$

Then, standard deviation is

$$\sigma_p = \sqrt{0.0363} = 0.1905 = 19.05\%$$

■

In the above example, we have a correlation coefficient, $\rho_{x,y}$, of 0.30, meaning that the prices or returns on the two assets making up the portfolio are less than perfectly positively correlated. The best way to interpret this correlation coefficient of 0.30 is to say that if the price of Asset 1 goes up by 1%, the price of Asset 2 will go up by 0.30%. One may ask, 'Has there been any diversification (reduction of risk) benefit by combining Assets 1 and 2 in this portfolio?' One way of checking is by comparing the portfolio's standard deviation with the average weighted

standard deviation of the portfolio, where the weighted average standard deviation of a portfolio is simply

$$W_1\sigma_1 + W_2\sigma_2 = (0.55)(0.20) + (0.45)(0.28) = 0.236.$$

Since the standard deviation of the portfolio of 19.05% is less than the weighted average standard deviation of the portfolio of 23.6%, we can conclude that by combining Assets 1 and 2 into a portfolio, we have reduced risk. Whenever the standard deviation of a portfolio is less than the weighted average standard deviation of the portfolio, it will always be the case that risk has been reduced by constructing the particular portfolio.

It is also the case that whenever the correlation coefficient between a pair of assets in a portfolio is less than +1 the standard deviation of a portfolio comprised of the two assets will be less than the weighted average standard deviation of the portfolio and, again, there will be diversification of risk and risk reduction.

7.6 Portfolio with more than two assets

So far we have dealt with portfolios constructed of only two assets. However, in the real world most investment portfolios will be comprised of more than two assets. In such cases, in order to calculate the variance and standard deviation of the portfolio, it is necessary to measure the co-movement between the prices or returns of each pair of assets making up the portfolio. For example, for a three asset portfolio comprised of Assets 1, 2, and 3, it would be necessary to take into consideration the relationship between Asset 1 and Asset 2, Asset 1 and Asset 3, and Asset 2 and Asset 3. To calculate the variance and standard deviation of returns on a three asset portfolio the formulae are as follows:

$$\begin{aligned}\sigma_p^2 = & W_1^2\sigma_1^2 + W_2^2\sigma_2^2 + W_3^2\sigma_3^2 + 2W_1W_2\rho_{1,2}\sigma_1\sigma_2 \\ & + 2W_1W_3\rho_{1,3}\sigma_1\sigma_3 \\ & + 2W_2W_3\rho_{2,3}\sigma_2\sigma_3.\end{aligned}$$

Then, standard deviation is measured as follows.

$$\sigma_p = \sqrt{W_1^2\sigma_1^2 + W_2^2\sigma_2^2 + W_3^2\sigma_3^2 + 2W_1W_2\rho_{1,2}\sigma_1\sigma_2 + 2W_1W_3\rho_{1,3}\sigma_1\sigma_3 + 2W_2W_3\rho_{2,3}\sigma_2\sigma_3}.$$

Varying the Correlation Coefficient

We know that having a correlation coefficient closer to -1 between a pair of assets in a portfolio will increase the diversification and risk-reduction benefits of the portfolio. In order to prove this, here we see the effect of reducing the correlation coefficient between the returns or prices of a pair of assets in a portfolio on the risk of the portfolio, as measured by standard deviation. This is best explained via an example.

■ **Example 7.8** We will assume that our two asset portfolio is made up of Asset 1 and Asset 2. Asset 1 has a standard deviation of 20% and Asset 2 has a standard deviation of 28%, with Asset 1 making up 55% of the value or weight of the portfolio and Asset 2 comprising 45% of the portfolio's value or weight.

Suppose that the correlation coefficient between Assets 1 and 2 is +1, meaning that the prices or returns of these two assets are perfectly positively correlated. This will give us a standard deviation on the portfolio as follows.

$$\sigma_p = \sqrt{0.55^2 \times 0.20^2 + 0.45^2 \times 0.28^2 + 2 \times 0.55 \times 0.45 \times 1 \times 0.20 \times 0.28} = 0.236 = 23.6\%.$$

This gives the same result as the weighted average standard deviation of the portfolio calculated earlier as $(0.55)(0.20) + (0.45)(0.28) = 0.236$ and this example proves that where a correlation coefficient is +1, there are no diversification benefits achieved by combining the two assets in a portfolio.

Next, we assume that the correlation coefficient between Assets 1 and 2 is 0.50, meaning that the prices or returns of these two assets are less than perfectly positively correlated. This also means that if the price of Asset 1 goes up by 1%, the price of Asset 2 will go up by only 0.5%. This gives us a standard deviation on the portfolio as follows.

$$\sigma_p = \sqrt{0.55^2 \times 0.20^2 + 0.45^2 \times 0.28^2 + 2 \times 0.55 \times 0.45 \times 0.50 \times 0.20 \times 0.28} = 0.21 = 21.0\%.$$

This standard deviation of 21% is obviously less than the standard deviation we had when the correlation coefficient between the two assets was +1 (23.6%) and supports the theory that having a correlation coefficient less than +1 and closer to -1 between a pair of assets in a portfolio will increase the diversification and risk-reduction benefits of the portfolio.

Next, we will assume that the correlation coefficient between Assets 1 and 2 is 0 (zero), meaning that the prices or returns of these two assets are less than perfectly positively correlated. This also means that if the price of Asset 1 goes up by 1%, the price of Asset 2 will not change, indicating that the prices or returns of these two assets are independent of each other. This gives us a standard deviation on the portfolio as follows

$$\sigma_p = \sqrt{0.55^2 \times 0.20^2 + 0.45^2 \times 0.28^2 + 2 \times 0.55 \times 0.45 \times 0 \times 0.20 \times 0.28} = 0.17 = 17.0\%.$$

This standard deviation of 17% is less than the standard deviation we had when the correlation coefficient between the two assets was +1 (23.6%), and is also lower than when the correlation coefficient was 0.50.

Next, we will assume that the correlation coefficient between Assets 1 and 2 is -0.50, meaning that if the price of Asset 1 goes up by 1%, the price of Asset 2 will fall by 0.50%. This gives us a standard deviation on the portfolio as follows.

$$\sigma_p = \sqrt{0.55^2 \times 0.20^2 + 0.45^2 \times 0.28^2 + 2 \times 0.55 \times 0.45 \times (-0.5) \times 0.20 \times 0.28} = 0.12.$$

Again, we observe a further reduction in standard deviation. Finally, we will assume that the correlation coefficient between Assets 1 and 2 is -1, meaning that the prices or returns of these two assets are perfectly negatively correlated. This gives us a standard deviation on the portfolio as follows.

$$\sigma_p = \sqrt{0.55^2 \times 0.20^2 + 0.45^2 \times 0.28^2 + 2 \times 0.55 \times 0.45 \times (-1) \times 0.20 \times 0.28} = 0.016.$$

This standard deviation of 1.60% is the least possible that can be achieved, given the standard deviation of the constituents of the portfolio and their weights. ■

7.7 Opportunity Set and Efficient Frontier

Generally speaking, and as previously discussed, the lower the correlation between asset prices or returns, the greater the potential reduction of risk in having the two assets in a portfolio. Additionally, the amount of risk reduction achieved is also dependent upon the proportions or weightings in which the assets are combined in the portfolio. Potentially, there is an infinite number of asset combinations available to an investor and, theoretically, it is possible to make use of all assets and combine them in every possible combination and then to plot the risk-return

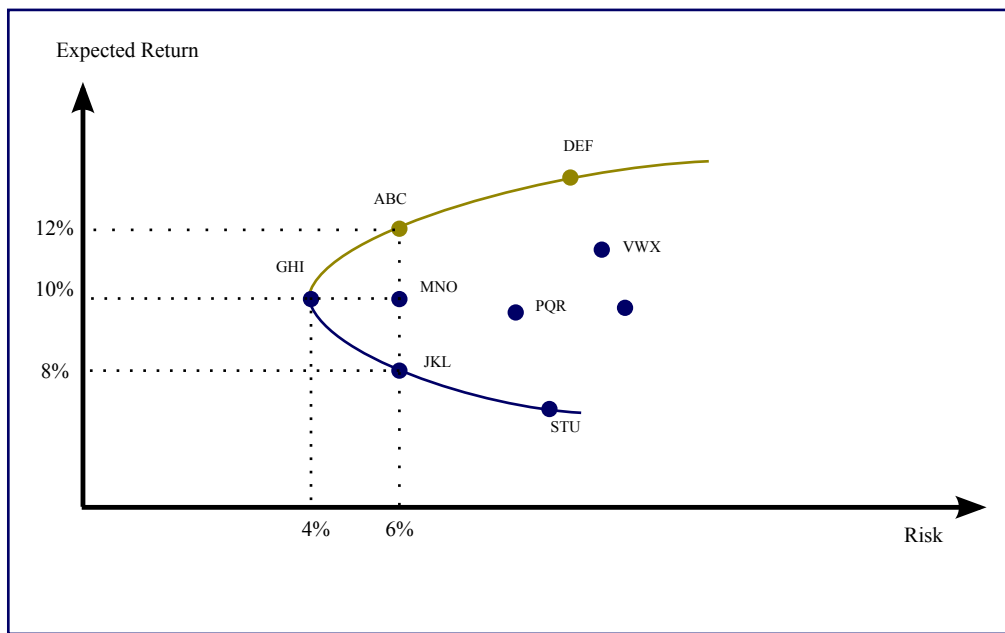


Figure 7.3: This opportunity set is showing the risk and return combinations for a number of different portfolios, with each portfolio made up of a different combination and weighting of assets.

outcomes. The resulting graph is called the opportunity set, which is the set of all possible portfolios that one may construct from a given set of assets, and presenting investors with an opportunity set may help them in making investment decisions.

An example of an opportunity set is Figure 7.3, where different combinations of assets are plotted (black dots) against risk, as measured by standard deviation, and expected return. Looking at the opportunity set in Figure 7.3, we can see how an individual investor chooses among possible investments.

7.7.1 Rational Investing

In finance, we assume that all investors are rational, therefore, they will choose investments or portfolios that provide the highest return for a given level of risk, or lowest risk for a given level of return. Such portfolios are known as efficient portfolios, and are found on the upper boundary of the bullet shaped graph of the opportunity set. In particular, the upper boundary is known as the *efficient frontier* (highlighted in red in Figure 7.3). Efficient portfolios on the efficient frontier provide maximum return for a given level of risk and minimum risk for a given level of return. If an investor is operating on the efficient frontier, he or she will be maximising their utility subject to their resource or finance constraints.

By closely examining Figure 7.3, we can prove how operating on the efficient frontier will maximise an investor's utility. Firstly, compare portfolios ABC, MNO, and JKL. Each of these portfolios has the same level of risk, a standard deviation of 6%. However, they do not have the same expected return. Portfolio ABC, which is on the efficient frontier, has an expected return of 12%, while Portfolio MNO has an expected return of 10%, and Portfolio JKL has an expected return of only 8%. Now, remember that there is a positive relationship between expected return and an individual's utility or happiness. Hence, since each of these portfolios has the same level of risk, meaning that with each of the three we have the same chance of losing our money, but

Portfolio ABC has the highest expected return, a rational person would choose Portfolio ABC out of the three possible investments. This means that to maximise utility, an investor should aim for the highest possible return for a given level of risk.

Now, compare Portfolios GHI and MNO. Both of these portfolios have the same expected return of 10%, but they do not have the same level of risk, as measured by standard deviation. Portfolio GHI, which is on the efficient frontier, has a standard deviation of 4%, while Portfolio MNO has a standard deviation of 6%. Now, remember that there is a negative relationship between risk and an individual's utility or happiness. Since both of these portfolios have the same expected return, but Portfolio GHI has lower risk, a rational person would choose Portfolio GHI over Portfolio MNO. This means that to maximise utility, an investor should aim for the lowest possible risk for a given level of expected return.

It is important to emphasise, at this stage, that any point underneath the efficient frontier, such as portfolios MNO, JKL, PQR, VWX, and STU in Figure 7.3, represent points of investment inefficiency. This means if an investor is operating at such a point he or she is not being rational because they are not maximising their utility. In other words, they are not achieving the highest expected return for a given level of risk, or not minimising risk for a given level of expected return.

7.7.2 Risk–Return Preference

For an individual investor, the point on the efficient frontier that they will choose to operate at will depend on their own attitude towards risk and return, i.e. their risk-return preference. For an investment advisor it is important to discuss with a client their attitude towards risk in order to determine their risk tolerance and risk-return preference.

Returning to Figure 7.3, we can compare the two portfolios ABC and GHI in order to explain how risk preference may influence where on the efficient frontier an investor operates. For a relatively conservative investor, who is an investor that generally prefers to avoid too much risk with their investments, Portfolio GHI would be the better option as it has a lower level of risk (standard deviation of 4%) as compared to Portfolio ABC with a standard deviation of 6%. However, the cost of achieving this lower level of risk is a lower expected return, which is 10% for Portfolio GHI against 12% for Portfolio ABC.

An example of an investor who would probably prefer Portfolio GHI over Portfolio ABC is an older investor, who would generally avoid risk. That is because, if they do lose money, they have less time and opportunity to get it back. For example, for a 70 year old retired person to lose money on investments is a problem, because he or she is no longer working and has a significantly reduced capacity to recover from financial losses. So, generally, older people are more conservative, risk-avoiding investors.

For a relatively risk-seeking investor, who does not mind a higher level of risk hoping to achieve a higher expected return, Portfolio ABC would be the better option as it has a higher expected return (12%) as compared to Portfolio GHI (10%), but it comes at the cost of a higher standard deviation. That is 6% for Portfolio ABC compared to 4% for Portfolio GHI.

Finance theory assumes that all investors are rational, which means they will choose investments that provide the highest return for a given level of risk or lowest risk for a given level of return. Such investors will choose investments on the efficient frontier of the opportunity set. However, finance theory also assumes that all investors are risk-averse.

Highlight 7.3 — Risk Averse Investor. A risk averse investor is one that, as the level of risk in an investment increases, the investor requires ever-increasing amounts of expected return

to compensate for each additional unit of risk, in order to keep the investor at the same level of utility. For example, assume initially that standard deviation increases from 5% to 6% and an investor's expected return increases from 10% to 12%, an increase of two percentage points in expected return for a one unit increase in risk. Then, if standard deviation increases from 6% to 7%, the investor's expected return must increase from 12% to 15%, an increase of three percentage points in expected return for a one unit increase in risk.

7.8 Revision Problems

Problem 7.1 The characteristics of two perfectly positively correlated assets are as follows: Asset A has a standard deviation of 20% while asset B has a standard deviation of 30%. The standard deviation of a portfolio consisting of an equal weighting of Asset A and Asset B is:

- (a) 50%
- (b) 25%
- (c) 75%
- (d) 20%

Problem 7.2 The standard deviations of two assets are 10 and 20 percent respectively. If an equally weighted portfolio made up of the two assets produced a standard deviation of 14%, we can deduce the prices or returns on the two assets are:

- (a) Negatively correlated
- (b) Perfectly positively correlated
- (c) Uncorrelated
- (d) Less than perfectly positively correlated

Problem 7.3 A risk-averse investor owning shares in White Corporation decides to add the shares of either Black Corporation or Green Corporation to her portfolio. All three stocks offer the same expected return and total risk. The covariance of returns between White shares and Black shares is -0.05 and White shares and Green shares is $+0.05$. Portfolio risk is expected to:

- (a) Decline more by buying Black Corporation Shares
- (b) Decline more by buying Green Corporation
- (c) Increase by buying either Black or Green Corporation
- (d) Decline or increase, depending on other factors

Problem 7.4 A risk-averse investor is faced with a selection between Asset A with a standard deviation of 20% and an expected return of 15%, and Asset B with a standard deviation of 25% and an expected return of 20%. Which asset would the investor prefer?

- (a) Asset A
- (b) Asset B
- (c) Depends upon the investor's level of risk aversion
- (d) Indifferent between A and B

Problem 7.5 The benefits from diversification are maximised when the prices or returns on two assets are:

- (a) Perfect positively correlated
- (b) Perfectly negatively correlated
- (c) Uncorrelated
- (d) Either (a) or (c)

Problem 7.6 You are given the following information about the possible returns from an investment:

Return	Probabilities
12%	0.15
9%	0.60
6%	0.25

Required:

- (a) Calculate the expected return.
- (b) Calculate the variance of the return.
- (c) Calculate the standard deviation of the return.

Problem 7.7 An investor invests 40 per cent of her funds in Company A's shares and the remainder in Company B's shares. The standard deviation of the returns on A is 20 per cent and on B is 10 per cent. Calculate the standard deviation of return on the portfolio assuming the correlation between the returns on the two securities is:

- (a) +1.0
- (b) +0.5
- (c) 0.0
- (d) -0.5

Problem 7.8 The return and associated probabilities of two assets in each of three possible states is given below. The probabilities are of each state occurring.

State	I	II	III
Asset A	10%	7%	6%
Asset B	5%	8%	9%
Probability:	25%	50%	25%

The expected return and standard deviation of returns for Assets A and B are:

- (a) $R_a = 7.5$; $R_b = 7.5$; $SD_A = 1.5$; $SD_B = 1.5$
- (b) $R_a = 7.5\%$; $R_b = 7.5\%$; $SD_A = 1.50\%$; $SD_B = 1.50\%$.
- (c) $R_a = 7$; $R_b = 7.5$; $SD_A = 2.25$; $SD_B = 2.25$.
- (d) none of the above

Capital Market Line

Borrowing and Lending at the Risk-Free Rate

Capital Asset Pricing Model

Systematic vs Non-Systematic Risk

Model Setup

Security Market Line

Calculation of Systematic Risk

CAPM and Equilibrium

Efficient Market Hypothesis

Types of Market Efficiency (EMH)

Informational Efficiency

Requirements for Informational Efficiency

EMH and Information Sets

Market Anomalies – 'How To Beat The Market'

Revision Problems

8. Asset Pricing Models

In this chapter you will learn about the risk-free asset, capital market line, and the new efficient frontier; borrowing and lending at the risk-free rate; the capital market line formula for expected return; the capital asset pricing model; non-systematic risk and systematic risk; required rate of return; security market line; equilibrium rate of return; the efficient market hypothesis; informational efficiency; information sets; and market anomalies.

8.1 Capital Market Line

The *Capital Market Line* may be achieved by introducing a risk-free asset to the opportunity set, where the risk-free asset (R_f) is usually considered to be a central government bond, such as a Commonwealth Treasury Bond in Australia. When introducing the risk-free asset to market analysis a number of assumptions are made, which are:

1. that asset markets are perfect, which means there are no taxes and no transaction costs;
2. quantities of assets are fixed;
3. all assets are marketable (can be sold at any time) and divisible (can be sold in any quantity); and
4. all investors can borrow and lend at the risk-free rate.

Once the risk-free asset is introduced to the analysis, the opportunity set is enlarged, with many more risk-return combinations possible, and we have a new efficient frontier called the capital market line (CML), with the CML representing all linear combinations of the risk-free asset (R_f) and the market portfolio (M).

Therefore, introduction of the risk-free asset enables investors to create portfolios that combine the risk-free asset with a portfolio of risky assets, where the risky asset is M or the market portfolio, which represents the most diversified portfolio in the economy. Each asset weight in the market portfolio will reflect its relative importance in the economy as a whole. For example, if the retail sector makes up 60% of the national economy, retail sector assets will make up 60% of the market portfolio. In this set up, the market portfolio is considered as being one asset.

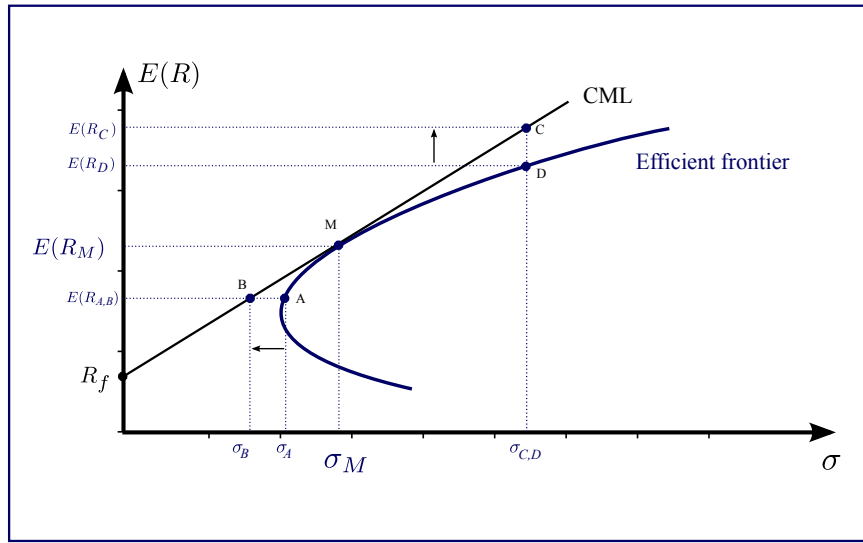


Figure 8.1: Capital Market Line.

Highlight 8.1 — Market portfolio. A theoretical bundle of investments that includes every type of asset available in the world financial market, with each asset weighted in proportion to its total presence in the market. The expected return of a market portfolio is identical to the expected return of the market as a whole, because a market portfolio is completely diversified.

Figure 8.1 provides an illustration of the CML and the new efficient frontier. When plotting the CML the horizontal axis measures standard deviation (or total risk) of the portfolio, while the vertical axis measures expected return. The intercept for the CML is at R_f , which is the risk-free rate of return, such as the return on a central government bond. Since a central government bond is a risk-free asset, the standard deviation corresponding to the vertical intercept is zero, meaning that there is no risk at this point. Point M on the CML is the market portfolio and is at the point of tangency with the old efficient frontier.

The straight line connecting R_f to M , the CML, is the new efficient frontier, and at any point on the CML, other than at the point of tangency with the old efficient frontier, point M , investors are better off. That is because they can receive a higher expected return for each unit of risk, or they can have lower risk for each unit of expected return. For example, comparing points C and D in Figure 8.1, both of these points have the same level of risk, with standard deviation of σ_{CD} . Here, point D is on the old efficient frontier and has an expected return of $\bar{r}_D = E(r_D)$, while point C is on the CML and at a higher level of expected return of $\bar{r}_C = E(r_C)$. Remember, there is a positive relationship between expected return and an investor's utility, hence, point C must give an investor higher utility than point D . Another example of an investor being better off by operating on the CML can be seen by comparing points A and B in Figure 8.1. Both points A and B have the same expected return of $\bar{r}_{AB} = E(r_{AB})$, with point A on the old efficient frontier and having a standard deviation of σ_A , while point B is on the CML and at a lower level of standard deviation of σ_B . Remember, there is a negative relationship between standard deviation and an investor's utility, hence, point B must give an investor higher utility than point A .

Once the risk-free asset is introduced and the CML has become the new efficient frontier, all investors will hold some combination of the risk-free asset and the market portfolio M . Investors will choose a portfolio of assets somewhere on the CML, but where exactly on the CML each investor operates will depend on each individual investor's attitude towards risk, i.e. it will

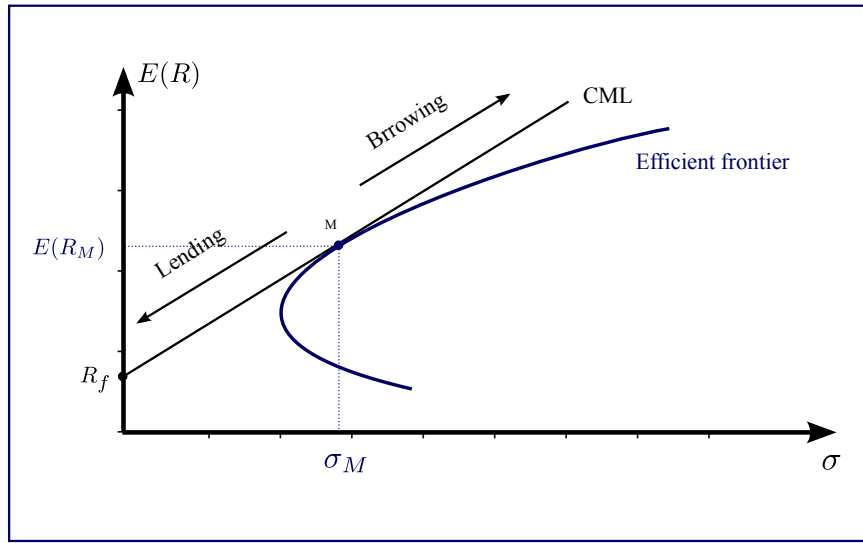


Figure 8.2: Capital market line with borrowing and lending at R_f .

depend on their risk preferences. Returning to Figure 8.1, for a relatively risk-averse investor, i.e. an investor who has a particular dislike for investment risk, the point chosen on the CML would more likely be to the left of M . For example, at point B , where the standard deviation or total risk is relatively low. Concurrently, expected return is relatively low. However, for a relatively risk-tolerant investor, the point chosen on the CML would more likely be to the right of M , for example, at point C , where the standard deviation is relatively high, but at the benefit of a relatively higher expected return.

8.1.1 Borrowing and Lending at the Risk-Free Rate

The introduction of the risk-free asset allows investors to borrow and lend at the risk-free rate of interest (R_f). If an investor is operating to the left of M on the CML, say point B in Figure 8.1, then the investor is lending at R_f , through purchasing government bonds. On the other hand, if an investor is operating to the right of M on the CML, say point C in Figure 8.1, then the investor is borrowing at R_f at the same interest rate that the government pays on government bonds. Borrowing and lending at R_f is also illustrated in Figure 8.2:

The CML has a formula, which is used to work out expected return for portfolios comprised of a linear combination of the risk-free asset and the market portfolio, and the equation for the CML is as follows.

$$\bar{R}_p = R_f + (R_m - R_f) \frac{\sigma_p}{\sigma_M},$$

where,

$\bar{R}_p$ is the expected return on the investor's portfolio;

R_f is the risk-free rate of return;

R_m is the return on the market portfolio;

$(R_m - R_f)$ is the market risk premium;

σ_p is the risk of the investor's portfolio;

σ_M is the risk of the market; and

$\frac{\sigma_p}{\sigma_M}$ is the measure of relative risk.

The CML and its formula for expected return reflect the positive relationship between risk and expected return, i.e. higher risk leads to a higher expected return, and vice-versa. It also reflects that the investor's risk is proportionate to the risk of the market.

With the CML, if investors want less risk than the risk of the market they can satisfy their preference by investing in a combination of M and lending at R_f . In this case, they will have a lower expected return than the expected return on the market portfolio M . On the other hand, if investors want a higher return than the return on the market, they can satisfy their preference by investing in a combination of M and borrowing at R_f , because they will have higher risk than the risk of the market portfolio. In this way, investors' risk-return preferences are satisfied by borrowing and lending at R_f .

Lending Portfolio

If an investor is relatively risk-averse, he or she would, generally, want a level of risk (as measured by the standard deviation) in his/her portfolio less than the level of risk of the market portfolio. Hence, this investor could satisfy their risk-return preference by investing in a combination of M and lending at R_f . Using the CML expected return formula, we could work out exactly what return this investor could expect on his or her portfolio.

■ **Example 8.1** Assume that R_f is 10%, $\bar{R}_m$ is 12%, and σ_m is 8%. The individual investor, who is assumed in this case to be risk averse, would like her portfolio to have a level of risk of only $\sigma_p = 6\%$. What will be this investor's expected return? Using the CML expected return formula we would find the investor's expected return as follows:

$$\begin{aligned}\bar{R}_p &= R_f + (R_m - R_f) \frac{\sigma_p}{\sigma_M} \\ &= 0.10 + (0.12 - 0.10) \frac{0.06}{0.08} \\ &= 0.10 + 0.02 \times 0.75 \\ &= 0.10 + 0.015 = 0.115 = 11.50\%.\end{aligned}$$

In this example, it makes sense that the expected return of the investor (11.50%) is less than the expected return on the market portfolio (12%), because the investor wishes to have less risk in her portfolio ($\sigma_p = 6\%$) than the risk of the market portfolio ($\sigma_m = 8\%$). Therefore, she must expect a lower return than the expected return of the market portfolio. Remember, there is a positive relationship between risk and expected return, meaning that lower risk leads to lower expected return. ■

Given that the investor above requires a standard deviation in her portfolio less than the standard deviation of M , this must mean that the investor is operating to the left of M on the CML, which, in turn, means that the investor must be lending at R_f . The question we can now ask is *what proportion of the investor's portfolio should be in the risk-free asset R_f , and what proportion should be in the risky asset R_m , so that the investor can achieve her desired level of risk of $\sigma_p = 6\%$?*

We know that this investor can tolerate less risk than the market risk, i.e. $6\%/8\% = 0.75$ times the risk of the market. Therefore, she must invest only 75% of her available funds in the market. It is always the case that an investor will begin with exactly 100% of her available funds to invest. Let us assume that this investor has \$100 of her own money to invest, meaning that \$75 of the investor's funds will be invested in the market portfolio M , and the remaining amount, \$25, will be invested in the risk-free asset R_f .

By transferring some of her funds to the risk-free asset, and away from the risky asset M , the investor is able to bring down the average risk of her portfolio. If the investor had kept all of her

money, \$100, in the market portfolio, the standard deviation of the investor's portfolio would be equal to the standard deviation of the market portfolio of 8%. However, in this example, the investor transfers \$25 of her funds to the risk-free asset, which has a standard deviation of zero, and this brings down the average risk of the investor's portfolio to the desired level of 6%.

Borrowing Portfolio

If an investor is relatively risk-seeking, he would, generally, accept a higher level of risk in his portfolio than the level of risk of the market portfolio in the hope of achieving a higher expected return. Hence, this investor could satisfy his risk-return preference by investing in a combination of M and borrowing at R_f . Using the CML expected return formula, we could work out exactly what return this investor could expect on his portfolio.

■ **Example 8.2** Assume that R_f is 10%, R_m is 12%, and σ_m is 8%, while the individual investor, who is assumed in this case to be risk tolerant, is prepared to accept a level of risk in his portfolio of $\sigma_p = 14\%$. What will be this investor's expected return? Using the CML expected return formula we would find the investor's expected return as follows:

$$\begin{aligned}\bar{R}_p &= R_f + (R_m - R_f) \frac{\sigma_p}{\sigma_m} \\ &= 0.10 + (0.12 - 0.10) 0.14 / 0.08 \\ &= 0.10 + (0.02) 1.75 \\ &= 0.10 + 0.035 = 0.135 = 13.50\%.\end{aligned}$$

In this example, the expected return of the investor (13.50%) is higher than the expected return on the market portfolio (12%), because the investor is prepared to accept more risk in his portfolio ($\sigma_p = 14\%$) than the risk of the market portfolio ($\sigma_m = 8\%$). Therefore, he can expect a higher return than the expected return of the market portfolio. Remember, there is a positive relationship between risk and expected return, meaning that higher risk leads to higher expected return. ■

Given that the investor in the above example has a standard deviation in his portfolio higher than the standard deviation of M , this must mean that the investor is operating to the right of M on the CML, which, in turn, means that the investor must be borrowing at R_f . The question we can now ask is what proportion of the investor's portfolio should be in the risk-free asset R_f , and what proportion should be in the risky asset R_m , so that the investor can achieve his desired level of risk of $\sigma_p = 14\%$?

We know that this investor can tolerate more risk than the market risk, i.e. $14\% / 8\% = 1.75$ times the risk of the market. Thus, the investor will invest 175% of his funds in the market and zero percent in R_f . Here, we must remember that whenever money is borrowed, risk rises, so the fact that this investor is borrowing money at R_f , automatically increases the risk (standard deviation) of his portfolio.

It is always the case that an investor will begin with exactly 100% of his available funds to invest. For example, if the investor has \$100 of his own money to invest, he would allocate all of that in the market portfolio, M , and then borrow a further \$75 to invest in M , which makes the total amount invested in the risky asset, M , \$175.

By borrowing at R_f and investing 175% of his funds in M , the investor increases the average risk of his portfolio, but also increases the expected return on his portfolio above that of the expected return on M , to 13.50%, because the investor is prepared to have more risk in his portfolio than the risk of the market portfolio.

8.2 Capital Asset Pricing Model

8.2.1 Systematic vs Non-Systematic Risk

The actual return received on an investment is made up of two components, the *expected return*, and the *unexpected return*. The latter part of return is the true risk of any investment because it results in uncertainty and variability of returns in the sense that the actual return may be less than or more than expected.

The risk inherent in an investment comes from the probability of receiving a return different from expected and there are two main types of risk, these being

Systematic risk, which affects a large number of securities, or the whole system.

Non-systematic risk, which affects only a single firm or asset or a small number of firms or assets, i.e. it does not affect the whole system.

Business Snapshot 8.1 — Non-Systematic Risk Factor: Steve Jobs And Apple Corp.

The following article by John Talty of the International Business Times, published soon after the death of Steve jobs in early-October 2011, highlights the significant contribution that Jobs made to Apple Corp. and how his passing had the potential to seriously affect the company.

Steve Jobs Dead: How Will His Death Affect Apple?^a

Apple co-founder and former CEO, Steve Jobs, sadly passed away at the age of 56 on Wednesday night after a battle with pancreatic cancer. Jobs, the ultimate salesman and visionary, had a hand in the creation of the iPod, iPhone, iPad, Macintosh computer, along with countless other immeasurable contributions. He made Apple go and despite his resignation of his CEO position in August, his sad death could have an impact on the company he built.

Investors heavily tied the success of Apple to Steve Jobs and his departure creates uncertainty for the technology powerhouse. The star power of Jobs and investor confidence in him helped propel the company to the world's largest market cap valuation, but that same confidence will now drop the company's stock price.

Jobs helped lead the company to resurgence in the mid-1990s when he returned back to the company he helped co-found following a 1985 falling out. Jobs built back his reputation through incredible innovation, including introducing iMacs, iPods, iPhones, and the newest addition – iPads. That incredible innovation and hands-on approach to the company could lead investors to either sell or take a wait and see approach to whether the company can keep up its current hot streak with its products.

Whether Apple can recreate the hit process without Jobs will ultimately be the key to if it will be able to rebound from the oncoming stock sell-off. Apple diehards appear to put faith in new CEO Tim Cook's capabilities, but he's no Steve Jobs. After his stunning resignation some analysts expected a major sell-off of Apple stock, but the resignation announcement stated that Jobs would continue to be an integral part of the company. At the time Jobs sickness was widely known so a step away wasn't unexpected and could have already been factored into the stock price.

Why 'Apple Corp'?

Why is it that the name 'Apple' was chosen for the computer company partly founded

by Steve Jobs? The story goes that the two founders of Apple Corp., Steve Jobs and Steve Wozniak, were big fans of the band the Beatles, whose own recording label was called Apple Records. So, when Jobs and Wozniak were establishing their computer company in the mid-1970s they sought permission from Beatles management to name their own company 'Apple'. Permission was granted, so Apple Corporation was born.

"Steve Jobs Dead: How Will His Death Affect Apple?", John Talty, International Business Times, 05/11/2011

The total risk inherent in an investment is comprised of the asset's systematic and non-systematic risk. Another term for systematic risk is market risk, since it affects an entire system, market, or economy, including all assets or investments in that market. Examples of systematic risk factors include government tax rate changes, interest rate changes, and changes in the price of essential, widely-consumed commodities such as oil.

Non-systematic risk factors are also referred to as non-market risk and do not affect the whole system or market, only a small component of the system, such as one type of asset, a small collection of firms, or one particular industry. See Business Snapshot 8.1, for example.

As non-systematic risk factors affect only a small number of investments, firms, or an industry, they can be eliminated by diversification of assets and are sometimes referred to as *diversifiable risk*. However, systematic risk affects the whole system and all assets and cannot be diversified away, so it is sometimes referred to as *non-diversifiable risk*.

Highlight 8.2 — Diversification. The process of diversification, spreading investments across different assets and countries, can reduce risk. Also, the gain from diversification is closely related to the correlation coefficient of each pair of assets in a portfolio, with portfolio risk reduced by combining assets that are less than perfectly positively correlated ($\rho_{x,y} < +1$), through reduction in the variability of cash-flows. However, not all investment risk can be eliminated.

8.2.2 Model Setup

The *capital asset pricing model* (CAPM) is an equilibrium model of the relationship between risk and return, which is based on the premise that says that the required rate of return on an asset is equal to the rate of return on a risk-free security plus a risk premium. If the expected return on an asset does not meet or exceed the required rate of return, then the investment should not be undertaken.

The CAPM states that an asset's required rate of return is mainly determined by three factors, these being

- the risk-free rate of return (R_f), which reflects the time value of money;
- the market risk premium, which is an investor's reward for bearing systematic risk; and
- an asset's beta coefficient, which is a measure of the amount of systematic risk present in a particular asset.

The *security market line* (SML) is the graphical representation of CAPM and plots the risk-return relationship for all levels of systematic risk.

Highlight 8.3 According to the CAPM's systematic risk principle, the required rate of return on a risky asset depends only on the asset's systematic (non-diversifiable) risk and investors are not rewarded for having non-systematic (diversifiable) risk in their portfolio.

This means that if an investor has not properly diversified his or her portfolio and some non-systematic remains, the total risk inherent in the portfolio will be relatively high. However,

the investor should not expect to receive a higher return for having the high risk. The investor will only be rewarded for the systematic risk inherent in his or her portfolio and not for any non-systematic risk.

The amount of systematic risk in an asset relative to the average risky asset, where the average risky asset is the market portfolio, is measured by the asset's beta coefficient (β_i).

In finance theory, investors require a risk premium for holding risky assets, and with the CAPM, the risk premium built into the model is the market risk premium, defined as $R_m - R_f$, i.e. the return on the market portfolio minus the risk-free rate of return. How much market risk premium an investor receives depends on the extent of their systematic risk exposure (as measured by β_i).

Beta is a measure of the sensitivity of an asset's return relative to changes in the market return and the beta of an individual asset (β_i) could have any value, although the beta of the market is always 1. For example, $\beta_i = 0$ means that the asset's returns are independent of the market, $\beta_i = 0.5$ means the asset's returns are half as responsive as the market, $\beta_i = 1.0$ means that the asset's returns will move exactly with the market, and $\beta_i = 1.5$ means the asset's returns will increase by 50% more than the market for any given increase in the market returns. With the same token, if $\beta_i = -1.5$, this means that the asset's returns will decrease by 50% more than the market for any given increase in the market returns.

It is possible for an individual asset to have a beta of any value. In some cases an asset will even have a negative beta, meaning that if the market rises (falls), the price of this asset will fall (rise). In the real world for some assets, such as shares, do have a negative beta. For example, following the terrorist attacks of September 2011, the US stock market was down, but the share prices of companies related to aerospace, defence, and logistics were up in anticipation of increased orders from the US government. The higher the degree of systematic risk, as measured by β_i , the higher the return required by investors.

Business Snapshot 8.2 — Betas of Selected Australian Companies. The table here lists betas of selected Australian companies and shows that the Commonwealth Bank (CBA) moves very closely with the market (beta of 0.97, very close to the market beta of 1.0), that Westpac Bank is significantly more risky than the market (with a beta of 1.53), and that Telstra is a good defensive stock with a very low beta 0.34.

Company	Beta	Company	Beta
CBA	0.97	Woolworths	0.54
Westpac	1.53	Telstra	0.34
NAB	1.46	BHP-Billiton	1.22
ANZ Bank	1.37	Rio Tinto	1.39
Wesfarmers	0.71		

■ **Example 8.3** If we have two assets, Security A and Security B, and Security A has a standard deviation of 30% and a beta of 0.60, and Security B has a standard deviation of 10% and a beta of 1.20, we can say that Security A has more total risk than Security B, but less systematic risk than Security B. So, according to CAPM, an investor would require a higher return for investing in Security B than for investing in Security A, as Security B has more systematic risk (it has a higher beta). ■

The formula for required rate of return, $\bar{R}_i$, with CAPM is as follows

$$\bar{R}_i = R_f + \underbrace{(R_m - R_f)}_{\text{market risk premium}} \beta_i.$$

total risk premium

At equilibrium, the actual rate of return on an asset will be equal to the required rate of return calculated with CAPM, so the CAPM required rate of return can also be referred to as the equilibrium rate of return. This is the return that the asset needs to yield in order for it to be willingly held by an investor, given its systematic risk.

■ **Example 8.4** Assume that R_f (the risk-free rate of interest) is 5% and that R_m (the expected return on the market) is 10%, while Share Z has a $\beta = 1.54$. What will be the required rate of return on Share Z? Using CAPM

$$\begin{aligned}\bar{R}_Z &= R_f + (R_m - R_f)\beta_Z \\ &= 0.05 + (0.10 - 0.05) \times 1.54 = 12.5\%.\end{aligned}$$

■

8.2.3 Security Market Line

The *security market line (SML)* is the graphical representation of the CAPM and is a positively sloped straight line displaying the relationship between the required rate of return on an asset and beta. CAPM is the equation that describes the relationship between required return and beta and CAPM is the equation of the SML. The capital market line (CML) gives the risk return relationship for portfolios consisting of some combination of the market portfolio and the risk free asset, where risk is total risk (σ_p). The security market line, on the other hand, is used for working out the risk and return relationship for individual assets and portfolios, other than those comprised of a combination of the market portfolio and the risk free asset, and where risk is measured only by systematic risk (β_i).

8.3 Calculation of Systematic Risk

Beta (β_i) is the measure of systematic risk for an individual asset i , measuring the risk inherent in an asset relative to the market. In order to calculate β_i for an individual asset the formula used is as follows:

$$\beta_i = \frac{\rho_{i,m}\sigma_i\sigma_m}{\sigma_m^2} = \frac{\text{cov}(i,m)}{\sigma_m^2}$$

where,

β_i is the beta of individual asset i ;

$\rho_{i,m}$ is the correlation coefficient between the individual asset i and the market;

σ_i is standard deviation of the individual asset;

σ_m is standard deviation of the market; and

σ_m^2 is the variance of the market.

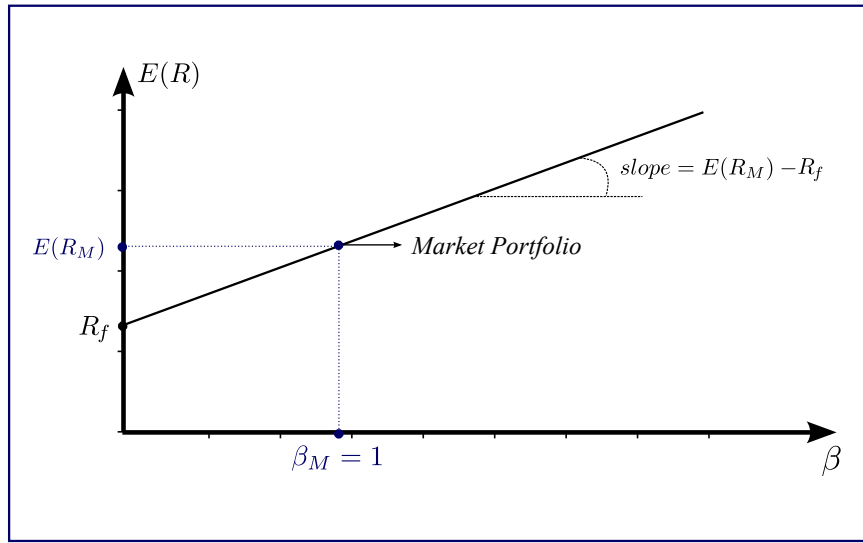


Figure 8.3: The *Security Market Line* plots the risk and return relationship for individual assets and portfolios (other than those comprised of a combination of the market portfolio and the risk free asset). With the SML, risk is measured on the horizontal axis as systematic risk (β) and required rate of return is plotted on the vertical axis, with the vertical intercept of the SML being the risk-free rate of return (R_f), where $\beta = 0$.

In order to calculate beta for a portfolio of assets (β_p), we simply calculate the beta of each of the individual assets making up the portfolio, then calculate the beta of the portfolio as the weighted average beta of the assets in the portfolio, as follows:

$$\beta_p = W_1\beta_1 + W_2\beta_2 + \dots W_n\beta_n$$

where,

W_i is the weighting of individual Asset i in the total value of the portfolio.

■ **Example 8.5** Assume that the correlation coefficient between Asset i and the market ($\rho_{i,m}$) is 0.30 and that the standard deviation of asset i (σ_i) is 7%, while the standard deviation of the market portfolio (σ_m) is 5%. This means that the beta of Asset i will be as follows:

$$\begin{aligned}\beta_i &= \frac{\rho_{i,m}\sigma_i\sigma_m}{\sigma_m^2} \\ &= \frac{0.30 \times 0.07 \times 0.05}{0.05^2} = 0.42.\end{aligned}$$

Assume that a portfolio is made up of two assets, Asset 1 and Asset 2, and that the beta for each of these assets are 0.54 and 1.20, respectively. Further, the weighting of each of the assets in the portfolio is 63% for Asset A and 37% for Asset B. Therefore, the beta of this portfolio will be as follows.

$$\beta_p = (0.63)(0.54) + (0.37)(1.20) = 0.3402 + 0.4440 = 0.7842.$$

■

8.4 CAPM and Equilibrium

CAPM is a model of required returns based on markets in equilibrium and it is also a model of how share prices are formed. With CAPM, price and return are inversely related. This means that if the actual return¹ on an asset is greater than its required rate of return, it becomes an attractive investment, because it is relatively cheap and giving a relatively high return. In this case, the asset is termed *under-priced*. Subsequently, investors will buy the under-priced asset, which will increase the demand and force the price of the asset up, until such time as the actual return falls to equal the required rate of return; at this point the market for the asset will be in equilibrium.

Similarly, if the actual return on an asset is less than its required rate of return, the asset is said to be *over-priced*. In this situation, investors will sell the asset and demand for the asset will fall, because it will be relatively expensive, giving a relatively low return. This fall in demand will force the price of the asset down until such time as the actual return rises to equal the required rate of return; at this point the market for the asset will be in equilibrium. Therefore, with CAPM, we assume the required rate of return to be the correct, equilibrium rate of return, and if the actual return on an asset does not equal the required rate of return, then there is a dis-equilibrium in the market. This creates an arbitrage opportunity, which will be exploited by the market participants. Subsequently, the miss-pricing will disappear.

■ **Example 8.6** Assume that market analysts have forecast that the expected (actual) return on Share X will be 7%, on Share Y will be 12.5%, and on Share Z will be 14%. Given that the risk-free rate of return is 5% and the expected return on the market is 10%, what would be the required rate of return on Share X if it has a beta of 0.8, on Share Y if it has a beta of 1.2, and Share Z if it has a beta of 1.8? Are these shares fairly priced and, if not, what will happen to return their markets to equilibrium?

In order to answer the above question we will firstly use the CAPM required rate of return formula to determine the equilibrium return for each of the shares and then we will compare each share's actual return to its required rate of return to determine whether the market for the share is in equilibrium. If the market for any of the shares is not in equilibrium, we will then explain the process that will occur in the market to achieve equilibrium for the asset.

Using the CAPM formula, we can calculate

$$\bar{R}_i = R_f + (R_m - R_f)\beta_i,$$

then

$$\bar{R}_x = 0.05 + (0.10 - 0.05)0.8 = 0.09,$$

$$\bar{R}_y = 0.05 + (0.10 - 0.05)1.2 = 0.11, \text{ and}$$

$$\bar{R}_z = 0.05 + (0.10 - 0.05)1.8 = 0.14.$$

Share X has a required rate of return of 9% , while the actual rate of return we are told is 7%. This means that the actual return on Share X is too low, as it should be 9%, so, the price of Share X must be too high, i.e. Share X is over-valued, and the market for Share X is in dis-equilibrium.

The reason why investors in Share X are receiving too low a return on it (7%) is because it is too expensive (over-valued). Consequently, since Share X is more expensive than it should be, and is giving a return lower than what it should be, it is an unattractive investment, so demand for Share X in the market will fall. This will push its price down. As the price of Share X falls, the actual return on the asset will rise, and demand for and the price of Share X will continue to fall until the actual rate of return on the asset rises to equal the required rate of return of 9%.

For Share Y the required rate of return is 11%, while the actual rate of return we are told is

¹ Same logic applies for expected return or forecast return of assets

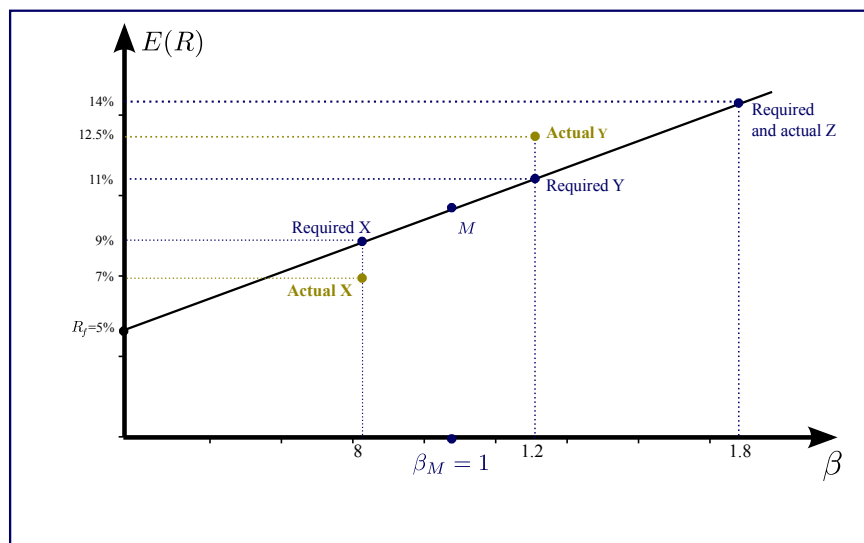


Figure 8.4: SML – Graphical representation of CAPM For Shares X, Y, & Z.

12.50%. This means that the actual return on Share Y is too high, and the price of Share Y must be too low. Thus, Share Y is under-valued, and the market for Share Y is in dis-equilibrium.

The reason why investors in Share Y are receiving such a relatively high return on it (12.50%) is because it is cheaper than it should be (under-valued). Consequently, it is giving a return greater than what it should. Hence, the demand for Share Y in the market will rise and this will push its price up. As the price of Share Y rises, the actual return on the asset will fall, and demand for and the price of Share Y will continue to rise until the actual rate of return on the asset falls to equal the required rate of return of 11%.

Figure 8.4 gives a graphical representation of the situation with Shares X, Y, and Z using the SML.

■

8.5 Efficient Market Hypothesis

The *efficient market hypothesis* (EMH) is a cornerstone theory of finance that is based on the premise that an investor cannot “beat the market”, because market efficiency means that asset prices always incorporate and reflect all relevant information and that assets are always priced at their correct value. The main implication of the EMH is that investors cannot purchase *mispriced* (undervalued or overvalued) securities as all prices are correct. Hence, it is pointless to search for undervalued stocks or to try to predict trends in the market, through either fundamental or technical analysis, as it is impossible to outperform the overall market.

8.5.1 Types of Market Efficiency (EMH)

The EMH postulates that there are different types of market efficiency, which when achieved together contribute to an overall efficient market. The first type of efficiency is *operational efficiency*, which is a market condition whereby investors can execute transactions and receive services at a price that equates fairly to the actual costs required to provide them. An operationally efficient market is also referred to as an *internally efficient market*, and, in effect, requires that financial services are produced at minimum cost.

The second type of efficiency required for an overall efficient market is *allocative efficiency*, which requires that investment funds be allocated to those investments that provide the highest return for a given level of risk. The idea of allocative efficiency is consistent with the overall notion that investors are rational as they aim to maximise their own utility, and to do this they seek the highest possible return for a given level of risk, or for a given return seek the lowest possible risk.

Thirdly, the EMH requires *dynamic efficiency*, which means that the market is able to adapt to changing needs and generate innovations in financial services. Certainly, over the last one hundred years, markets have been dynamically efficient, with developments in finance that have enhanced the ability of participants to achieve their investment objectives and made markets more efficient. Examples of this include futures; options; swaps; and forwards (forward contracts), which have been beneficial in, for example, hedging risk in share and foreign exchange trading and with interest rates. However, there have also been innovations in finance that have hindered market efficiency. A good example of this is collateralised debt obligations (CDOs), trading of which (mainly in the form of toxic (sub-prime, low grade) mortgage-backed debt securities) significantly contributed to the global financial crisis.

Finally, an efficient market requires *informational efficiency*, meaning that market prices should reflect all relevant information regarding assets and adjust fully and quickly to any new information that becomes available. As per the definition of the EMH above, informational efficiency is the key component of the hypothesis that means that it is impossible to *beat the market*, as asset prices will incorporate and reflect all relevant information and be priced at their correct value, so that investors cannot purchase mispriced securities and profit from this.

In Business Finance it is informational efficiency that we are most interested in and which will be discussed in more detail.

8.5.2 Informational Efficiency

The informational efficiency component of the EMH requires that market prices of financial assets reflect all available information and adjust fully and quickly to any new relevant information. For this to happen, individual traders and investors must process the information available to them and trade in securities in response to that information. This, in turn, means that markets will aggregate this diverse information and reflect all relevant available information.

8.5.3 Requirements for Informational Efficiency

To achieve capital market informational efficiency there are a number of important requirements. Firstly, there must be a large number of well-informed investors and analysts who continually evaluate all available information regarding any particular asset. Secondly, informational efficiency requires large, well-functioning security markets with significant competition between participants, in order to ensure that securities are fairly and accurately priced. Thirdly is the requirement for speed, meaning that there must be instantaneous adjustment of prices to any new information so that investors cannot trade at old prices and earn abnormal profits when prices adjust.

If markets are informationally efficient, investors cannot consistently earn abnormal profits because prices of securities adjust instantaneously to fully reflect all relevant information. By abnormal profits we mean that the investor cannot consistently earn a return on an investment greater than the risk-adjusted return i.e. greater than the required rate of return predicted by CAPM.

8.5.4 EMH and Information Sets

With the EMH, information sets are used to categorise all types of relevant information, with the type of informational efficiency related to the information set specified.

In 1976, Professor Eugene Fama² specified three levels of market informational efficiency by setting-out three different types of information sets, these being

1. Weak Form,
2. Semi-Strong Form, and
3. Strong Form.

with each form incorporating and building-upon the previous form.

Weak Form Informational Efficiency

Weak form informational efficiency is associated with Eugene Fama's random walk hypothesis, which states that price changes represent random departures from previous price changes and that there is no pattern in price changes. The assertion is that market prices of securities fully reflect all the information contained in the historical sequence of past prices, and as such, all past prices of an asset are reflected in today's stock price.

Weak form efficiency means that investors cannot devise an investment strategy to yield consistently abnormal profits on the basis of analysis of past price patterns. Importantly, this means that the use of the widely-popular technical analysis investment method is of no benefit, as any new information from such an analysis will have already been factored into current market prices and could not be used to predict prices and beat the market. However, because weak form informational efficiency deals only with past prices, if a market is only weak-form efficient, then consistently abnormal profits could still be earned by analysis of other types of information, such as company reports (fundamental analysis).

Highlight 8.4 — Technical Analysis. Charting or technical analysis, is the study of historical price patterns and trends of publicly traded assets using tools such as bar or candlestick charts and trading volumes to determine the future behaviour of an asset's prices. Much of this practice involves discovering the overall trend line of an asset's price movements.

Semi-Strong Form Informational Efficiency

Semi-strong form market informational efficiency is based on the notion that publicly available information is incorporated into an asset's current market price, therefore, neither fundamental analysis, nor technical analysis can be used to consistently beat the market. Thus, with semi-strong form efficiency current market prices reflect not only historical price data, but also all publicly available information relevant to a security. If a market is semi-strong form efficient, which also means that it is weak form efficient, then, regardless of the amount of fundamental and technical analysis undertaken, only information that is not publicly available can be used by investors to consistently beat the market.

Highlight 8.5 — Fundamental analysis. Analysis of publicly available information, such as company financial statements (balance sheets and P&L statements) or news announcements.

²Eugene Fama is a Professor of Finance at University of Chicago, most famous for his 'random walk hypothesis' (1965). He is the person most responsible for the development and popularity of the efficient market hypothesis.

Strong Form Informational Efficiency

Strong form market informational efficiency asserts that all information available in a market, whether public or private, is already factored into the current market price of an asset. The consequence of this is that, regardless of the amount of research or information an investor has access to, not even insider information (private information) could be used by the investor to consistently beat the market. Under such conditions, it is impossible, even with insider trading, to ever consistently earn a return on an asset greater than the risk-adjusted return. For investors, the implication of strong form informational efficiency is that the optimal investment strategy is to simply invest in a well-diversified portfolio.

8.5.5 Market Anomalies – ‘How To Beat The Market’

The EMH postulates that it is impossible to do better than hold a well-diversified portfolio, as one is not able to consistently earn above-normal returns. The hypothesis has been extensively tested in the real world, usually by university finance academics, and the conclusion is that, generally, there is considerable evidence suggesting that markets perform well in reflecting available information and respond quickly to new information. The Australian market, for example, is considered to be at least semi-strong form efficient.

However, around the world, every day millions of investors, traders, and analysts are attempting to “beat the market” in the hopes of making themselves rich. There are many examples of investors who have consistently beaten the market, the best example probably being US investment guru Warren Buffet, who perennially beats market averages. This points to the fact that there is evidence of market anomalies and market informational *inefficiencies*.

Business Snapshot 8.3 — Insider Trading Scandal at ABS. In May 2014, it was revealed in the media that an employee of National Australia Bank (NAB) and a friend working at the Australian Bureau of Statistics (ABS) had been arrested by the Australian Federal Police for allegedly using unreleased economic data to reap a \$7 million profit from foreign exchange trading^a. The Sydney Morning Herald reported in May that it was alleged that the NAB employee used the ABS employee’s access to unreleased jobs, retail, and international trade data to make profitable bets in currency markets, using the unreleased (private, insider) information to predict movements in the Australian dollar and used the proceeds to procure residential property, bank accounts, and a motor vehicle.

The two accused men appeared in court in May and were charged with various offences, including counts of insider trading; conspiring to commit insider trading; offering to pay a bribe; using false documents to commission an offence; dealing with the proceeds of crime; and receiving a corrupt benefit.

^aNAB And ABS Staff in Insider Trade Bust, Georgia Wilkins and Ben Butler, Sydney Morning Herald: Business Day, 10/05/2014.

The first market anomaly we will cover is the *small-firm effect*. The small firm effect postulates that the returns on shares of smaller companies (companies with smaller market capitalisations) are generally higher than the returns for larger companies. This usually occurs because smaller firms typically are able to grow at higher percentage rates than larger companies, so the share prices of smaller firms reflect this. The implication for investors of this anomaly is that to earn higher (excess) returns, investors should target small-cap firms.

The *January effect* is another market anomaly in which share prices are found to generally rise more in January. This anomaly is mainly relevant to the US and other markets, where the financial year aligns with the calendar year and tax laws give incentives and disincentives to buy

and sell at the end of the calendar year, unlike Australia where the financial year ends in June.

With the January effect assets that under-perform in the December quarter of the prior year tend to outperform markets in January because investors, for tax reasons, often sell under-performing assets late in the year, so that they can use the losses to offset capital gains taxes. This end-of-year selling, which may not be fully justified by fundamentals, can push these assets to price levels where they become attractive to buyers in January, thus increasing demand and pushing-up their prices in the first month of the year. The implication of the January effect for traders wanting to beat the market is to buy the affected under-performing shares at the end of December, then sell them at the end of January when prices should have risen.

The *Tuesday effect* is a market anomaly that is said to occur in Australia, whereby there exists a bias toward negative market performance on Tuesdays, with share-market prices tending to fall on this day of the week. A possible explanation for this anomaly is that, perhaps, beginning-of-the-week optimism permeates the market on Mondays as traders and investors look forward to the week, but by Tuesday investors may have begun to worry about the market and develop pessimism going into the rest of the week. Of course, if this anomaly holds true, the profitable strategy is to short-sell on Monday afternoons when prices are higher, then undertake a reversal and buy the shares back at lower prices at the end of the day on Tuesday.

Highlight 8.6 — Short selling of securities. Short selling is a trading technique, most commonly used with shares, whereby an investor sells borrowed securities (shares) in anticipation of a price decline and is required to return an equal number of shares at some point in the future.

The logic behind short selling is the opposite of what is common with the buying and selling of assets, which is taking a long position and buying when prices are low then selling at a later stage when prices have, hopefully, risen. With short selling, the investor is anticipating that the market price of the asset is going to fall, so to take advantage of this, the investor will sell the asset when its price is still high, and before the investor has bought the asset, then when the price of the asset does fall at a later stage, the investor will undertake a reversal transaction and buy back the same number or volume of the asset, as was sold previously at the higher price. In this way, the investor is still able to buy low, sell high, but the ordering of the transaction is reversed.

The final market anomaly we will discuss is the *turn-of-the-month effect*, whereby share prices generally rise at the beginning of a month. A possible explanation for this effect is that, perhaps, beginning-of-the-month optimism permeates the market, with traders and investors looking forward to the month ahead and leaving behind any negative investment sentiment generated in the previous month, but by the end of the month, the optimism has disappeared and prices are lower. Obviously, the profitable strategy for the astute investor is to buy towards the end of the previous month, when the prices of affected stocks are relatively low, then sell early in the new month when the turn-of-the-month effect has pushed share prices higher.

If these market anomalies are true, then they imply that markets are not even weak-form efficient and, therefore, knowledge of such effects can be used to earn abnormal returns over an extended period of time. There is evidence from both Australian and overseas markets that some market professionals, for example, stockbrokers, dealing-room traders, and fund managers, who may have access to private information, consistently earn abnormal above-average returns. However, the question that needs to be asked here is whether these investors are able to consistently beat the market because they are insider-traders or because they are better skilled traders?

Business Snapshot 8.4 — Warren Buffet– Investor King. Warren Buffet, also known as the “the Oracle of Omaha”, is Chairman of Berkshire Hathaway and arguably the greatest investor of all time. Buffet’s net wealth fluctuates with the performance of the market, but as of 2008 his net worth was estimated at \$62 billion, making him the richest man in the world. Buffett is a value investor. His company Berkshire Hathaway is basically a holding company for his investments. Major holdings he has had at some point include Coca-Cola, American Express and Gillette. Critics predicted an end to his success when his conservative investing style meant missing out on the dotcom bull market. Of course, he had the last laugh after the dotcom crash because, once again, Buffet’s time tested strategy proved successful.

8.6 Revision Problems

Problem 8.1 Beta can be described as a measure of:

- (a) The sensitivity of a share's return to the return on the market
- (b) The share price sensitivity to changes in interest rates
- (c) The combination of business risk, liquidity risk, and purchasing power risk inherent in a share
- (d) The return earned on a share relative to its risk

Problem 8.2 If the capital market is in equilibrium then for an asset:

- (a) it's expected return will be equal to the risk-free rate
- (b) it's expected return will equal it's required rate of return
- (c) it's expected return will be equal to the market return
- (d) none of the above

Problem 8.3 The slope of the capital market line is:

- (a) the expected return divided by beta
- (b) beta
- (c) the market risk premium divided by the market standard deviation
- (d) the market risk premium

Problem 8.4 Which of the following statements are true?

- i. Total risk is measured by the standard deviation
 - ii. Systematic risk is measured by beta (β)
 - iii. Diversifiable risk is measured by the standard deviation
 - iv. Systematic risk is also called market risk
- (a) i, ii and iii
 - (b) i, ii and iv
 - (c) ii, iii and iv
 - (d) i and ii only

Problem 8.5 Which of the following statements is most correct?

- (a) An efficient portfolio is one that provides the highest expected rate of return for a given amount of risk.
- (b) An efficient portfolio is one that has the lowest amount of risk for a given expected rate of return.
- (c) The set of efficient portfolios is the same whether or not a risk-free asset is considered.
- (d) Both a and b are correct.

Problem 8.6 Which of the following statements is most correct?

- (a) Portfolio diversification reduces the variability of the returns of the individual stocks in the portfolio.
- (b) If an investor buys enough stocks diversification will eliminate all of the non-market (or company-specific) risk. Indeed, if the portfolio contained all publicly traded stocks, it would be riskless.
- (c) The required return on a firm's ordinary shares is determined by its systematic risk plus its non-systematic risk.
- (d) A security's beta measures its non-diversifiable (systematic, or market) risk relative to that of an average stock. Note: average stock = market portfolio

Problem 8.7 Adding a security to an existing portfolio will reduce:

- (a) Systematic risk as long as the correlation coefficient between the new asset and the existing assets is less than +1
- (b) Non-diversifiable risk as long as the correlation coefficient between the new asset and the existing assets is less than +1
- (c) Market risk as long as the correlation coefficient between the new asset and the existing assets is less than +1
- (d) None of the above

Problem 8.8 Which of the following risks will form part of the systematic risk of a company?

- (a) The risk that the company's managing director could experience a serious health setback
- (b) That the research and development department's latest project will be unsuccessful
- (c) Of a strike in the company's marketing and distribution department
- (d) That oil price increases will slow demand for its products

Problem 8.9 The cost of capital for a project is estimated to be 14%, the risk free rate is 8% and the return on the market is 20%. The project's must be:

- (a) Half as volatile as the market
- (b) Equally as volatile as the market
- (c) 50% more volatile than the market
- (d) Twice as volatile as the market

Problem 8.10 If the expected return on the market is 12% while the risk-free rate is 5%, does the following description of securities constitute an equilibrium situation? And for those securities that are not in equilibrium are the securities expected to rise in price or fall in price?

Security	Actual Return	Beta
1	15.50%	1.5
2	6.40%	0.2
3	1.50%	-0.5
4	9.50%	0.8
5	14.80%	1.2

Problem 8.11 Both Portfolio X and Portfolio Y are well diversified. The risk-free rate is 8%, and the return on the market is 16%. The following information is available:

Portfolio	Analysts Forecast Return	Beta
X	16%	1.00
Y	12%	0.25

In this situation, which of the following about Portfolio X and Portfolio Y is true?

- (a) Portfolio X is Overvalued and Portfolio Y is Properly valued
- (b) Portfolio X is Properly valued Portfolio Y is Undervalued
- (c) Portfolio X is Undervalued Portfolio Y is Properly valued
- (d) Portfolio X is Properly valued Portfolio Y is Overvalued.

Problem 8.12 Consider two investors, Henry and John, each of whom have \$100,000 to invest. Both investors wish to hold a well-diversified portfolio which comprises some cash and fixed interest and some growth assets. They have both selected a growth fund (market portfolio) which has an annual expected return of 16% and a standard deviation of 12%. The return on 90-day treasury bonds is a proxy for the risk free rate of interest and is currently 6%. Their concern is in determining the proportion of their investment that they will place in the growth fund and the proportion they will place in cash and fixed interest ("risk-free" assets). Henry is relatively risk tolerant and you have been able to determine that he would be comfortable with his total portfolio having a standard deviation of 10%. John is relatively risk averse. He is only comfortable with a standard deviation in his portfolio of 7%.

Problem 8.13 You have obtained forecasts of the following data on three securities, the market portfolio, and the risk-free asset in a large and well-traded securities market.

			Correlation Matrix				
	$E(R_i)$	$\sigma(R_i)$	A	B	C	M	R_f
Security A	0.08	0.14	1	0.3	0.7	0.5	0
Security B	0.1	0.16	0.3	1	0.4	0.6	0
Security C	0.13	0.2	0.7	0.4	1	0.8	0
Market Portfolio M	0.1	0.1	0.5	0.6	0.8	1	0
Risk-Free Asset R_f	0.05	0	0	0	0	0	1

You are contemplating investing in a portfolio with the following asset weights:

Asset	Weight
Security A	40%
Security B	25%
Security C	35%

Required:

- Calculate the standard deviation of the portfolio.
- Calculate the beta coefficient of Securities A , B and C .
- Calculate the equilibrium returns for Securities A , B and C based on the Capital Asset Pricing Model.
- State whether Securities A , B and C are correctly priced, underpriced or overpriced based on the CAPM.



9. Weighted Average Cost of Capital

In this chapter you will learn about the weighted average cost of capital, or WACC. We will cover issues such why does a company calculate its WACC; what is capital and the cost of capital; simple WACC calculation; the treatment of interest expense when calculating WACC for a company; the need to calculate the market value of the firm when calculating WACC; how to work out the cost of equity and debt when calculating WACC; caveats in the use of the WACC for project evaluation; and the 7-step process for calculating WACC.

9.1 Introduction

Weighted average cost of capital, WACC, is a calculation of a firm's overall cost of capital in which each category of capital is proportionately weighted according to the percentage of the total capital. In Business Finance there are up to a maximum of six capital or finance items in a firm's balance sheet that are included in calculating a firm's WACC, which are ordinary shares and preference shares, representing equity funding, as well as mortgages, overdrafts, debentures/bonds, and term loans, representing debt financing.

Generally, a company's asset purchases are financed by either debt or equity and WACC is the average of the costs of these sources of financing, where each cost is weighted by its respective use in the given situation. A firm's WACC is the overall required return of the firm as a whole and, as such, it is often used internally by company management. For example, in the corporate finance department it is used to determine the economic feasibility of expansionary opportunities and mergers. Securities analysts often employ WACC when valuing and selecting investments and many investors employ WACC as a tool to decide whether to invest in a company or project, and WACC represents the minimum rate of return at which a company produces value for its investors.

9.2 Capital and the Cost of Capital

The definition of *capital* in a business or corporate setting depends on the context in which it is used. For example, capital could be the financial assets or the financial value of assets, such as cash, of a company, or it could be the physical assets of a company, such as factories, machinery and equipment that are used to produce goods and services. In terms of WACC, it is the former

definition that we are interested in, the financial resources, in the form of equity and debt funding, that are available for use by the company to purchase real assets and to generate wealth through investment.

Business Snapshot 9.1 — Balance Sheet - Woolworths Limited 2013-14. The following is the balance sheet of Woolworths Limited for 2014 versus 2013 which lists the Real Assets, the Current Assets and Fixed Assets of the company.

	2014	2013
	\$m	\$m
Asset		
Current assets	5,560.4	5,400.6
Non-current assets	16,702.7	15,836.3
Total assets	22,263.1	21,236.9
Liabilities		
Current liabilities	10,340.1	9,728.8
Non-current liabilities	5,165.6	5,264.8
Total liabilities	15,505.7	14,993.6
Equity		
Issued capital	4,850.1	4,522.7
Shares held in trust	(218.9)	(180.5)
Reserves		
Hedging reserve	(99.7)	(36.2)
Remuneration reserve	303.1	290.6
Equity investment reserve	(3.5)	(7.6)
Retained earning	1,926.3	1,654.3
Total equity	6,757.4	6,243.3
Total liability and equity	22,263.1	21,236.9

From this information, we can extract the information about the financial assets, or capital, of Woolworths Limited, which is the funding provided by investors to invest in the real assets of the company.

In simple terms capital is a firm's stock of funds, and is represented in a balance sheet on the right-hand side as the financial assets of the firm, the debt (liabilities) and the equity (proprietorship). Capital can be viewed as one of the inputs or factors of production of the firm's operations in the same way as salaries and wages; raw materials; rent; fuel and power; and just like these other inputs capital has a cost and must be paid for.

The cost of capital for a given firm is the cost of funds used for financing the business, i.e. it is the cost of funds used to finance the activities of the firm; the compensation for providers of capital; the opportunity cost foregone by providers of capital for investing funds in the real assets of the firm; and the rate of return the firm must generate from investment in real assets to compensate suppliers of capital.

A firm's cost of capital will depend on the mode of financing used. It refers to the cost of equity if the business is financed solely through equity, or to the cost of debt if the firm is financed solely through debt. However, in reality most companies use a combination of debt and equity to finance their activities.

Generally, the cost of capital is determined by the use to which it is put. Capital is used to finance investment in real assets, with these real assets in turn generating the cash-flows used to service the capital. The risk of the cash-flows used to service the capital will be determined by the nature of the real assets. The higher the risk of the cash-flows, the higher will be the return required by the suppliers of the capital, so it is the nature of the investments or real asset that determines the cost of capital.

Highlight 9.1 — Cost of equity and cost of debt. In corporate finance, the cost of equity (CoE) funding is the return that shareholders, both ordinary shareholders and preference shareholders, require from a company. CoE represents the compensation that the market demands in exchange for owning the asset and bearing the risk of ownership. The cost of equity funding is the dividends that are paid to shareholders of the company. The cost of debt (CoD), on the other hand, refers to the rate of return that a company pays on its current debt, measured as either the before or after-tax return. CoD is the interest expense that is paid to those who have lent money to the company.

9.3 Simple formulation of WACC

As a company draws finance (capital) from several sources, the firm's overall cost of capital or required rate of return, r , is the sum of the weighted average cost of each source of capital, i.e. it is the WACC of the company. WACC is one cost, representative of the cost of all sources of finance of the firm, where each cost is weighted by its relative importance to the total finance of the firm. The simple WACC formula (with no taxation) is

$$WACC = r = \sum r_i w_i \quad ,$$

where,

$WACC, r$ is the weighted average cost of capital of the firm;

r_i is the cost if the i th source of funds of the firm; and

w_i is the weighting of the i th source of funds in the firm's capital structure.

The cost of capital before tax for a firm, that is the WACC before-tax, is the total revenue of the firm less the firm's operating costs, which equals the *net operating income* (NOI) of the firm. From the NOI, we then subtract the interest expense of the firm, which is the cost of debt, and then subtract taxation, which is the cost of government. This leaves us with net income (NI), which is the funds left over that are available to distribute to the shareholders of the company in the form of dividends. Often in reality, part of the NI of the firm is kept for future investment as retained earnings. The dividends paid by the firm is the firm's cost of equity.

Assuming two sources of funding for the firm, debt and equity, and no taxation, the simple WACC becomes

$$WACC = r = r_e \frac{E}{V} + r_d \frac{D}{V}$$

where,

r_e is the cost of equity capital for the firm,

$\frac{E}{V}$ is the proportion of equity in the capital structure of the firm,

r_d is the cost of debt capital for the firm,

$\frac{D}{V}$ is the proportion of debt in the capital structure of the firm,

E is market value of equity of the firm,

D is market value of debt of the firm, and

V is the total market value of the firm (i.e. $D + E$).

The cost of capital after-tax assumes that taxation, or compensation to the government, has been paid by the firm, so the WACC after-tax is equal to the cost of debt plus the cost of equity, and it is the rate of return that the firm needs to earn on its assets/projects in order to compensate debt and equity holders given the government has already been compensated.

9.3.1 Interest Payments and Tax

In calculating the WACC of a firm it is important to remember that interest payments on debt are tax deductible, while dividend payments on equity are not. Therefore, when calculating WACC, the after-tax cost of debt is $(1 - t)$ of the interest payments, where t is the corporate tax rate. Therefore, because of the tax-deductibility of interest expenses, the firm receives some “help” from the government in covering the cost of debt, which reduces the overall return that needs to be earned on the firm’s assets.

When calculating WACC after-tax the, cost of servicing debt is $r_d(D)(1 - t)$, and the cost of servicing equity is $r_e(E)$. Further, the firm’s assets have to generate a return sufficient to satisfy both debt and equity holders. Letting r be the after-tax required rate of return the firm must earn on its assets (capital), the after-tax WACC becomes

$$WACC = r = r_e \frac{E}{V} + r_d(1 - t) \frac{D}{V}.$$

9.3.2 Market Value of the Firm

In order to calculate the WACC of a firm we need to find the *market value* of the firm. A firm is a collection of real assets that generate cash-flows. Therefore, when finding the market value of a firm, in order to calculate the firm’s WACC, we should value the current income claims of those having provided the finance or capital to purchase the real assets of the firm.

The real assets of a firm are the current and fixed assets that are used to produce goods and services that are sold in the market place and, in turn, generate cash-inflows for the firm. With the real assets of the firm, the firm’s value is given by the present value of the net cash-inflows generated by the real assets, with part of these cash-inflows available for distribution to the capital investors of the firm.

The financial assets of the firm are the firm’s liabilities and proprietorship. These financial or capital assets represent claims on the net cash-inflows of the firm by those who have provided funding to the firm, i.e. by the lenders and shareholders. With the financial or capital assets of the firm the firm’s value is given by the present value of the claims upon the net cash-inflow stream, where these claims are the repayment of debt, including interest expense, and dividends paid to shareholders.

■ **Example 9.1** Zoom Company Ltd currently has 78.26 million ordinary shares on issue with a par value of \$22.40 per share and a current market price of \$58 per share. The required rate of return on the company’s equity is 11.65%. The market value of Zoom Company Ltd’s debt is \$1.474 billion and the before-tax cost of debt is 7.15%, while the corporate tax rate applicable to Zoom Company Ltd is 30%. Calculate Zoom’s WACC.

The current market value of Zoom Company Ltd's equity is

$$E = 78.26 \text{ million shares} \times \$58 = \$4.539 \text{ billion.}$$

The current market value of Zoom Company Ltd's debt (D) is \$1.474 billion, and, so, the total market value of the company is

$$V = E + D = \$4.539 \text{ billion} + \$1.474 \text{ billion} = \$6.013 \text{ billion}$$

The weighting of equity (E/V) in the total capital structure of Zoom is $\frac{\$4.539 \text{ billion}}{\$6.013 \text{ billion}} = 75.50\%$, and the weighting of debt (D/V) in the total capital structure of Zoom is $\frac{\$1.474 \text{ billion}}{\$6.013 \text{ billion}} = 24.50\%$. Hence, the WACC is

$$WACC = r = (0.1165)(0.7550) + (0.0715)(1 - 0.30)(0.2450) = 0.1002 = 10.02\%.$$

■

9.3.3 Cost of Ordinary Shares

When calculating WACC the capital asset pricing model (CAPM) required rate of return formula is used to determine the cost of ordinary shares, as this represents the return required by investors to purchase the ordinary shares of the company.

■ **Example 9.2** If the risk-free rate of return in the market is 6% p.a., the return on the market portfolio is 13% p.a., and the company's beta is 1.40, the cost of ordinary equity for this company would be found as follows.

$$\begin{aligned}\bar{R}_i &= R_f + (R_M - R_f)\beta_i \\ &= 0.06 + (0.13 - 0.06) \times 1.40 \\ &= 0.06 + 0.098 \\ &= 0.158 = 15.8\%.\end{aligned}$$

■

9.3.4 Cost of Preference Shares

When calculating WACC, dividend yield is used to determine the cost of preference shares, as this represents the return required by investors to purchase the preference shares of the company.

■ **Example 9.3** If the dividend paid on the preference share is \$8 and the current market price of the preference share is \$120, the cost of preference equity for this company would be found as follows

$$\begin{aligned}\bar{R}_{\text{pref}} &= \frac{D}{P_0} \\ &= \frac{\$8.00}{\$120} = 0.0667 = 6.67\%.\end{aligned}$$

■

9.3.5 Cost of Debt

The cost of debt, r_d , in calculation of WACC is the effective interest rate that a company currently pays on its borrowings and is equivalent to the interest rate on new borrowings. The cost of debt, r_d , can be measured in either before- or after-tax returns and is observable as the yield (current interest rate in the market) on currently outstanding debt or as the yield on newly-issued similarly-rated debt.

Highlight 9.2 — Historical vs market value. When calculating the cost of debt, the current value (market value) of debt is used, rather than the historic cost (book value). This is because book values represent historical values and, consequently, fail to recognise the concept of the cost of capital as an opportunity cost. A company will use various forms of debt financing, for example, bonds, debentures, etc. Generally, riskier companies will have a higher cost of debt.

■ **Example 9.4 — Current cost of debt.** Ishta Co. sold a 20 year, 12% p.a. bond 10 years ago at par of \$100, and the bond is currently yielding 14% p.a. What is its current price?

In finding the current market value of this bond, we must use the current market cost of the bond, its yield to maturity, which in this case is the current market rate of 14% p.a. The relevant information in pricing this bond is as follows:

- Face value = \$100;
- coupon rate = 12% p.a.;
- coupon PMT = \$12 p.a.;
- number of periods to maturity, $n = 10$ (years); and
- r_d (current yield) = 14% p.a.

Then the bond price is

$$\begin{aligned}\text{Bond price} &= PMT \left[\frac{1 - (1 + r_d)^{-n}}{r_d} \right] + \text{Face Value}(1 + r_d)^{-n} \\ &= \$12 \left[\frac{1 - (1.14)^{-10}}{0.14} \right] + \$100(1.14)^{-10} \\ &= \$62.59 + \$26.97 = \$89.56\end{aligned}$$

We can see that based on the current cost of this debt in the market, the current yield, of 14% p.a., the market value of the bonds has fallen from the face value of \$100 to the current price of \$89.56. ■

As mentioned above, the cost of debt used in calculating the WACC is the effective interest rate. As a reminder, the formula for calculating an effective interest rate is as follows.

$$EFF = \left[1 + \frac{NIR}{m} \right]^m - 1$$

9.4 Caveats in Use of the WACC

In order to be able to use the WACC of a company for project evaluation, there are two important criteria that must be met, which are listed below.

1. The risk of the project the firm is considering must be similar in nature and risk to the projects that the firm normally takes on.
2. The manner in which the project is financed, in terms of the debt-equity mix, must be similar to the way in which the firm is financed overall; this will be the case if the firm currently has the optimal capital structure.

Business Snapshot 9.2 — BHP-Billiton Challenges Project WACC^a. BHP-Billiton (BHP-B) made a submission to AER and discussed in its submission, amongst other things, the rate of return that was being used by APA GasNet to evaluate the project. In terms of the rate of return BHP-B stated the following.

BHP-B submits that the 9.06% (nominal post-tax) weighted average cost of capital (WACC) proposed by APA GasNet in the Proposed Access Arrangement is not reasonable, has not been justified by APA GasNet, is substantially overstated, and should be between 5 and 6%.

^aThis is a public submission by BHP-Billiton in response to the proposed revisions to the Victorian electricity transmission system access arrangement.

9.5 Detailed WACC Problem

In a detailed WACC question you are required to calculate the market value, weighting before-tax and after-tax, as well as weighted costs of the relevant capital items, rather than these values being given to you, as is usually the case with a simple WACC question. In calculating the WACC, we follow a 7-step process, as below:

- Step 1:** Identify the relevant components of the **Capital Base**;
- Step 2:** Calculate the **Market Value** of each component and of the firm;
- Step 3:** Determine the **Weighting** of each component;
- Step 4:** Determine the **Before-Tax Cost** of each component;
- Step 5:** Determine the **After-Tax Cost** of each component;
- Step 6:** Calculate the **Weighted Cost** of each component; and then
- Step 7:** Sum the weighted costs to obtain the WACC.

When calculating a detailed WACC you will only have to include the following six items, namely, 1. ordinary shares, 2. preference shares, 3. mortgages, 4. overdrafts, 5. debentures/bonds, and 6. term loans. The first two are classified under the equity side of the balance sheet, and the rest under debt.

■ **Example 9.5 — Calculating WACC - A detailed question.** The following is an extract from the balance sheet of Stratco Ltd. as at 1st of January 2016:

Equity	\$'000	Debt	\$'000
Issued and Paid-Up Capital		Mortgage	5,300
75m 50c ordinary shares	37,500	Debentures (\$1,000 each)	6,250
9.5m \$1 3.5% preference shares	9,500	Bank overdraft	14,804
Reserves and Retained Profits			
Retained profits	11,000		
Share premium reserve	13,800		
General reserve	4,000		
	71,050		26,354

Additional Information as at 1st of March 2016:

- The current market price of the firm's ordinary shares is \$1.00.
- The firm's preference shares are currently selling for 35 cents each.
- The debentures will mature in two years and they have a coupon rate of 10% per annum.
- If the company currently sought long-term finance it would have to pay an interest rate of 12% per annum on debentures. An amount of \$5,300,000 is owing on the mortgage and has 6 years remaining. It was taken out at 12% and current rates are 14%
- The firm is currently paying 11% per annum on its bank overdraft loan.

6. Interest on debentures, mortgage and the bank overdraft is paid half-yearly.
7. The corporate tax rate is 50 cents in the dollar.
8. The estimated Beta of the company's ordinary shares is 1.
9. The current yield on a 10-year government bond is 12% per annum
10. The expected return on the market portfolio is 17% per annum.

Before starting the 7-step process, notice the following

Note 1: In this balance sheet, we have five of the six relevant items, namely, ordinary shares, preference shares, mortgage, debentures, and overdraft. Notice also that there are other items listed in the balance sheet apart from the relevant items, which should be ignored. This includes retained profits, share premium reserve, and general reserve.

Note 2: The values listed in the balance sheet are par values, or the original issue values of the listed items. For example, 50c for the ordinary shares tells us that when the ordinary shares were first listed on the stock exchange they were listed at 50 cents each. Also, the mortgage value of \$5.3 million tells us that when the mortgage was taken out \$5.3 million was borrowed.

Note 3: For a detailed WACC question you will be provided with additional information which will give you the information you need to calculate current markets values, as well as before and after-tax costs of the relevant items.

Step 1: Identify the relevant components of the capital base. The five components of the capital base that are relevant in this question are ordinary shares, preference shares, mortgage, debentures, and overdraft.

Step 2: Calculate the market value of each component and of the firm

1- Ordinary shares: In the balance sheet extract we are told that there are 75 million ordinary shares. From point 1. of the Additional Information, the current market price of the firm's ordinary shares is \$1.00. Then,

Ordinary shares: $75 \text{ million shares} \times \$1.00 = \$75,000,000.00$

2- Preference shares: From the balance sheet, there are 9.5 million preference shares, and from point 2. of the Additional Information the firm's preference shares are currently selling for 35 cents each.

Preference shares: $9.5 \text{ million shares} \times \$0.35 = \$3,325,000.00.$

3- Mortgage: In the balance sheet extract, the par value of the mortgage is listed as \$5.3 million. From the second part of point 4. of the Additional Information, there are six years remaining on the mortgage which was taken out at an interest rate of 12%, while the current interest rate is 14%. Since the yield (current interest rate) on the mortgage has changed, the market value of the mortgage has also changed, so it must be revalued. Also, from point 6. of the Additional Information interest on the mortgage is paid half-yearly.

Revaluation of the mortgage is a two-step process. First, work out what the regular periodic repayment, PMT is. We know that \$5,300,000 was originally owing and that there are six years remaining to pay off the loan, with a 12% p.a. historical interest rate and a new interest rate of 14% p.a., and with interest compounded twice a year. To find the unknown PMT on the mortgage, we use the PV of an ordinary annuity formula, the original amount owing as the present value, and the original interest rate as follows.

$$\begin{aligned}
 PV_0 &= PMT \left[\frac{1 - (1 + r_d)^{-n}}{r_d} \right] \\
 \$5.3\text{m} &= PMT \left[\frac{1 - (1.06)^{-12}}{0.06} \right] \\
 \$5.3\text{m} &= PMT [8.3838] \\
 PMT &= \$632,171.57
 \end{aligned}$$

Second, work out the current market value of the mortgage by using the PMT repayment and the current interest rate of 14% p.a. The new present value of the mortgage is as follows.

$$\begin{aligned}
 PV_0 &= PMT \left[\frac{1 - (1 + r_d)^{-n}}{r_d} \right] \\
 &= \$632,171.57 \left[\frac{1 - (1.07)^{-12}}{0.07} \right] \\
 &= \$632,171.57 [7.9427] = \$5,021,149.13.
 \end{aligned}$$

4- Debenture: In the balance sheet extract, the par value of the debentures in total is listed as \$6.250 million. From point 3. of the Additional Information, we are told that debentures will mature in two years and they have a coupon rate of 10% p.a. From point 4. of the Additional Information, we know that if the company currently sought long-term finance it would have to pay interest rates of 12% p.a. on the debentures. Also, from point 6. of the Additional Information, we are told that interest on the debentures is paid half-yearly.

The current yield on the debentures of 12% p.a. is different to the original yield of 10% p.a., therefore, it is necessary to revalue the debentures.

$$\begin{aligned}
 \text{Bond price} &= PMT \left[\frac{1 - (1 + r_d)^{-n}}{r_d} \right] + \text{Face Value}(1 + r_d)^{-n} \\
 &= \$50 \left[\frac{1 - (1.06)^{-4}}{0.06} \right] + \$1,000(1.06)^{-4} \\
 &= \$50 [3.4651] + \$1,000(0.7921) \\
 &= \$965.36
 \end{aligned}$$

Hence, the total market value of debentures is $\$6,033,500.00 = 6,250 \times \965.36 .

The face-value of each of the debentures is \$1,000 and the coupon rate is 10.0% p.a. Since coupons are paid half-yearly, the interest payments are $(10\% \times \$1,000) \div 2 = \50 . Also, as the current yield on the debentures is 12% p.a. with half-yearly coupon payments, then the discount rate is $12\% \div 2 = 6.0\%$. Further, $n = 4$ as there are two years remaining on the debentures with two coupon payments per year.

From the balance sheet extract, the total par-value of the debentures is \$6,250,000.00 and the face-value of each debenture is \$1,000, meaning that the total number of debentures issued by Stratco Ltd. is $6,250 = \$6,250,000.00 \div \$1,000$.

5- Bank Overdraft: From the balance sheet extract the par-value of the bank overdraft is \$14.804 million, and apart from point 6. of the Additional Information which states that interest on the bank overdraft is paid half-yearly, we have no other information relevant to the bank overdraft. As such, we must assume that the yield on the overdraft has not changed, meaning that its market value has not changed from its par-value, and that the current market value of the overdraft is still \$14.804 million.

6- Total Market Value: Given that we have now worked out the current market value of each of the relevant capital items, we can now calculate the total market value of Stratco Ltd. as follows:

Ordinary Shares		\$75,000,000.00
Preference shares	+	\$3,325,000.00
Mortgage	+	\$5,021,149.00
Debentures	+	\$6,033,500.00
Bank overdraft	+	\$14,804,000.00
Total		\$104,183,649.00

Step 3: Determine the weighting of each component.

In order to work out the weighting of each of the relevant components in the capital base of Stratco Ltd., we must divide into the current market value of each of the relevant items the total market value of the company, as follows:

Ordinary Shares	\$75,000,000.00 ÷ \$104,183,649.00	=	0.7200 (72.00%)
Preference shares	\$3,325,000.00 ÷ \$104,183,649.00	=	0.032 (3.20%)
Mortgage	\$5,021,149.00 ÷ \$104,183,649.00	=	0.048 (4.80%)
Debentures	\$6,033,500.00 ÷ \$104,183,649.00	=	0.058 (5.80%)
Bank overdraft	\$14,804,000.00 ÷ \$104,183,649.00	=	0.142 (14.20%)
Total		=	1.00 (100%)

Step 4: Determine the before-tax cost of each component.

1- Ordinary shares: To determine the cost of the ordinary shares we must use the CAPM required rate of return formula, taking into account the risk-free rate of interest, the return on the market, and the Beta of Stratco's ordinary shares, as follows.

$$\begin{aligned}\bar{R}_i &= R_f + (R_m - R_f)\beta_i \\ &= 0.12 + (0.17 - 0.12) \times 1 \\ &= 0.17 = 17\%\end{aligned}$$

From point 8. the estimated Beta (β) of the company is 1, and from point 9. the current yield on a 10-year government bond, which is the risk-free rate of return, is 12% p.a. Additionally, from point 10. the expected return on the market portfolio, R_m , is 17% p.a.

2- Preference shares: To determine the cost of the preference shares we must find the dividend yield on the preference shares, as follows:

$$DY_i = \frac{D}{P_0} = \frac{\$0.035}{\$0.35} = 0.10 = 10\%$$

From the balance sheet the par-value of the preference shares is \$1 and they have a dividend rate of 3.5%. Hence, the fixed dividend paid on each preference share is 3.5% of \$1, which is 3.5 cents. From point 2. the firm's preference shares are currently selling for 35 cents each, meaning that the current dividend yield on the preference shares is 10%.

3- Mortgage, debentures, and overdraft: From point 6. the interest on the debt items is paid half-yearly, so it is necessary for us to work out the effective cost of the debt items before-tax. To do this we use the effective rate of interest formula as follows.

$$EFF = \left[1 + \frac{NIR}{m} \right]^m - 1$$

Then, for the Mortgage the effective rate is:

$$EFF = \left[1 + \frac{0.14}{2} \right]^2 - 1 = 14.49\%$$

For the Debentures the effective rate is:

$$EFF = \left[1 + \frac{0.12}{2} \right]^2 - 1 = 12.36\%$$

For the Mortgage the effective rate is:

$$EFF = \left[1 + \frac{0.11}{2} \right]^2 - 1 = 11.30\%$$

Notice that because with these debt items we are given annual rates of interest but are also told that interest is calculated more than once per year, the effective rate is greater than the nominal, or annual rate.

Step 5: Determine the after-tax cost of each component.

In calculating the after-tax cost of the relevant components, we must remember, again, that it is only the cost of debt that is tax-deductible. The cost of equity is not tax-deductible. Hence, we calculate the after-tax cost of the debt items as follows, remembering that from point 7, the corporate tax rate is 50 cents in the dollar:

The cost of the Debentures after-tax is

$$0.1236(1 - 0.50) = 0.1236(0.50) = 0.0618 = 6.18\% p.a.$$

The cost of the Bank Overdraft after-tax is

$$0.1130(1 - 0.50) = 0.1130(0.50) = 0.0565 = 5.65\% p.a.$$

The cost of the Mortgage after-tax

$$0.1449(1 - 0.50) = 0.1449(0.50) = 0.0725 = 7.25\% p.a.$$

Step 6: Determine the weighted cost of each component.

In order to calculate the weighted (after-tax) cost of each of the relevant components of the capital base, we must multiply the weighting of each of the components in the total capital base of the firm (Step 3) by the after-tax cost of each component (Step 5). Thus,

Ordinary shares	0.7200×0.1700	$= 0.1224$
Preference shares	0.0320×0.1000	$= 0.0032$
Mortgage	0.0480×0.0725	$= 0.0035$
Debentures	0.0580×0.0618	$= 0.0036$
Overdraft	0.1420×0.0565	$= 0.0080$

Step 7: Sum the weighted costs to obtain the WACC.

$$WACC = r = 0.1224 + 0.0032 + 0.0035 + 0.0036 + 0.0080 = 0.1407 = 14.07\%.$$

■

Highlight 9.3 — Systematise your approach. The best practice in answering detailed WACC questions is systematising your solution by setting out the following table, at the beginning of your solution. Then, fill in the numbers as you go through each step.

Item	Market Value	Weight	Cost Before Tax	Cost After Tax	Weighted Cost
Ordinary Shares					
Preference Shares					
Mortgage					
Debenture					
Term Loans					
Overdraft					
Total					

■ **Example 9.6** For the Stratco Ltd. question the table would look as follows:

Item	Market Value	Weight	Cost Before Tax	Cost After Tax	Weighted Cost
Ordinary Shares	75,000,000	0.72	0.17	0.17	+ 0.1224
Preference Shares	3,325,000	0.032	0.10	0.10	+ 0.0032
Mortgage	5,021,149	0.048	0.145	0.0725	+ 0.0035
Debenture	6,033,500	0.058	0.1236	0.0618	+ 0.0036
Overdraft	14,804,000	0.142	0.1130	0.0565	+ 0.0080
Total	104,183,694	1.00		WACC=	= 0.1407

9.6 Revision Problems

Problem 9.1 The weights used to compute the weighted-average required rate of return (WACC) should be obtained using:

- (a) the book value of a similar risk company
- (b) the market value of debt and equity
- (c) the book values of debt and equity
- (d) the CAPM

Problem 9.2 If the risk of a firm is identical to the risk of the project, the required rate of return for the firm may be applied to the project. Where the risk of the firm and project differ a solution is to use:

- (a) the return for the market
- (b) the WACC of the firm
- (c) the return for the industry identical to that of the project
- (d) none of the above

Problem 9.3 Given the following information on S & G Inc.'s capital structure, compute the company's weighted average cost of capital. The company's tax rate is 40%.

Type of Capital	Percent of Capital Structure	Before-Tax Component Cost
Bonds	40%	7.50%
Preference shares	5%	11%
Ordinary shares	55%	15%

- (a) 13.3%
- (b) 7.1%
- (c) 10.6%
- (d) 10%

Problem 9.4 If:

$$WACC = 11.4\%$$

$$r_d = 10\%$$

$$V = 100$$

$$D = 40$$

$$t = 40\%$$

$$E = 60 \text{ (i.e. } V - D = 100 - 40 \text{)}$$

What is the required rate of return on equity (r_e)?

- (a) 15%
- (b) 30%
- (c) 40%
- (d) 50%

Problem 9.5 Calculate the weighted average cost of capital (WACC) for the Peach Computer Company. Notes

- The book value of Peach's outstanding debt is \$10 million.
- Currently, the debt is trading at 90 percent of book value and is priced to yield 12 percent (r_d).
- The 1 million outstanding shares of Peach stock are selling for \$20 per share.
- The required return on Peach stock is 20 percent (r_e).
- The tax rate is 30 percent.
- Market value of outstanding debt = \$9million, i.e. 90% of \$10 million
- Market value of equity = 1 million shares x \$20 = \$20million

8. Therefore, V (total value) = \$9m + \$20m = \$29million

Problem 9.6 Baker's Deluxe, a national baking franchise, has decided to professionalise its project evaluation. Previously, franchise locations had been selected on an ad hoc basis, with mixed results. The managing director has decided that to conduct a project evaluation he requires you to determine the WACC of the firm. You are told that the firm has used the following securities to finance its operations.

Note 1: Four hundred thousand (400,000) ordinary shares, with a current market price of \$5 per share. The beta of Baker's Deluxe is 1.5, the risk-free rate is currently 4% p.a., and the expected return on the market is 10% p.a.

Note 2: An overdraft of \$300,000. The current interest rate on the overdraft is 9.5% p.a., with interest calculated on a monthly basis. Debentures with four years to maturity.

Note 3: The debentures offer an 8% p.a. coupon paid half-yearly. (The coupon is calculated on the face value.) The face value of the debentures is \$300,000, and the current market interest rate for these securities is 10% p.a., calculated half-yearly.

Required:

If Baker's Deluxe pays tax at 30 % p.a., what is the WACC of Baker's Deluxe?

Problem 9.7 Calculate the WACC of AOI Ltd, using the following information:

Balance Sheet Extract	
Liabilities	
10% debentures (\$100 par)	\$10,000,000
Shareholders' funds	
Paid-up capital - ordinary shares (\$1 par)	\$25,000,000
12% preference shares (\$10 par)	\$5,000,000

Note 1: The ordinary shares are currently trading at \$4.20 per share.

Note 2: Commonwealth government bonds trade at 5%.

Note 3: The return on the market portfolio is 13.5%.

Note 4: AOI's beta is 1.25.

Note 5: Its debentures are priced at \$96.50 and its preference shares are trading at par.

Note 6: The current return AOI debentures is 2% above the government bond rate.

Note 7: The existing capital structure is unlikely to change.

Note 8: Company tax rate is 30%.

10. Capital Structure

In this chapter you will learn what is capital structure, is there an optimal capital structure for firms, and the effects of financial leverage on capital structure; the effects of capital structure on shareholders, as well as WACC and the value of the firm; why debt is cheaper than equity for a firm; the traditional approach to capital structure; Modigliani and Miller's theories related to capital structure; all debt capital structures; financial distress costs and conflict of interest costs; and the capital structures of Australian firms.

10.1 Capital Structure

Capital structure refers to the mix of a company's debt and equity and it reflects how a firm finances its operations and growth by using different sources of funds. Generally, a firm's capital structure is reflected in its debt-to-equity ratio and is the combination of debt and equity used by the firm to finance investment in real assets. The capital structure of a firm is found on the right-hand side of the balance sheet as liabilities (debt) and proprietorship (equity) and is established by the financing decisions of the firm.

10.2 Optimal Capital Structure

The *optimal capital structure* for a firm is the combination of debt and equity that minimises WACC such that the value of the firm is maximised. Determining optimal capital structure is one of the main responsibilities of any firm's corporate finance department.

Highlight 10.1 — Financial leverage. Financial leverage refers to the use of debt financing by a firm, in part or whole, to fund the purchase of its real productive assets. The greater the percentage of debt financing in the firm's capital structure, the higher the firm's debt-to-equity ratio. A firm which uses debt financing is commonly referred to as a levered firm.

In this chapter, we will examine the effect of introducing financial leverage into the firm's capital structure on the shareholders of a company. In particular, we explore the effect of introducing debt into a firm's capital structure on the return received by the shareholders of the company. We must remember that a firm earns revenue by selling goods and services produced

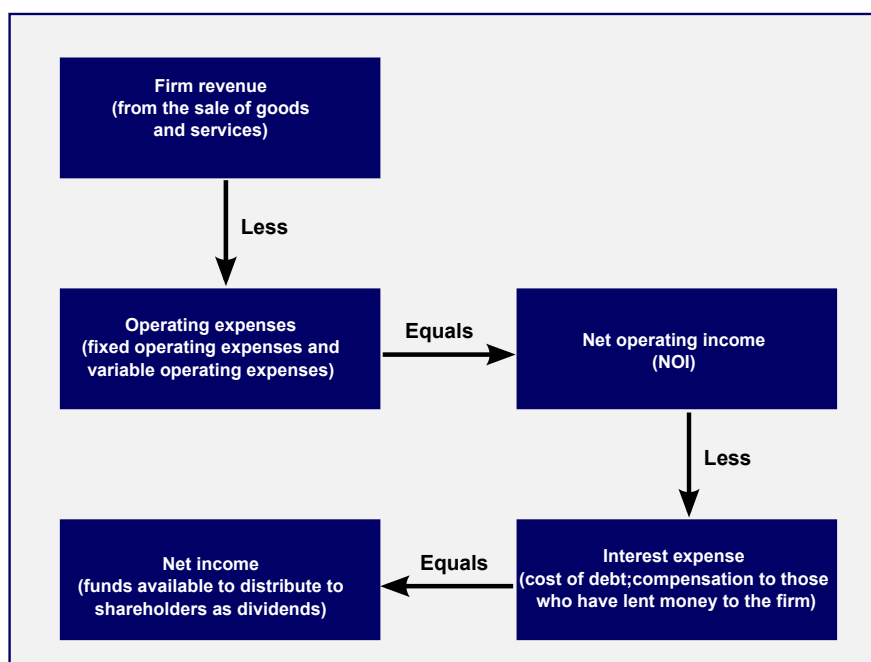


Figure 10.1: Diagram of a firm's cash-flows, in the absence of taxation

with its real assets. Then from these cash-inflows, operating costs, both fixed and variable, are subtracted to get the net operating income (NOI) of the firm. From the firm's NOI the cost of debt is then subtracted, with this cost being interest expense paid as compensation to those who have lent money to the firm. Assuming no taxation, what is left over is net income (NI), which is the funds available to distribute to the shareholders of the firm as dividends, representing shareholders' dollar return on equity investment in the company. This process is reflected in Figure 10.1.

10.2.1 Debt and capital structure

What effect does introducing debt into a firm's capital structure have on the return received by the shareholders of the company? What happens to shareholders' return when leverage (debt) is introduced?

The best way to answer these questions is to use a numerical example which shows possible outcomes when leverage (debt) is introduced into a firm's capital structure. In this example, we are initially assuming that the return on total assets (ROA) and NOI of the three firms is relatively high. Thus, each of the three firms is, initially, performing relatively well.

Assume firms A, B, and C are identical in all respects, apart from the way in which each firm is financed. Each firm has total real assets of \$400,000, but the mix of debt and equity used to finance these real assets differs. We assume that each firm earns a good return on total assets (ROA) of 15% p.a. Therefore, neglecting the effects of taxation, NOI for each of the three firms is relatively high at $0.15 \times 400,000 = \$60,000$. The firms that use debt pay 10% p.a. interest.

In Table 10.1, we can see the effect of financial leverage on the companies' returns. The ROA and NOI are relatively high, so each firm is doing relatively well. Firm A has no financial leverage, and has used \$400,000 worth of equity funding to purchase its \$400,000 worth of real assets, giving Firm A a debt-to-equity ratio of 0. Firm B has some leverage as it has taken on \$200,000 in debt to partly finance the purchase of its \$400,000 worth of real assets, giving Firm B

	Firm A	Firm B	Firm C
	100% equity	50% debt & 50% equity	75% debt & 25% equity
	(unlevered firm)	(levered firm)	(levered firm)
	\$	\$	\$
Equity	400,000	200,000	100,000
Debt	0	200,000	300,000
Total Assets	400,000	400,000	400,000
Debt-to-equity ratio	0	1:1	3:1
Net Operating Income	60,000	60,000	60,000
Interest Expense (10% p.a.)	0	20,000	30,000
Net Income (NI)	60,000	40,000	30,000
Return On Total Assets	0.15	0.15	0.15
=NOI÷Total Assets	= 60,000÷400,000	= 60,000÷400,000	= 60,000÷400,000
Available Return on Equity	0.15	0.20	0.30
= NI÷Equity	= 60,000÷400,000	= 40,000÷200,000	=30,000÷100,000

Table 10.1: Effects of Leverage on Shareholder Return: In this example ROA and NOI are relatively high at 15% and \$60,000, respectively.

a debt-to-equity ratio of 1:1 (\$200,000 : \$200,000). Firm C is the most highly levered firm as it has taken on \$300,000 worth of borrowings to partly finance the purchase of its \$400,000 worth of real assets, giving the highest debt-to-equity ratio of 3:1 (\$300,000 : \$100,000).

Note that the two levered firms B and C, are being charged interest at a rate of 10% p.a. on their borrowings, while Firm A, the unlevered firm, has no interest expense. Firm B has an interest expense on its \$200,000 worth of debt of \$20,000 and Firm C has an interest expense of \$30,000 on its total debt of \$300,000.

Given that we are assuming that the NOI of each firm is \$60,000, and that interest expense must be subtracted from NOI in order to arrive at NI, the Net Income of the three firms is calculated as follows.

$$\text{NI(Firm A)} = \$60,000 - \$0 = \$60,000,$$

$$\text{NI(Firm B)} = \$60,000 - \$20,000 = \$40,000,$$

$$\text{NI(Firm C)} = \$60,000 - \$30,000 = \$30,000.$$

Yet, refer to the bottom row of Table 10.1, which lists the available return on equity for each of the three firms. We see that the shareholders of Firm C, which has the lowest NI, do best, with an available return on equity of 30%. Then the shareholders of Firm B come in second-best at 20%, while the shareholders of Firm A, which has the highest NI, do worst with an available return on equity of only 15%. Why do the shareholders of Firm C, with the lowest NI, do best, while the shareholders of Firm A, with the highest NI, do worst?

The reasons for this are two-fold. Firstly, for each firm we are assuming that the business is going well, with a relatively high ROA of 15% and a relatively high NOI of \$60,000. Secondly, although the NI available to distribute to the shareholders of Firm C is lowest at \$30,000, the shareholders of that company have only had to invest \$100,000 to get the \$30,000 return, giving

	Firm A 100% equity (unlevered firm)	Firm B 50% debt & 50% equity (levered firm)	Firm C 75% debt & 25% equity (levered firm)
	\$	\$	\$
Equity	400,000	200,000	100,000
Debt	0	200,000	300,000
Total Assets	400,000	400,000	400,000
Debt-to-equity ratio	0	1:1	3:1
Net Operating Income	36,000	36,000	36,000
Interest Expense (10% p.a.)	0	20,000	30,000
Net Income (NI)	36,000	16,000	6,000
Return On Total Assets	0.09	0.09	0.09
=NOI÷Total Assets	= 36,000÷400,000	= 36,000÷400,000	= 36,000÷400,000
Available Return on Equity	0.09	0.08	0.06
= NI÷Equity	= 36,000÷400,000	= 16,000÷200,000	=6,000÷100,000
Change in Available	-40%	-60%	-80%
Return on Equity	= (9% - 15%)÷15%	= (8% - 20%)÷20%	= (6% - 30%)÷30%

Table 10.2: Effects of Leverage on Shareholder Return: In this example ROA and NOI are relatively low at 9% and \$36,000, respectively.

them a percentage return on their \$100,000 investment of 30%. In contrast, the shareholders of Firm A, which has no leverage, have had to invest \$400,000 to receive a return of \$60,000, giving them a percentage return on their \$400,000 of only 15%. Thus, the results highlight an important point.

Highlight 10.2 — High NOI and leverage. If a highly levered levered is generating a relatively high ROA and a relatively high NOI, and, consequently, a relatively high NI, shareholders in that company will do exceptionally well. This is because in order to receive the relatively high NI they have to invest a relatively smaller amount of money. However, in a firm with relatively low leverage, even at the same high level of NOI, shareholders must invest more money in the equity of the firm, which will lead to a lower percentage return. Therefore, when return on assets and NOI are relatively high, introducing leverage (debt) into the capital structure will increase the available return to shareholders.

However, it will not always be the case that shareholders of more highly-levered firms will do better. In fact, if a firm is not performing particularly well and its ROA and NOI are relatively low, shareholders of the company tend to do much worse. To explore this, we will continue with our previous example. Now assume that each of the companies are performing relatively poorly, and consider the following two scenarios. First, suppose that ROA has fallen to 9% p.a., giving a NOI of \$36,000 for each of the three firms. The results of this can be seen in Table 10.2. Second, assume that each company is performing even worse, with ROA falling to 0% and, hence, NOI being \$0. The results of this can be seen in Table 10.3. We will see in these two scenarios that when business is not going well for a firm, the higher is debt-to-equity ratio, the greater the drop in the available rate of return to its shareholders.

In Table 10.2 we see that when a firm is not performing well and its ROA and NOI are

	Firm A 100% equity (unlevered firm)	Firm B 50% debt & 50% equity (levered firm)	Firm C 75% debt & 25% equity (levered firm)
	\$	\$	\$
Equity	400,000	200,000	100,000
Debt	0	200,000	300,000
Total Assets	400,000	400,000	400,000
Debt-to-equity ratio	0	1:1	3:1
Net Operating Income	0	0	0
Interest Expense (10% p.a.)	0	20,000	30,000
Net Income (NI)	0	-20,000	-30,000
Return On Total Assets	0	0	0
=NOI÷Total Assets	= 0÷400,000	= 0÷400,000	= 0÷400,000
Available Return on Equity	0	-0.10	-0.30
= NI÷Equity	= 0÷400,000	= -20,000÷200,000	= -30,000÷100,000
Change in Available	-100%	-225%	-600%
Return on Equity	= (0-9%)÷9%	= (-10% - 8%)÷8%	= (-30% - 6%)÷6%

Table 10.3: Effects of Leverage on Shareholder Return: In this example ROA and NOI are lowest at 0% and \$0, respectively.

relatively low then the shareholders of more highly-levered firms do worse than the shareholders of firms with lower leverage or unlevered firms. The Net Income of the three firms is calculated as follows.

$$\text{NI(Firm A)} = \$36,000 - \$0 = \$36,000,$$

$$\text{NI(Firm B)} = \$36,000 - \$20,000 = \$16,000,$$

$$\text{NI(Firm C)} = \$36,000 - \$30,000 = \$6,000.$$

Subsequently, the available ROE can be calculated, as shown in the table.

For the unlevered Firm A, the amount available to distribute to shareholders as dividends, and the return available to shareholders is 9%. This represents a fall of 40% in the rate of return available to shareholders from the previous example in Table 10.1 (from 15% to 9%). However, for the shareholders of the levered firms B and C, the situation is worse. This is mainly because the levered firms must still make the same interest payments on their borrowings, despite the fact that for both those firms NOI has fallen to \$36,000. For Firm B, NI falls to \$16,000, and given that the shareholders of Firm B have invested \$200,000 in the company, the return available to the shareholders of Firm B is only 8%, having fallen from 20% in Table 10.1, a fall of 60%. For the shareholders of Firm C, the most highly-levered firm, the situation is even worse as the NI of this company has fallen to only \$6,000, meaning that the return available to the shareholders of Firm C falls to 6% from the 30%, a fall of 80%.

In the second scenario, each company is doing even worse than the first scenario. Looking at Table 10.3 we see that ROA and NOI are very low at 0% and \$0, respectively. For the unlevered firm, NI and the return available to shareholders are also 0%, which represents a fall of 100% in available ROE compared to Table 10.2. For the shareholders of the levered firms, Firms B and C, the situation is worse, mainly because the levered firms must still pay interest expense on their borrowings, despite the fact that for both those firms NOI has fallen to \$0. For Firm B NI is -\$20,000, and given that the shareholders of Firm B have invested \$200,000 in the company, the

return available to the shareholders is now negative, at -10%, having fallen from 8% in Table 10.2, a fall of 225%. For the shareholders of Firm C, the most highly-levered firm, NI has fallen to -\$30,000, meaning that the return available to the shareholders has also become negative at -30%, from 6% in the example in Table 10.2, a fall of 600%.

Highlight 10.3 — Low NOI and leverage. If a highly levered company is generating a relatively low ROA and NI, the return available to shareholders would be very low. When a group of firms are not performing well, the more highly levered the company, the lower the available return to shareholders.

Based on the above numerical examples, the effects that the substitution of debt for equity in capital structure has on a firm has been summarised here. Firstly, increasing the leverage of a firm increases the available return to equity-holders on their investment. For example, this can be seen by comparing the available returns to equity-holders for Firm A and Firm C. The range of returns available to the shareholders of Firm A, the unlevered firm, is from 15% to 0%. The range of returns available to the shareholders of Firm C, the most highly-levered firm, is from 30% to -30%. Therefore, when things are going well and ROA and NOI are high, shareholders in levered firms do better, with higher return available on their equity investment. When things are not going well and ROA and NOI are low, shareholders in levered firms do worse, with lower returns on their equity investment.

Highlight 10.4 — Leverage and risk. Another effect of introducing leverage into the capital structure of a firm is that it increases the risk of equity investment, holding all other factors constant. In the previous examples, we saw the range of possible returns available to the shareholders of Firm C is much greater than Firm A. The greater the range of possible returns on an investment, the greater the variability of those returns, because it is more likely that the return received will be different to what is expected. Therefore, the more debt in a firm's capital structure the riskier that firm becomes, *ceteris paribus*.

10.3 Capital Structure, WACC, and the Firm's Value

Having examined the impact on shareholders of introducing debt into the capital structure of a firm, we now turn to examine how it could impact on the value of the firm.

As the net operating income (NOI) of a firm increases, so does the value of the firm. NOI reflects the investment (capital budgeting) decisions of the firm. Recall from Chapter 6, if a firm invests wisely in real assets that generate good net cash-inflows, then the NPV of these assets will be higher, and so will be the market value of the firm. Further, recall from Chapter 9 that as WACC, which reflects the firm's financing decisions, decreases the market value of the firm increases. That is because the discount rate used in NPV analysis will be lower and the NPV of the firm's projects or investments will be higher, hence, the market value of the firm will be higher.

In regards to capital structure, which influences WACC, we will explore if it is possible to alter the capital structure of a firm so as to minimise the firm's WACC, and increase the market value of the firm. In other words, can changes in the firm's financial assets (capital structure) be used to increase the firm's market value, rather than it being only the quality of investment in real assets (capital budgeting) that solely influences the market value of the firm?

Another important point to remember, which has been discussed previously, is that, generally for any given firm debt finance will be cheaper than equity finance. Debt-holders have priority over equity-holders in their claims on the firm's cash-flow stream. Debt-holders also have a

contractual right to receive payment on their investment in the firm, whereas equity-holders have a *residual claim*¹ on the cash-flows of the firm. Moreover, equity-holders do not have a contractual right to receive payments on their investment in a firm, meaning they cannot seek legal redress if they are not paid a dividend or the value of their shares falls. Consequently, debt-holders or lenders face lower risk on their investment in a firm than equity investors, so the return required by debt-holders (lenders) is usually less than the return required by shareholders.

Thus, the above knowledge leads us to this important question - Does substitution of cheaper debt finance for equity in the capital structure of a firm reduce the overall weighted average cost of capital (WACC) of the firm? If so, it would increase the NPV of the firm's projects, investments, products and assets, thereby, increasing the market value of the firm. In answering this question we will begin by looking at the traditional approach to capital structure, which assumes that changes in capital structure can be used to affect the market value of a firm.

10.4 Traditional Approach to Capital Structure

The traditional approach to capital structure was based on the notion that there is an optimal capital structure for a firm and that management of the firm can increase its market value, through the financing method. This traditional approach implies that the cost of capital is dependent on the capital structure of the firm. However, it is a practical approach without a theoretical basis and the determinants of, or path to, the optimal debt-to-equity ratio or capital structure were never specified. This concept comes from the fact that for a firm, debt is generally cheaper than equity, therefore, in order to reduce its costs and cash-outflows, a firm should use more of the cheaper resources of debt, and less of the more expensive resources of equity funding.

The traditional approach to capital structure states that, as the amount of debt used by a firm relative to equity funding increases, WACC is reduced because debt is cheaper than equity. However, the model also states that the cost of debt is not constant, but at some point will start to increase. Furthermore, the model assumes that, as the level of debt of a firm rises, the required rate of return of the firm's shareholders will increase, as the shareholders now facing greater risk on their equity investment and this offsets the initial reduction in WACC. The traditional approach says that there is an optimal capital structure and it is where WACC is at a minimum and, consequently, the value of the firm is at a maximum.

■ **Example 10.1** Suppose we have a firm, Lev Co., that initially has a debt-to-equity ratio of 1:9, i.e. 10% of its funding comes from debt financing and 90% of its funding comes from equity financing. We will also assume that the market value of Lev Co. is \$100 million, meaning that its initial capital structure is comprised of \$10 million in debt funding and \$90 million in equity funding. Remember that the formula for WACC (assuming perfect markets and no taxation) is as follows.

$$WACC = R_d\left(\frac{D}{V}\right) + R_e\left(\frac{E}{V}\right)$$

We will alter the capital structure of Lev Co. a number of times, each time increasing the percentage of debt in the firm's capital structure, in order to see how we can attain the optimal capital structure where WACC is at a minimum and the market value of the firm would be maximised. The table below contains WACC and debt-to-equity (D/E) ratios for Lev Co. with different combinations of debt and equity.

¹A residual claim means shareholders only receive payment in the form of dividends if there is any cash left after debt-holders' interest and taxes have been paid.

WACC	=	$R_d(D/V) + R_e(E/V)$	D/E Ratio
0.086	=	$0.05(10/100) + 0.09(90/100)$	1:9
0.084	=	$0.05(25/100) + 0.095(75/100)$	1:3
0.075	=	$0.05(50/100) + 0.10(50/100)$	1:1
0.077	=	$0.07(90/100) + 0.14(10/100)$	9:1

Initially, the capital structure of Lev Co. is comprised of \$10 million in debt and \$90 million in equity funding, giving a D/E ratio of 1:9. The cost of debt is $R_d = 5\%$, the cost of equity is $R_e = 9\%$ ², which results in a WACC of 8.60%.

Next, we will assume that the capital structure of Lev Co. is changed, with an increase in the percentage of debt financing and a reduction in the percentage of equity funding. The new D/E ratio of the firm is 1:3, so the percentage of debt funding is 25% (\$25 million of the total market value of \$100) and the percentage of equity funding falls to 75% (\$75 million of the total market value of \$100 million). Additionally, the cost of debt, R_d , stays at 5%, but the cost of equity, R_e , rises to 9.5% as the shareholders of the company now face higher risk due to the increase in the use of debt funding by the firm, so demand a higher rate of return. The effect of these changes in the capital structure of Lev Co. is to reduce the WACC to 8.40%, from 8.60%. This is because Lev Co. has substituted cheaper debt for equity in its capital structure and the cost of this debt has not risen, even though the cost of equity for the firm has risen (to 9.50%).

The assumption above, that the cost of debt has not risen as Lev Co. has increased its D/E ratio to 1:3 is very important for the traditional theory of capital structure. As a result of this assumption, as the percentage of debt in the capital structure of the firm increases, it allows the WACC of the firm to be lower.

We will again assume that the capital structure of Lev Co. is changed, with a further increase in the percentage of debt financing and reduction in the percentage of equity funding. The D/E ratio of the firm now rises to 1:1, the cost of debt, R_d , stays constant at 5%, but that the cost of equity, R_e , rises to 10.0% as the shareholders of the company now face even higher risk and demanding an even higher rate of return. The effect of these changes in the capital structure of Lev Co. is to reduce the WACC to 7.50%, from 8.40%. This, also, is because Lev Co. has substituted cheaper debt for equity in its capital structure and the cost of this debt has not risen, even though the cost of equity for the firm has, again, risen to 10.0%.

At this stage we have assumed that Lev Co. has increased its D/E ratio and the percentage of debt in its capital structure twice and that the cost of debt has not risen, while the cost of equity has increased. More importantly, WACC has fallen on both occasions, thus facilitating an increase in the market value of Lev Co., holding all other factors constant. e.g. assume that the quality of Lev Co.'s real assets and net cash-inflows has not changed.

We will now assume that Lev Co. increases its D/E ratio again, this time to 9:1. However, rather than assuming the cost of debt, R_d , stays constant, at 5%, we now assume that the cost of debt starts to rise, going up to 7%. Concurrently, the cost of equity, R_e , rises (this time to 14%), as the shareholders of the company face even higher risk due to the further increase in the use of debt funding by the firm, so demanding an even higher rate of return. The effect of the above changes in the capital structure of Lev Co. is to now increase the firm's WACC to 7.70%, from 7.50%. The key question to ask now is why has Lev Co.'s WACC increased, even though it has further increased the substitution of cheaper debt for equity in its capital structure.

The reason why Lev Co.'s WACC has increased as it increased its D/E ratio to 9:1 is that as a firm takes on more debt, eventually the debt-holders start to become worried that the company

²This is consistent with finance theory as the cost of debt for a firm should be lower than the cost of equity

has too much debt and is therefore becoming riskier. Concurrently, the equity-holders of the company will increase their required rate of return as the company takes on more debt, thus increasing the cost of equity. Once the level of debt in a company goes above a certain level the lenders will start charging a higher interest rate on the money they lend to the company. Then if the company takes on more debt and uses less equity funding, and even though debt funding is cheaper than equity financing, the WACC of the firm will start to rise. ■

In terms of the traditional theory of capital structure, the key is to find the D/E ratio of the firm that is just right, so that the cost of debt has not yet started to rise, thus maximising the market value of the firm. In the example of Lev Co., the best D/E ratio is 1:1 (50% debt financing and 50% equity financing), although, in the real world this ratio of 1:1 may not necessarily be the optimal capital structure.

10.5 Modigliani and Miller's Approach

Modigliani and Miller (Franco Modigliani and Merton Miller; M & M) were two university finance professors in the 1950s who examined the relationship between firm value and financing choice, and questioned the traditional view of capital structure. M&M's analysis was based on perfect market assumptions and they developed the capital-structure irrelevance proposition. Essentially, the proposition states that it does not matter what capital structure a company uses to finance its operations. Additionally, the market value of a firm is determined by its earning power and the quality and risk of its real assets. Therefore, the firm's value is independent of the way it is financed.

M&M undertook empirical studies of US electric utilities and oil companies. The evidence they found supported the conclusion that there is no relationship between a firm's D/E ratio (capital structure) and the cost of capital in these industries.

Highlight 10.5 — M&M perfect market assumptions. In their analysis M&M had five perfect market assumptions, which were as follows:

1. a perfectly competitive market in which all investors have perfect knowledge and act rationally;
2. investors are perfectly certain about the future profitability of any company;
3. all companies can be divided into homogeneous (of the same kind or equivalent) risk classes;
4. no personal or company tax; and
5. individuals and companies can raise unlimited debt funds at the same rate of interest.

Obviously, in the real world all or most of these assumptions do not hold, but initially in their analysis M&M assumed these perfect market assumptions.

From their analysis M&M came up with three propositions related to the irrelevance of capital structure, these propositions being as follows:

Proposition 1: In perfect capital markets the value of the firm is independent of its capital structure, i.e. capital structure is irrelevant. Thus, $V_u = V_L$, which means the value of an unlevered firm (V_u) will be the same as the value of an equivalent levered firm (V_L).

Proposition 2: Required rate of return on equity, R_e , varies linearly and positively with changes in financial leverage and offsets the effect on r (WACC) of the lower cost of debt.

Proposition 3: The appropriate discount rate for a particular investment proposal is completely independent of how the investment is financed, but depends on the features of the investment proposal, especially the risk of the cash-flows. Thus, the financing decision is irrelevant from the

point-of-view of maximising shareholders' wealth.

10.5.1 M&M Proposition 1

M&M's Proposition 1 is that, in perfect capital markets, the value of the firm is independent of its capital structure. From the point of view of maximising the market value of the firm capital structure is irrelevant, meaning that the value of an unlevered firm will be the same as the value of an equivalent levered firm ($V_u = V_L$). In effect, this means that WACC does not vary with changes in the D/E ratio of a firm, as long as individuals and corporations can borrow at the same rate of interest. Also it means that a firm's market value is determined by the quality and risk associated with its real assets, not by the securities it issues, i.e. not by its financial assets or capital structure.

10.5.2 M&M Proposition 2

M&M's Proposition 2 states that the required rate of return on equity, R_e , for a firm varies linearly and positively with changes in financial leverage and offsets the effect on WACC of the lower cost of debt. Proposition 2 shows how the required rate of return on equity in a levered firm changes as the degree of leverage changes.

For a levered firm the required rate of return for equity-holders can be shown to be equal to the required rate of return for unlevered equity, plus a *financial risk premium* that is a function of the levered firm's D/E ratio. We will assume that

$$NOI_u = NOI_L,$$

where NOI_u denotes the NOI of the unlevered firm and NOI_L denotes the NOI of the equivalent levered firm. Further, from M&M's Proposition 1 we know that $V_u = V_L$. Additionally,

$$V_u = \frac{NOI_u}{R_e}.$$

In other words, the market value of an unlevered firm will be equal to its NOI divided by its overall required rate of return. Also, for the unlevered firm

$$WACC = R_e = R_{eu}\left(\frac{E}{V}\right),$$

where R_{eu} is the initial required rate of return on equity in the unlevered firm, and will be equal to R_e . E is equity, which will equal 100% of the value of the firm as the firm has no debt. For the levered firm

$$V_L = \frac{NOI_L}{R},$$

which means the market value of a levered firm will be equal to its NOI divided by its overall required rate of return and R is equal to $WACC = R_d(D/V) + R_e^*(E/V)$. Since we know from M&M that $V_U = V_L$, we require

$$R_e = WACC = R_d\left(\frac{D}{V}\right) + R_e^*\left(\frac{E}{V}\right).$$

Additionally, in order to prove M&M's Proposition 2, we solve for R_e^* , the required rate of return on equity in a levered firm, as follows.

$$R_e^* = R_e - (R_e - R_d) \times \frac{D}{E}.$$

This proves that, as the degree of leverage in a firm increases, the required rate of return on equity in the firm increases. This increase will be exactly in line with the increase in the available rate of return that occurs through the use of cheaper debt finance and, thus, negating any effects on R . Therefore, WACC remains unchanged despite any change in the firm's capital structure.

For capital structure to be irrelevant to the market value of a firm the return equity-holders in a levered firm require on their investment must increase in line with the return available to them. The return available to the shareholders of a levered firm increases as 'cheaper' debt is substituted for equity in the capital structure, but at the same time the return required by the shareholders in the levered firm increases in response to the increased risk associated with their investment, thereby, exactly offsetting the effect on the overall cost of capital (WACC) of the debt finance, with no resulting change in WACC and no change in the market value of the firm.

■ **Example 10.2 — M&M's proposition 2.** In this example we will assume that we have a levered firm, the capital structure of which is comprised of 40% equity and 60% debt. We will also assume that the cost of debt, R_d , for this firm is 8% p.a. and that the cost of equity, R_e , for an equivalent unlevered firm is 8.8% p.a. To find the cost of equity, R_e^* , for the levered firm we use the formula $R_e^* = R_e + (R_e - R_d)D/E$, which becomes

$$\begin{aligned} R_e^* &= 8.8\% + (8.8\% - 8.00\%) \frac{0.60}{0.40} \\ &= 8.8\% + (0.8\%)1.5 = 10\%. \end{aligned}$$

Then,

$$\begin{aligned} WACC_L &= (10\% \times \frac{40}{100}) + (8\% \times \frac{60}{100}) \\ &= 4\% + 4.8\% = 8.8\%. \end{aligned}$$

Now, we can see that the WACC of the levered firm of 8.8% is exactly equal to the required rate of return on equity in the unlevered firm, R_e , of 8.8%. This is also the WACC of the unlevered firm, since this firm has no debt and its WACC is equal to $WACC = R_e = R_{eu}(E/V)$. Therefore, despite the fact that the levered firm has a different capital structure to the equivalent unlevered firm and that the levered firm has partly used cheaper debt finance to fund its operations, the WACC of the levered firm is still exactly the same as the WACC of the equivalent unlevered firm, so capital structure has proved to be irrelevant.

Further, assume that the capital structure of the levered firm changes so that the percentage of debt in its capital structure increases to 70%. M&M argue, in their Proposition 2, that this increased use of debt will increase the risk faced by the shareholders of the levered firm and, so, their required rate of return will increase. Thus, assuming that the capital structure of levered firm is now comprised of 30% equity and 70% debt, R_e^* for the levered firm will be as follow.

$$R_e^* = 8.8\% + (8.8\% - 8\%) \frac{70}{30} = 10.67\%$$

So, after the capital structure change

$$\begin{aligned} WACC &= (10.67\% \times 30/100) + (8\% \times 70/100) \\ &= 3.2\% + 5.6\% = 8.8\%, \end{aligned}$$

which shows no change to the WACC of the levered firm. This has come about because, while on its own increased use of cheaper debt financing would lower the WACC of the firm, as the

firm has taken on more debt shareholders of the firm, faced with greater risk, have increased their required rate of return from 10% to 10.67%. This has offset the benefit of using cheaper debt, thus leaving the WACC of the levered firm unchanged and still exactly the same as the WACC of the equivalent unlevered firm.

■

10.5.3 Debt and Risk

Generally, there are two types of risk that a business faces. The first of these is *business risk*, which is the possibility that a company will have lower than anticipated profits. Business risk is influenced by numerous factors, including sales volume; per-unit price; input costs; competition; overall economic climate; technology; and government regulations. Put simply, business risk is the risk faced by a firm due to the nature of the products sold by the company. For example, risk posed by new competition or technological improvements. All businesses face business risk.

The second main type of risk faced by companies is *financial risk*, which is the risk of bankruptcy a company faces due to using debt in its capital structure. Financial risk is directly and positively related to the amount of debt in the capital structure of a firm. It is only businesses that use debt financing that face financial risk, and all-equity financed firms do not face financial risk.

As we know, because debt is an additional source of risk, shareholders of levered firms increase their expected return as the firm uses more debt. In other words, the composition of shareholders' required rate of return in a levered company is

$$R_e^* = R_f + BRP + FRP,$$

where

R_e^* is the required rate of return of equity investors of a levered firm,

R_f is the risk-free rate of return, which reflects the time-value of money,

BRP is a business risk premium, which reflects the risk inherent in any company's operations,

FRP is a financial risk premium, which represents compensation for the additional risk introduced through the use of debt financing by a firm.

Referring back to M&M's theory of capital structure irrelevance, their Proposition 2, that the required rate of return on equity, R_e^* , for a firm varies linearly (and positively) with changes in financial leverage (and offsets the effect on WACC of the lower cost of debt), means that the expected return to equity-holders in a levered firm is equal to the required rate of return for unlevered equity plus a financial risk premium that is a function of the debt-to-equity ratio, which we saw previously as the formula used to find the cost of equity, R_e^* , for a levered as:

$$R_e^* = \underbrace{R_e}_{\text{Business risk premium}} + \overbrace{(R_e - R_d) \frac{D}{E}}^{\text{Financial risk premium}},$$

where

R_e^* is the rate of return required by shareholders in a levered firm,

R_e is the rate of return required by shareholders in an unlevered firm,

R_d is the rate of return required by debt-holders (in a levered firm).

The conclusion that we reach here, in regards to M&M's Proposition 2, is that, assuming perfect markets, increasing or decreasing leverage does not affect a firm's cost of capital. The cost of debt (R_d) is an explicit cost included in the cost of capital, while the financial risk created by

leverage is an implicit cost of debt, which increases the cost of capital by increasing the required rate of return of equity-holders (R_e^*).

10.5.4 M&M Proposition 3

Given that we now understand M&M's Propositions 1 and 2, this leads to an understanding of their Proposition 3. It states that the appropriate discount rate for a particular investment proposal is completely independent of how the investment is financed. Hence, a firm's financing decision is irrelevant from the point-of-view of maximising shareholder wealth. The appropriate discount rate to be used in project evaluation depends on the features of the investment proposal, especially the risk, and does not depend on how the investment is financed.

10.6 Introducing Market Imperfections

In their seminal study on the effects of leverage on the market value of firms and capital structure irrelevance, M&M then went on to relax the assumption of perfect markets and introduced markets imperfections, specifically, taxation. Given that for a firm interest expense is tax-deductible expense, once they had introduced market imperfections into their model, M&M concluded that for a company subject to taxation, borrowing provides a company with tax savings. Therefore, the value of a levered firm will be greater than the value of an equivalent unlevered firm by the present value of the tax saving on the levered firm's interest expense, i.e. $V_L > V_U$. More formally,

$$V_L = V_U + \text{the present value of the tax savings on interest expense}$$

$$V_L = V_U + (D)(T_c),$$

where

V_L is the value of the levered firm,

V_u is the value of the unlevered firm,

D is the total debt of the levered firm, and

T_c is the corporate tax rate.

10.7 All Debt Capital Structures

From M&M we learned that, once market imperfections are introduced, specifically taxation, levered firms have additional tax deductions associated with interest payments. Therefore, the value of a levered firm will be greater than the value of an equivalent unlevered firm by the present value of those tax benefits. This implies that the optimal capital structure is obtained when a company is all debt.

However, in the real world all debt capital structures are rare, as there are factors that offset the tax advantages of debt finance when borrowing becomes too high, namely, 1. Personal Income Taxes, 2. Financial Distress Costs, and 3. Conflict of Interest Costs.

In regards to personal income taxes, the impact of the introduction of these in relation to capital structure is beyond the scope of the Business Finance course and will not be discussed any further.

10.7.1 Financial Distress Costs

Financial distress refers to a situation where a firm cannot meet its creditor commitments and is

Company	Percentage of Debt
BHP-Billiton	40.83
Rio Tinto	45.18
QANTAS	226.20
Woolworths	43.84
Wesfarmers	19.49
Westfield	75.19
Myer	47.62
Telstra	117.52

Table 10.4: Debt As Percentage of Firm Value - Selected Australian Firms

in danger of bankruptcy. With the same token, *financial distress costs* are the costs of managerial time devoted to avert financial failure and bankruptcy, as well as fees paid to insolvency specialists and lawyers. More specifically, financial distress costs can be broken up into direct financial distress costs and indirect financial distress costs. *Direct financial distress costs* are those costs directly associated with actual bankruptcy. For example, the legal, accounting, and administrative expenses incurred when winding-up a company. *Indirect financial distress costs*, on the other hand, are those costs associated with attempting to avoid bankruptcy. For example, restructuring costs and redundancy costs that are incurred in order to attempt to keep a company running. In regards to financial distress costs, the *Static Theory of Capital Structure* states that a firm should only borrow up to the point where the tax benefit from an extra dollar in debt is exactly equal to the cost that comes from the increased probability of financial distress. If a firm goes beyond this point and continues to take on more debt, then the costs of financial distress will become greater than any tax benefit that the firm receives on the interest expense on the debt, resulting in the market value of the firm decreasing.

10.7.2 Conflict of Interest Costs

Debt can bring on *conflict of interest costs* for a firm, where these costs form a component of financial distress costs and which arise because debt-holders may fear management of a firm will transfer wealth from the debt-holders to the shareholders of the company by undertaking actions that benefit the shareholders, but put at risk the repayment of the debts of the firm. Therefore, in such a situation, the debt-holders of the company will need to protect themselves from erosion of their wealth. The way that they do this is to increase R_d as the level of debt in the firm rises, or by imposing covenants or restrictions on what activities, actions, and investments the firm can and cannot do. These represent conflict of interest costs for the levered firm, which generally increase as a firm's D/E ratio increases.

10.8 Capital Structures of Australian Firms

Companies in Australia have an average debt-to-equity ratio of 0.835³, but D/E ratios are significantly different between firms. The Financial Services Council (FSC) reports that for large firms in Australia, the average debt-to-equity ratio is 1.417. For medium and small firms this is 0.927 and 0.459, respectively, reflecting the greater access to external debt financing available to

³Report For The Financial Services Council (FSC) - Capital Market Structure Comparisons, p. 15, Dr Andrew Lepone and Dr Danika Wright, Capital Markets Consulting, 19/03/2014.

large companies.

For selected individual companies, debt as a percentage of firm values are listed in Table 10.4. We notice that the figures range from a very high 226.20% for QANTAS (a highly-leveraged firm) down to 19.49% for Wesfarmers, owners of Coles Supermarkets and the Bunnings hardware store chain. For the biggest company listed on the Australian Stock Exchange, the mining and resources company BHP-Billiton, the percentage of debt in its capital structure is 40.83%, while for Rio Tinto, the figure is slightly higher at 45.15%.

Business Snapshot 10.1 — Capital Structure - QANTAS Group. Here we have a consolidated balance sheet for the QANTAS Airline Group as at 30 June 2014, which lists the real assets of QANTAS as Current Assets and Non-current Assets, where the Liabilities (Debt) and Proprietorship (Equity), together, reflect the capital structure (or debt-to-equity ratio) of QANTAS and its financing decisions. We can see from the balance sheet that, as at 30 June 2014, the total real assets of QANTAS are \$17.318 billion, while the funding that QANTAS has used to purchase these real assets is \$14.452 billion in debt (liability) financing and \$2.866 billion in equity (shareholder) financing, giving QANTAS a *D/E* ratio of 5.04:1.

QANTAS Group Consolidated Balance Sheet, 30 June 2014			
Current Assets	\$m	Current Liabilities	\$m
Cash and cash equivalents	3,001	Payables	1,851
Receivables	1,196	Revenue received in advance	3,406
Other financial assets	172	Interest-bearing liabilities	1,210
Inventories	317	Other financial liabilities	182
Assets classified as held for sale	134	Provisions	876
Other	112		
Total current assets	4,932	Total current liabilities	7,525
Non-Current Assets		Non-Current Liabilities	
Receivables	158	Revenue received in advance	1,183
Other financial assets	34	Interest-bearing liabilities	5,273
Investments accounted for under the equity method	143	Other financial liabilities	66
Property, plant and equipment	10,500	Provisions	405
Intangible assets	741		
Deferred tax assets	548		
Other	262		
Total non-current assets	12,386	Total non-current liabilities	6,927
Total assets	17,318	Total liabilities	14,452
		Equity	
		Issued capital	4,630
		Treasury shares	-16
		Reserves	-81
		Retained earnings	-1,671
		Equity attributable to the members of Qantas	2,862
		Non-controlling interests	454
		Total equity	2,866

10.9 Revision Problems

Problem 10.1 Booster Ltd has a net operating income of \$30,000 p.a. If the required rate of return of equity holders is 10%, and the company tax rate is 34%, what is the value of Booster Ltd?

- (a) \$340,000
- (b) \$300,000
- (c) \$198,000
- (d) \$102,000

Problem 10.2 The increase in risk associated with the leverage of a firm is called:

- (a) systematic risk
- (b) business risk
- (c) financial risk
- (d) none of the above

Problem 10.3 Modigliani and Miller (1958) proposed that in a perfect capital market:

- (a) the firm's weighted average cost of capital does increase or decrease as the firm's capital structure changes
- (b) the required rate of return on equity of a levered firm is lower than the required rate of return on equity of an unlevered firm
- (c) the value of a levered firm is lower than that of an unlevered firm
- (d) none of the above

Problem 10.4 Thelma Durandt is about to start an earth-levelling operation for which she requires capital of about \$1m to buy the equipment and operate the company. Earnings before tax and interest are expected to amount to \$200 000 (tax rate is 30%). The funds can be raised by issuing shares for the full amount or, alternatively, half the requirements may be raised through the issue of 15% debentures. Calculate the return on equity under both alternatives. Which factors should be considered in deciding on which financing alternative to adopt?

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